Strength Training and Coordination: An Integrative Approach

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Introduction

Background

Training effects are the result of highly complex processes – so complex that there is now a great need for simplified models in sport-specific training and sports injury rehabilitation. Simplified models have systematically classified the many underlying aspects in a number of separate basic components of the training process. The underlying mechanisms can be evidenced by research and serve as a basis for appropriate practical application. This makes the various components more 'workable'.

The division of training processes into distinct components has led to the emergence of specialists in a number of areas: technique (technical coaches), mental processes (sports psychologists), speed (sprinting coaches), endurance (fitness or conditioning coaches), strength (strength coaches), recovery (recovery trainers and sports physical therapists), and so on. Such specialization is more marked in some sports cultures than in others, and perhaps most of all in the American sports culture, where for example the profession of 'athletic trainer' exists – somewhere between a physical therapist and a fitness coaches. In America, professionalism has become synonymous with the presence of specialists.

Yet there is an opposite tendency in the world of sports preparation: the integrative holistic approach, which sees the whole as more than the sum of the parts. Advocates of this approach are fond of the term 'functionality'. Here the quality of training processes lies in how the various aspects of the training processes affect one another. A highly systematic way of working is considered scarcely possible, and the structure of training theory seems to be constantly collapsing under its own weight and returning to the amorphous clay from which meaningful structures were so hopefully built. In this sense, training theory is a building that is constantly in need of extensive renovation. The vague terminology generated by such a holistic and essentially frustrating perspective is, to the say the least, unsatisfactory.

So training theory is far from complete, and must from time to time be rebuilt from scratch as an exercise in disciplined thinking. It is occasionally useful to review the primary basic elements of thinking about training theory: basic motor properties. Distinctions are made in strength, speed, agility, stamina and coordination. How justified are these distinctions – in other words, to what extent are they grounded in reality? Is there little or even no reason for strength training to take account of other basic elements, since these are more or less independently functioning quantities that each have their own separate significance in training theory?

This book sets out to demonstrate that this is not so, and that the various basic motor properties can hardly exist in isolation. Strength and coordination are thus closely related, and should in fact be treated as a single unit. Strength and coordination are basically one and the same thing. This notion is a fascinating one, for it implies that various areas of research are brought together in a single systematic approach to strength training. Knowledge about improving coordination (motor control and motor learning) must be applied to strength training. Knowledge of motor learning processes has so far had little or no impact on strength training. As a result, most literature about strength training is highly mechanical in its approach, and Isaac Newton seems to have contributed more to strength training theory than all the neurophysiologists in history.

This book can be seen, if you will, as an attempt to take the clay of training theory and create a new structure that is more useful in actual practice than previous structures. Instead of approaching strength training in terms of its mechanical manifestation, an attempt is made to produce a model geared to what is known about the underlying processes, particularly in the field of neurophysiology. In this book, sport-specific strength training means coordination training against resistance. This in itself is an admission that the book is inadequate. Knowledge from research is still too limited to allow a clear, consistent translation from theory to practice. Hypothetical models will always be needed. At the same time, this book does not attempt to integrate knowledge of coordination and exercise physiology, for that would be too complex for a workable approach.

Would all this make Newton turn in his grave? Probably not – he wasn't keen on sport.

The route

Chapter 1 describes the organization of complex biological systems. These are characterized by perhaps surprising mutual influence between components such as decentralized control and phase transitions. A 'classic' reductionist approach to systems ignores such influence. Standard training theory is assessed in terms of the special structure of this complex organization. The conclusion is that much of this basic theory, such as the concept of strength as a distinct entity, is inadequate because it is based on reductionist and hence oversimplified models of thinking. This has a major impact on, for example, the design of sports injury rehabilitation protocols, which should focus on the relationship between strength and coordination. Examples are given of protocols that take fuller account of how complex systems function.

Chapter 2 looks more closely at the anatomical and neurophysiological links between strength and coordination. It describes how the production of force is determined by a number of anatomical details at muscle–tendinous level, by threshold values in the neuromuscular transition, by exciting and inhibiting circuits at spinal cord level and by central nervous system influences. Production of force turns out to be regulated at various levels at once, and the central nervous-system's contribution can be described as a coordinative influence.

Chapter 3 analyses the structure of complex contextual movements, such as the movement during sporting competition. Using dynamic systems theory, the distinction between open and closed skills is specified in terms of the interplay of attractor and fluctuator components of the movement. This division between attractors and fluctuators is the basis for the relationship between strength exercises and athletic movement, and is essential for the design of strength training systems based on coordination.

Chapter 4 starts by considering whether strength training should follow a physiological rather than a coordinative track. An approach specifically based on physiological adaptation only makes sense in endurance sports in which coordination is less important. We then look at how non-linear control of overall contextual movements in which coordination plays a key role can limit production of force.

Since force has a strongly coordinative component and motor control limits production of force, the laws of motor learning are important within strength training. Both motor control and motor learning processes are highly intention-based, and principles of intention-based learning in strength training are described. The importance of variation in the learning process is emphasized, and types of variable learning are described.

Chapter 5 looks at specificity and transfer. Specificity between different types of exercises is a precondition for transfer, since motor control develops through underlying matrices and exercises must conform to the structure of the matrix. The matrix is fine-meshed, and general categories of strength exercises such as maximal strength and generation of power are not subtle enough to cope with this. Strength training must therefore help to improve in performance through carefully described specificity. Six categories of specificity are identified, and their characteristics are described. To guarantee specificity, the design of strength training must meet many conditions – to ensure not only that strength training has a positive impact on athletic performance, but also that it does not have a negative impact.

One example, the function of hamstrings during running, is used to analyse how the specificity requirement is applied in strength training, and a rehabilitation protocol is drawn up on the basis of that analysis. The theory of attractors and fluctuators plays a key role here.

Chapter 6 discusses the counterpart of specificity: overload. Overload and specificity are opposites, and this is reflected in the central/peripheral model. The term 'overload' is highly quantitative, which is not how the learning system responds to training stimuli. The substitute term 'variation' implies a qualitative assessment that is more in keeping with the principles of motor learning. Using a qualitative assessment means that overload is no longer automatically equated with physical load. This means that heavy strength training is not necessarily the same as good training.

To provide meaningful variation in strength training, use is made of the constraints-led approach, involving variation in the task, the environment and the organism. In particular, variation in the organism by targeted use of fatigue is a new and relatively unused concept that may have major potential.

Chapter 7 translates all this into practice. The strength training system is based as much as possible on the contextual coordinative adaptations that will occur. This means abandoning the standard division of types of strength. Categories such as strength endurance and explosive power are now dismissed as one-dimensional. Of the remaining categories, reactive strength is not customary, but essential for all sports in which movements must be performed under time pressure.

Finally, the specificity system is applied to the theoretical concept of attractors and fluctuators. A systematic approach to relevant sport-specific strength exercises at the intramuscular level, at the level of elementary intermuscular cooperation and at the level of larger contextual movement patterns is discussed. An example is used to demonstrate how this can be translated into exercises.

The resulting book gives coaches and physical therapists (sports physiotherapists and others) tips for designing a coherent approach based on the laws of coordination. However, that does not mean that coaches and physical therapists no longer need to be creative —this is certainly needed when making the transfer from theory to the design of a tailor-made individual training plan.

Additional knowledge

Although the book is largely the result of thinking models, I have attempted to link up its content with what has been identified by researchers. In doing so, I have frequently made use of types of research that are seldom used when analysing how strength training works. Prior knowledge of these theories makes it easier to understand the text, and here and there it may be useful or even essential to consult other sources. In the context of this book, however, translating the information into practical situations is more important than a full mastery of the underlying theories.

Indeed, this translation to practice is the real challenge for scientific theorizing. However innovative and interesting some of these theories may be, the translation to practice is often rather disappointing, whereas there are major implications for practice —translation of theory may result in a substantially different approach to, say, training interventions and rehabilitation.

Chapter 1 describes the implications of dynamic systems theory. This theory is based on —or rather confirmed by —the work of Nikolai Bernstein. The word 'confirmed' is more appropriate here, for dynamic patterns theory had already developed to some extent in the Western world by the time Bernstein's work became known after the fall of the Berlin Wall. His manuscripts focused on the problem of degrees of freedom, including the role of variability, when identifying the structure of motor control. This is entirely in keeping with theorizing on complex biological and other systems, chaos theory and so on, which were being studied by researchers such as Kelso. These theories have a strong mathematical foundation. However, it is not necessary to know the underlying maths in order to understand this book. Understanding a number of exemplary elements of decentralized control, such as the meaning of the attractor-fluctuator landscape, prereflexes and phase transitions is sufficient in order to grasp the implications of these theories for functioning in practice. Readers who want to find out more about the relationship between motor skills and dynamic systems could look more closely at such terms as variability in movement and synergies in movement.

Chapter 3, for instance, provides a practical translation of dynamic patterns theory into analysis of open and closed skills. This analysis is the basis for distinguishing between the incidental and the generic in sporting movements, which in turn is a crucial starting point for positioning sport-specific strength training.

Chapter 2 looks at standard neurophysiology, which can easily be found (for purposes of further study) in numerous textbooks. This is also true (but less so) of knowledge about central pattern generators, about which relatively little is yet known, and knowledge of central governor (and related) theories, which are also in a relatively early stage of development.

Chapter 4 starts by elaborating on the practical implications of the dynamic systems theory discussed in Chapter 1. It then looks at theories on motor control and motor learning. A key basis is the intention-action model, which unfortunately can only found fragmentarily in texts on neurophysiology and motor control. Readers who want to find out more about it have no choice but to plough through the available literature. However, if this somewhat abstract model is translated into motor skills, we find a considerable body of literature on the role of attention in movement, with Gabrielle Wulf as the unmistakable champion of insights into internal and external attention. More detailed specification of the role of intention and attention in theories of feedback again yields a considerable body of literature that provides more in-depth information. However, this body of literature is still far from complete, as witnessed by the speculative additional reasoning in Chapter 4 in the direction of intrinsic feedback, which is result-oriented and for which no terminology is yet available.

To find out more about the role of variation in motor learning, readers are referred in particular to German research. More and more is now known about the role of variation in learning, but insight into the underlying mechanisms again means searching through the literature, partly in neurophysiology (e.g. on the role of chaos in the development of new neural networks) and partly in empirical studies (e.g. on the role of variation in elite athletes).

Chapters 5 and 6 offer alternatives to the standard classifications in training theory. Of course, knowledge of this standard training theory, on which there is substantial literature, makes it easier to read critically and weigh up the various factors – for training theory is not an exact science, but floats somewhere between science and belief.

Chapter 6 examines the constraints-led approach. This theory, founded by Newell, attempts to bring together the existing theories on motor control. After reading the associated literature, we can really only conclude that the theory has greater potential as an aid to translation into practical application than a model for explaining the underlying theoretical principles of motor control. The theory is rather 'linguistic', which may mean it will be unable to fulfil its promises.

The same basically applies to Chapter 7 as to Chapters 5 and 6. It is a translation into practice, which seeks an alternative to what is customary. Additional knowledge of what is customary is therefore extremely useful for a critical study of the potential of practice models.

Frans Bosch, October 2015

1
The basic concepts of strength and speed

1.1 Reductionism versus complex biological systems

In an ideal world, all training theory would be evidence-based. However, since ours is not an ideal world, especially where knowledge of training processes is concerned, we speak of 'training theory' rather than 'training science'. The term 'theory' indicates that training is only partly based on scientific knowledge – much of it based on models, such as models of physiological process, biomechanical models and so on. These models are assumed to be reflections of reality. Where possible they are supported by the available scientific evidence, and they continue reasoning from the point where the evidence leaves off. This reasoning beyond scientific evidence is necessary because training theory must be applicable and usable in practice. Directing training is therefore not just a science but also an art: intuition – fed by experience – always plays an important part in guiding the training process.

Scientific support for training models is rather limited. This is partly because little money is invested in training research. However, it is also because a large number of factors play a part in training and they influence one another in such complex ways that it is extremely hard to analyse what actually goes on during the training process. There are simply too many factors to cover them all in a single research project. A choice must be made about which factors will and will not be studied. Of course, attempts are made wherever possible to include those factors that have a major impact on the training mechanism to be researched. It goes without saying that such choices will always be arbitrary. At worst, they will largely determine the results of the research. There is therefore a great need for insight into which factors are critical in a given training setting, and which are not. This calls for a sound knowledge of training practice. With help from experienced coaches, who usually have a better sense of what is actually going on, researchers can gain a somewhat better idea of the mechanisms that play a key part in the reality of training. This will enable research to move closer to reality. In short, in order to achieve deep insights, research requires not only facts, but also thinking models based on practical experience that can provide a framework for gathering more evidence.

1.1.1 *The reductionist approach*

Apart from the fact that researchers' choices about the aspects of training to be measured cannot be based entirely on facts, but also partly on thinking models, researchers who may want to interpret their measurements in the light of statistics must make a second important assumption – namely that, after studying parts of the (biological) system, valid statements can be made about how the entire system will actually behave. This assumption is based on the idea that reality is shaped by abstract underlying principles that can be captured in physical and mathematical formulas. In scientific and other philosophy this is often labelled 'reductionism' or 'anti-inductionism' (Anderson, 2001; Ladyman, 2002). The basis for a reductionist approach to training process mechanisms research is as follows: the impact that numerous factors have on the training process can be captured fairly accurately in such physical and mathematical principles, and hence is constant. According to this way of thinking, major factors have a major impact, and minor factors do not. Omitting the factors that have little, or only occasional, impact on the process ('noise') makes the measurement manageable (Figure 1.1). These manageable measurement results are then used in an attempt to make statements about the underlying mechanisms that actually shape the training process. Even if a very large amount of data is gathered during such research, the basic assumption that reality is built up from a number of underlying abstract constants, which moreover remain equally dominant as the system becomes more complex, is still unproven. In particular, the components of standard training theory that are meant to have a predictive function – what the effect of training will be in the (near) future — are based on the assumed stable dominance of underlying principles, and hence depend on whether or not that theory is accurate and complete.

Research has proved barely capable of grasping the predictive aspects of training theory. As yet there is little or no scientific evidence for the mechanisms which according to traditional training theory underpin the changes (adaptations) that result from training. Many physiological variables (parameters) can be measured, but we cannot predict what will happen to those physiological parameters if they are perturbed (training). That is why there is little or no scientific literature on how adaptations take place (for instance physiologically). This is partly due to the limited amount of research that is carried out, and perhaps partly because research generally focuses on the major parameters.

In practice, the way in which training leads to adaptation turns out to differ greatly from individual to individual, and even from time to time within a single individual. Monitoring an individual's training history may reveal a trend in how adaptations occur, but this will seldom yield reliable information about the impact of training. As a result, the aspect of training theory that deals with predicting the impact of training —in training planning and the available periodization models – is vulnerable, for it is based on the assumption that identifying a limited number of factors can provide a meaningful reflection of the reality of the training process. Examples of such periodization models are the models developed by Matveyev, Verkhoshansky, Tschiene, Bompa, Issurin and so on. These models sometimes even indicate in detail how training can best be organized. Although attempts are made to back up these models with as much research data as possible, the practical evidence for their effectiveness and predictive value remains very thin. All that has been proved about the operation of these advanced periodization models is that they work better than periodization with little or no variation in training. So training with variation is better than training without variation; but it is still far from clear to researchers why variation in one model should work better than variation in another. Adaptations as a result of training thus remain incidental and relatively unpredictable events, and hence are the most vulnerable aspect of training theory. Omitting the noise (the minor influences on the impact of training) that occurs in complex systems does not appear to benefit research results (Kiely, 2011).

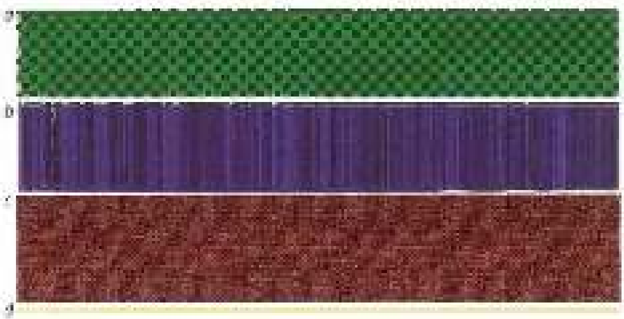


Figure 1.1
The underlying paradigm in reductionist research: major parameters (a, b and c) have a major impact, and minor parameters (d) have a minor impact.

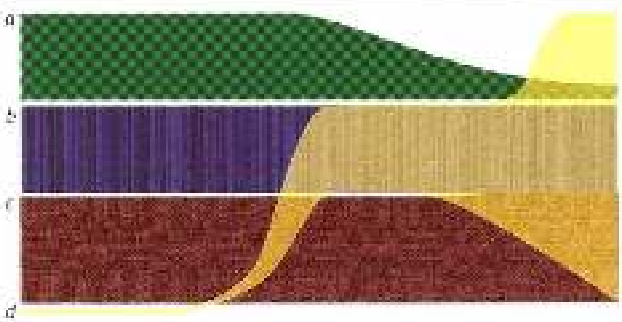


Figure 1.2
The underlying paradigm in complex biological systems: under some circumstances both major and minor parameters (a, b, c and d) may have a major impact, but in other circumstances little or no impact, on what happens in the system.

1.1.2 *Complex biological systems*

The reductionist approach to research has been greatly criticized. The criticism comes not only from practitioners, who indicate that there is a great difference between research and reality in various areas, not only in the theory of training planning; researchers now also strongly criticize reductionist approaches that omit the noise in the system. This criticism is mainly based on the theory of complex biological systems, which is founded in dynamic patterns theory (Kelso, 1995). The term 'dynamic systems' refers to the overall structure of complex systems and its implications for how the system behaves. The term 'dynamic patterns' refers to underlying structures on the basis of which changing behaviour occurs. For clarity's sake, the term 'dynamic systems' will be used in the remainder of this book. This theory posits that the principles of the reductionist approach are only valid for relatively simple systems. In other words, only the behaviour of systems built up from a limited number of parameters can be understood in the light of underlying abstract principles. With highly complex systems, the linear approach and the omission of noise are inadequate. Complex systems, in which there is a great deal of noise and the various aspects of the system interact in complex ways, have different dynamics, so that a reductionist approach no longer has predictive value. Examples of complex systems include the weather, the economy and urban infrastructure. Another is the training athlete. In such complex systems, small and seemingly insignificant factors may, in interaction with other influences, have a major impact on the adaptations that occur. A 'major' physiological system, such as the energy system that we wish to influence, will respond differently in each case to training stress through the impact of other, even much smaller systems such as diet, hormonal changes, sleep, mood, social environment, motivation, ambient temperature, familiarity with the training and so on. These 'minor' influences — noise — may have a crucial impact on the adaptations that occur (Figure 1.2). Owing to the interaction between all these factors, the system may appear to behave chaotically. Whereas the established periodization models may claim a degree of predictability for the training effect, in reality the large number of factors involved will produce a far more uncertain response to training stimuli. The seemingly chaotic nature of adaptations resulting from training stimuli may be awkward for researchers, but it has a key function in the ability of the organism to adapt to the greatly changing demands of the environment, for a biological system must be able to respond flexibly to the changing demands of its environment. A rigid system that adapts in rigid ways will not survive.

According to the theory of complex biological systems, an organism that is essentially controlled by the central nervous system and also works from blueprints (such as a dominant brain) is such a rigid system. Central control will not provide the necessary flexibility. Flexibility requires 'chaos caused by noise', and such noise is decentralized — it occurs throughout the organism. This means that processes in the organism are not directed from a dominant command centre, but are shaped everywhere at once. Decentralized processes are like a flock of starlings in the autumn; the birds seem to fly in organized patterns, but — despite appearances — these are not centrally controlled. Each starling responds to a number of signals around it, and because each starling receives slightly differently signals (noise), we see spectacular changes in the shape of the whole flock. So the physiological response to training does not arise because a single centrally controlled stimulus for adaptation is transmitted, but because more or less independent influences that shape the eventual adaptations occur throughout the organism. This means that noise cannot simply be omitted in research, and hence adaptation processes are non-linear.

Additional Information

'Non-linear behaviour' and 'phase transition' are key terms in the theory of complex systems. A system that behaves in a non-linear manner displays not only gradual transitions from one arrangement of the system to another, but also sudden, abrupt ones. The system jumps, as it were, from one state to another — a phase transition. We are all familiar with such sudden transitions in physics: the sudden transition from liquid to gas, and from water to water vapour. The transition when water freezes is equally abrupt – there is no intermediate form or gradual transition between water and ice. Similarly abrupt transitions are found in physiology and coordination. Sudden changes in patterns of physiological organization can be seen in, for instance, the chaotic fluctuations in hormone (e.g. cortisol) levels and the irregularity of the heartbeat (a healthy heartbeat is an irregular heartbeat).

The transition between walking and running is an example of such a phase transition in coordination. The transition is sudden: there is no intermediate form or gradual change from one type of gait to another. In addition, the difference between walking and running is much greater than may at first be thought. When we walk, the body's centre of mass moves up and down, and is at its highest point when the legs are side by side and at its lowest point when they are far apart. When we run, the opposite is true: the body's centre of gravity is at its highest point when the legs are far apart and at its lowest point when they are side by side. In the transition from walking to running, the phase of up-and-down movement of the body's centre of gravity is suddenly reversed. There is also a sudden change in the way kinetic energy is preserved in the system. When walking, the pendular motion of the free leg ensures that no energy is lost when the centre of gravity rises after the foot placement by pivoting upwards, as it were, against the stance leg. When running, the energy is preserved in the system by storing it during the foot placement in elastic stretch of structures in the movement apparatus (Blazevich, 2005; Ker et al., 1987). The transition from walking to running or vice versa is thus a very drastic change in the movement pattern that occurs from one moment to the next (Figure 1.3).

Figure 1.3 Left: walking – the body's centre of gravity is at its lowest point when the legs are apart. Right: running – the body's centre of gravity is at its lowest point when the legs are side by side. This means that the two types of gait are organized in two essentially different ways.

Such a drastic difference in the organization of a movement pattern can also be analysed in the upper limbs. Thrusting movements, such as a shot put, a punch in boxing, the explosive push-off just before passing the bar in pole-vaulting or the push-off in the vault in gymnastics, are essentially different from throwing movements such as pitching a baseball, a tennis service or a volleyball smash. Furthermore, there are no effective intermediate forms of these two categories of movement, so there is no transition from thrusting to throwing. A thrusting movement is based on a movement of the shoulder towards an internally rotated position. A technically well-performed thrusting movement ends in simultaneous extension of the elbow and twisting of the trunk round its longitudinal axis, so that the shoulder remains in a relatively abducted position. The movement is performed by concentric muscle action. A throwing movement, by contrast, is based on first externally rotating and then internally rotating the shoulder joint, with the muscles (eccentric-concentric) operating elastically and energy being transported from a large mass (the trunk) to a small mass (the hand). Although some of the same muscles (e.g. pectoralis major) are involved in thrusting and in throwing, the (self-)organization of the two movement patterns is so fundamentally different that they are seldom if ever combined in sport: there are no athletes that excel in both javelin-throwing and shot-putting.

Strength training for throwing is likewise essentially different from strength training for thrusting. However, the combination of shot-putting and discus-throwing does occur, for the basic organization (the basic building blocks) of the two movements is similar. Such sudden transitions occur constantly in a complex biological system such as the athlete, not only in large overall movement systems like walking to running or thrusting to throwing, but also in countless small subsystems of movement that contribute to this type of overall movement, e.g. the interaction between systems that regulate the transition from stance to flight phase, and so on. The influence of all these subsystems of movement can easily be observed when we see the change in gait that occurs when we pick up an object while walking and carry it under our arm, or the surface suddenly changes from, say, grass to loose sand.

Phase transitions occur when the organization of the physiological or coordinative system loses its stability. Below a given stability threshold the organization of the system suddenly shifts to a different stable state. If we walk faster and faster, the movement eventually becomes less stable. A minor perturbation may then make the movement pattern too instable, and the movement pattern will then shift to running. Such a transition is not usually deliberate: phase transitions in types of gait also occur in animals.

Phase transitions thus occur at many levels in the biological system, from the smallest subsystems to overall patterns of the organism, and they are essential to the functioning of the system (Figures 1.4 and 1.5). The mechanisms responsible for phase transitions in complex biological systems have two key features:

- There is little relationship between the 'size' of the influencing factor and the size of the phase transition it may cause (minor causes may have a major impact – noise).
- The phases are self-organizing, i.e. they are not directed from a central planning system or command centre.

Such decentralized organization, in which the size of the parameters operating in the system is unrelated to their possible impact, means that the processes in the system are unpredictable. However, it also means that the system is flexible and does not respond in a stereotyped manner to the influences in its environment (e.g. the heavy object under one's arm, or the transition from grass to loose sand), so that it can often achieve carefully planned results.

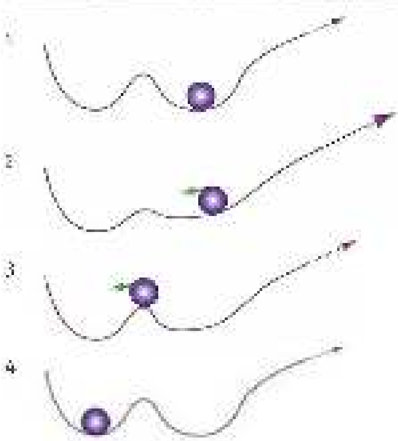


Figure 1.4
Phase transition: An element (the sphere) is in a stable position (1). This stability is perturbed (the purple arrow) (2). The element's position becomes unstable (3), and the element then jumps to a different stable position (4).

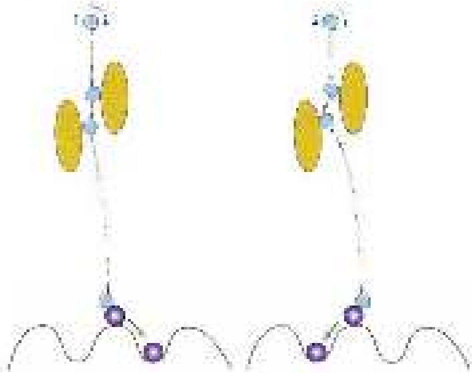


Figure 1.5
Phase transition triggered by a small, seemingly insignificant factor. The small difference between the rotation on the left (a) and the rotation on the right (b) results in different paths after the collision with the two ovals. These different paths result in a difference in the collision with the unstable sphere, which therefore ends up in different stable positions.

Besides being flexible, the organization of the system must be meaningful, i.e. it must generate meaningful problem-solving behaviour. The unusual thing about the theory of complex biological structures is that actions can be intentionally organized as long as the system is sufficiently complex. A single termite cannot build a termite mound, and has no idea how to. Even a hundred termites working together cannot display meaningful overall behaviour. And even if a few million termites work together, not one of them has the slightest idea what the goal is, but changes in each termite's behaviour in response to signals from its environment will produce an ingenious structure. Our brains work in the same way. Not one of our neurons has any idea what this book is about when we read it, but if large numbers of our neurons are sufficiently active we will understand it, and if neural activity is linked to simulations in our own bodies we may recognize the situation and make decisions to do things differently (Damasio 2006).

This unusual property of complex biological systems obviates the need for a command centre for launching actions, and provides a new explanation of how actions take place. The self-organization of countless subsystems dynamically generates behaviour that appears coherent and intentional to the outside world. Of course, such a model of movement behaviour based on self-organization has far-reaching implications for exercise rehabilitation and training, especially if the goal is a result that must also be visible beyond the immediate practice situation.

Not only physiological aspects of training are organized in a complex way. The biomechanical and coordinative aspects also have a complex, non-linear structure. This means that a highly reductionist approach to these aspects will not take proper account of the dynamics that shape movement patterns. Complexity is the logical consequence of the need to design movement patterns that are efficient under a wide range of different environmental influences. Athletes must be able to run on hard, soft, uneven and smooth surfaces, with a heavy object under one arm or on our shoulders, and so on. The same applies when zigzagging our way between defenders to pick up a long pass in soccer. Not only must athletes be able to move appropriately in all these situations, but they must also be able to do it with a minimum of control mechanisms (see also Section 3.2.6). If a different movement control system was required every time the surface changed while running through woodland, running would be physically and mentally very strenuous. In addition, there would often be movement and foot placement misjudgements, creating numerous disadvantages and hazards. The adjustments that the constantly changing environment requires of the moving system cannot be made if a central control mechanism has to keep switching from one control program to another. All these environmental influences must be absorbed into the movement pattern (within limits, of course). This can be done if the rules for control are flexible and multi-purpose, and that is only possible if the rules are shaped simultaneously throughout the movement system.

The control of contextual movements must therefore satisfy two criteria:

- It must be effective, i.e. it must realize the intention of movement.
- It must be as non-incidental as possible, and hence suitable for solving several movement problems.

This means that there are probably no fairly rigid motor programs stored in the brain, but that movement is composed on the basis of flexible sets of movement rules that are generally applicable and can filter and shape incidental adjustment to the demands of the environment. Running on an athletics track, running on an uneven surface, running while carrying an object and even changing direction during a ball sport are thus all variations on a single theme that ultimately takes shape through the interaction of several factors in and around the moving system (Figure 1.6).

Figure 1.6 Moving in changing environments (unpredictable opponents, unstable support surfaces etc.) is only possible if the number of central mechanisms used to realize the intention of the movement is kept to a minimum (see Section [illegible]).

To find control mechanisms that can be used in many different environments, movement must be designed in a plastic rather than linear manner (Van Cranenburgh, 2002). This can be done because movement is designed by constant interaction between all the aspects that influence the movement. We can see this from how the learning process works. Learning a new movement does not involve moving from not mastering a movement to being able to manage it slightly, then mastering it better and better and eventually mastering it completely. Constant interaction between the numerous factors results in a winding path in which temporary skills appear and disappear again, to be replaced by other skills – a seemingly chaotic path that eventually results in a both stable and flexible movement pattern (Figure 1.7). In other words, learning a movement does not mean learning how to perform it in an ideal manner which is fragile and only usable in a single incidental environment, but how to apply numerous variations on a theme in order to create a movement plan that can withstand a variety of environmental perturbations.

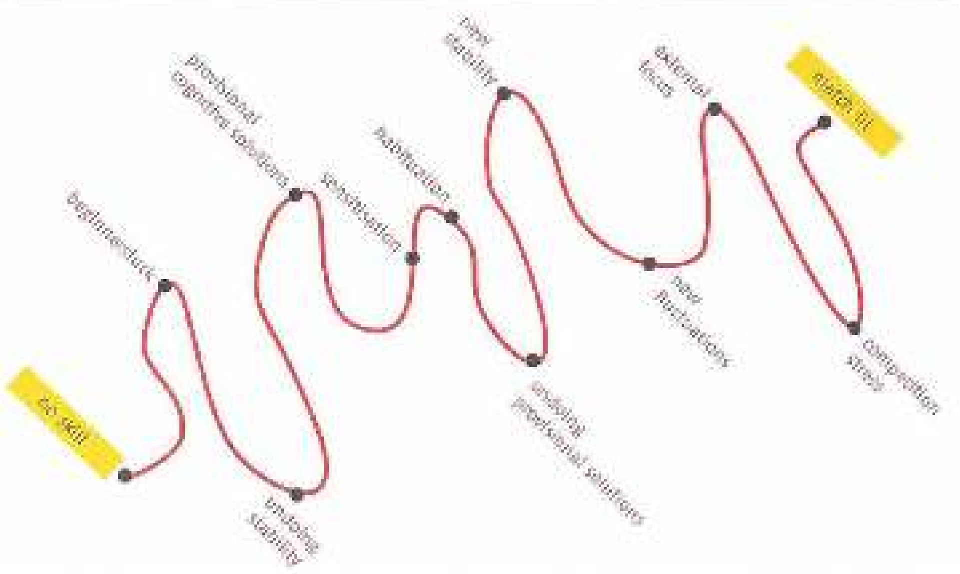


Figure 1.7 The process of learning a motor skill does not follow an ascending line from poor mastery to better and better mastery and eventually optimal performance, but a seemingly illogical winding path (shown here in a fictitious form) in which skill may temporarily decline during the learning process (see also Figure 1.8).

Seen in that light, the precise movement corrections that, for example, physical therapists, golf teachers and coaches in oriental martial arts are so fond of making serve little purpose. These are incidents that the learning system will not recognize as universally applicable and will therefore also dismiss as incidents; in other words, the system will not want to learn much from them. The precisely taught lifting technique will not be remembered, for it is not universally applicable, if only because the objects that are lifted in everyday life all differ in shape and weight. Precise corrections to the position of the pelvis to improve trunk control will have little positive impact in everyday life or in sport, for stability can develop differently (in a self-organizing manner) in different situations. Precisely learned judo techniques will not be of much use in a competition setting if their performance cannot be changed at will – and so on. Stable yet flexible movement patterns do not develop by learning techniques precisely, but through self-organization from complexity.

An example of flexibility

The fact that flexibility is a basic precondition for movement is apparent from how we keep our balance. Research has shown that healthy young people who have no trouble in keeping their balance have greater ankle range of motion than people with impaired balance (e.g. older people). The body needs this shift in foot pressure ('postural sway', Figure 1.8; Davids et al., 2003; Van Emmerik & Van Wegen, 2002) to gather information on how well it can keep its balance (it is more useful to know how much longer it will take to reach the boundary of the support surface than to know exactly where in the support surface the pressure point is at any given movement). Healthy young people can compensate for this faster, broad range of motion in good time by moving their trunk and shoulders, whereas in older people this reaction is delayed and hence less adequate. Older people are therefore less able to keep their balance, because their postural sway is delayed, there is less variation/flexibility in the position of the body and hence less information is gathered about the limits of balance. Variation is crucial – it is not just noise in the system, but an essential aspect of accurate movement.

Figure 1.8 Postural sway: in healthy people, the projection of the body's centre of gravity onto the ground is not immobile, but constantly sways out to the boundaries of the support surface (left). In older people (right) not only do the boundaries of the support surface become more vague, but the range of motion of the sway is reduced.

An example of coordinative complexity

Most scientific biomechanical measurements and studies are based on externally measurable biomechanical data. Reasoning from this external biomechanics, it seems a good idea to focus on the toe-off instant at high-speed running, for a favourable horizontal component can be added to ground reaction force by thrusting firmly posteriorly especially towards toe-off. Elite sprinters have therefore been trained in accordance with this idea of efficient thrust based on external mechanics. Kinematic measurements of many sprinters' running technique at maximal speed then revealed that their stance-leg knees were fully, or almost fully, extended even at toe-off (Yessis, 2000). Today's sprinters extend their knees much less in the second part of the stance phase, for it is now realized that full extension is not efficient. That is because there are many more influences that determine a runner's top speed than just a favourable direction of thrust. If various other factors besides kinematic data are taken into account in studies of the ideal knee angle at toe-off – such as EMG of muscle activity or changing moment arms in muscles – it is already less obvious what the ideal knee angle should be. If even more factors are taken into account, including aspects that research has as yet scarcely identified, such as increasing stability by internal and external rotations in the stance leg (Kizzler et al., 2006), neural influences on movement patterns (e.g. reflexes and influences from central pattern generators and so on; Stuart

& McDonagh, 1998) and the impact of preflexes (see Section 4.5.3), the reality may be that there is no such thing as an ideal knee angle at toe-off. For instance, a very minor change in local fatigue in a muscle group such as the hamstrings or the abdominals may have a relatively major impact on the eventual ideal knee angle at toe-off. Owing to the complexity of the system, it is therefore not possible to draw up rules about ideal joint angles or angle velocities in the joints. Furthermore, the example of a sprinter on an athletics track is a relatively 'simple' setting for a movement pattern as compared with other sports.

A reductionist approach may easily create a notion of technique that ultimately proves not to be the most efficient one. A good description of technique in a sporting movement is therefore not one that prescribes ideal joint angles, but one that describes universally valid underlying principles of the movement and leaves room for variants that develop from self-organization and are related to the individual properties of the body.

1.1.3 *Basic motor properties*

The reductionist paradigm on which traditional training theory is based – major parameters have a major impact – has led to training usually involving division into 'basic motor properties': coordination, agility, stamina, strength and speed. These are seen as the building blocks of the athlete's performance. This division is a typically reductionist approach to what happens in training and in adaptations. It assumes that a clear and at least workable distinction can be made between the various properties. In making this distinction, it is thought that the whole training process can be managed more effectively. If, for example, the 'strength' building block is inadequate, it makes sense to focus on strength training; if the 'speed' building block is inadequate, to focus on speed; and so on.

If we are really to gain better control of the training process by assuming basic motor properties, the latter must satisfy at least two criteria:

- Independent entity: each of the basic motor properties must to some extent be an independent entity. This means that a clear distinction can be made between features that are part of such a property and those that are not. Only if such a distinction can be made, with arguments to support it, can such a property be dealt with separately as an independently operating aspect of the athlete's performance.
- Automatic transfer: there must be more or less automatic transfer of the quality of the basic motor property between various movement patterns (see also Section 5.1). In other words, if a basic motor property is trained using a given exercise type, another type of exercise in which the same property also plays a role will automatically have to change and improve.

If either of these criteria is not satisfied, the concept of basic motor properties becomes less workable, particularly in sports in which performance depends on the complex interplay of numerous factors. In that case there is little point in organizing training according to basic motor properties.

If the 'independent entity' criterion is not satisfied, an approach is unworkable because it is impossible to define sufficiently which performance variables one is attempting to improve in training within that basic motor property. For example, it is then impossible to train stamina as an independent entity (i.e. without other basic motor properties having an impact on the adaptations). This can be seen, for example, in middle distance running. There are coaches that attribute loss of running speed at the end of an 800-metre race to exhaustion of the glycolytic power production, others attribute it to increased energy costs owing to deterioration in technique and hence decreasing efficiency. There is no simple answer to the question of whether the final sprint in the 800 metres can best be approached by training shorter distances ('speed') or longer distances ('endurance'), for training shorter or longer distances inevitably has an impact on running technique and efficiency. These performance-determining factors (speed, stamina and coordination) may be so closely linked that no clear, generally applicable training strategy can be devised on the basis of distinctions between basic motor properties.

If the second criterion is not satisfied, there can be no guarantee that training the particular property is sport-specific. In practice it is clear that there is barely any automatic transfer within, for example, the property 'speed'. An elite javelin thrower may also be able to hit a fast tennis service, but not necessarily a fast backhand. Speed does not easily transfer from an overhand to a backhand technique. A track-cycling sprinter cannot usually sprint well on an athletics track. Sprint training on an athletics track may even have a negative impact on speed-skating sprinters, and so on. Positive, strengthening transfer can only occur if a number of specific criteria that go beyond the limits of the basic motor property have been satisfied (see Chapter 5 on specificity).

In practice, then, neither criterion is satisfied. Nor can an approach based on basic motor properties be justified in terms of the theory of complex biological systems, in which the dynamics of the interaction between the various components of the performance are neither linear nor mechanistic. To put this more simply, so many different aspects of performance —which, moreover, interact in complex ways —are involved in the development of adaptations that we cannot clearly distinguish building blocks of the performance that can be used separately in a training strategy. More specifically, there is not enough difference between, say, strength, speed and coordination to allow a meaningful division of these building blocks. The strength-speed-coordination triangle may well be the most forced (reductionist) structure in training theory.

1.1.4 The basic motor property 'strength'

The amount of force that a person can produce is largely determined by the way in which muscles are controlled from the brain. Even in seemingly simple movements, such control is not automatically optimal, but has to be trained. Sporting movements are usually complex, and performing such a movement with high production of force is a difficult task for the brain. 'Switching on' a muscle to produce strength does not just involve a strong signal from the central nervous system, but the effect of numerous interconnected enhancing and inhibiting neural stimuli, and hence is complex, just as coordination is complex (for a more detailed discussion of this, see Chapter 2).

The training of force production by isolated muscles appears very simple. Sufficient motivation and repetition against high/maximal resistance would seem enough to obtain the intended progression. The complex interplay of excitation and inhibition (e.g. tensioning and relaxation) is simply dictated by 'higher' functions such as goal orientation and motivation. The central nervous system can thus achieve an output that is closer to the optimal output of the muscle. Yet training individual isolated muscles does not contribute to the complexity-based dynamics of contextual movement patterns. The complexity of the moving biological system only develops when muscles have to generate strength in cooperation and so produce a contextual movement pattern. During this intermuscular cooperation, the amount of force produced is no longer the most important feature of good movement – what is crucial is the timing of the production of force. Just as in an orchestra the point is not who plays loudest but how to coordinate the tempo and volume of all the instruments, in contextual movements the point is to make force production by each muscle group a perfect part of the whole. In other words, the more complex and contextual a movement pattern becomes, the less strength can be seen as a separate phenomenon. The more contextual a movement pattern becomes, the more strength and coordination become a single entity.

This can also be seen in everyday practice. Practice may even indicate that the rules of complexity develop much sooner that we are at first inclined to think. If we tend to see strength as a separate phenomenon, we will overlook issues that matter in order to train efficiently. One example is the frequent use of a double-leg squat with barbell weights to improve single-leg contextual movements such as running, jumping on one leg, etc.). It is then simply taken for granted that there will be transfer – but in reality this is not the case. In fact, it is easy to explain why transfer does not automatically occur. There is an important difference in coordination between the two movement patterns, and this may well be controlled quite differently by the central nervous system. A single-leg toe-off involves not only knee and hip extension but also abduction in the hip (so that the free/swing side of the pelvis is elevated during toe-off). This abduction requires a great deal of strength and does not occur in a double-leg toe-off. In practice, well-trained athletes (such as professional rugby players) often turn out to be much stronger in a double-leg squat than, for instance, elite high jumpers, but to perform significantly less well in a single-leg squat or 'step-up'.

If strength training or sports injury rehabilitation are to enhance the quality of athletic movement (if specific transfer is to occur), account must be taken of the complexity of contextual movement patterns, and oversimplification of such patterns in strength exercises (such as isolation of muscles or isolation of range of motion in joints) is unlikely to be of much use.

1.1.5 The basic motor property 'speed'

Coordination also has a substantial part to play in the basic motor property speed. One aspect of speed that is more or less separate from the influence of coordination is the speed at which muscle fibres can contract. As one might expect, rapid muscle fibres can do this more quickly than slow ones. However, the speed of action of muscle fibres only partly determines the eventual speed of contextual movements. If we zoom out from the action of a single muscle fibre to the operation of the entire system, there is a whole series of other factors in a sporting movement pattern, just as in strength, that determine the eventual speed of movement – for example, the way in which muscles cooperate and the way in which their elastic properties can be used. Many of these factors may be summed up as 'timing' (and hence coordination).

Linear translation of speed of action of muscle fibres to contextual speed is reductionist in the extreme. According to this reasoning, speed is mainly determined by the athlete's percentage of fast-twitch (FT) fibres. Coaches and researchers who are convinced of this close relationship often go out of their way to classify athletes' muscle fibres by taking muscle biopsies, perhaps in the hope of predicting talent.

However, this highly reductionist reasoning becomes meaningless if we consider the influence of numerous other aspects when determining how speed develops. There is a paradox in the phenomenon of speed. Great external speed often develops by limiting internal speed. In other words, there is no connection between the average angular velocities that are achieved in the joints and the speed at which the body moves when sprinting. The best 100-metre sprinters do not achieve higher angular velocities than less successful ones (Weyand *et al.*, 2000) —if anything, the opposite. Sprinters who as a result of their inferior technique require a larger range of motion in their joints and hence higher average angular velocities (especially in the hip joint) will therefore achieve a lower horizontal speed of movement.

Indeed, this paradox is far more apparent in other sport disciplines. In speed skating the highest speeds are not achieved by making wild movements, but by being patient and performing the movements efficiently rather than quickly (to quote world champion speed skater Jeremy Wotherspoon, 'I do not try to do things fast – I just try to be patient in getting the pressure on the ice'). The same applies to swimmers (they have to be patient and complete their stroke) and high jumpers (they above all have to stay rigid and immobile at take-off), and so on. Speed is not simply generated by performing the fastest possible (concentric) muscle action, but above all by distributing the movement over as many joints as possible, so that limitation of performance in one joint is postponed as long as possible. Speed is thus a function of coordination. In sport-specific strength training it is therefore essential to make a distinction between speed, strength and coordination as late as possible when constructing thinking models. The longer the interconnections between the various components of the training design are maintained, the better.

1.2 Strength training characteristics required for effective transfer to athletic movement

1.2.1 Sensorimotor function

The focus of sport-specific strength training is transfer to athletic movement. This transfer takes place according to mechanisms that are anchored in the learning system. These mechanisms (see Chapters 5 and 6) ensure coherence between movement patterns. In other words, movement patterns are interrelated, and this provides the matrix for developing new movement patterns. This is necessary in order to make the right decisions when selecting movement patterns, for the choices we make must relate not only to the environment in which we are moving (e.g. how steep and smooth the slope is, and whether or not we can run uphill – should we perhaps crawl on all fours instead?), but also to the body's properties (whether there is enough mobility in our hips and enough strength in our arms to climb up the rocky slope). If movement patterns were not interrelated, we would never be able to gain a clear picture of what our bodies can and cannot do. We would constantly run into trouble, because in the absence of a frame of reference we would be unable to assess whether we could solve a new movement problem, or to predict the results of our decisions to move. Movement would become a reckless undertaking.

We can only properly assess how movements will develop, and whether our bodies can cope with them, if we assess both the motor and the sensory aspects of the movement to be made. Someone who wants to lift and control a weight above his head assesses not only the required motor component but also the sensory component (e.g. the tensile force acting on the tendons and the change in muscle length). This combined sensorimotor package of the whole movement can be linked to the existing arrangement of movement patterns that have been mastered. In the light of experience, we can assess whether or not a movement will be performed successfully, and hence whether or not it should be made. We constantly make such assessments: when jumping off a wall, when dashing across the street to avoid an approaching car and then rapidly decelerating so as not to run into a shop window, when trying to carry a heavy bag up a flight of stairs, and so on. We also constantly adjust these assessments to the changing state of our bodies. For instance, the maximal height from which we can jump off a wall will be assessed differently if we have had a knee injury or are very tired. Such 'calculations' are performed in both feed-forward and inverse models of movement design (Franklin & Wolpert, 2011).

In order to function well in a changing environment, movement patterns must therefore be recognized as a single sensorimotor package: the movement, and the sensory information it releases. This means we perceive movements as related not just in motor terms, but in sensorimotor terms. This is important to know when deciding whether a preparatory exercise helps improve a sporting movement (a sport-specific strength exercise is a preparatory exercise for the sporting movement we want to improve): if a preparatory exercise greatly resembles the sporting movement in terms of the movements made but is very different from it in sensory terms, the moving system will not easily recognize it and there will be little or no transfer. This means that many of the preparatory exercises that at first sight seem useful will be far less relevant for improving the sporting movement than is generally assumed. For example, it is highly questionable whether aquajogging is of any use to an injured runner who is trying to keep making movements related to running despite the injury. Although the range of motion may resemble actual running (although the most important motor feature – elastic muscle use – is absent), the sensory impact (registration of gravity and registration of water resistance) is so different that the body will hardly recognize the similarity, and hence there will only be very limited transfer.

The connections that the body must be able to make between numerous sensorimotor links are thus essential to the functioning (and survival) of the organism. It therefore makes sense to design the learning process so that not only sensorimotor links develop, but also – and above all – connections between the various links. In other words, the underlying rules that allow sensorimotor links to develop must be part and parcel of the learning process. This means that the non-functional elements of a movement pattern guide the learning process; and these are discovered not by exploring the core of the pattern (perfect technique), but by exploring its limits – by understanding what perturbation of the pattern implies. Such perturbation is the 'noise' in the pattern, which is so important in dynamic systems (Figure 1.9). A reductionist approach to movement – whether this means thinking in terms of categories such as strength and speed, or pursuing perfect technique in isolation – does not focus on the limits of sensorimotor links, and so cannot teach us anything about the crucial transfer quality of contextual movement. Nor can it teach us when movements are sufficiently similar for effective transfer to occur, and when they are not. Put more simply, a reductionist approach cannot teach us which types of exercise do or do not help to improve the sporting movement, and hence are or are not effective. To understand the connections (the transfer), we need to grasp the complexity of the biological system and the self-organization that takes place within it.

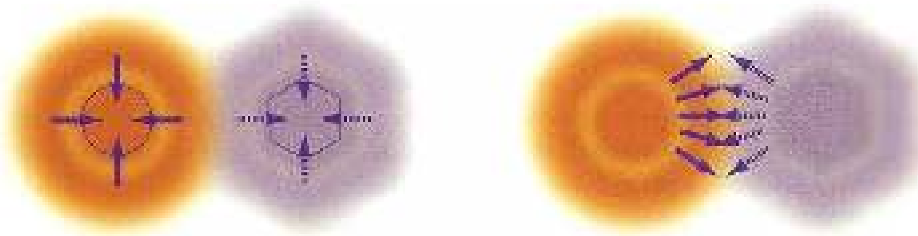


Figure 1.9 Transfer between movements is not made clear by exploring the core of the movement (perfect technique) (left), but by perturbing the movement and exploring its limits (right).

1.2.2 *Whole practice and part practice*

It is useful to choose types of exercises that as far as possible guarantee a combination of sensorimotor components similar to those in the sporting movement. A series of exercises in which the sensorimotor combination of exercises is as similar as possible to those in the sporting movement is known as 'whole practice'. This uses simplified (often highly simplified) versions of the sporting movement, with the overall picture and the intention of the sporting movement left as intact as possible. If the intention and the overall picture remain intact, the sensory aspect probably will too. A 'whole practice' series of exercises therefore always focuses on the whole sporting movement, the trick being to make this movement so simple at the start of the learning process that the movement can be performed successfully.

The alternative is 'part practice' (Schmidt & Lee, 2008). Here only one or more parts of the overall movement are practised, rather than all of it. They are taken out of the context of the sporting movement and practised separately, always ensuring that the chosen parts are performed as similarly as possible to the corresponding parts of the sporting movement. The emphasis is usually only on motor skills. This often overlooks the fact that in part practice the sensorimotor impact may change greatly, because the sensorimotor information that is then released is often very different from the sensorimotor information in the sporting movement. As a result, in part practice there will be far less transfer from the part to the whole than is generally thought. Part practice is essentially reductionist, so it cannot be taken for granted that part-practice preparatory exercises in sport and physical therapy will be effective. This notion is very much reflected in the rather questionable focus on the detailed practice of various striking, kicking or parrying positions in oriental martial arts, or the minute, isolated corrections to posture that physical therapists tend to make. Such precise corrections to posture in part-practice settings are often seen as a sign of great expertise. However, not only are such precise instructions about movement of questionable value, but the sensory impact of these precise partial exercises is often so different from the sensory impact of the sporting movement that the exercise will only have a limited transfer effect, however precisely it is performed and however relevant it may be in motor terms. Since the intended effect of preparatory exercises is transfer to the sporting movement, part practice is less useful than may at first be thought, even when performed precisely.

In contrast, whole practice automatically guarantees a combination of sensorimotor factors that is relevant to the sporting movement. That is why whole-practice training is preferable. Sometimes, however, it is difficult to ensure progression in the series of exercises using whole practice only, for some components of the movement will not improve in whole practice. It is then necessary to practise those components separately, without interference from other components of the movement. This means that part-practice exercises will occasionally be required in the series of whole-practice exercises, and hence that sport-specific training will include both part-practice and whole-practice components. It is important here to strike a satisfactory balance between the two, while keeping the amount of part practice to a minimum (Figure 1.10, Magill, 2006).

Figure 1.10 Part practice (left) and whole practice (right) [illegible] when running

Additional information on sensory function

Ecological theory of motor control, based on the work of James J. Gibson, focuses on the theory of 'direct perception' and the function of 'affordances'. In this theory, sensory function plays a crucial part in the development of actions. There is a great deal of sensory information in the environment, and we need it in order to move in a meaningful way. This may be simple, meaningless information (e.g. colours, spots or lines), but it may also be complex, high-order information (e.g. information provided by the change of speed in our locomotion). According to ecological theory, humans (like all animals) are good at picking up not only simple information, but also information that is organized in a complex manner. Direct perception theory also states that we do not process this complex information by converting meaningless information in the brain into the required high-order complex perception, but that the system (body) can observe and process high-order information directly from the environment – i.e. without having to convert it in the brain. The body can link this information directly to meaningful movements, again without having to convert it in the brain. Direct perception is thus a guiding factor in our movements.

The most 'prominent' form of direct perception in ecological theory is 'time to contact' (= τ, the Greek letter 'tau'; figure 1.1), which allows us to observe directly how long an object that is heading for us will take to reach us. To do so, we do not need to register where an object is in space and how fast it is heading for us, and then calculate how long it will take to reach us. Direct perception occurs by measuring the ratio at which the projection of the approaching object increases in size on our retina. This is a direct measure of the time the object will take to reach us. This is how we catch a ball, assess when landing will begin when we jump off a table, and so on. 'Time to contact' is a measure that is directly derived from information gathered by the retina, and does not depend on the dimensions or texture of the observed object. Besides tau, there are many other types of high-order complex information that we observe directly. Peripheral flow outside the foveal area of the eyes provides a good deal of additional information about the environment. An example of this optical flow is how we intercept a long pass. The link between our own movement and the movement of the ball to be intercepted is a complex one. We must not only assess the speed of our body and that of the ball, but also calculate the angle between the two directions of movement, in order to determine whether the mover and the ball will reach the same point at the same time. Direct perception allows this complex calculation to be replaced by a single observation: if the ball appears stationary in relation to the background while I am moving, we will always arrive at one and the same point – the point at which I intercept the ball – at the same time. We use such observation of the movements of objects in relation to their background daily in order to avoid collisions. If a car comes out of a side street onto the intersection I am approaching, I can tell from the movement of the car in relation to the most distant background whether I will reach the intersection earlier than the car (if it moves backwards in relation to the background), later than the car (if it moves forwards in relation to the background) or at the same time as the car (if it remains stationary in relation to the background).

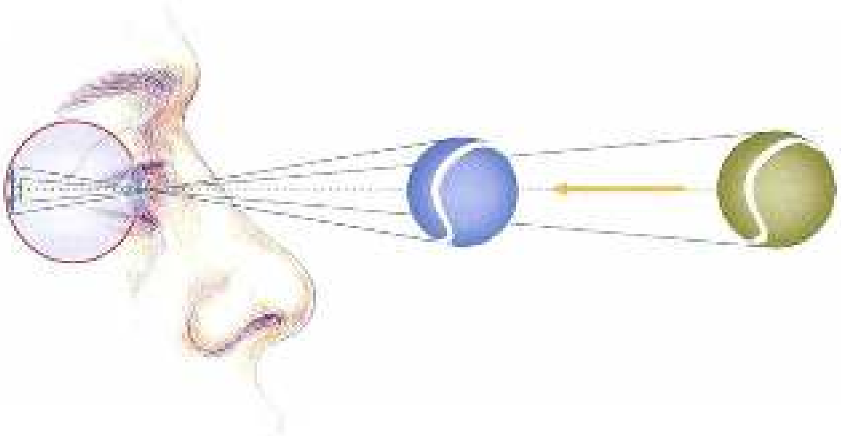


Figure [illegible] Time to contact (tau) is determined by reading the ratio at which the projection of the approaching object increases in size on the retina. The eye registers this ratio directly and so can directly perceive the value of tau.

Direct perception includes the possibilities of moving, or 'affordances', defined by Gibson (1977) as 'all "action possibilities" latent in the environment, objectively measurable, and independent of the individual's ability to recognize those possibilities'. An affordance is in fact the link between the observation and the body's possibilities, creating a distinction between movement possibilities that can and cannot succeed. We thus pick up meaningful, successful movement patterns from the environment.

In direct perception theory, the function of cognition is greatly reduced. The theory thus more or less explains why animals with very limited cognitive capacity can still move meaningfully in a complex environment.

The difference between whole practice and part practice is, moreover, not only relevant when partial movements are compared with total contextual movements. In a sense, practising the whole movement in a setting that is only partly the competition setting is also part practice. When a tennis instructor hits controlled balls so that the learner can alternate practising forehand and backhand, that is a type of part practice compared with an actual tennis match. The sensory information that can be used to determine tactics is absent. Practising a combination of punches on a punching bag is part practice in relation to a boxing match against an opponent. Even as regards the setting in which practice takes place, there is plenty of evidence that a whole-practice approach is more effective than one based on part practice (whole practice: learning to hit a forehand by playing a match with an adapted racket and adapted balls so that the technique is easier to perform, Reid, 2009). In physical therapy, too, it is important for transfer to be setting-specific. Gait analysis in a laboratory-type setting ('walk five metres, then turn round and walk back five metres, so we can see how you walk') is not a daily-life situation ('will you get me a beer from the fridge?'), and it cannot simply be taken for granted that a person's way of walking will be the same in both cases.

The great advantage of whole practice is therefore that the combination of sensorimotor information more or less automatically remains the same in the exercise as in the sporting movement. The main reason why this is so important is that it is not possible, or scarcely possible, to 'read' from outside which components of the available sensorimotor information are selected as relevant components in the contextual movement. This means it is scarcely feasible to work out whether the sensory information that plays a role in a part-practice approach is related to the information in the sporting movement. Using part practice in an attempt to improve the sporting movement is thus a matter of guesswork.

If, then, the purpose of practice and training is not to practise a movement in isolation ('mastering a trick') but applicability in numerous settings – i.e. transferability of a movement pattern (a skill) – we must focus on which approach best guarantees transfer. Whole practice probably does this better than part practice, because it is more in line with the dynamics of the complex biological system.

1.2.3 Sport-specific strength training is part practice

Clearly, strength training to improve sport performance is in fact a type of part practice. And this is the main problem with sport-specific strength training: although the whole purpose of strength training is to transfer the qualities that are trained to the sporting movement, the fact that it is part practice means transfer cannot be guaranteed. In particular, sensory information from the environment will be very different in strength training and in the sporting movement. Sensory information from the body (proprioception) will be somewhat more like the information in the sporting movement, but will still often imperceptibly differ from it. This is because sensory information from the body, especially registration of tensile force acting on the muscles, is strongly influenced by the resistance used, and that influence is usually absent in the sporting movement.

Since there is no guaranteed transfer, we must make a thorough analysis of the relationship between strength exercises and the sporting movement. This cannot be done if the analysis is based on an oversimplified thinking model. A reductionist approach, such as thinking in terms of basic motor properties, is therefore unworkable as a means of identifying transfer of sport-specific strength training to the sporting movement – for in such an approach strength is a more or less separate entity, which means that sensorimotor observation by the body is irrelevant and transfer mechanisms can be disregarded.

If transfer *is* to be identified, this requires in-depth analysis of the sensorimotor features of both the strength exercise and the sporting movement. Only when both have been sufficiently identified can we assess (to some degree) to what extent a strength exercise will help improve performance (see Section 5.3). This is particularly true if a sporting movement is already mastered at a higher level and is to be improved by training. The higher the level of mastery, the less obvious it is that a preparatory exercise will lead to improvement in the sporting movement. Besides possible performance-improving transfers from the preparatory exercise to the sporting movement, there may also be performance-reducing transfers, such as the negative impact of differing rhythms in the preparatory exercise that perturb those in the sporting movement. At a high level of mastery in which more transfer occurs than at a low level of mastery (in both motor and situational terms) and the margins between better and worse performance of the movement are reduced, exercises must therefore be selected more carefully.

1.3 Tradition of sport-specific strength training resulting from the reductionist approach

Historically, sport-specific strength training has mainly been emphasised in sports – bodybuilding, powerlifting and weightlifting – in which transfer plays little or no part. These sports are based on influences that precede transfer, and disregard the complexity of transfer. The resulting systems of sport-specific strength training are marked by limited interest in the sporting movement, and in countries such as Australia, Britain and the United States have led to the emergence of specialized strength coaches who often ignore the sporting movement and focus instead on isolated strength-production mechanisms, such as hormone response due to strength training (Kraemer & Ratamess, 2005) and post-activation potentiation (PAP): the effect of improvement in performance in a contextual movement (e.g. vertical jumping or sprinting) as the result of a previous maximal or submaximal strength exercise (French *et al.*, 2003; Hamada *et al.*, 2000). Such mechanisms are then seen as the best ways to improve performance through strength training (see also Section 2.2.1).

The main historical influences on sport-specific strength training come from bodybuilding, physical therapy and physiology. We should note here that by no means all sports base their strength training practice on such influences that do not focus on transfer. Gymnastics coaches, for instance, make particular use of methods that base strength exercises on similarity of movement, and in athletics many leading coaches do not use strength coaches that lack specific knowledge of athletics.

1.3.1 *Influences from exercise physiology*

Approaching sport-specific strength training from a purely physiological angle disregards the way in which the learning system organizes movements and transfers between them.

The main physiological aspects of strength training are:

- protein synthesis and muscle work;
- energy and force production;
- metabolism in the neuromuscular synapse;
- hormonal response as a result of strength training;
- influence of diet and supplements on strength training.

Of course, these are all important aspects of strength training that may be of value in, for example, its timing. However, focusing sport-specific training entirely on such physiological parameters is only useful in sports in which the quality of the movement pattern (the efficiency of the movement) plays a marginal role. One sport to which, at first sight, this particularly seems to apply is cycling. The movements are simple, and are guided by outside influences (the revolving pedals). As a result, coordinative transfers will hardly matter. When training track cycling sprinters, attempts are therefore made to focus strength training mainly on measurable physiological parameters. However, this approach has its limitations even in cycling – first and foremost because many physiological parameters may be measurable but the measurements cannot simply be used to predict which adaptations will occur as a result of the training. Not much is yet known about the underlying mechanisms of adaptation, and so adaptations cannot yet be predicted. Furthermore, coordinative aspects, which are complex and must be learned, play a part even in such seemingly simple movements as turning the pedals during a sprint. In particular, producing power in an aerodynamic posture and quickly building it up at the start of the pedal movement are aspects that affect performance. These aspects can be improved by designing special bicycles that optimize aerodynamics and power transmission, and strength training that focuses on coordinative aspects may enhance this. That is why the world's leading track cycling coaches seek to improve coordination by devising exercises that train power production in movement patterns that are relevant to cycling.

1.3.2 *Influences from bodybuilding*

Bodybuilding and strength training were long connected before strength training became a part of sports such as athletics, swimming, judo and so on. In the past, sport-specific strength training was therefore greatly influenced by ideas from fitness training and bodybuilding. In bodybuilding, specific efforts are made to bring about hypertrophy in certain parts of the body. The result is a strong focus on training isolated muscle groups (the 'body-part approach'). Under the influence of bodybuilding, isolation of muscle groups has been further extended to isolation of individual muscles, and even separate parts of muscles. Exercises have thus been devised that attempt to isolate the deep or lowest parts of the abdominal muscles. There are many exercises for the shoulder girdle that focus closely on loading parts of muscles, such as the clavicular head of *pectoralis major*, the long head of *triceps brachii*, the posterior portion of *gluteus medius* or the lower fibres of *trapezius*. In this philosophy, efforts are thus made to make strength training more efficient for throwers, swimmers and rowers. For sprinters in athletics, efforts are made to find ways of isolating *gluteus maximus*, and similar precisely targeted loading is also sought in track cycling and speed skating. The bodybuilding strategies focusing on hypertrophy are not normally adopted here, but are replaced by other strategies such as training for maximal force production.

The body-part approach has a number of major shortcomings as regards the sport-specific function of strength training. Cooperation between muscles does not play a major part in the training. Bodybuilding exercises therefore never involve performing complex contextual patterns. Apart from a wish to control where hypertrophy is to occur, this is also understandable because hypertrophy only occurs when a muscle is exhausted. If exhaustion occurs in complex coordinative patterns, control over the patterns will be so greatly reduced that there will inevitably be serious errors in performance, and hence injuries. Coaches who mainly think in terms of influence from bodybuilding avoid complex intermuscular strength exercises, because these are likely to cause injuries. However, what they do not realize is that the injuries are due to exhaustion of muscles rather than the complexity of the exercises.

Complex training that targets neural adaptations is therefore incompatible with such a body-part approach. This means that strength training based on the body-part approach is rather one-sided, involving only very simple movements. Isolation of muscles, or parts of them, has nothing to do with improvement of neural qualities, and hence such strength training does not improve the coordination of athletic movements. On the contrary, such body-part training can reduce coordination, especially if it takes place in the hypertrophy-sensitive zone (see Section 7.2.1), and hence should be avoided in sports in which coordination and/or high levels of neural drive matter greatly. Such isolating strategies are therefore becoming less popular in the practice of sport-specific strength training. However, the body-part approach is a stubborn one, and is still often used for types of strength exercise other than hypertrophy training, such as maximal strength training and power training.

As stated in the introduction to this book, 'complete' training theory combines all the aspects of physiological adaptation and all the possible contributions from part practice into a coherent system. For the time being the resulting puzzle is too complex as a basis for the practice of strength training, and an approach based on conviction will have to be adopted.

1.3.3 *Influences from physical therapy*

Muscles not only get the body moving, but also protect joints and various passive tissues and keep them healthy. That is why muscle training has become a key part of physical therapy and why physical therapy has come to influence thinking about sports strength training. Increased stability in the body has become the guiding principle for practical applications. Stability problems are attributed to poor functioning of the muscles that are meant to guarantee protective stability of the joint. The deeper muscles, those close to the joint to be stabilized, are considered of particular importance to stability. That is why there is usually a build-up from low to high force production, from controlled performance of isolated ranges of motion in the joints to contextuality, and from control of small muscles near the joint to larger, more distant muscles. The training of small muscles located near the joint depends on proprioception. The improved proprioceptive feedback resulting from training is proposed to be essential for proper protection of the joints and injury-free movement. For example, trunk control is often approached in physical therapy in terms of precise control of muscle groups located close to the joints, such as the *transversus abdominis* and the *multifidi*. Force production is stepped up during the training process, the range of motion in which force production is applied become greater, and more and larger muscles are recruited. A similar system is also used in training the shoulder and the pelvic girdle: first the small muscles located close to the joint (the deep hip muscles in the pelvic girdle and the rotator cuff muscles in the shoulder) are subjected to low-intensity loads. The larger, more distant muscles are then loaded when training intermuscular coordination based on proprioception. In this way an attempt is made to achieve better stability in the body when making complex, high-intensity movements (Figure 1.12).

Figure [illegible] Three phases of stabilisation training ([illegible] local muscle system). Athletes' trunk control built upon the traditional assumption that control of the trunk when low forces are acting on it takes place in the same way as when large forces are acting on it ([illegible], 2003; [illegible], 2006).

Dynamic systems and stability training

Such an approach to stability training, which is pervasive in physical therapy, assumes a more or less linear, gradual shift from low force and control by small muscles to high force production and the recruitment of bigger and bigger muscle groups. However, it is highly questionable whether there is a gradual transition from movement patterns with small ranges of motion and small use of strength to large ranges of motion and large use of strength. There is no scientific evidence for this assumption in physical therapy. There may also be sudden phase transitions in which coordination changes abruptly and fundamentally, for instance because different muscles suddenly become important within a pattern (though there is no scientific evidence for this either). This is quite possible within the dynamics of complex systems. Low impact control may be a different phase in this process than high impact control, and there may be a sudden transition from phase to phase. In other words, it is questionable whether the activity of the *transversus abdominis* and the *multifidi* is important when controlling the trunk during an extended somersault with twist or take-off in pole vaulting, i.e. a contextual movement in which large forces act on the body. Especially when such forces have to be processed elastically, the organization of trunk control may develop in a fundamentally different way than with low impact control. In that case, low impact control, such as core balance on a physio ball, may produce little transfer to high impact control as applied in jumping, running and throwing, and the influence of the *transversus abdominis* and the *multifidi* is limited in high intensity movement (Lederman, 2010).

Besides in the theory of complex biological systems, there are also neurophysiological reasons to assume that control is fundamentally different when processing (slow) low impact and (rapid) high impact. In physical therapy, trunk control in low impact control is mainly based on proprioceptive feedback. Signals from muscle spindles, joints, tendon sensors, skin senses and so on are processed to correct posture and movement. Depending on the path used (spinal or supraspinal), this takes from 25 (spinal) to 100 (supraspinal) milliseconds. In sporting movements, however, perturbations must often be processed much faster. In addition, they are often too large to be absorbed by feedback correction. For example, the stance phase in high-speed running is too short to respond to the direction of the ground reaction force in relation to the knee. A different control mechanism must be used, one with a delay so brief that perturbations can be compensated for in good time during the landing. This rapid control is based on cocontractions of agonists and antagonists, and ensures the necessary stiffness and spring action round the joint. The function of cocontractions can be compared to the action of shock absorbers in a car (see also Section 4.8.4), has a response time of zero milliseconds and can compensate for the shortcomings of proprioceptive feedback. The afferent part of the nervous system (registration from senses) plays no part in such control. The movement is controlled by efferent feed forward (open-loop control; Figures 1.13 and 1.14).

Thus, since there is no guarantee of transfer from low-intensity to high-intensity mechanisms, it is not advisable to take transfer from low-intensity to high-intensity movements for granted in rehabilitation and sport-specific strength training. The self-organizing effect of intermuscular coordination that occurs in high impact may be a better starting point for sport-specific strength training and sports injury rehabilitation in the event that phase transitions occur. In the case of ankle stability after an injury, for example, this means that simply practising proprioception with low-intensity forces will not automatically restore control of rapid impact with high-intensity forces. Control by means of cocontractions must also be trained to ensure, for example, that a twisted ankle (which is often, and probably wrongly, only associated with malfunctioning proprioception) does not recur more often after recovery. Simply training on balance boards and so on will then not suffice to recover correct function. At each stage of rehabilitation, the athlete must choose which type of motor control should be practised, of course depending on the extent to which load capacity allows intensity of practice. Similar phase transitions from low-intensity to high-intensity movements may occur anywhere during sporting movements (Figures 1.15 and 1.16, see also Section 5.6).

Figure 2.9 Left: proprioceptive feedback. Response to changing ground reaction forces through facilitation in the muscle spindle and via the golgi tendon organ (see also Section 2.5.4). Right: preflex through co-contraction. Stiffness control helps to respond to the changing ground reaction forces, and through the force/length, force/velocity and elastic properties of the muscles (see also Section 2.4.3).

Figure 2.10 Left: trunk control based on the processing of proprioceptive information. Right: trunk control based on co-contraction of all the muscles that influence the spine. The co-contraction is generated by pushing the weight as far away as possible.

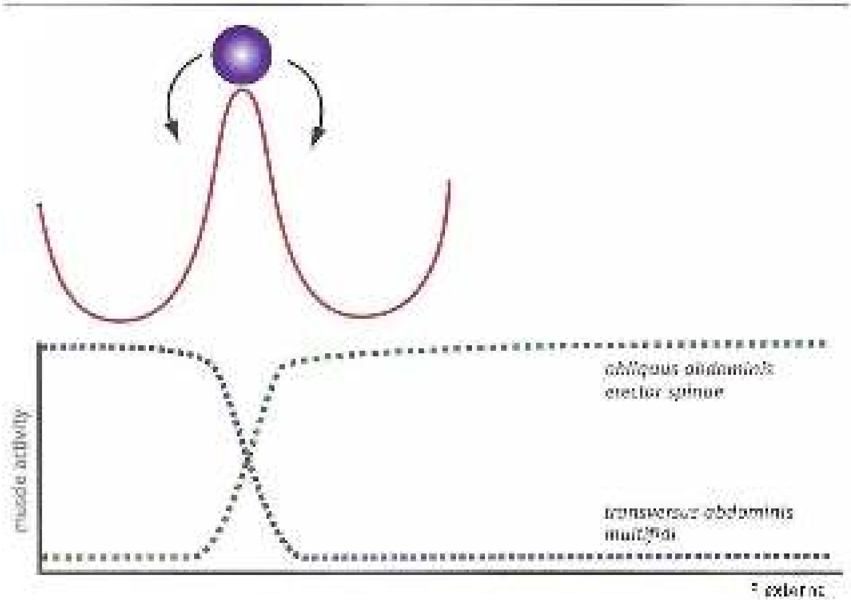


Figure 1.15 Graphic representation of a phase transition in trunk control. After the transition, stability is regulated by different muscles. This means that the similarity between control with low-intensity forces and control with high, opposing forces is very limited.

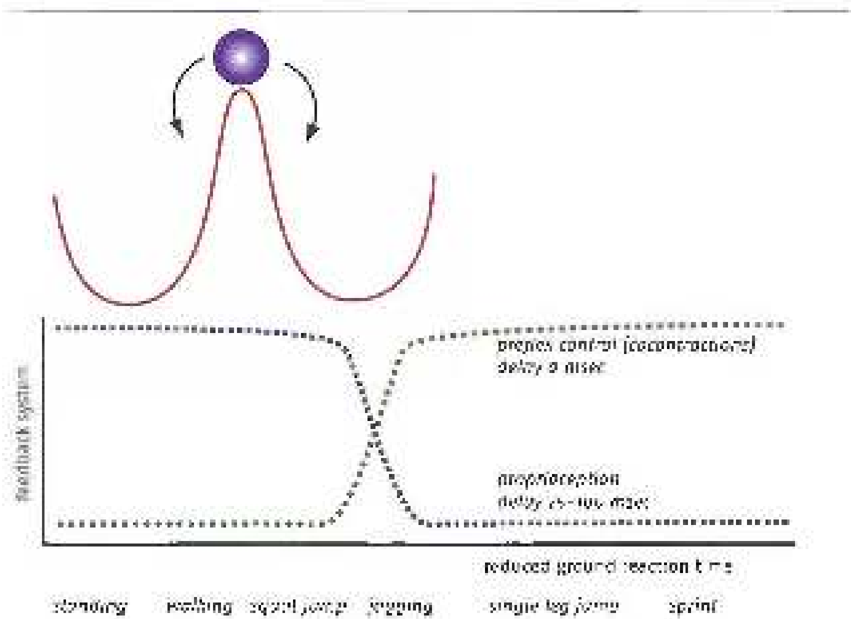


Figure 1.16 Graphic representation of a phase transition in stance-leg stability. As the contact times are reduced, proprioception will play a smaller and smaller role, and cocontractions and stiffness will become more and more dominant. There may be a sudden transition between the two types of control. There is still no evidence about where the transition (in movement patterns that are more intensive than jogging) actually takes place.

Practice: a groin injury

Conflicts between approaches by physical therapists and fitness and strength coaches mainly arise in rehabilitation after injuries, especially in elite sport, where coaching is intensive. Physical therapists tend to make their part of the rehabilitation last longer than strength and fitness coaches would like. They see practice time as essential to prevent problems arising later on in the rehabilitation process, whereas strength and fitness coaches often do not see the relevance of the exercises to the remainder of the process. They think physical therapists are far too cautious, and they sometimes even see physical therapists' actions as counterproductive. This difference in outlook can be traced back to the debate about continuum versus phase transition. The debate about the optimal rehabilitation process, which in practice often results in stalemate, could perhaps be relaunched by no longer building up from low-intensity to high-intensity movement, but by first analysing the biomechanics of high-intensity movement and then applying the analysis to the low-intensity movements at the start of the rehabilitation process. This would create a better connection between low-intensity and high-intensity movement. The rule would then be 'rehabilitation that doesn't resemble regular training is no good'.

An example: rehabilitation after a groin injury

Opinions differ as to the causes of groin injuries. Some seek the main cause in hip pathology (Bradshaw et al., 2008); others mainly associate it with adductor problems (Hölmich, 2007). At the start of the rehabilitation process, after initial recovery, there is usually cautious training with ranges of motion in every plane and direction in the hip joint and the lumbar spine, and the load is then gradually increased. A contrary route would be as follows: a healthy athlete can place a heavy load on the groin without causing problems. This is possible because muscles use strong cocontractions to conduct opposing forces round the vulnerable passive tissues, and so protect them. Running and jumping on one leg cause strong forces to act on the groin. A posture (the left position, Figure 1.7) in which muscles can use cocontractions to protect the groin while running and jumping on one leg is by raising the free/swing side of the pelvis while standing on one leg, and slightly rotating the pelvis anteriorly and with slight internal rotation of the hip. This is accompanied by flexing the hip and knee of the free leg, while attempting to move the heel towards the hamstring. This posture is a key part of the movement pattern in maximal acceleration, sprinting at full speed and take-off in a single-leg jump. It can be practised early on the rehabilitation process at low intensity, for example by leaning against a wall, and can then be developed in more intensive ways, for instance by running up stadium stairs while carrying a weight over one's head with outstretched arms. In this way the intermuscular organization of the high-intensity movement pattern, with its intrinsic protective mechanism, is practised from the start of the training process. Later in the rehabilitation process the similarity of movement to running and jumping can be increased by means of such exercises as a single-leg clean and a single-leg snatch (see Figure 1.8). A possible cause of injury may be poor control of the cocontractions round the pelvis at toe-off during running and jumping. In these single-leg versions, the cocontractions can be practised in a body posture that greatly resembles the posture at toe-off. In a single-leg snatch, an attempt is made to delay for as long as possible the landing of the foot on a box placed in front of the athlete. The longer the landing is delayed while maintaining the correct body posture, the better the cocontraction of all the muscles round the pelvis is maintained. In this way, relevant coordination of high-intensity movement can be trained in the injury prevention programme and relatively early in the rehabilitation process. The single-leg snatch exercise can be performed in various versions. Finishing with a barbell above the head will lengthen the abdominals, so that control of this muscle group will come under pressure within the overall pattern of cocontraction. Ending with a barbell plate behind the head will put even more pressure on the abdominals.

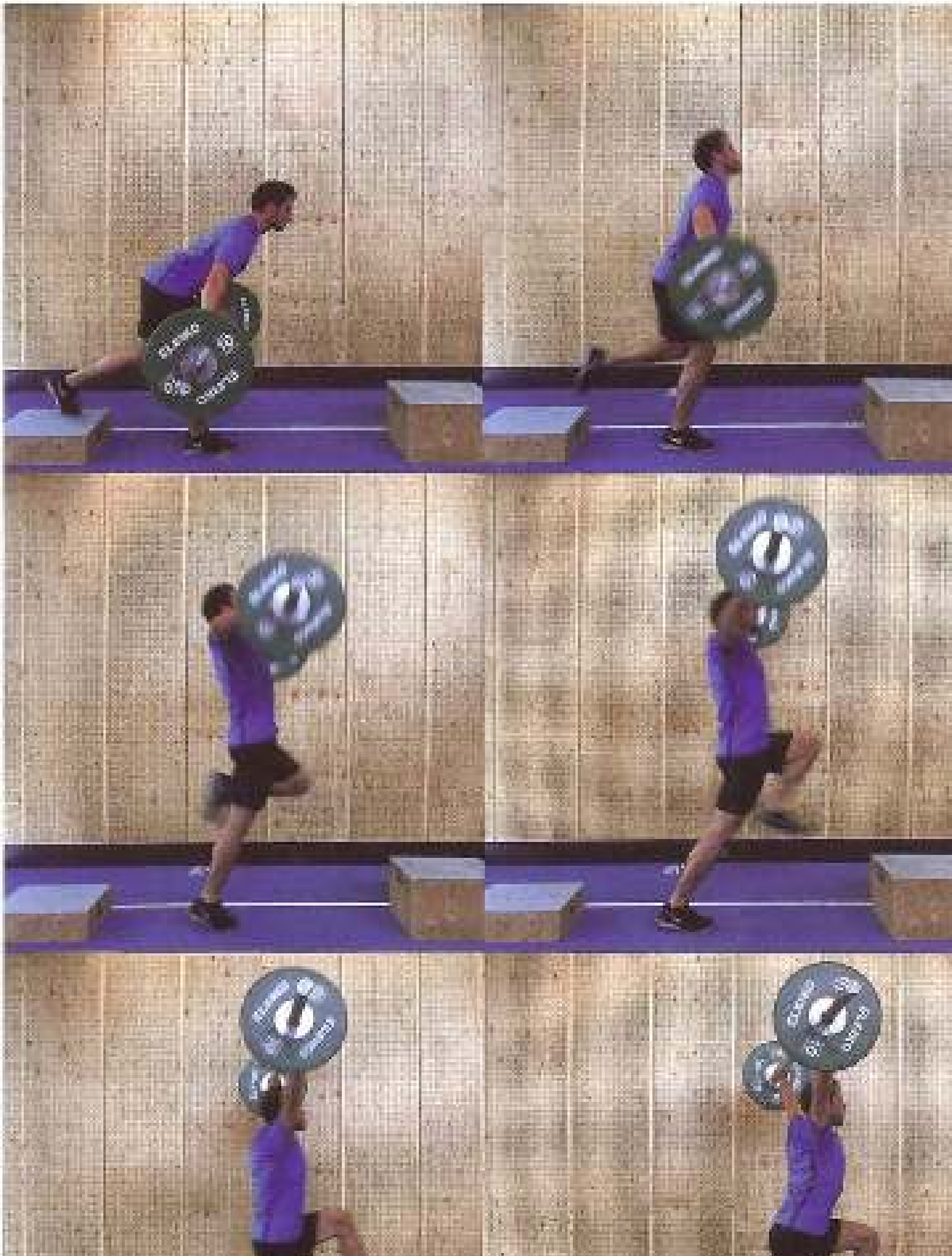
Variation in load is also possible. A tube partly filled with water can be used instead of a barbell. The unpredictable movements of the water will put further pressure on the trunk muscles and the muscles round the hip.

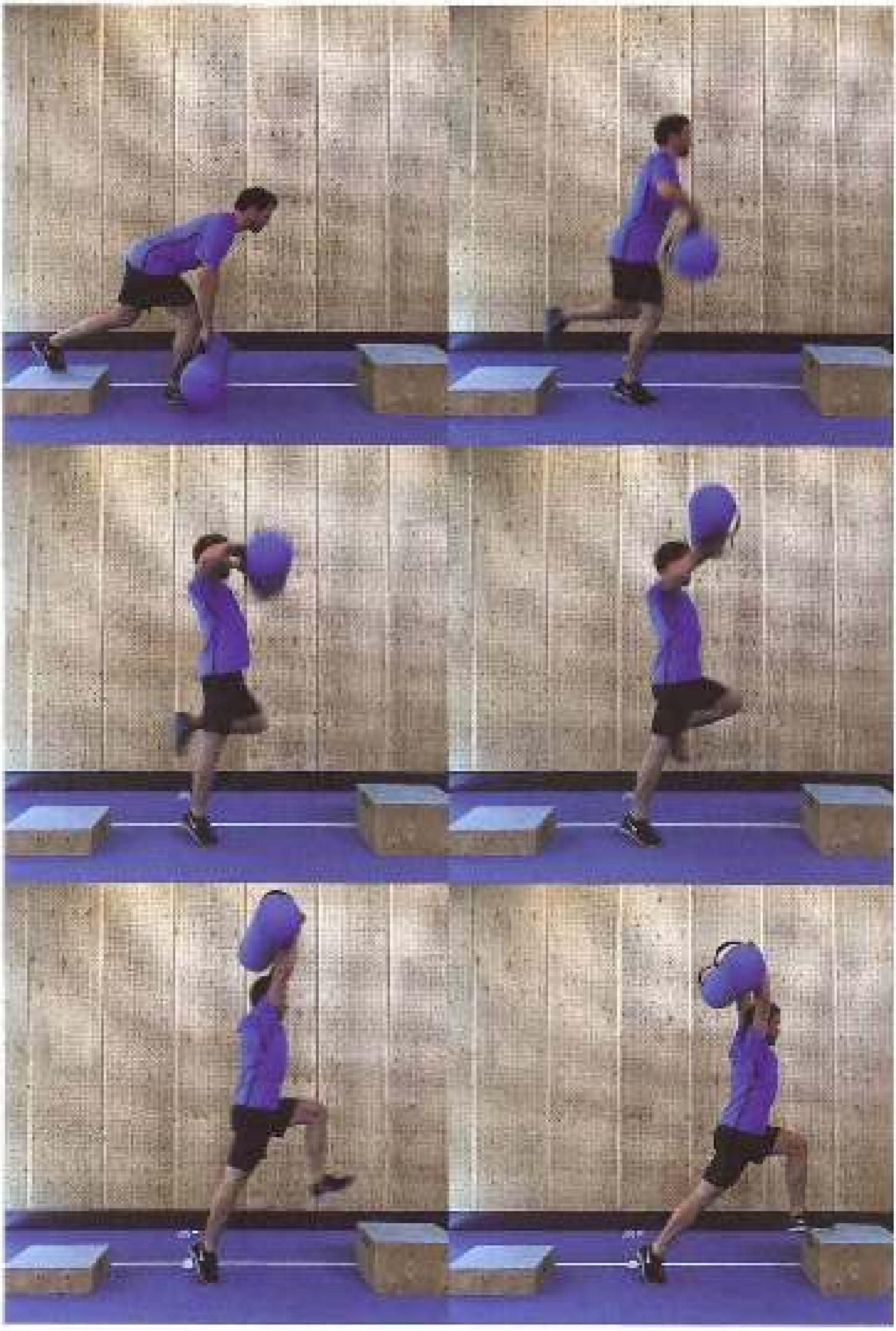
Complex exercises such as these can be used for various purposes. In section 9.4 the same basic movement pattern is used to improve patterns supported by reflexes. Rehabilitation and conditioning thus merge, and contextuality – which is a relevant concept in the rehabilitation process – is thus built into rehabilitation as early as possible. Phase transitions are of course also important in injury prevention sessions that athletes include in their programmes (figure 1.18). Here again, we need to consider whether low-intensity movement has any impact on high-intensity movement.

An example: calf injuries

Practice shows that it is important to go through a sound rehabilitation process after a calf or ankle injury, so that the injuries do not recur or that problems do not arise elsewhere in the body. Unlike traditional programmes (such as protocols in which running with a highly reduced load is placed fairly early in the rehabilitation process), the rehabilitation process shown in

Figure 1.17
[illegible] position the feet [illegible] head and [illegible] forward; the pelvis rotates slightly anteriorly. In the resulting position of the pelvis, [illegible] area.





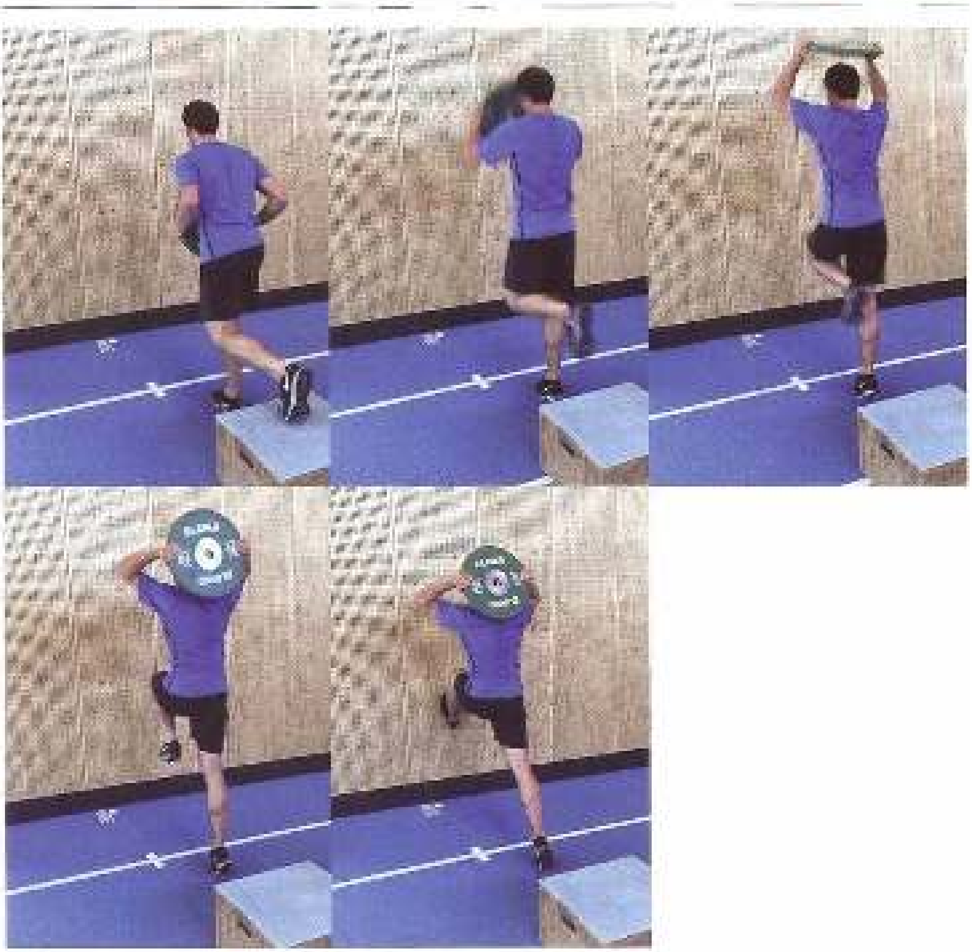


Figure 1.18 Versions of a single-leg snatch in which cocontractions round the pelvis are practised at take-off. The foot placement on the box in front of the athlete is postponed for as long as possible.

Figure 1.19 is based on a gradual increase in load. There will always be sudden (or phasic) increases in load during the process, and gradual build-up is not feasible in practice. What matters is to try and keep control of these sudden increases, which occur because other stressors – such as elastic loading of structures – suddenly come into play while the load is being built up. Elastic load cannot be built up gradually, but will have a major impact on the system as soon as it is introduced. It is therefore a good idea to incorporate these sudden increases into the rehabilitation strategy, so that they can be properly controlled. Stressors that act on the calves and ankles in sporting movements (the sporting movement) can be divided into the following categories (see Figure 1.19):

- movements (mobility) of the ankle joints, combined with proprioception training by introducing new stressors;
- energy transport from the knee to the ankle by the gastrocnemius, in combination with reflex action of the lower leg and foot muscles;

- distortion round the longitudinal axis of the foot (pronation/supination), and hence transverse forces acting on the Achilles tendon and calf muscles;
- elastic stretch owing to opposing forces.

Such influences must be avoided after the injury during the first acute phase of rehabilitation (1) and brought together during the last phase (6) in, for example, running exercises. Rehabilitation must take account of those control mechanisms that play a part in high-intensity movement, and must prepare the athlete as well as possible, in a safe environment, for the stress in the sporting movement. This means that the aforementioned stressors of high-intensity movement must be included at an early stage of the rehabilitation process. The abrupt increases in load and the control mechanisms in the sporting movement are combined in the stage-by-stage model in Figure 1.19. The basic idea here is that the stressors are no longer all trained at once, as in the traditional approach, but that the stressors to be mastered again after rehabilitation are introduced one by one (a "single-stressor model"). However, the previous stressor must continue to be offered in a sustaining dose. This ensures that the athlete is at slightly greater risk of overloading the system only when the next stressor is introduced (e.g. in the transition from phase 2 to 3), and that once a stressor has been introduced its load can gradually be increased. The great advantage of this separate introduction of stressors is that, once a first stressor has been

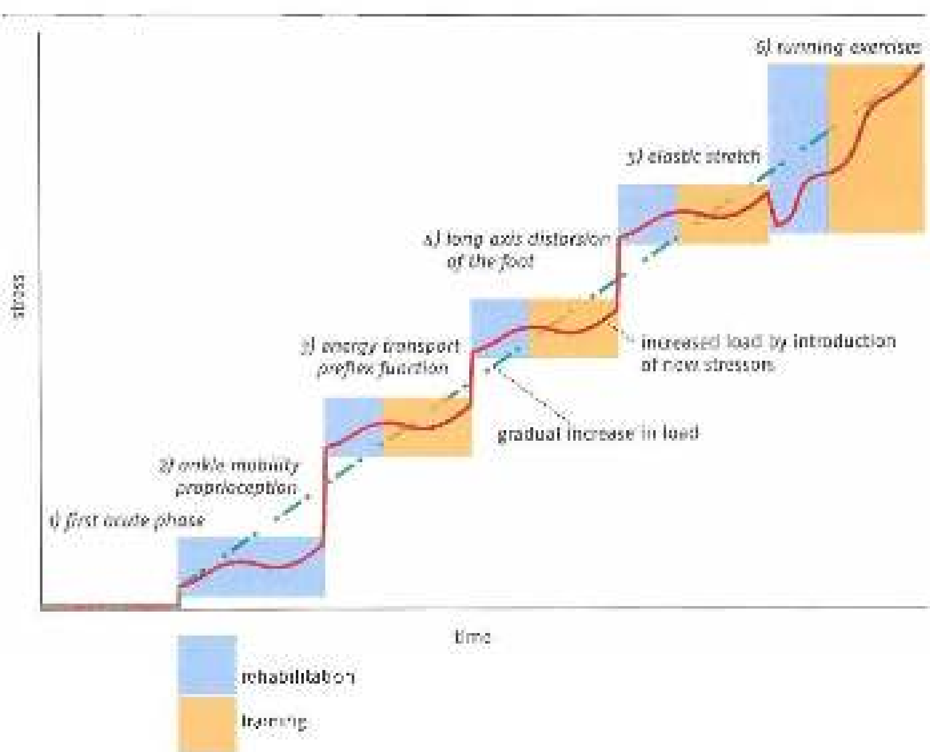


Figure 1.19: Diagram of a rehabilitation strategy in lower-leg injuries (see the main text for further explanation). This strategy was developed for the Welsh national rugby team in cooperation with Craig Ranson, a Wales Rugby Team sports physical therapist.

successfully introduced, that aspect of the sporting movement can soon start to be trained at a higher level of intensity. This blurs the boundary between rehabilitation and training. Depending on the injury and the sporting movement, some stressors in the rehabilitation process will be more important than others. The process will therefore have to be designed differently according to the situation. The order of the rehabilitation phases (1–6) is the same regardless of the type of injury. In this approach, physical therapy and improvement of fitness largely merge, and rehabilitation resembles regular training. Rehabilitation may not always proceed faster than in a traditional approach, but will usually be much easier to control, for in the event of a relapse it will immediately be clear which stressor is to blame. Some features of the stages in the rehabilitation of a grade 2 gastrocnemius strain (Figure 1.20):

- Phase 1: day 1–3. Acute protection phase with regular physiotherapeutic action.
- Phase 2: after day 3. Ankle and calf mobility with proprioception training and eccentric/concentric muscle action, as in calf raises.
- Phase 3: after day 6. Energy transport from knee to ankle – but only if there is no pain when walking with a sled and no pain at toe-off.
  - step-up movements;
  - double-leg vertical acceleration to explosive vertical jumping;
  - double-leg vertical acceleration to explosive horizontal jumping;
  - single-leg explosive jumping;
  - running movements up stairs;
  - running movements up stairs under conditional pressure;
  - preflex training.
- Phase 4: after day 12. Torque round the longitudinal axis of the foot – but only if full loading has been achieved in the phase 3 exercises.
  - starting and acceleration with sled from high to low weight.
- Phase 5: after day 15. Elastic stretch owing to opposing forces – but only if full loading has been achieved in the phase 4 exercises.
  - low impact, as in running training with low horizontal speed;
  - higher impact, as in double-leg ankle bounces.
- Phase 6: after 18 days. Running.
  - 60% speed over short distance (forty metres);
  - build-up to 100% speed over short distance (sixty metres);
  - longer distances.

This rehabilitation framework can of course be fine-tuned and specified for various types of injury, with additional strategies from other therapeutic disciplines. The main principle is to bridge the gap to contextual high-intensity movement as early as possible.

The details of each phase will of course vary according to the type of injury. In the case of a gastrocnemius injury, phase 3 will be of critical importance and hence will be emphasized. In the case of an Achilles tendon injury, phase 5 will need to be practised particularly carefully, and perhaps for a longer period.

Grade 2 Gastrocnemius Strain		Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Phase		Date																												
1 Acute Protection			P	P	P																									
RICE/POLICE, Boot & Calf Raise			P	P	P																									
Partial Weight Bearing & Crutches?			P	P																										
Stationary Bike					P	P/T	P/T	T	T	T	T	T																		
2 Ankle Mobility																														
Mobility Exercises	1				P	P/T	P/T	T	T	T	T	T																		
Balance Activities	2				P	P/T	P/T	T	T	T	T	T																		
Concentric/Eccentric Triceps Surae	3						P	P	T	T	T	T	T																	
Proprioception in functionality	4						P	P	T	T	T	T	T																	
3 Knee to Ankle Energy Transfer																														
Check – Painfree Walking								P																						
Check – Painfree Sled Walking								P																						
Double Leg Hang Cleans	5								P	P	T	T																		
Box Jump Progression	6								P	P	T	T	T	T	T	T	T	T												
Single Leg Hang Cleans	7										P	P	T	T	T	T	T	T	T	T										
Plyo/Sled training	8										P	P	T	T	T	T	T	T	T	T	T	T	T	T						
Ramp Runs	9										P	P	T	T	T	T	T	T	T	T										
Stadium Stair Progression	10										P	P	T	T	T	T	T	T	T	T										
4 Long Axis Foot Distortion																														
Check – Painfree Sled Running														P																
5a Elastic Work Impacts																														
Low Impact Speed Jumps															P	P	T	T	T	T	T	T	T	T						
Skipping															P	P	T	T	T	T	T	T	T	T						
High Knee High Cadence Runs															P	P	T	T	T	T	T	T	T	T						
High Impact Low Speed Jumps	11															P	P	T	T	T	T	T	T	T	T					
Off boxes																P	P	T	T	T	T	T	T	T	T					
Over hurdles																P	P	T	T	T	T	T	T	T	T					
5b Elastic Work Impacts																														
Stair Running																		P	T	T	T	T	T	T	T	T				
6 Return to Running																														
Sub-Max Accelerations																			P	T	T	T	T	T	T	T	T	T	T	T
Uneven Surface Accelerations																				P	T	T	T	T	T	T	T	T	T	T
Lateral runs & side-stepping																						P	T	T	T	T	T	T	T	T
Non-Contact Rugby drills																							P	T	T	T	T	T		
Endurance																												T	T	T
Full Rugby Training																														T
7 Return to Play																														

Figure 1.20 Checklist for rehabilitation of a gastrocnemius injury.
P = physical therapy, P/T = both physical therapy and regular conditioning, T = regular training (note, on feet day and off feet day strategy is not in this table).
RICE = Rest, Ice, Compression, Elevation; POLICE = Protection, Optimal Loading, Ice, Compression, Elevation.

A number of exercises from the protocol:

1 Mobility exercises, as part of standard physical therapy practice.

2 Balance exercises for proprioception, as part of standard physical therapy practice.

3 Concentric/eccentric action of the biceps muscle

4. Foot dorsiflexion in contact.

5. Double-leg clear from above the knee.

5 Progression in squat jumps onto a box

Single-leg clean from above the knee.

6 Pole training: keeping one's balance under the pressure

9 Step-ups in series, focusing on energy transport from knee to ankle.

10 Progressions in walking up stairs, focusing on energy transport from knee to ankle under time pressure.

1.4 Sport specific strength training and motor control

1.4.1 Strength and coordination

Many coaches, especially in complex-coordination sports such as gymnastics, intuitively sense that strength training will produce the best transfer if the movements are performed in movement patterns similar to those in the sporting movement. There appears to be a close connection between strength and coordination. This can be seen, for example, when a person with no experience of strength training does some for the first time (2-3 times a week, with standard barbell exercises such as squats, step-ups and so on). During the first few weeks the individual muscles will not get stronger, nor will they increase in size. Performance will improve when there is improved cooperation between the agonists, synergists and antagonists (improved intermuscular coordination). After several weeks individual muscles will start to perform better when they have to produce force in isolation (improved intramuscular coordination), and only later (after about eight weeks) will the muscles increase in size (hypertrophy). Research has provided ample evidence of this pattern of improved performance through strength training (Figure 1.21, Huijbregts & Clarijs, 1995).

Figure 1.21 The three phases of strength development over time. Strength development through training can be divided into three phases over time. In the first phase the increase in strength can be attributed to improved intermuscular coordination, in the next phase there is also an improvement in intramuscular coordination, and in the final phase there is also hypertrophy.

Such a pattern of improved performance also occurs when experienced athletes include a new, complex exercise such as the clean in their strength training. Performance will then rapidly improve during the first few weeks, because the exercise technique is mastered more effectively. This makes sense, because the overall interplay of forces in the clean is particularly involved and hence the clean is complex-coordination exercise. After some time the improvement in performance will level off, and performance will be influenced by the increasing strength of individual muscles as well as by cooperation between muscles. A coach who wants to achieve further improvement in the level of strength using the same exercise must then consider whether, and if so how, to improve the limiting factor in the exercise (the first muscle group to reach the limits of its capabilities). Can this best be done by repeating the exercise and slowly increasing the load, or by adding other related exercises that alter the emphasis in the load?

Since in this approach strength training is basically a type of coordination training, and since training transfer between the two movement patterns is due to the coordinative similarity between the two movements, the main guide in sport specific strength training is technique. This means a great deal of attention must be paid to the way in which strength training exercises are performed. It also raises questions about the habit of allowing the training to be done in accordance with a printed paper plan, without a coach having to be present.

1.5.2 *The biomechanics of strength training and transfer*

Those who see coordination as the main vehicle for transfer look for a usable classification of similarities in movement. The usual focus here is on the outward appearance of the movement. Speed skaters will want to use the same angles in strength training as in skating. Javelin throwers will look for the same extreme position of the shoulder joint in various strength exercises as in throwing the javelin, and will not confine themselves to bench pressing. In the case of elastic jumping, strength exercises are sought in which the contact times can be very short, and low barbell weights are used in order to achieve this.

Performing the movements against resistance means they are performed in a controlled setting with high force production. This more closely resembles the internal structure of the sporting movement (its intramuscular and intermuscular features) than if only the outward form of the movement is imitated in low-intensity exercises. Since there may be phase transitions in the internal structure of the movement when shifting from low-intensity to high-intensity movement, training with external resistance may be a useful strategy for working in a controlled manner on the same phase(s) as in the sporting movement. This is because of the self-organization generated by the high force production.

In thinking about the function of strength training, there may thus be a shift from seeing strength and strength training as separate entities to seeing strength training as a way of staying close to the high-intensity coordinative patterns of the sporting movement. This may be of importance not only to sport specific strength training but also, as we have seen, to sports injury rehabilitation.

An example: gymnastics

Perhaps more than in any other sport, the emphasis in gymnastics is on technique, and so strength training for gymnasts focuses greatly on coordination. This also applies to ring exercises, which are the most strength-oriented of all. Even when learning an 'iron cross', gymnasts therefore attempt to merge technique and force production, for the technical aspect of the exercise is considered very important. When we study an iron cross we are struck by the fact that not only the muscles that prevent the body from sagging – muscles that provide adduction – are highly active, but also muscles that produce the opposite effect must work hard to keep the ball and the socket of the shoulder joint together (Figure 1.22). To ensure stability, many muscles surrounding ball and socket joints such as the shoulder and hip have a fan-like architecture in which their various parts may produce opposite effects. This guarantees the stabilizing function of the muscles more effectively. Coordination round the shoulder joint is therefore always complex. To make such complex coordination part of the learning process at an early stage, an aid such as the 'spuso' may, for example, be used: a belt with an attached counterweight, allowing the amount of strength needed for an iron cross to be greatly reduced. This enables gymnasts to learn the required coordination for an iron cross before they have sufficient strength to perform the exercise with their full body weight.

Figure 1.27 Left: not only the muscles that provide adduction, but also antagonists such as the deltoid, are active in an iron cross. This means that an iron cross involves both strength and technique. To allow gymnasts to develop the technical aspect at an early stage, aids that shorten the moment arm and allow the exercise to be performed with less production of force are used.
Right: a 'Maltese cross' also includes a technical aspect, and so is often trained with less deployment of strength by using a 'spuso' belt.

1.4.3 *Contextuality, intentionality and transfer*

As we have seen, transfer from strength exercise to athletic movement is increased not only by considering the outward form of the movement, but also by ensuring that sensory factors are similar and by integrating anatomical aspects of the movement into the approach. However, these are not the only possible ways to increase transfer. The influence of motor control and motor learning patterns also plays a part in transfer. There has been plenty of research into this, but so far the resulting knowledge has scarcely been incorporated into thinking about sport-specific strength training. These patterns lie largely beyond the range of mechanical comparison between strength exercises and the sporting movement. If optimal use is made of these aspects during training, transfer will be substantially improved. Some aspects of motor control that affect transfer:

- When controlling and learning motor skills, the body focuses not only on how a movement is performed, but also on the function of the movement. The function of throwing a ball may be to make the ball hit a target at a given speed. Hitting the target is the intention of the movement: the future state the athlete wants to achieve. The learning body focuses closely on the goal of the movement, and hence will also be sensitive to the link between two movement patterns with the same goal.
- The body tries to be economical with its control capacity and so will seek to learn movement patterns that can be used in many situations. Conversely, movement patterns that can only be used in one situation are not of interest and so will be learned with difficulty.
- The body tries above all to adapt in response to stimuli that it perceives as new and that it cannot yet respond to appropriately. Stimuli that have been processed frequently are perceived as monotonous, and hence will lead to less adaptation and transfer. Such influences on transfer lie beyond the range of mechanical similarity. These aspects of transfer will be discussed in more detail in Chapters 4, 5 and 6.

Sport-specific strength training is often referred to as 'contextual strength training' so that the non-mechanical influences (sensory function, intention, generalization and so on) are also included in the training design strategy. Contextual training, in which the movement and the intention of the movement merge, as is more or less guaranteed in whole practice, thus attempts to further optimize transfer. Seen in terms of the theory of complex biological systems, however, it is obvious that such transfer can never be considered complete or universally applicable. There will always be influences (minor influences may result in major differences) that cause the transfer to proceed differently than expected.

The close link between strength and coordination means that no sharp distinction can be made between them in training practice. The boundaries between strength and other components of movement are blurred. No clear distinction can therefore be made between strength training and technique training. Especially when strength training is used to support a ball sport, it is important to be aware of this blurring.

The fact that strength training cannot be distinguished from technique training has major implications for the choice of types of training. The most difficult choices that coaches have to make are in the grey area between strength training and technique training. Should an exercise be approached as a form of strength or a form of technique? A good example is horizontal jump training for speed skaters with added resistance from an elastic band, so that the push-off requires more strength. Is this a form of strength training, and should the resistance of the elastic band therefore be continually increased? Or is it a form of technique, and is the resistance simply a means of improving coordination? If the main problem when performing these jumps is producing force, the exercise will be more difficult if the resistance is increased. But if the essence of a good horizontal jump

is technical performance, the exercise may be more difficult if the resistance is *reduced*. Coaches need to know what the secret of a good skating jump is, and design the type of training so as to optimize improvement in performance.
This link between strength and coordination is a major problem when measuring and testing strength. If the aim is to make the measurements as unambiguous as possible, the influence of coordination must be minimized. This can be done by making the movement as simple as possible. Muscle strength is then often measured isometrically (a static muscle action in which the muscle does not lengthen or shorten) or isokinetically (the muscle shortens at a predetermined speed). Such measurement is very different from what happens in the sporting movement. If an attempt is made to measure strength in a situation that is very like the sporting movement, so many coordinative and other factors play a part that the measurement becomes too complex and the result cannot be properly analysed. As a result, there are scarcely any good measurements that can predict level of performance in the sporting movement.

1.4.4 The purpose of this book

'Strength training is coordination training against resistance' is a fair attempt to define the purpose of this book, which is to emphasize the close links between the many different aspects of competition performance. In traditional approaches to strength training, these links are abandoned at an early stage of the thinking process. This book attempts to keep them intact for as long as possible. By incorporating knowledge from many different fields of research, an attempt is made to create a practical model of contextual sport-specific strength training that is guided by coordinative motor learning patterns and mechanisms. Of course, the book does not claim that the mechanisms occurring in training can be comprehensively described; but it does attempt to shift the boundaries of useful reasoning. The emphasis is on the term 'models', since even after studying this book coaching will still be partly 'knowledge' and partly 'art'.

1.5 Summary

A reductionist approach is unsuitable for understanding a complex biological system such as the training and adapting human being. Since complex biological systems do not behave in a linear manner, adaptation is less predictable than the reductionist approach, and above all training planning protocols, would have us believe. This applies not only to the physiological aspects of performance, but also to adaptations in the field of coordination. Because it is so complex, coordination is likewise non-linear. Movement patterns must be designed in a non-linear manner because movement has to be efficient, effective and flexible all at once. Linear, central control is too rigid to ensure this.

Training is often based on 'basic motor properties'. A distinction is made between these performance categories in an attempt to make training controllable and predict adaptations.

This would work if the basic motor properties met two criteria: (1) a basic motor property must be a separate entity, and it must be clear what is and is not part of it; (2) there must be automatic transfer of the quality in that property between the various movement patterns. In practice, however, these criteria are not satisfied. In sporting movements, strength is not a separate entity because contextual movement is made up of complex intermuscular patterns and hence has a restriction that is more complex than the mere sum total of the muscles' maximal capabilities. Speed is also so closely linked to coordination that neither criterion can be met.

Since movements do not occur in a linear manner and links must be created between related movement patterns, thinking about training needs to focus on transfer. Transfer to the sporting movement is a particularly serious problem in sport-specific strength training. Transfer occurs if the combination of sensorimotor factors in two movements is similar. Sensory information in strength exercises is very likely to differ from sensory information in the sporting movement. The main reason for this is that strength exercises are part-practice exercises. Whole-practice exercises more or less guarantee similarity of sensorimotor information; part-practice exercises do not.

The practice of sports strength training has traditionally been based on exercise physiology, bodybuilding and physical therapy. Coordinative transfer does not play a major part in any of these. Bodybuilding has led to the body-part approach (part practice *par excellence*), and physical therapy has led to an oversimplified approach to the problem of specificity that fails to take account of such phenomena as phase transitions. That is why sports injury rehabilitation makes use of protocols that are not very efficient in making the body robust for high-intensity movements.

2
Anatomy and limiting influences on force production

In Chapter 1 we explained why an approach in which strength is treated as a separate entity (one of the 'basic motor properties') is not workable in practice. Not only are there always links with other aspects of performance, but these links are an essential part of how athletes' bodies function. 'Strength training is coordination training against resistance'.

Force production is influenced at many different levels of the organism, including the central nervous system. The neural components of strength show that strength requires a major coordinative component. In the case of sport-specific strength training and rehabilitation it is important to identify some of the levels involved:

- muscle level: mechanical and anatomical aspects of force and power production;
- neuromuscular transition: the all-or-nothing principle of muscle stimulation;
- spinal-cord level: links that process outside influences to further adapt initial force production;
- central nervous system level: the brain (brain stem, cerebellum and cerebrum).

All these factors generate the complex interplay of coordination, part of which is regulation of force production. An interesting question here, and one that is of crucial importance to training, is which of the four levels truly limits maximal force production during athletic movements. There may be no clear answer to this question. Traditional strength training strategies focus on improving the qualities in the contractile parts of the muscle. More modern approaches put far more emphasis on the role of the central nervous system in force production. The resulting sport-specific strength training strategy is fundamentally different from the traditional one – so different that a synthesis of the two approaches, which would seem useful, does not occur and there is an almost religious controversy between members of the two camps. Yet such a synthesis is necessary in order to develop an effective system of sport-specific strength training that includes both transfer to the athletic movement and physiological adaptations.

2.1 Influences at muscle level

2.1.1 *Influence of sarcomeres arranged in parallel and in series*

The amount of muscle mass in the body is genetically limited. The benefit of large muscle mass (great strength) is counterbalanced by the fact that it impedes rapid movement and has a high energy cost – disadvantages that threaten the survival of the species. An optimal balance must therefore be struck between the costs and benefits of muscle mass.

Economical use of the limited amount of muscle mass has resulted in a complex, ingenious muscle architecture in which the available sarcomeres can in principle be arranged either in parallel or in series. All the sarcomeres arranged in parallel in a muscle are sometimes called the muscle's 'physiological cross section' (Figure 2.1). The greater this is, the stronger the muscle. Thick muscles are therefore stronger than thin ones. Sarcomeres can also be arranged in series (in a line); they then exert traction on each other. Just as in a chain, the whole series is as strong as its weakest link. So its total length does not affect the amount of force it can produce (Figure 2.2).

Whereas arrangement in parallel allows greater force production, arrangement in series allows greater speed of muscle action. If each sarcomere can shorten by a given amount within a given unit of time, the total shortening of the muscle within that unit of time will be the sum total of the shortening of the sarcomeres arranged in series. The longer the chain, the faster the whole muscle will contract. Muscle architecture can thus take the form of a structure that allows high force production, or one that is more suitable for rapid shortening (with less force) (Wilmore & Costill, 2005).

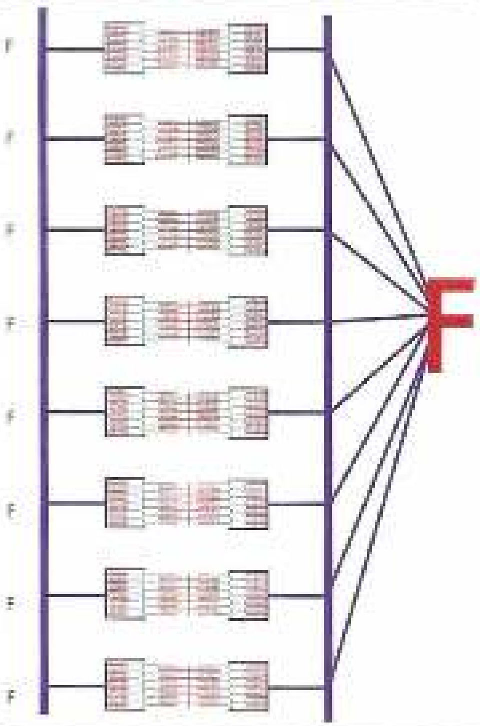


Figure 2.1
If sarcomeres are arranged in parallel, each sarcomere produces a force (F) that acts on the attachment points. The total force equals the sum total of the forces produced by the individual sarcomeres.

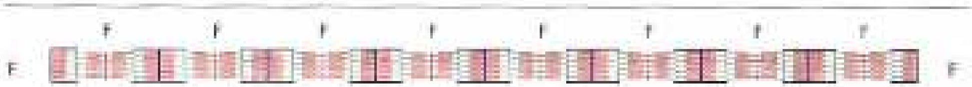


Figure 2.2 A contracting sarcomere produces force that acts on another sarcomere arranged in series. The force acting on the attachment point equals the force of one sarcomere.

The cross-section of a muscle depends not only on genetic predisposition but also on the load acting on the muscle. Training (including strength training) may cause the physiological cross-section to increase, and thereby increases the number of sarcomeres arranged in parallel. The customary view on the increase in a muscle's physiological cross-section is that the number of muscle fibres remains the same (motor unit: a quantity of muscle fibre composed of myofibrils innervated by the same motor neuron), but that the number of myofibrils in the muscle fibre increases – resulting in what is known as hypertrophy. This creates more links between actin and myosin proteins (the building blocks of the myofibril), so that each muscle fibre produces more force. However, animal studies have shown that the increase in physiological cross-section may also involve hyperplasia: an increase in the number of muscle fibres. Such an increase is difficult to measure in human beings (Gonyea, 1980; Sjöström *et al.*, 1991).

2.1.2 *Force/length and force/velocity characteristics of muscles*

It is easier to study isolated muscle fibres than whole muscles, and a number of key muscle properties have therefore been identified by studying muscle fibres in isolation. The properties thus discovered are a useful starting point for describing characteristics of the whole muscle. The two main properties of muscle fibres that can be measured in mechanical terms are the force/length (F/L) and force/velocity (F/V) characteristics. These two muscle properties are so important that training in many sports is mainly based on improving them. Particularly in sports whose aim is to cover a given distance as fast as possible, such as the sprint events in speed skating, running or cycling, coaches seek to improve the ratio between force production and speed of muscle action (force x speed of muscle action = power).

The F/V ratio in muscle fibres

Muscle fibres consist of overlapping actin and myosin filaments. The more they overlap, the shorter the muscle; the less they overlap, the longer. This means that the amount of force the muscle fibre can produce is not always the same. It is greatest when the filaments overlap the most (the 'optimal length'). If the fibres lengthen (outer range), the overlap is reduced and hence there are fewer 'cross-bridges' (links between actin and myosin filaments) between the fibres (Figure 2.3), so that the muscle can produce less force. Even if the muscle shortens considerably (inner range), the overlap between the fibres is reduced, for the actin filaments are no longer aligned but slide over each other and so the total overlap between actin and myosin is reduced. This reduces the amount of force that can be produced. The force/length chart that can then be produced for a muscle fibre displays a minimal/optimal/maximal curve (Figure 2.4; Bergmann *et al.*, 2006; Van Cranenburgh, 2002), which is often near the mid-range.

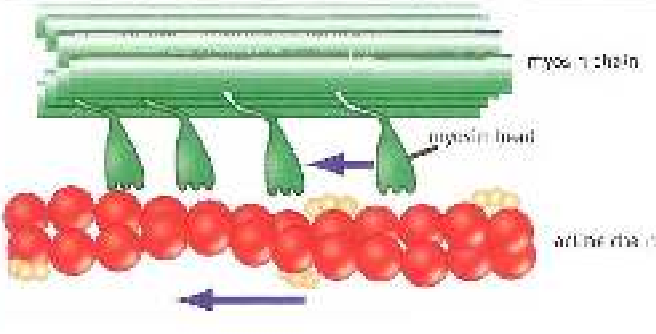


Figure 2.3
Sliding filaments: the actin chain lines up with the heads of the myosin chain ('cross-bridges'). In the powerstroke, the actin chain shifts in the direction of the blue arrow, and the muscle fibre shortens.

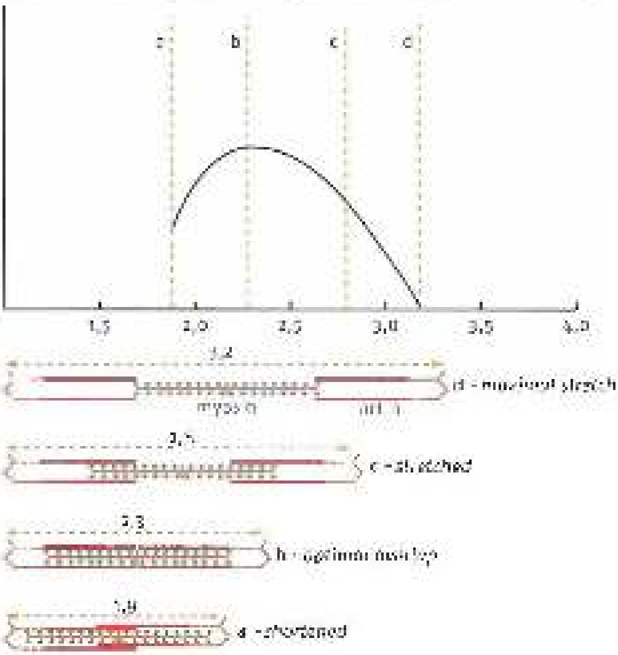


Figure 2.4
In optimal overlap the force production of the fibres is greatest. In the case of lengthening or shortening, overlap and force production are less. The chart shows an optimal length, which allows the highest level of force. When fibres are shortened from the optimal length, the actin filaments slide over each other (a).

The F/V ratio in muscle fibres

A muscle fibre is not able to produce high force and shorten rapidly at the same time. The mechanism can be compared to what happens in a tug of war contest. If there is no movement in the rope (i.e. both teams are equally strong), there are many hands holding the rope, and so plenty of force can be produced. If one team is much stronger, the rope can be pulled across to their side. However, this means that hands must constantly be released to take hold of the rope further along, and hence that less force can be produced. The faster the rope is pulled across, the less force can be produced. The same thing happens with muscle fibres. When they shorten quickly, many cross-bridges have to be released at once, and so less force can be produced (Figure 2.5). This means that high-force athletic movements are performed relatively slowly, whereas movements requiring little force can be performed quickly.

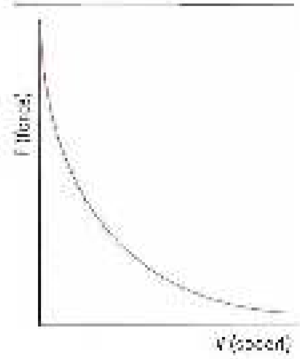


Figure 2.5
The muscle fibre can produce the most force at zero shortening speed, and can produce less and less force as shortening speed increases.

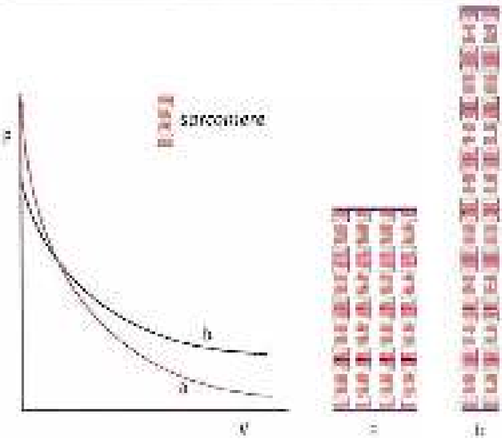


Figure 2.6 The total quantity of sarcomeres is the same for muscle a and muscle b, the F/V charts for muscle a and muscle b are different, because muscle a has more sarcomeres arranged in parallel than muscle b.

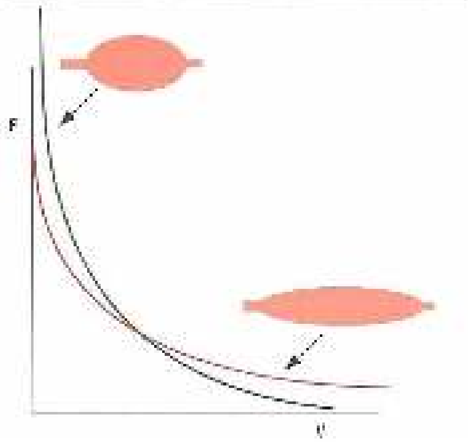


Figure 2.7 Thick muscles specialize in force production, and long muscles specialize in speed.

Long, thick muscles (*e.g. gastrocnemius*) are the most suitable for all muscle action types (see Figures 2.6 and 2.7). They can generate both force and rapid muscle actions. However, if all muscles were built like this, the body would have a huge mass, with resulting disadvantages. That is why muscles specialize. Some muscles (short, thick ones such as *gluteus maximus*) are specialized to produce high force but are not able to act at high speed. Others (long, thin ones such as *rectus femoris*) specialize in high shortening speeds without being able to produce much force (Figure 2.8).

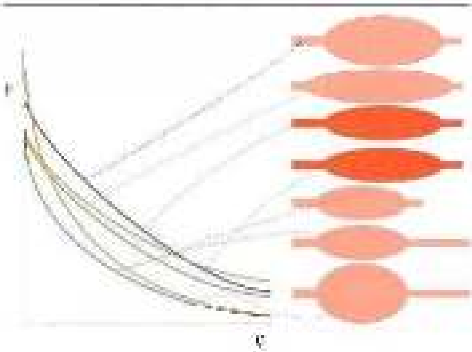


Figure 2.8
If the F/V characteristics of muscles are shown in a single chart, we get a 'zigzag' pattern. Clearly, such a pattern cannot easily be used to design a detailed training plan for strength and speed. The two dark coloured muscles look outwardly the same but have a different internal structure, and hence different F/V characteristics (see also Figure 2.9).

Pennate muscle structure and influence on F/L and F/V

The F/L and F/V characteristics of muscle fibres are important starting points for the design of training, but that is all they are – starting points. The characteristics of a single muscle fibre cannot simply be translated into the action of the whole muscle. Owing to the complex architecture of the muscle, the characteristics of individual muscle fibres may differ greatly from those of the whole muscle, and differently structured muscles will have different F/V and F/L properties. As a result, the contractile properties of muscle fibres are not indicative of the properties of the whole muscle. The differences in the architecture of the whole muscle thus make one muscle more suitable for one task, and another more suitable for another. An athlete's overall performance will therefore depend on the quality of intermuscular cooperation. If athlete A can produce more force on a bicycle than athlete B, this does not mean that athlete A will also produce more force on a rowing ergometer—simply because athlete A's intermuscular cooperation may well produce more force on a bicycle than on a rowing ergometer, and vice versa in the case of athlete B.

So muscles differ in their capabilities. Specialization in strength develops with large physiological cross-section, and specialization in speed with great muscle (sarcomere chain) length. However, such muscle specialization has been taken further within the muscle architecture. The specific suitability of muscles is further differentiated by 'parallel' and 'pennate' fibre arrangement.

In parallel-fibre muscles (e.g. *sartorius*), the direction of the fibres is parallel to the line of action of the muscle. The line of action runs from one attachment point to the next, and hence along the line along which the muscle's force is produced. In muscles that are structured in this way, the above analysis of thick and short muscles certainly applies even in relatively long muscles (Figure 2.9).

fibre orientation

Figure 2.9
Parallel-fibre muscle structure: the muscle fibres are parallel to the line of action.

In pennate muscles such as *extensor digitorum longus* or *peroneus longus*, the direction of the muscle fibres is at an angle to the line of action of the muscle. The angle may be as much as 30°. In some muscles it is somewhat less, and such muscles are somewhat less pennate in structure (Figure 2.10; Rozendal & Huijing, 1998).

Figure 2.10
Pennate muscle structure: the direction of the muscle fibres is at an angle to the line of action.

Pennate muscles are less suitable for rapid shortening than similar muscles that are parallel in structure, for if the muscle fibres shorten over a given length, the attachment points of the whole muscle will shorten over a smaller length (a pennate muscle can be compared to a pair of toothbrushes – if the brushes are pressed completely together, the heads will not come together). In addition, pennate muscles have a shorter in-series arrangement of muscle fibres than outwardly similar parallel-fibre muscles. This makes pennate muscles even less suitable for rapid shortening (Figure 2.11).

Figure 2.11
Pennate muscle structure. Maximal shortening of the muscle fibres results in limited shortening of the whole muscle.

tube
toothbrushes

So pennate muscle cannot shorten rapidly. The purpose of the pennate structure is to make muscles whose attachment points are often a long way apart suitable for great force production without the muscle requiring a great deal of mass. Owing to the diagonal structure of the muscle, the physiological cross-section (a measure of the maximal force a muscle can produce) is greater than in an equal-sized parallel-fibre muscle. It seems there are muscles that 'do not care' about how fast they can shorten, but above all want to be able to 'brace themselves' by producing plenty of force (Figures 2.12 to 2.14).

b

Figure 2.12
Two muscles with more or less the same volume and mass. Muscle a [illegible] has a parallel fibre structure and is suitable for rapid shortening with low force production. Muscle b (peroneus longus) has a pennate structure and is suitable for bracing with plenty of force without necessarily shortening.

Figure 2.13
Pennate (a) and parallel-fibre (b) muscle structures with corresponding PCSA [illegible]. The values are presented in relation to one another.

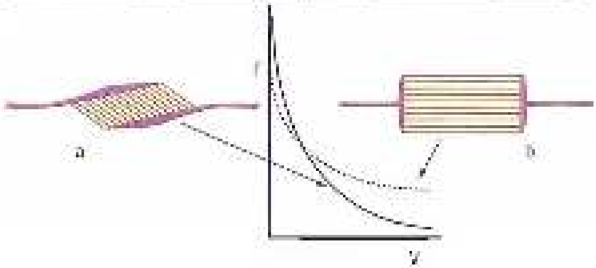


Figure 2.14
Pennate (a) and parallel-fibre (b) muscle structures with corresponding F/V characteristics. The values are presented in relation to one another.

Other muscles that exchange speed for strength

Apart from pennate structure, there is another way in which muscle architecture can exchange speed of muscle action for force. Muscles do not consist entirely of muscle fibres and tendons, and there may be a great deal of collagen tissue between the muscle fibres. Like the tendons, this belongs to the passive parts of the muscle. In these muscles the muscle fibres are not attached to each other, but to the passive elements, meaning that the fibres do not exert force on each other. The strength of one fibre is conducted past the other fibre, even though the fibres appear optically in series. The muscle can thus be long and thin, even though the total cross-section of the muscle is relatively large and the in-series arrangement relatively small. Examples of such muscles are the *rectus abdominis* and *erector spinae* (Figure 2.15). Optically they are long and thin, but their F/L characteristics are more typical of short, thick muscles. This means they are not suitable for rapid shortening. It also means that they are very ability to produce force at their optimal length, but rapidly lose their strength when they shorten. That is why sit-ups are performed with relatively low force production. The same applies to the *erector spinae* muscles, the main consequence being that muscles rapidly lose their strength in trunk flexion (for example in the speed skating posture). That is why trunk flexion is taboo when carrying barbell weights, for the abdominal and *erector spinae* muscles cannot perform their protective function properly in that posture.

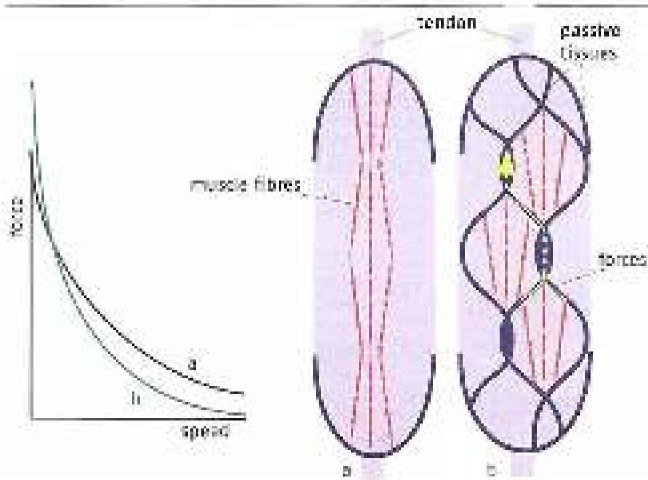


Figure 2.15
A parallel-fibre muscle (a) compared with the structure of rectus abdominis and erector spinae (b). The forces (dotted yellow arrow) are diverted through the passive tissues (dark blue), thereby increasing the physiological cross-section. As a result, rectus abdominis and erector spinae have F/V characteristics that are somewhat more similar to those of a pennate muscle.

Implications of F/V and F/L for contextual movements

The population of muscles in the human body is like the population of a zoo. A zoo with just one species of animal will not be very successful. Diversity of species is the key to success; and in a motor skill, too, diversity of muscle structure and muscle function is the key to successful movement.

Complex structure of muscles with differing characteristics is a precondition for efficient and stable movement. Efficient movement calls not only for the generation of kinetic energy by muscles but also for the transport of that energy through the body, as happens in pennate biarticular muscles (see also Section 5.2). Stability of movement likewise requires a differentiated muscle structure in the body. If all its muscles had a similar structure, and hence could make a similar contribution to movement, there could be no fixed cooperation between muscles, for there would always be several combinations of muscles that would be effective in a given movement pattern. Since the differing structure and suitability of muscles greatly reduces the number of effective combinations within a movement pattern, there are only a limited number of ways to move efficiently. Diversity in muscle architecture thus creates generic building blocks for movement whose efficiency make the movement stable and provide a generic basis for related movements (see Section 7.3.2).

The zoo metaphor can be taken even farther. A zoo director who decides that all the animals should be fed on the same diet will run into trouble. The rabbits will die if they have to eat meat, and so will the lions if they have to live on lettuce. Training all the muscles in the same way will likewise cause problems. Muscles that have to perform unsuitable work will not function in an optimal, i.e. sufficiently differentiated, manner. In the zoo, guinea pigs will be fed grass, dolphins fish and lions meat. Good coaches and sports physical therapists similarly make sure to load muscles so that they perform work suitable to their structure and function, in useful movement patterns.

Additional information

Muscles have force/velocity and force/length characteristics. This means that rapid shortening at a less than optimal length is always at the expense of maximal force production and stability round the joints. If muscles are to be used at the limits of their potential, it is undesirable that they should shorten very fast or end up a long way from their optimal length. To minimize this, the general principle is to distribute the range of motion over as many joints as possible, so that this undesirable effect is limited and the joints (and hence the muscles) remain as much as possible within a stable range of motion. An example is a tennis service, in which the muscles may be fully abducted. In this position, some muscles are extremely shortened (deltoid, supraspinatus), whereas others are extremely lengthened (*pectoralis major*, *latissimus dorsi*, *infraspinatus*). This means they cannot provide optimal accompaniment for the service movement. By laterally rotating the scapula and bending the spine sideways, the amount of abduction required in a tennis service can be reduced to about 90°, which places the shoulder muscles in their mid-range and enhances striking power and stability. Such a strategy of distributing a range of motion over as many joints as possible is largely self-organizing, and of course affects how, for example, sport injury rehabilitation exercises should be designed (Figure 2.16). The habit of eliminating compensatory movements as much as possible in rehabilitation and attempting to improve isolated ranges of motion may therefore not be such a good strategy. It may be better to allow compensatory movements and gradually remodel them during rehabilitation into an effective strategy of distributing the movement over several muscles.

Figure 2.8
The final position in a tennis service is largely self-organizing, so that muscle properties distribute the movement over several joints and the arm does not have to be in an extreme abduction position.

2.1.3 *Moment arms*

The torque that muscles can produce in relation to a joint depends not only on the force produced by the muscle, but also by the muscle's moment arm (the shortest distance from its line of action to the axis of rotation of the joint) in relation to the joint. In many muscles, this is not the same in every joint position. There is clearly an important relationship between the muscle's F/L characteristics and its (changing) moment arm. In the case of monoarticular muscles, this relationship is fairly evident. A given joint position goes with a given muscle length and a given moment arm. In biarticular muscles (muscles that run over two joints whose axes of movement are a long way apart and parallel to each other) there is no such simple relationship. In each position of the two joints involved, the length of the muscle may vary according to the position of the other joint.

A number of conclusions can be drawn from the properties of muscle moment arms:

- The changing size of the moment arm of a muscle means that the effectiveness of muscle action is partly, or largely, determined by the moment arm. In the case of monoarticular muscles this means that the muscle can only function optimally in a particular angle of the bridged joint. In the specific case of biarticular muscles, this means that an effective technique is a well timed combination of the moment arms of both joints, with the optimal length of the muscle maintained as much as possible. This means that biarticular muscles operate more or less isometrically during contextual movement, which corresponds well to the potential of their pennate structure (Van Ingen Schenau & Bobbert, 1988; Jacobs & Van Ingen Schenau, 1992).

- The moment arms of the muscles in relation to the joints that they bridge will almost always be far smaller than the moment arms of the external forces. The strength of the muscles in overcoming an external resistance will therefore usually be much greater than the force produced by the external load.
- The moment arms of the external forces are highly variable, whereas those of the muscles are much less. Good movement technique therefore primarily means being able to deal properly with the moment arm of the external force. In single-leg jumping from a run-up, for instance, this means that it is more important to control the moment arm of the ground-reaction force in relation to, say, the knee joint than to create opportunities to produce considerable power. A single leg take-off from a run-up is therefore performed with the take-off leg extended as much as possible and with the trunk, if possible, parallel to the take-off leg. This minimizes the moment arm of the ground reaction force in relation to the various joints. Production of force from positive (concentric muscle action) work plays no part in a single leg take-off from a run-up.

2.1.4 *Elastic properties*

Halfway through the previous century, Hill (1950) proposed an eponymous behavioural model for muscles. Unlike an anatomical model, the Hill model does not show which structures are located where in the muscle, but only how a muscle behaves during activation (Figure 2.17).

The Hill model distinguishes between

- CE: the contractile element, the part that ensures muscle action
- SEC: the serial elastic component, passive elastic parts that are an extension of the CE parts
- PEC: the parallel elastic component, passive elastic parts that are parallel to the CE parts.

Muscles can vary greatly in the quantity of passive parts (SECs and PECs) in the muscle. Muscles such as *semimembranosus* and *semitendinosus* owe their names to the large quantity of passive tissues within the muscle–tendinous unit. In contrast, *gluteus maximus* has relatively little passive tissue.

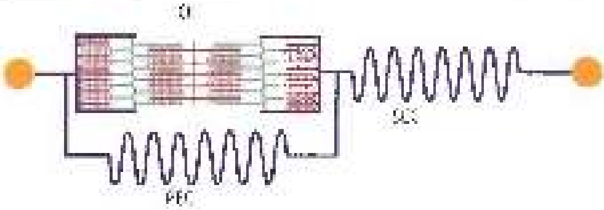


Figure 2.17
The Hill model is a behavioural model for the action of muscles. It can above all be used to draw important conclusions about how muscles work.

The muscle's elastic components have a major impact on the muscle's force production. In this model, the contractile strength of the CE components do not act directly on the attachment points of the muscle, but through absorption of force by the SEC components. The system of sliding filaments (actin and myosin) is not very appropriate for a gradual flowing development of power. In a sense, muscle action jumps from one position to another (Van Ingen Schenau & Van Soest, 1996). The SEC component more or less dampens the spasmodic character of the muscle action and ensures that the development of force at the attachment points is 'smoother'. This is very important to the regulation of force in contextual movements.

The SEC components have a particularly important function in resisting opposing external forces. The elastic components act not only as shock absorbers for muscle action in the CE parts, as well as for the opposing forces. This makes it possible to control movements in an unfamiliar, changing environment, just as a car's shock absorbers make it possible to drive on a poorly maintained, unpaved road.

Even more important than shock absorption is the function of SEC elements in storing the energy of opposing forces during elastic stretch. This energy is then used in the discharge of elastic stretch to produce force in the opposite direction and shorten the muscle. An important factor here is that this storage and return of energy will only be optimal if the CE parts remain as isometric as possible. The elastic character of muscles is one of the movement apparatus's most important energy-saving characteristics.

Additional information

Elasticity (reactivity)

Apart from the types of muscle action in which muscle fibres lengthen and shorten, elastic muscle use is also a part of contextual movement. This form of muscle action differs greatly from concentric and eccentric action. Elastic muscle use is predominant in many sports. Coaches seldom have a good understanding of what exactly happens during an elastic muscle action. Even researchers have still not sufficiently identified this muscle action, especially in contextual movements. This is because elastic muscle use is very hard to interpret, owing to the extreme speed and the small range of motion of the muscle action. Invasive techniques in which sensors are placed in tendons and muscle bellies (for example in running turkeys) can measure elasticity somewhat more accurately (Roberts et al., 1997). However, for a true understanding of elasticity, we must still rely on very simplified models.

The Hill model and elasticity

The main model for explaining reactivity is the Hill model. What is important in the model is that the passive tissues connecting the attachment points are not seen as static structures but as elastic structures that can change in length. The Hill model is a one-dimensional representation of reality. However, to identify elastic action properly we need a three-dimensional model that indicates in detail the direction of the action of force within the muscle (Williams et al., [illegible]). For muscle groups such as *triceps surae*, a three-dimensional model is not really sufficient, and a four-dimensional one that includes the time factor may in fact be needed. The addition of the fourth dimension is meaningful, for there is a very strict time limit on *triceps surae*, within which the loading and unloading of structures has to take place. Take the following example. A solid rubber ball made of optimally elastic material will bounce in free fall almost as high as it falls before bouncing, but a cube made of the same rubber will hardly bounce if dropped from a great height. The shape and manner in which elastic energy is transmitted three-dimensionally in the object evidently has considerable influence. If the rubber ball lands on a surface whose bounce time is longer than that of the ball (the fourth dimension) – for instance, a springboard that is used for gymnastics – the ball will not as bounce as high as when it lands on a hard surface. A simplified Hill model is sufficient to understand the basis of elasticity. The PEC (parallel elastic component) parts are omitted, and there is only a SEC (serial elastic component) and a CE (contractile element) between the attachment points. The behaviour of both components in elastic muscle use is then described (Figure 2.18).

Opposing forces

Most rapidly performed movements generate large external forces that load the muscle with eccentric torque and so try to move the attachment points further apart, e.g. during foot placement when running and the take-off in a gymnastics vault. The opposing eccentric loading forces may also be caused by muscles in the body – for instance the stretch on abdominals caused by action of leg and pelvic muscles when throwing, or the stretch on forearm muscles when hitting a forehand. The Hill model can be used to identify how a muscle deals with such forces. Depending on how the CE parts of the muscle behave, the SEC parts will or will not be stretched. A muscle may behave superelastically in response to large external forces. How great the elastic effect of a muscle is will depend on the size of the external force that is trying to stretch the muscle. However, this force may not exceed the maximal isometric force in the CE parts. If the opposing torque requires more force than the muscle can produce isometrically, the CE part will lengthen at the expense of the stretch in the SEC parts (Figures 2.19 and 2.20). Elasticity and isometry therefore belong together. It is therefore very disappointing to see how carelessly elastic muscle action is described in much of the literature on training. The sloppy use of such terms as eccentric-concentric, plyometric and so on presupposes eccentric action in the muscle fibres, and fails to provide a clear picture of what elasticity actually is. Before the criteria that exercises must satisfy in practice can be identified, the behaviour of the CE parts must be accurately described.

Application of reactivity in jumping

In bounding jumps the height of maximal jumps is not achieved by concentric (motor or positive) muscle action, but by elastic muscle action. Elastic muscle use and concentric explosive muscle use are completely different. This means they are aspecific to one another. There is therefore little point in explosive sport athletes practising elastic muscle use in order to increase the speed of their explosive movements. A speed skater who wants to push off faster or a swimmer who wants to leave the starting block faster will not gain much benefit from practising bounding jumps. There are barriers between these two types of muscle use, and so there will be little transfer of training between them. In contrast, throwing (javelin-throwing, pitching and so on) at maximal effort is very much based on elastic muscle action, and so there is no eccentric-concentric action in the muscle fibres when loading and unloading elastic energy during throwing. A good (and fast) throw is performed with the muscle fibres acting isometrically during the phase in which the musculo-tendinous units lengthen and shorten through stretching of the elastic parts. It is elasticity that transports kinetic energy from the trunk to the arm, just as it conveys energy from the handle to the end of a whip. In this process of energy transport through elasticity, the speed of movement in the joints gets larger and larger because the mass to be moved gets smaller and smaller, from the trunk (large mass) to the hand (small mass), again just as in a whip (Figures 2.21 and 2.22; van Ingen Schenau & Bobbert, 1996; LaStayo et al., 2003).

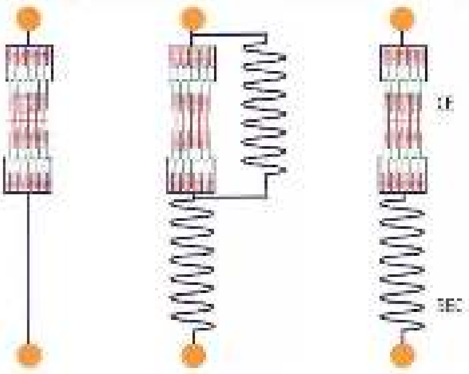


Figure 2.13
Left: a classic illustration of the tendon-belly-tendon complex; center: the extensive Hill model (with PEC parts); right: the simple Hill model, which is suitable for a description of reactivity.

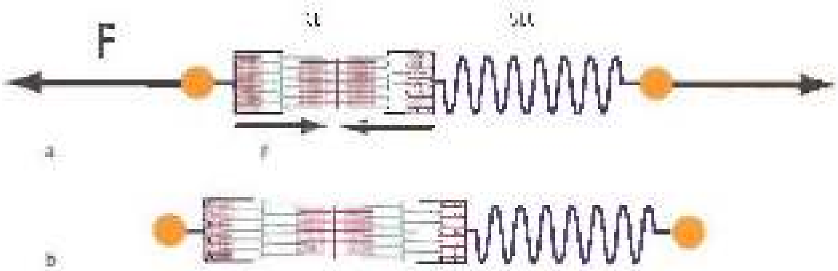


Figure 2.14 The opposing force is extremely large. The CE parts are not strong enough to provide sufficient counterforce, and the whole muscle is stretched (a). Lengthening of the CE parts reduces the stretch of the SEC parts (b). This occurs, for instance, when jumping down from a great height and coming to a stop by landing with bent knees.

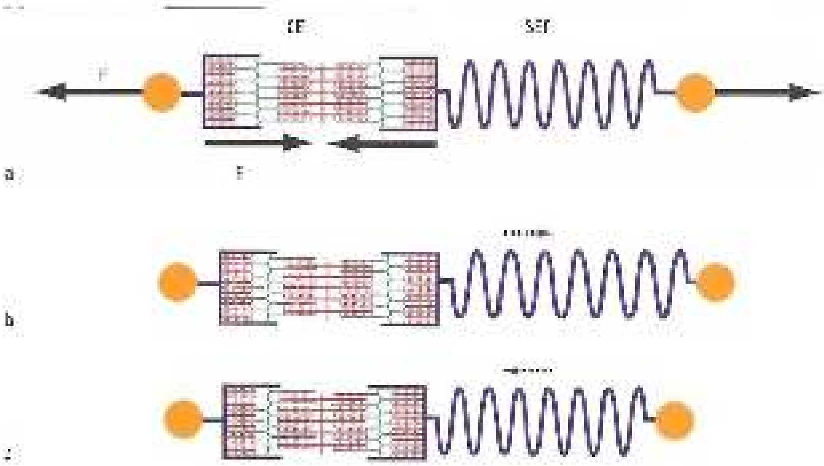


Figure 2.15 The opposing force is precisely as large as the CE parts can cope with (a). The CE parts tense to the maximum, and the SEC parts are stretched and loaded with a great deal of energy (b). Once the external force ceases to be applied, the energy in the SEC parts is unloaded and the muscle shortens vigorously (c). This occurs, for instance, when jumping down from a limited height and performing a bounce on landing.

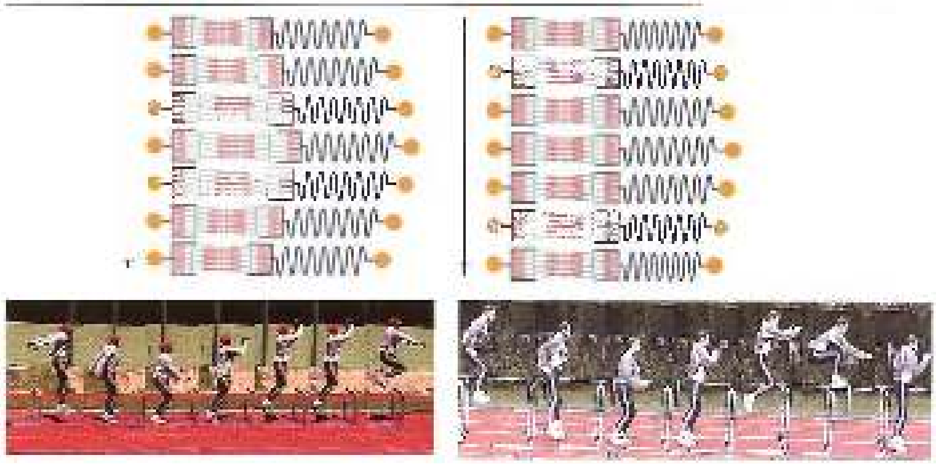


Figure 2.27 Left: the muscle as motor, and below it a jump in which the muscle action is mainly eccentric-concentric. Right: the muscle as a superelastic band, and below it a bounce in which the muscle action is mainly elastic. If the change in knee angle in the stance phase exceeds 20–25°, there is little opportunity for short contact time and elastic muscle action.

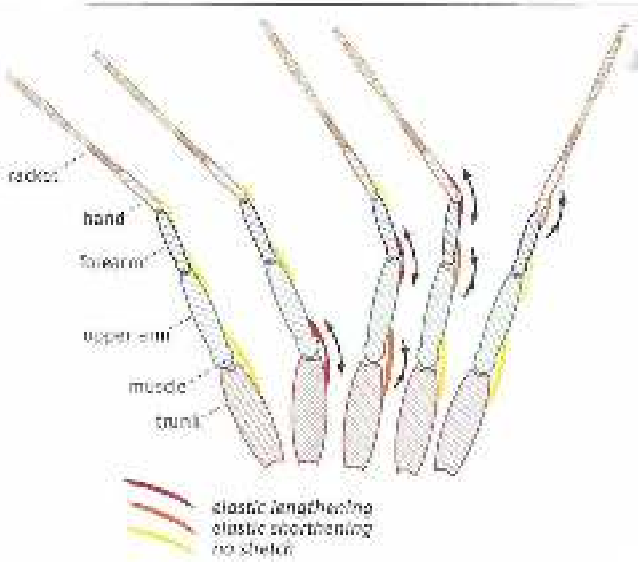


Figure 2.28 When throwing and hitting (in this case a tennis service), kinetic energy is transferred by elasticity from a large mass to a small one (from the trunk to the arm to the hand). The smaller the moving mass, the greater the speed of the moving segment. When throwing this can generate considerable angular velocity in the wrist joint.

Additional information

Athletes can hang from a high bar either actively or passively. The way in which the muscles can produce the necessary force is explained by the extended Hill model. Active hanging involves the FE parts, and passive hanging the PEC parts (Figures 2.23 and 2.24).

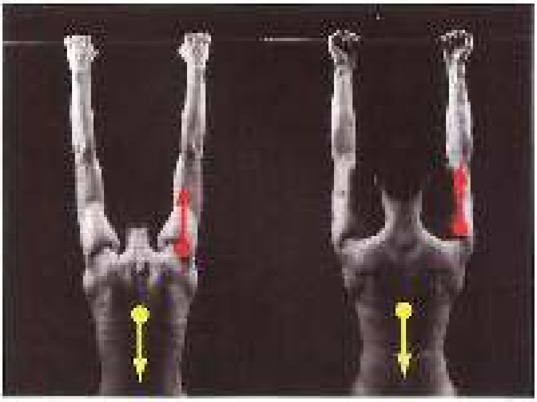


Figure 2.23
In passive hanging (left) the body weight is the same as in active hanging (right). The muscles that have to ensure a balanced posture must therefore produce the same amount of force in both cases (red arrow).

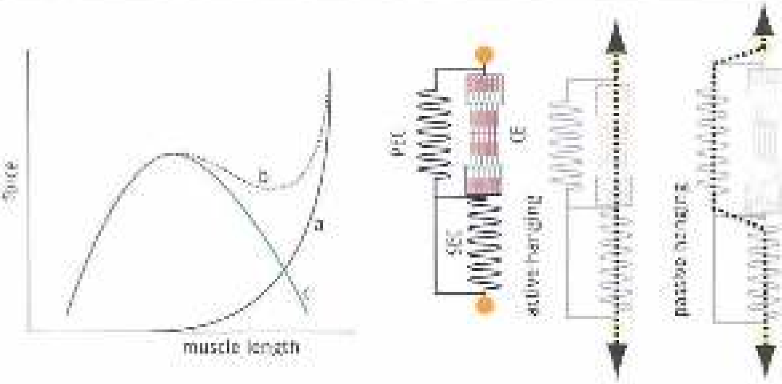


Figure 2.24 Left: the force-length ratio of the whole muscle; right: the extended Hill model in active and passive hanging.

The Hill model can be used to explain in detail the difference between active and passive hanging. Curve C in Figure 2.24 shows the FL ratio of the active parts of the muscles (the CE parts). This is a minimal–optimal–maximum curve, in which the minimum shows the greatest possible shortening of the muscle fibres (with little overlapping of the actin and myosin filaments), the maximum shows the muscle fibres in their maximal stretched position (again with little overlapping of the actin and myosin filaments) and the optimum shows the situation in which the filaments overlap as much as possible.

Curve A shows the force production of the PEC parts when they are stretched. Curve B – curve A + curve C – shows the total force curve when the whole muscle lengthens. The force of the CE parts is reduced, and that of the PEC parts is increased (the force passes through the PEC parts rather than the CE parts).

There are thus two ways of supporting and hanging: active (CE) or passive (PEC). A midway solution is scarcely possible, since the total amount of force that the muscles can produce in a state that is between active and passive hanging is lower.

2.1.5 *Muscle slack*

The Hill model shown in Figure 2.17 is not really correct. It assumes that the various parts of the muscle are neat extensions of its line of action, and hence that any shortening of the muscle will bring the attachment points closer together.

Figure 2.25 is a better picture of reality. Muscles are not located in the body 'all ready for action'. Before muscle action begins, the components of the muscle are not neatly aligned between the attachment points; muscles hang in the body like slack ropes, and must first be basically tensioned and then tautened before effective muscle action can occur. A muscle is like a guitar string. A guitar string that is attached at both ends but is still slack is useless – it must be tautened before any sound can be produced. The degree to which a muscle's operative tension must be increased in order to become taut is known as 'muscle slack' (Figure 2.26) – a key phenomenon in determining performance. During athletic movements there is very little time to perform the movement, among other things because of the relatively short acceleration path of the body or object. In many movements, building up the operative tension of the muscles is a problem, and the optimal level of tension will only be achieved some time after the movement commences. This means that the first part of the acceleration is largely ineffective, limiting the performance. The speed at which muscles can build up their tension (overcoming muscle slack) is therefore usually more important to performance than the amount of force they can eventually produce (Figure 2.27).

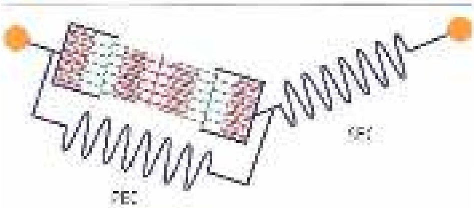


Figure 2.25
A more accurate illustration of the Hill model, with the muscle hanging between the attachment points rather like a slack rope. Before the muscle can exert its force on the attachment points, the CC and SEC parts must first be positioned in a straight line between the attachment points.

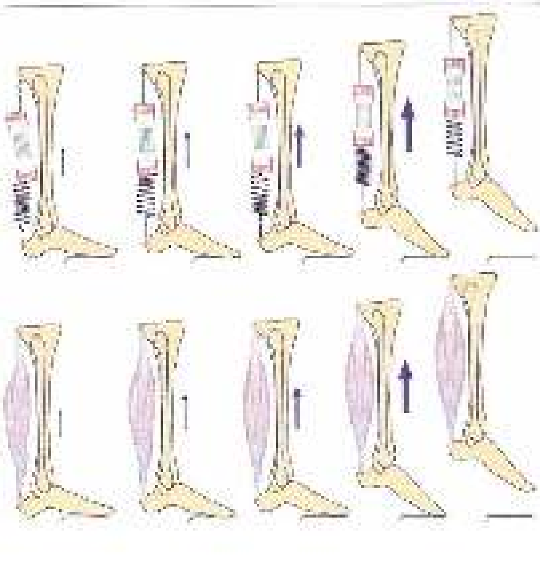
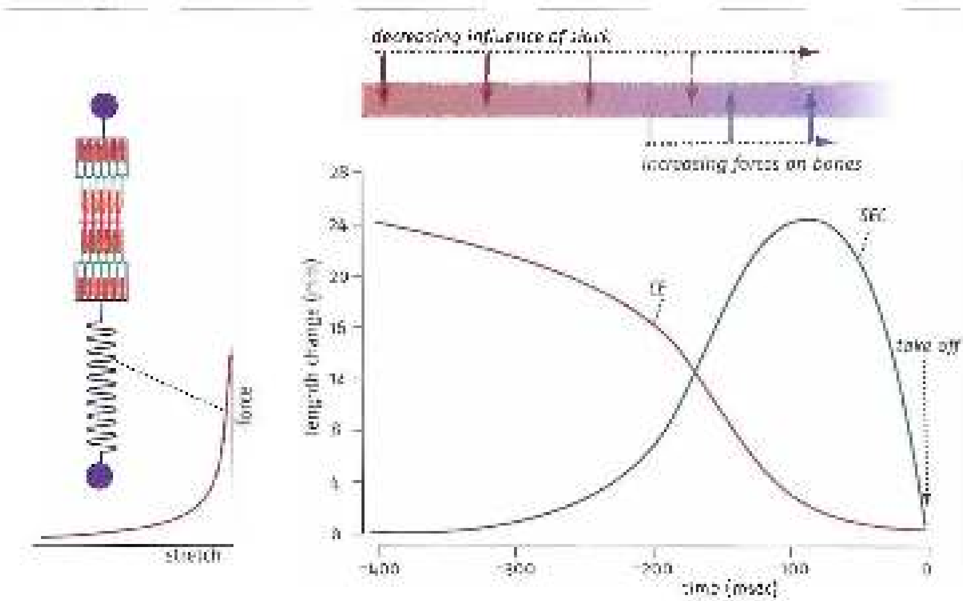


Figure 2.26
Muscle slack. At the start of the movement the muscle is still rather slack. It is only about halfway through the take-off that the muscle can produce peak force.



a b

Figure 2.27 (a) When SEC components are about to be stretched, little force will be needed for the first part of the stretch, but when they stretch further the required force will rapidly increase.
(b) Vertical squat jump. Before a muscle can actually shorten, not only must its actomyosin bridges be aligned, but the series of elastic components must be brought up to length so that they are sufficiently stiff and can actually exert force on the muscle's attachment points. This compliance of the series elastic components increases muscle slack (see Fukashiro et al., 2005).

The influence of muscle slack in concentric muscle action

In explosive muscle action the contractile part of the muscle shortens. Force is built up very slowly, and hence is a performance-limiting factor. Athletes intuitively search for ways to reduce the problem. In some situations (such as speed skating) this is very difficult, and can only be done by training the right technique for many years. In other situations it can be done by making a countermovement.

In a countermovement, a movement in the opposite direction to the intended movement is first made. The countermovement moves the attachment points of the muscles apart and pretensions the muscle. Countermovements may be useful in preparing for the movement, provided there is enough time to make them. In some sports, however, there is not much time to make a countermovement (e.g. a backswing in tennis or hockey, a kicking movement in soccer, a rapid block in volleyball or a rugby line-out jump that cannot be predicted by the opponent); and in other sports there is really no time at all for a countermovement, which will always impair performance. Swimmers would lose time if they had to move backwards before accelerating forwards at the starting signal. Similarly, baseball batters would be sure to miss the ball if they had to swing the bat a long way back before their muscles had sufficient tension for the batting movement. This means they must create pretension, which they often do by making small swinging movements with the bat. This makes it easier to increase the tension in the muscles of the shoulder girdle and the arm. We can see the same thing in a cat that is about to jump – it increases the pretension in its paws by making small stepping movements before jumping.

Additional information

Cocontractions (simultaneous action of agonists and antagonists) perform an important stabilizing function, which protects the joints from being damaged by external forces acting on them. However, it would be atypical of nature if cocontractions served just one purpose (i.e. only stabilizing the movement pattern). They also play an important part in regulating pretension and muscle slack.

Athletes are often unable to limit the range of the countermovement sufficiently because they lack the necessary mastery of the pretension technique. They are unable to generate sufficient pretension in the muscles by means of cocontractions and then stop the action of the antagonists at the point when the movement is to be made, so that the agonists can start the movement with a high level of initial tension. The execution and timing of this technique are fairly complex, and have to be learned.

In the literature, differences in the performance-enhancing effect of countermovements (e.g. when doing a squat jump) tend to be attributed to differences in how types of muscle fibre are distributed in individual athletes. However, it is questionable whether the mechanical properties of muscle fibres can fully account for the differences between the various types of athletes – the differences between the effect in endurance and explosive sport athletes are simply too great. Part (even most) of the explanation may therefore be that endurance athletes may 'by their nature' have somewhat poorer motor skills than explosive sport athletes and have less mastery of complex coordination, such as explosive acceleration from the pretension created by cocontractions (see also Section 5.3.2).

Muscle slack and its relationship to cocontractions are among the most performance-determining factors in sport. A great deal of attention should therefore be paid to them, and there should even be exercises (as in gymnastics) that are solely designed to learn effective cocontractions and hence body tensioning.

The influence of muscle slack in elastic use of muscles

The opposing external force that seeks to lengthen the muscle from outside plays a key role in elastic muscle action. It also ensures that the slow build-up of force owing to muscle slack is much less of an inhibiting factor. However, pretension also has an important part to play here, for the opposing forces always develop very rapidly. There is very little time to respond, so the muscle must already be sufficiently pretensioned at the point when the opposing force begins to act. Building up pretension and timing the absorption of the opposing forces (e.g. the moment of landing in a bounce) is a coordinative challenge that requires plenty of practice. We are not talking here about timing in tenths of seconds, but timing that is accurate to hundredths of seconds (Figures 2.28 to 2.31).

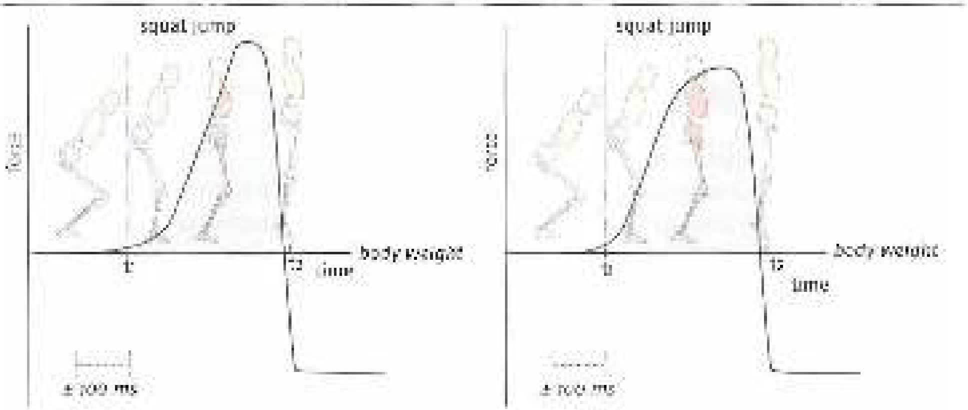


Figure 2.28 Force production in a squat jump (t1 = the start of upward acceleration, t2 = the moment of leaving the ground). Left: only at the end of the squat jump is there sufficient muscle tension for an effective take-off. The height of the jump is limited, even though peak force is high.
Right: a jump with pretension. Force production increases very rapidly, and total power (the content of the curve (graph)) is great. The jump is higher, even though the peak force production is lower.

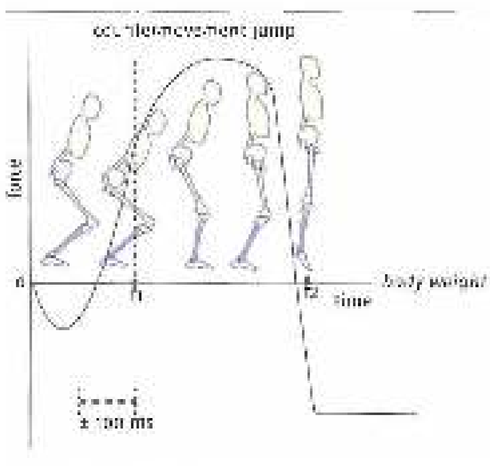


Figure 2.29 The effect of a countermovement in building up force for an explosive jump. At the start of upward acceleration (t1), the countermovement is already exerting great pressure on the surface, so the jump is higher than one without a countermovement.

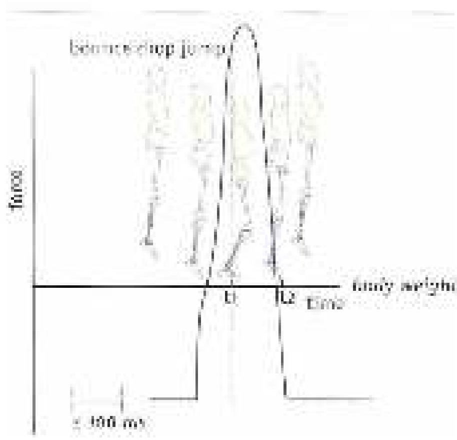


Figure 2.30 Muscle slack is avoided because the muscles are tightened by the opposing force when landing in a bounce.

Managing muscle slack is one of the most performance-determining factors in sport, and it is remarkable how little coaches and sports physical therapists know about it. Yet it does play a key role in fundamental research into the mechanical properties of muscles and muscle fibres

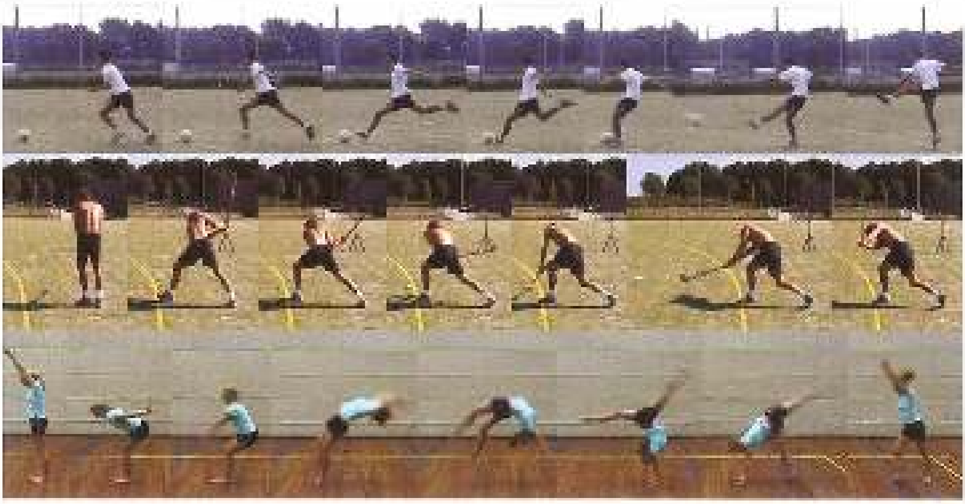


Figure 2.[illegible] A long backswing in soccer or hockey is not always effective, and may need to be reduced by controlling muscle slack. The movement will then be performed more quickly. A back handspring involves a large range of motion, and too much shortening of muscle slack would result in poorer performance.

The lack of attention to muscle slack within training concepts has led to a number of persistent misconceptions, and hence to training methods that may actually be counterproductive. The main misconception concerns the operation of the 'stretch shortening cycle', and particularly the associated ideas about prestretch. It is assumed that if an explosive concentric muscle action is preceded by a strong eccentric muscle action, the subsequent concentric muscle action will be more powerful. For example, when practising the technique for a single-leg take-off or a backswing for a hitting or throwing movement, athletes first attempt to make a rapid, large movement in the opposite direction, on the assumption that the bigger the range of the movement, the more effective the following muscle action will be.

Recent developments in movement technique in many different sports have shown this assumption to be false. In tennis, for instance, the backswing in a forehand or a service has been further and further reduced (to gain time) without any adverse impact on the stroke. In single-leg jumping, limited knee flexion in the first part of the take-off is a sign of quality. Elite volleyball players do not flex their knees more than poorer players at the take-off for a smash, but less. The best sprinters display considerable stiffness in the ankle joint (no dorsiflexion) when accelerating and when running at speed. In other words, the best athletes make smaller countermovements during athletic movement. Their management of muscle slack may account for this. The countermovement involves only limited eccentric action of the muscle fibres. Lengthening of the muscle is mainly due to the slack muscle becoming taut. Once tensioned, the muscle can produce plenty of force from the very start of the concentric muscle action, and so enhances performance. If the opposing forces to be absorbed by the countermovement are great, such as knee flexion in a single-leg jump with a run-up, there will be further stretching of the passive parts of the muscle after the muscles are aligned and tensioned, which may substantially improve the take-off. However, this additional stretch will not be increased by a larger countermovement. On the contrary, a large countermovement will make it technically very difficult to generate elastic stretch in the passive parts of the muscle. By limiting muscle slack, a small countermovement will allow storage of elastic energy (see Section 5.6.3 for a discussion of eccentric muscle action versus elastic stretch when performing a countermovement).

Training that is based on large countermovements aimed at will therefore increase muscle precension will therefore only lead to longer muscle slack and, in sports that require movements to be made under time pressure, poorer performance. What this means in sport-specific strength training is that all countermovements should be avoided. In sports physical therapy it means that rehabilitation should be based on exercises that enhance the protective function of muscles by reducing muscle slack.

In conclusion, muscle structure has far-reaching implications for coordination and force production. These properties are largely determined by muscle architecture. Since this may vary, muscles have different force/length and force/velocity characteristics. These differences in properties form an important basis for intermuscular coordination. Besides force/length and force/velocity characteristics there are mechanical properties such as muscle slack, which may be less dependent on muscle architecture but also help determine force production by muscles.

2.2 Neuromuscular transition

2.2.1 *The size principle*

The contractile parts of a muscle consist of a collection of motor units (motor unit: a quantity of muscle fibres made up of myofibrils innervated by the same motor neurone). The muscle fibres within a motor unit have identical biochemical and physiological properties (Brooks *et al.*, 1996). On the basis of these properties, muscle fibres can be divided into types: histochemically into Types I, IIa and IIb, and mechanically into slow ST (slow-twitch) and fast FT (fast-twitch) muscle fibres; the FT fibres can be further divided into FTa, FTb and FTc types. ST motor units contain clusters of only 10 to 180 muscle fibres, whereas FT motor units contain 300 to 800. However, this is a rather arbitrary classification – there is in fact a gradual transition between the various types of muscle fibre, and a distinction could thus also be made between, say, five types.

ST fibres are slowly contracting muscle fibres with oxidative energy production. Slow oxidative muscle fibres are suitable for long-duration, low-intensity efforts, and are almost immune to fatigue. It is noticeable that they have thicker Z lines (protein molecules which form the link between the sarcomeres) than FT fibres, which may mean that ST fibres are good at absorbing opposing external forces, for example in the elastic muscle action of, say, *soleus*.

FTa fibres are rapidly contracting, fatigue-resistant muscle fibres with an oxidative and glycolytic energy production, so that they can have both an anaerobic and an aerobic capacity. This means they can contribute to both low-intensity aerobic efforts and efforts at a higher-intensity anaerobic level.

FTb fibres are even more rapidly contracting fibres that have glycolytic energy production and are rapidly fatigued. Their high concentration of glycolytic enzymes makes this type of muscle fibre especially suitable for anaerobic energy production.

FTc fibres are seldom found in muscle fibre (less than 3% of the total), and little is known about their properties.

ST fibres thus build up their force slowest (under isometric conditions), and FTb fibres fastest. This is because of differences in the molecular structure of the myosin heads in the various types of fibre. FTb fibres also have a shorter relaxation time than smaller fibres. ST fibres thus contract more smoothly and slowly than FT fibres, which contract more quickly and spasmodically. As a result, FT fibres need a higher rate coding than ST fibres to achieve their maximal level of force, and they have a higher stimulus threshold than ST fibres. Since their high stimulus threshold makes FTb fibres the hardest to recruit, they only come into play when a high muscle action force is required. Besides shortening faster, motor units with FT fibres also produce greater contractile force. This is due to the larger number of myofibrils in a motor unit with FT fibres, for there is not much difference in the force produced by an individual ST or FT myofibril.

ST	FTa	FTb
slow	faster	fastest
oxidative endurance type	anaerobic	anaerobic rapidly fatigued
small less strength	larger	largest more strength

Figure 2.32
Slow-twitch (ST) muscle fibres build up force more slowly and relax more slowly, produce less force per fibre, and are seldom fatigued. Fast-twitch (FT) muscle fibres build up force fast and relax faster, produce more force per fibre, and are rapidly fatigued.

Since a muscle fibre is activated by action potential on the all-or-nothing principle (neuromuscular transition involves only excitation, not inhibition), the order in which the various motor units are recruited depends on the difference in their stimulus thresholds. Rate coding may therefore further affect contractile strength.

Motor units are therefore recruited according to the 'size principle' (Henneman *et al.*, 1974). This means that the order of recruitment depends on the size of the stimuli emitted by the central nervous system. As we have seen, each motor unit has a minimal stimulus threshold that must be exceeded in order for the fibres to contract. The threshold is closely related to the size of the cell body. This means that ST motor units will be activated first, because they have relatively small cell bodies. As the stimulus increases, motor units with increasingly large cell bodies (FTa fibres) will be activated, and finally fibres with the largest cell bodies (FTb fibres). Recruitment of motor units is thus related to contractile strength rather than speed of muscle action (Gollnick *et al.*, 1974). Besides building up contractile strength by increasing the strength and frequency of stimuli (which particularly benefits action of the larger, faster fibres), the eventual contractile strength is also influenced by better synchronization of actions in the muscle fibres. In the normal situation, motor units act asynchronously in order to achieve a fluent movement. With higher force production, trained athletes are better able to activate the motor units synchronously. The quality of this synchronization greatly depends on the movement involved, which means that athletes achieve the most effective synchronization within their own particular sport (Figures 2.33 to 2.35).

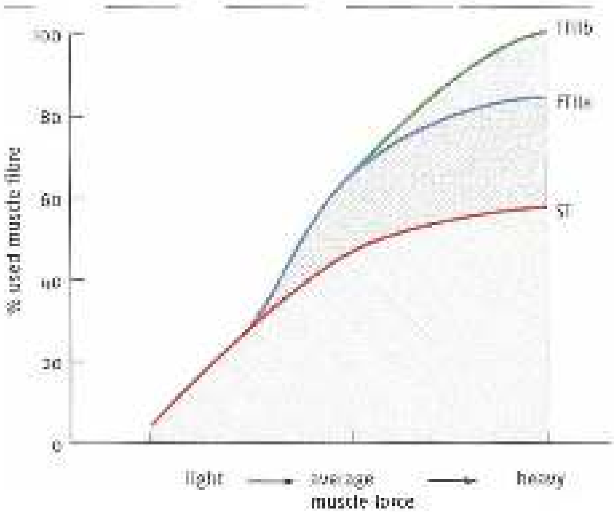


Figure 2.31 Size principle 1. Recruitment of motor units depends on the required size of the contractile force. If this increases gradually, the slow-twitch (ST) motor units are recruited first, followed by the fast-twitch (FTa and FTb) units if the fibres are stimulated more strongly.

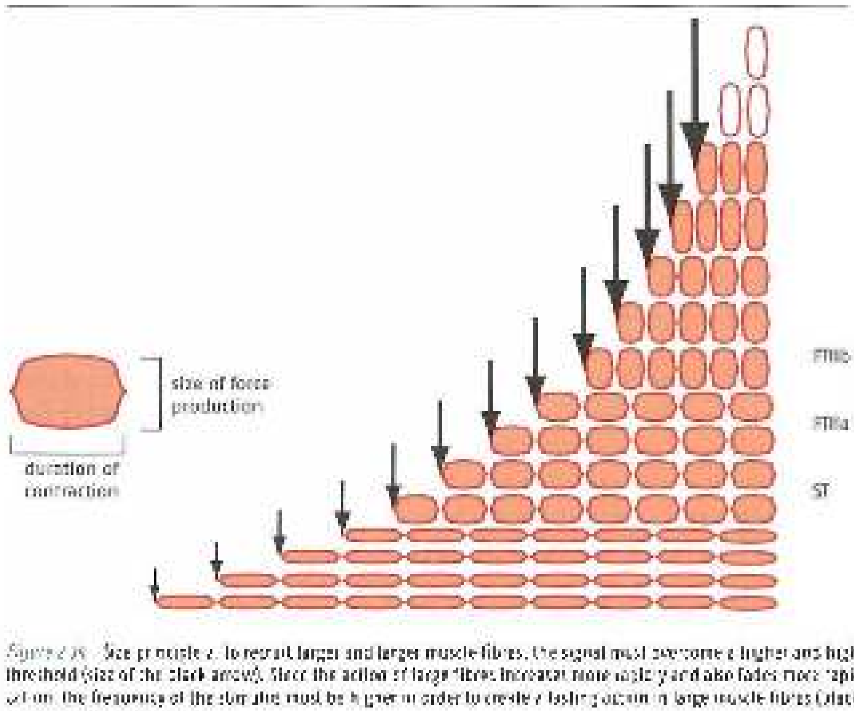
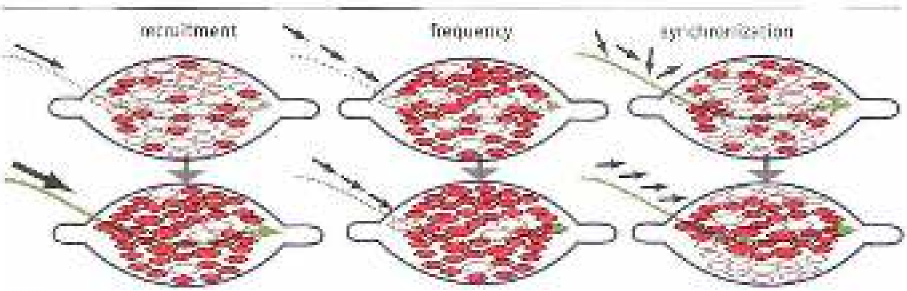


Figure 2.32 Size principle 2. To recruit larger and larger muscle fibres, the signal must overcome a higher and higher stimulus threshold (size of the black arrow). Since the action of large fibres increases more rapidly and also fades more rapidly after the action, the frequency of the stimulus must be higher in order to create a lasting action in large muscle fibres (black arrows closer together as force increases).



[illegible] Force production in a muscle depends on the size of the stimulus (i.e. recruitment), the height of the stimulus frequency and the synchronization of the muscle fibres.

The size principle is the first major step in ensuring that force production in muscles can be controlled by the central nervous system. The result – the amount of force – of a given stimulus from the central nervous system would be very unpredictable if there were no fixed pattern within which the various motor units with their various properties were activated. The predictability of force production is a substantial problem for the central nervous system, not just because of force/length and force/velocity ratios in muscle fibres, but also because the response of muscle fibres to a neural stimulus partly depends on those fibres' recent 'history': possible fatigue owing to previous muscle actions and the influence of previous potentiation (also known as post-activation potentiation, or PAP: the fact that explosive muscle action improves if muscles have been briefly loaded with high resistance in an exercise just beforehand: Hodgson *et al.*, 2005; Robbins, 2005). As a result of all these phenomena, the relationship between the size of the neural stimulus and the force eventually produced may become very unclear. As we have seen, the size principle is a first step towards solving this problem. Researchers are seeking other solutions within the complexity of the neuromuscular system.

Owing to stimulus size, stimulus frequency and synchronization of activity, contractile force is increased by recruiting larger and larger motor units. The larger the fibres, the sooner they are fatigued. This creates a relationship between how much force is produced and how long it can be sustained: a small amount of force can be sustained for a longer time, and a larger amount of force for less time. This relationship between fatigue and force production is a very useful one, for it means that fatigue can be much more effectively anticipated by the organism. If the various muscle fibres with their various characteristics were recruited at random, a fleeing animal would be unable to estimate how long it could sprint away from a predator and find a place of refuge before falling prey to fatigue, and its attacker. For instance, many rapid muscle fibres may have been recruited in the course of ordinary locomotion just before the animal flees, so there may already be a degree of fatigue in these fibres when it has to take flight. In that case the animal would no longer be able to estimate its maximal possible speed and its ability to maintain speed. The link between strength and fatigue through the size principle thus makes it possible to link intention (the state we wish to reach in the near future) and action, in other words to plan effective movement strategies.

2.3 Circuits at spinal-cord level

Force production in muscles is regulated by the central nervous system. In a motor skill both excitation and inhibition of stimuli play a part at all levels of the nervous system, except in neuromuscular transition, which only involves excitation (Bongaardt et al., 2006; Van Cranenburgh, 2002).

Much of the regulation of force production takes place at spinal-cord level, with major fundamental integration of interacting forces into contextual movements taking place within several complex circuits: forces that muscles exert on each other, forces exerted on muscles by the inertia of moving parts of the body, and external forces acting on the body. The integration of these forces can only be controlled to a limited extent by higher parts of the central nervous system, although even such limited control is still of substantial importance in optimizing movement patterns. In running, for example, there are clear, primary control components at spinal-cord level. These operate automatically, but can be further modulated and optimized from intention (higher parts of the central nervous system, e.g. the brain). This also means that training of a pattern that is so strongly controlled at spinal-cord level must involve the right balance of free execution of primary basic movement patterns and intentional control of execution of the movement – perhaps the starting point to the age-old debate about decisive personal style versus being able to learn the right or wrong technique. Such spinal-cord-based patterns are also of great importance to strength, force production and strength training.

2.3.1 *The stretch reflex and the Golgi tendon reflex*

The muscle spindle is a structure that is about five millimetres long and is located in the muscle parallel to the extrafusal muscle fibres. The spindle contains a sensor that transmits a signal in response to stretch. Its structure makes it responsive to passive changes of length in the muscle. The afferent Group Ia fibres register and transport information on the amount and speed of the increase in length. The afferent Group II fibres register and transport information on the increase in length. Since the muscle spindle is parallel to the extrafusal muscle fibres, changes of length in these fibres affect the length of the sensor in the muscle spindle. If the muscle stretches, the muscle spindle with its sensor will also stretch and transmit a stronger signal. If the muscle shortens, the muscle spindle with its sensor will also shorten and transmit a weaker signal. The sensors in the muscle spindle may also change in length under the influence of higher parts of the central nervous system via the efferent gamma motor neurons, which innervate the intrafusal muscle fibres located in the muscle spindle. If these intrafusal fibres contract concentrically, the muscle spindle sensor stretches, and if they relax, the sensor shortens. As a result, stretching of the sensor does not simply depend on changes in the length of the extrafusal muscle fibres, but may be linked to changes in the length of the muscle in a particular range of muscle length. If the sensor has to respond to changes of length in a greatly shortened muscle, and hence a greatly shortened muscle spindle, the intrafusal fibres will be greatly shortened through stimulus via the gamma motor neurons to set the sensor to its initial length for registering stretch. If the muscle then has to work in a more stretched state, the muscle spindles will also lengthen, and the sensor can be set to the same initial length by relaxing and passively lengthening the intrafusal muscle fibres. The stretch sensors can then measure the change in length from their initial length (Bruggeboom et al., 2006). The muscle spindle can thus respond to changes in the length of the muscle at every muscle length.

The stretch sensors pass on the registration of length and changes of length in the muscle to the spinal cord via the afferent fibres (Types Ia and II). From there the information is passed on to higher parts of the central nervous system, providing important information on the state of the body. The alpha motor neurons are also activated at spinal-cord level as a result of the afferent gamma information. This causes stimulation and muscle action. The muscle spindle therefore acts as a maintainer of length (a muscle that is stretched further than anticipated contracts in a reflex response to the signal from the muscle spindle). The muscle can thus be controlled in two different ways: directly through stimulus of alpha motor neurons, and indirectly through stimulus of gamma motor neurons, which alters the responsiveness of the muscle spindle and in turn supports the activity of the alpha motor neurons. The latter mechanism is known as 'alpha/gamma coactivation' or the 'stretch reflex'.

Besides monitoring of muscle action by the gamma loop, the muscle's contractile strength is influenced by another 'sensor', the Golgi tendon organ (GTO). The GTO is not controlled by the central nervous system, and hence is a passive sensor. When the muscle is tensed, the tendon stretches, activating the GTO. The GTO activates, among other things, muscle-inhibiting feedback at spinal-cord level. This system is far more sensitive than was long thought: the GTO can even register the increase in tension caused by the activity of just one motor unit.

Muscle spindle and Golgi tendon systems are essentially conservative systems – they attempt to counteract rapid, extreme changes in muscle length and tension (Figure 2.36 to 2.40).

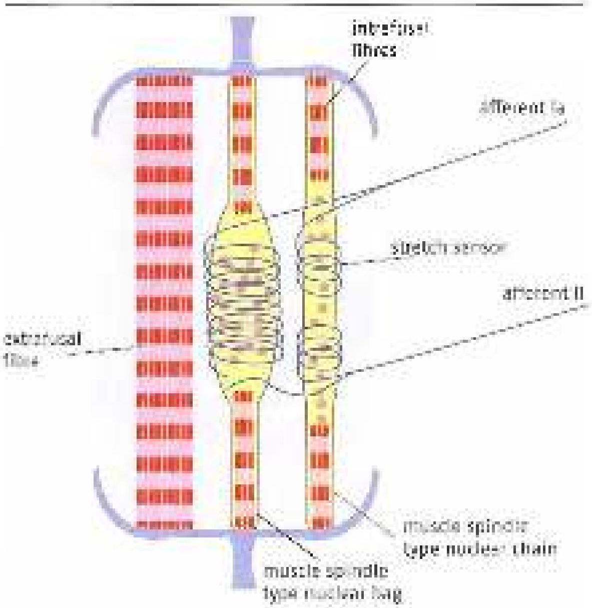


Figure 2.36 Structure of the muscle spindle. The muscle spindle is parallel to the extrafusal muscle fibres. In the muscle spindle the intrafusal muscle fibres are arranged in series with the stretch sensor. There are two types of sensor: the nuclear chain cell and the nuclear bag cell with afferent Group Ia fibres with afferent Group II fibres.

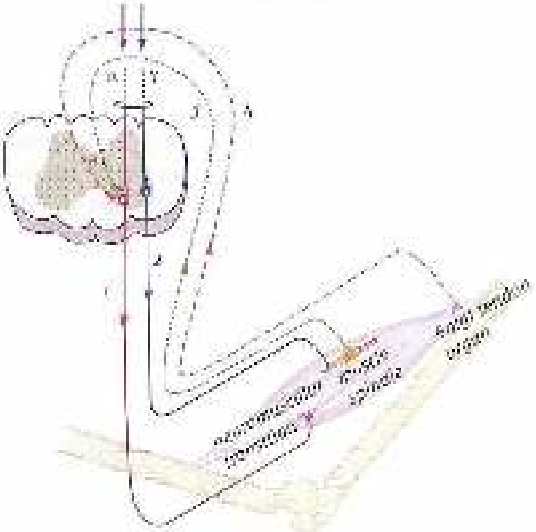


Figure 2.37
Alpha/gamma innervation:
1 Efferent innervation of the extrafusal muscle fibres by the alpha motor neurons via neuromuscular transition
2 Efferent innervation of the intrafusal muscle fibres by the gamma motor neurons
3 The afferent nerve path from the muscle spindle to the spinal cord, where it links up with the path of the alpha motor neuron
4 The afferent nerve path from the Golgi tendon organ to the spinal cord, where it links up with the path of the alpha motor neuron

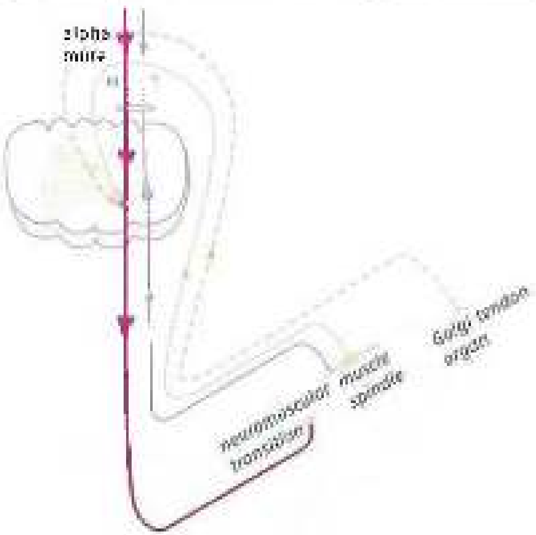


Figure 2.38
The alpha path. Efferent alpha innervation runs from higher parts of the central nervous system via the spinal cord to the extrafusal muscle fibres for in a force production.

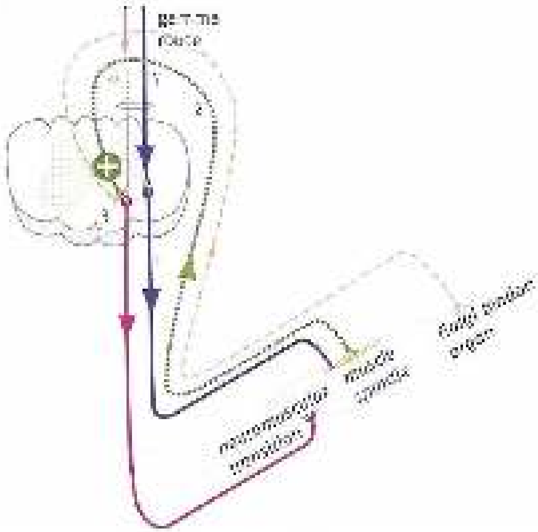


Figure 2.39
The gamma path.
1 Efferent gamma innervation runs from higher parts of the central nervous system via the spinal cord, to the intrafusal muscle fibres.
2 When the muscle stretches, an afferent signal is transmitted from the muscle spindle sensor to the spinal cord.
3 Linking up with it and reinforcing (+) the path of the alpha motor neuron to maintain the length of the muscle.

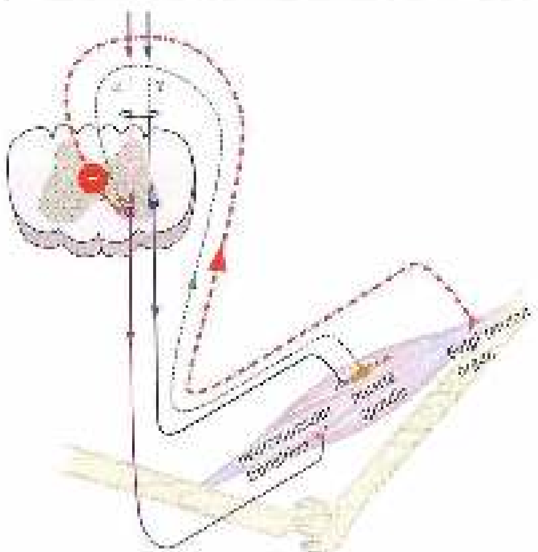


Figure 2.40
The afferent nerve path runs from the Golgi tendon organ to the spinal cord, linking up with and inhibiting (-) the path of the alpha motor neuron.

The way in which the muscle spindle and the GTO function is actually very complex. The activated muscle spindle is in contact with higher parts of the central nervous system, which can have a major influence on the entire effect of alpha/gamma coactivation and the stretch reflex through muscle spindle activity. In deliberate isometric muscle action, muscle spindle activity supports muscle tone. In deliberate concentric muscle action, muscle spindle activity ceases (through control by the central nervous system). This plays a key role in movements that are largely routine. The old idea that muscle spindles mainly, or only, operate through reflexes at spinal-cord level must therefore make way for a far more complex, interactive, flexible model, of which action at spinal-cord level is only a part.

More important than reflexes is the fact that the muscle spindle system (the indirect path) enables the muscle to attain a predetermined length, more or less regardless of the external forces involved. Seen in broader motor-function terms, this means that the muscle spindle system can ensure that perturbing external influences (forces) can to some extent be counteracted by motor-function design. The movement system is thus able to design motor skills in accordance with a predetermined plan that incorporates those adaptations to external factors (monitoring of muscle length by the gamma loop) that are necessary for successful execution of a given task (Figure 2.41).

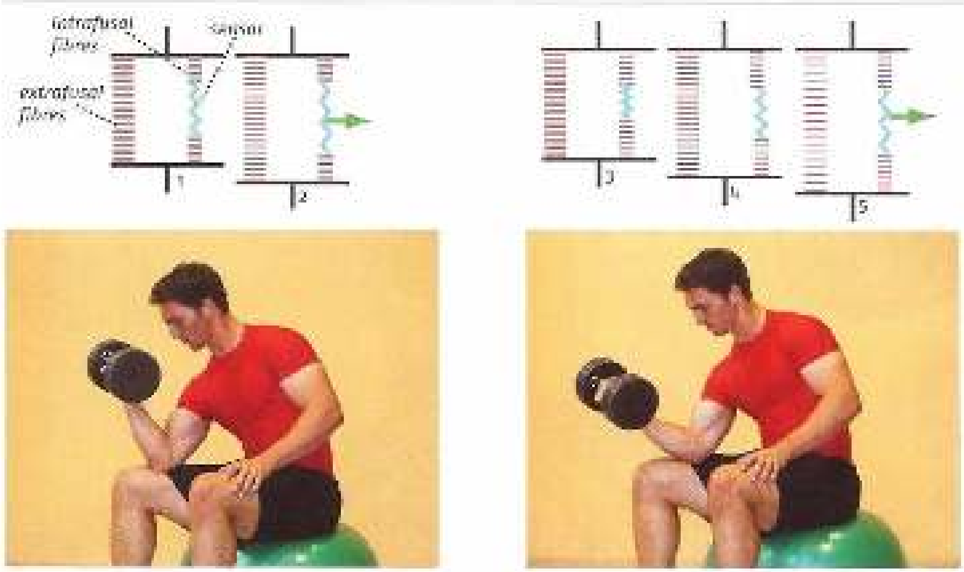


Figure 2.41 Left: biceps brachii at more or less neutral length (1). The intrafusal muscle fibres are set so that lengthening of the muscle immediately produces stretch in the muscle spindle sensor (2). The gamma loop maintains the length of the muscle and the wrist remains in the same position.
Right: the elbow extends (3 to 4) because the extrafusal fibres produce less force. At the same time the intrafusal fibres relax, causing the sensor to lengthen so far that it no longer responds. However, once the elbow has reached the intended position, the intrafusal fibres are fixed at such a length (5) that further lengthening of biceps brachii will cause stretch in the sensor and hence a stimulus that will increase the contractile force of the extrafusal fibres. If the elbow moves rapidly from the left-hand to the right-hand position, more force will be needed to inhibit the movement than if the movement is slow. The muscle spindle system ensures that the required force does not need to be estimated accurately.

Besides reflexive activation of their 'own' muscle, the muscle spindle and the GTO have an inhibiting influence on the antagonists, known as 'reciprocal inhibition'. In hip extension, activation of the muscle fibres in *gluteus maximus* will thus be accompanied by an inhibiting effect on the *iliopsoas* fibres. This interplay of excitation and inhibition is also of importance to intermuscular coordination. It will be altered by strength training. There will be a reduction in cocontractions: simultaneous tensing of the agonist and the antagonist (Huijbregts & Clarijs, 1995). The cocontractions stabilize the joint, but at the same time they reduce the joint torque. Reduction in cocontractions thus increases the joint torque. The action of the various agonists (e.g. the various extensor muscles round the hip joint) will also be better coordinated. Both changes resulting from strength training are movement-specific.

In many sporting situations it is very difficult to estimate how much alpha and how much gamma control are the optimal combination for a movement pattern. This has to be explicitly learned. The motor control system must, as it were, learn to estimate a bandwidth within which the size of the external forces has to remain in order to allow optimal correction of deviations by the gamma system. Downhill skiers must adjust their muscle spindle activity so that unexpectedly large external forces (due to such things as uneven terrain) can be adequately corrected for while maintaining more or less the same knee and hip angles. At the same time, gamma activity must not be so high that the knee and hip angles do not adjust at all to the external forces and the skis no longer slide but bounce over the snow. Boxers who want to learn a combination of punches must do so in such a way that their arm movements remain independent of the forces acting on the punching hand during the combination. An opponent who advances during such a combination will generate larger external forces than one who retreats. If the combination is controlled only by alpha activity, a misjudged movement by the opponent will perturb the pattern so badly that it will be very difficult to continue the combination. Boxers often train with punching balls which move up and down so quickly that it is impossible to estimate what external forces will be generated on contact. In this way they learn to perform punching movements regardless of the opposing external forces.

In fact, unfamiliar interplay of forces is far more common than might at first be thought. In every sport involving contact with opponents, unpredictable surfaces, balls that have to be caught and so on, athletes are constantly confronted with forces that cannot be accurately foreseen. Translated into sport specific strength training, this means that if the performance movement involves dealing with constantly changing, unpredictable forces and hence the gamma path plays a key part in regulating the force produced, it may also be useful to make the opposing force somewhat less predictable in strength training. For reasons of health and safety, this cannot be done when training against high resistance, and lower resistance will instead have to be used (e.g. by training with a water-filled medicine ball or physio ball, training on a less stable surface such as a mattress, and so on; see also Section 6.1.4).

The idea of training with lower resistance and somewhat less predictable external forces is important not only in sport-specific strength training, but also in exercise rehabilitation. Being able to cope with unexpected external forces may be more important when relearning how to function in everyday situations than learning to cope with large external forces that can easily be estimated. This means that a rigid exercise setting with standardized body postures and more resistance in a predictable exercise area will be probably be less effective (Figure 2.42).

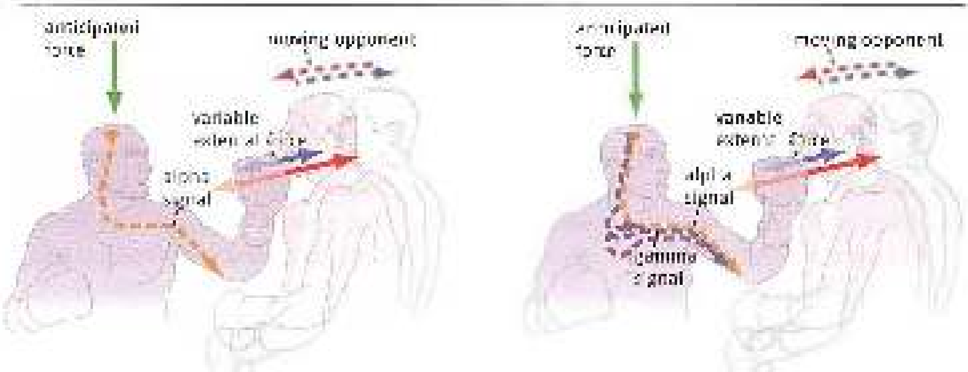


Figure 2.42 In a combination of punches, the force to be produced must be as similar as possible to the anticipated force. The combination is disturbed if the force produced does not match the external force.
Left: a boxer with only alpha control in a combination of punches. The external force that the boxer encounters while punching changes constantly. This means that the final position of the punching hand cannot be properly controlled and the combination cannot be performed properly.
Right: a boxer with alpha and gamma control in a combination of punches. If an external force is misjudged, the gamma loop ensures that the arm ends up in the right position so that the combination can be continued.

2.3.2 *Preferred movements and CPGs*

At spinal-cord level there are not only circuits such as the gamma loop and reciprocal inhibition of antagonists, which control individual muscles. Larger intermuscular patterns – the building blocks of contextual movement – can also be controlled. What are known as central pattern generators (CPGs) ensure primary interaction between these larger units, for instance how the movement of one leg influences the movement of the other when walking and running (Crook & Cohen, 1998). Animal studies have clearly demonstrated the existence of CPGs, although there are still many questions about how they operate. They are generally believed to act as oscillating rhythm generators. The patterns generated by these networks are flexible rather than rigid. Motor output may vary in response to sensory feedback. Thus the foot position in the swing leg when walking and sensory information from the sole of the foot in the stance leg influence how the walking pattern develops.

When thinking about movement patterns in sport, such control at spinal-cord level has little significance. It is still very difficult to translate this into intensive contextual movement, especially in the upper limbs. The punching pattern described in Section 1.3.2, in which extension of the arm is synchronous with rotation of the trunk around the longitudinal axis, may be such a basic movement that arises as a result of (among other things) the action of CPG circuits. Extension of the arm and rotation around the longitudinal axis may thus be a generic building block of movement that can be used in such varied patterns as shot putting, boxing and arm entry in freestyle swimming.

Primary rhythmic patterns in the lower limbs are somewhat easier to interpret, if only because leg movements are somewhat less variable. Two basic movements in walking, running and single-leg jumping are the stumble reflex and the crossed extensor reflex. In a stumble reflex, moving one leg backwards (e.g. the stance leg that moves backwards in relation to the trunk when walking and running) causes the other (swing) leg to move forwards (flexion at the hip). The pattern is most evident when the swing leg steps moving forwards because the foot has caught on something. The resulting stumble movement, in which the other leg shoots forwards, is a vigorous reflexive movement. In the crossed extensor reflex, flexion of the swing leg (flexion of the hip and knee, and dorsiflexion of the ankle) is linked to extension of the other leg and elevation of the free/swing side of the pelvis. Both basic patterns are primary components of running. Optimization of running patterns must therefore be based on optimal incorporation of both patterns into the running cycle. The fact that they are primary patterns is apparent from the fact that a better position of the free leg when running always leads to better elevation of the free/swing side of the pelvis, even if no instructions are given about this.

Sprinting is therefore an advanced way to carry out both the stumble reflex and the crossed extensor reflex (Figure 2.43). The peak forces that occur in reflexive support during sprinting, such as these two reflexes and the muscle spindle reflex, are greater than those that can be created by 'maximal voluntary contraction' (MVC) (Kyröläinen *et al.*, 2005). Excitation and inhibition (i.e. tension and relaxation) succeed one another so rapidly and with such great peak differences that proper reflexive support is a *sine qua non*. In technique training for running at speed, the push-off must therefore be properly coordinated with the action of the swing leg, so that reflexive support is optimal. This means, for example, that the position of the swing leg is very important when the stance leg is directly below the hip. The thigh must be just below horizontal and the heel must move towards the hamstrings (Figure 2.44).

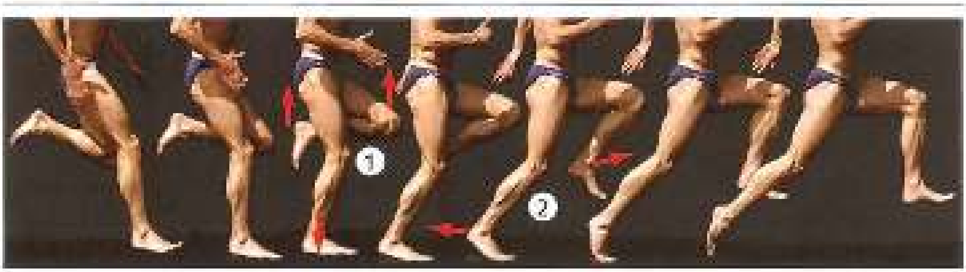


Figure 2.43 Stumble reflex and extension reflex. Running is a contextual arrangement of movements around the crossed extensor reflex (1) and the stumble reflex (2). Both movements are stable preferred movements for the running cycle.

Figure 2.44 Running technique in which too little use is made of the extension reflex, because the pendular motion of the lower leg is too early. This means that the vertical component of the push-off is too small, and the step frequency too high.

In strength training it is also well worth looking for exercises in which such basic movement patterns play a part. In particular, types of running in which speed is linked to strength can be usefully supported in this way. Since reflexive patterns are not rigid, independent mechanisms but part of a complex of central, peripheral regulating mechanisms and sensory feedback, there is no guarantee that they will be optimally expressed when running. In ball sports such as soccer, in which there is little running training or strength training based on running training, there are very few players even at the highest professional level that have optimally incorporated the basic patterns into their running patterns. This category of basic movement patterns with a crucial control component at spinal-cord level is important enough to give them a special place in strength training, known as 'reflexive strength' (Figure 2.45).

Figure 2.45 Basic patterns built into a strength exercise, showing the influence of the sensory stimulus from the sole of the foot (1), the crossed extensor reflex (2) and elevation of the free hip (3).

In practice, types of strength training that focus on reflex-controlled basic patterns are marked by abrupt fatigue (usually after about six repetitions), which is very severe but then quickly fades. Fatigue as a result of a strength exercise with these characteristics – known as 'neuromuscular fatigue' – may point to a type of training that will be effective for explosive sports (see also Section 7.2.4).

2.4 Central influences

The body/task relationship is described in Chapter 1. A task must be in accordance with the body's capabilities. Correctly estimating the body's capabilities, and hence the feasibility of the task, is essential for avoiding dangerous actions and for survival. Put more simply, if you don't want to land in the ditch, you have to be able to estimate correctly whether you can jump across it or should walk the long way round instead. We constantly have to make such estimates, and these are often difficult, for instance if we have to jump across a ditch and the other side is higher than the side we are on. An interesting question that arises here is whether in estimating the body's capabilities we are testing the limits of what is possible, or whether there is a built-in reserve and hence not all the body's capabilities are being used.

There are a number of important arguments suggesting the existence of mechanisms that make it difficult, or even impossible, to reach the limit of load capacity when moving. Peripheral factors that limit performance (the influence of cocontractions and muscle slack) are described in Chapter 4. There may also be performance-limiting influences from higher parts of the central nervous system (supraspinal control). The brain limits the stimulus to the muscles well before the limit of what the muscles can do is reached. This is already apparent from the fact that we are not normally able to recruit more than about 75% of our potentially available muscle strength. Reasoning from the size principle, we cannot simply recruit muscle fibres with the highest threshold value. Training can increase this percentage without the muscles having to increase in size. This improvement in performance is very movement-specific (see also Section 5.2). As well as by training, the percentage can be increased in stressful situations. There are many well-known cases of people turning out have unsuspected reserves of strength in life-threatening situations, such as lifting a small car to free someone trapped underneath. In training theory this is known as the 'autonomously protected reserve'.

It is not easy to determine how this reserve develops, for it is probably the result of complex interactions at many different levels, but it may have a highly central component. The brain underestimates the body's capabilities as a matter of course, and training seeks to counteract this. This means, among other things, that the mental component of strength training is very important. When providing a strength training programme, plenty of attention must therefore be paid to the mechanisms behind this restraint on the part of the brain (see Section 6.4.2).

Particularly when athletes are recovering from injury, such restraint is a crucial part of muscle control. Initially, of course, it is important in order to protect the injured part of the body. Later, however, it must be reduced by increasing the load. Physical therapists often make things worse by adhering for far too long to an extremely cautious exercise protocol in which the intensity of the load is very greatly reduced. It may therefore be wondered whether exercise therapy with very low loads is of much use in reactivating the system. Not only may irrelevant movement patterns be exercised (see Section 1.3.3), but the autonomously protected reserve may be increased. In view of this, a number of leading sports injury rehabilitators now start training the muscles round a dislocated shoulder within twenty minutes if possible, and, for example, start training recruitment of muscle fibres with high-force isometric muscle action just a few days after a significant hamstring strain injury.

The brain's underestimation of the body's capabilities sheds another light on the mechanisms of training adaptation. In physiology this is expressed in Tim Noakes's central governor theory. Measurements led Noakes to conclude that the brain reduces the strength of signals to the muscles before they reach the physiological limits of exhaustion, in order to prevent irreparable damage to the body (Noakes, 2011, Noakes *et al.*, 2001). An explanation for coordinative restraint, particularly the restriction of high-intensity movements, may be provided by probabilistic prognosis theory. This theory, which is an extension of Nikolai Bernstein's thinking (Feigenberg, 1998), states that in the environment we are moving in, before coming into action, we make an estimate of the future state we wish to achieve and whether or not the environment and the body have the necessary properties to achieve it. Of course, this estimate is very much based on past experience, and is stored in the memory. The result is a set of possible actions and movements that are likely to be successful. More cognitive theories focus on the memory for the exploration of interaction between the task, the organism and the environment in order to move successfully, whereas ecological theories consider the 'affordances' (see direct perception theory) that arise from interaction between the organism and the environment (Araújo, Davids & Hristovski, 2006) important.

A number of important conclusions about the effect of strength training can be drawn from this probabilistic prognosis theory. Perhaps the most important of these is that the predictability of what happens in training (monotony) may act as a brake on the intended training effects. Variation and alternation in types of training may keep the brain interested in adapting control (see also Chapters 4 and 6) and yield better results.

A second important conclusion that can be drawn from the theory is that the limitation of force production is linked to perceptual-motor anticipation (prior estimation of the combination of sensory and motor signals that will arise when an action is performed) and hence is highly movement-specific. In other words, there is no general measure of 'being strong' in athletic movement – on the contrary, it is highly variable, with improvement in force production being linked to changes in the perceptual-motor coupling, or (according to affordance theory) another link between movement capabilities in the environment and the body's capabilities. What this means in practice is, for instance, that the run-up speed that high jumpers can still cope with will vary. The higher the run-up speed that high jumpers can handle in terms of technique, the higher they can normally jump. Training can improve this, which means both that force production will increase and that the perceptual-motor coupling will be adjusted. This means in turn that training should focus both on increasing force production and on learning to 'understand' how the higher level of force in actual jumping should be sensed and anticipated. This last aspect of performance may be of decisive importance in elite sport, but also plays a key role in sports injury rehabilitation (see also Section 5.2).

Owing to the presence of the autonomously protected reserve, the maximal voluntary contraction (MVC) will not allow more than 75% of the total quantity of muscle fibres to be recruited. It is still a matter of conjecture what causes this limitation. This chapter has described a number of possible limitations, including central influences on the limit of performance. It should be noted here that each possible limitation has different implications for how best to design training (Figure 2.46).

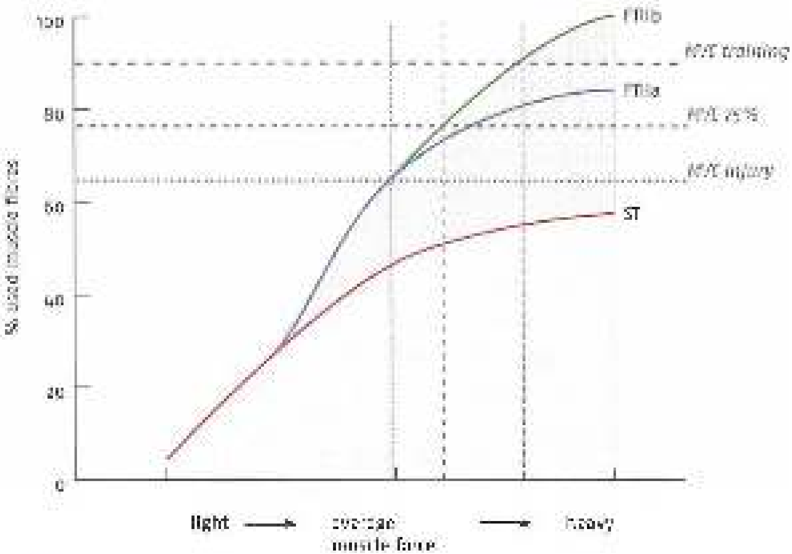


Figure 2.46 We are usually able to recruit about 75% of our maximal available muscle strength (75% MVC). Training can greatly increase this in a specific setting (training MVC), but injury can also reduce it (injury MVC).

2.5 Summary

Force production is influenced at many different levels. Starting at muscle level, factors such as muscle length and shortening speed play a part. The F/V and F/L characteristics differ in muscles with differing architectures. Force production is also affected by muscle properties such as elasticity, muscle slack and moment arms. As a result, there is no very direct or very strong link between signals from the central nervous system and the force eventually produced by the muscle.

However, to make the amount of input from the central nervous system and the implications for force production easier for the central nervous system to predict, muscle fibres are stimulated according to the size principle: the small fibres (ST) are recruited first, followed by the larger ones (FTIIa) and finally the largest ones (FTIIb). This ensures a link between the amount of force produced and increasing fatigue, allowing meaningful movement strategies to be devised.

Complex circuits at spinal-cord level create mechanisms whereby muscle force production is adapted to the opposing forces acting on muscles (the gamma loop) and flexible rhythmic basic patterns develop as building blocks for all types of contextual movements. Movements geared to these basic patterns are preferred movements that are both economical and stable and play an important part in explosive athletic movement.

Finally, force production is also influenced supraspinally. This influence may be so great that the limitation of force production ultimately takes place in the brain. However, this has not been proved, and the influence may vary from movement to movement.

3
Analysing the sporting movement

Anyone with practical experience of sports knows, if only intuitively, that an exercise cannot improve athletic movements unless the two movements are similar. The degree of similarity between the two is known as 'specificity'. If specificity is guaranteed, the exercise and the sporting movement influence one another – this is known as 'transfer'. It is also common knowledge that, in order for transfer to the sporting movement to occur, the body must be given a training stimulus it is not yet equipped to deal with – this is known as 'overload'. The adaptation process is highly individual, varying from athlete to athlete – this is known as 'individuality'. To continue the adaptation process, the system must be 'progressively loaded'. Moreover, the effect is only temporary; if athletes stop training, the training effects vanish – this is known as 'reversibility'. Finally, the degree of adaptation as a result of training (including strength training) is reduced as athletes' training histories lengthen and their level of training increases – this is known as 'the law of diminishing returns'.

Given the importance of these six training factors, one might expect their various aspects to have been studied in detail by researchers. Unfortunately, they have not, and there is still no systematic analysis of how 'specificity' and 'overload' actually operate within training processes. Such an analysis will be attempted in Chapters 4, 5 and 6.

Such systematic analysis must not only identify the qualities of preparatory strength training, but also look in detail at the sporting movement to identify the link between the preparatory exercise and the athletic movement. There is a particular problem when analysing the sporting movement, namely that many types of sport involve movements that are difficult to identify because they do not follow fixed patterns. This in turn makes it difficult to establish the link with strength training. Besides a systematic analysis of links between the preparatory exercise and the performance movement, there must therefore also be one to identify the sporting movement. We can then determine which components of the sporting movement can be improved by preparatory strength exercises, and which ones cannot. Such an analysis is above all important if we are to understand the coordinative transfer of strength training.

3.1 Open and closed skills

A key distinction is made in sport between open and closed skills. A closed skill is a movement pattern in which the movements to be made are predetermined, because the environment in which they are made is unchanging. With open skills, the environment is not unchanging, and the movement must therefore be adapted (improvised) in response to the demands of the environment at that moment. Closed skills are found in sports such as gymnastics, most track and field events, weightlifting, competition swimming and diving, long-track speed skating and figure skating. Open skills are found in such sports as soccer and other ball sports, whitewater canoeing, all martial arts and mountain biking. There are also sports in which closed and open skills alternate, such as tennis (services are closed, rallies are open), cricket and baseball (bowling and pitching are closed, batting and fielding are open). Finally, there are sports whose skills are considered more 'closed' by some and more 'open' by others, such as dressage. In practice there is a gradual transition from extremely closed to completely open skills, as demonstrated in Gentile's taxonomy, which ultimately identifies sixteen categories of skills ranging from extremely closed extremely open (Gentile, 2000).

The highly improvised nature of open skills makes it difficult to analyse the sporting movement, for in such an open skill situation its outward form constantly changes. It is therefore hard to draw conclusions about which types of training are most effective. In fact, some open-skill sports may have so few fixed movement patterns that there is little point in looking for similarities to particular types of strength training. An example of this view is the often-expressed opinion that running in open-skill sports such as soccer and rugby is so situation-specific that the principles applying to running in a closed-skill situation (running in a straight line, without influence from opponents, etc.) are meaningless in an open skill environment. In that case, what we know about running in athletics cannot be transferred to soccer or rugby, sports in which the demands of the environment call for constantly changing organization of the running pattern. Strength training and running training, whose specificity makes them very effective in athletics, thus become 'academic' and ineffective in the aforementioned ball sports. Running can then only be trained in 'game-like' types of training, and strength training should remain very general.

Yet there is good reason to believe that even the most open skills do not completely lack a fixed structure. Having to improvise a movement and adapt it to the constantly changing demands of the environment does not mean that *all* the components of the movement are constantly adapted – instead, some are adapted, but others remain unchanged. Effective movement is then a matter of changing the right components in response to the demands of the environment, while leaving others alone. This concept, which comes from dynamic systems theory, establishes links between closed and open skills, and allows useful links (which are effective in training situations) to be made between strength exercises and the sporting movement. Seen in this light, strength exercises are very suitable for improving unchangeable components of open skills.

3.2 Attractors and fluctuators within movement patterns

The organizational structure of both closed and open skills can be described in a number of steps. First the problem of degrees of freedom will be described. It will then be concluded that movement is non-linear, and rules of non-linear organization, which determine the subdivision of movements into improvised and non-improvised components (including in open skills) will be discussed.

3.2.1 *The degrees of freedom problem*

In the field of motor skills, a key starting point in applying dynamic systems theory is the degrees of freedom problem. This was described by the Soviet physiologist Nikolai

Bernstein long before long before other researchers began to develop insight into the mechanisms of complex systems (Bernstein, 1996).

If we have to move from position A to position B, there are very many different ways of making the movement, especially within contextual movements. There is such a vast choice of alternatives that it is difficult to select the most efficient one. When making a movement that involves several joints, each with its owndegrees of freedom, there are very many possible combinations of ranges of motion, which can all produce the same result. The degrees of freedom resulting from the combination of possible permutations of joint angles in several joints are further increased because the movements can usually be performed by more than one muscle. For example, the elbow can be flexed by *biceps brachii*, *brachialis* or *brachioradialis*, multiplying by seven the elbow flexion degrees of freedom. For other movements, such as hip extension, there are even more different muscles available, increasing the number of degrees of freedom even further. This entails a vast increase in the possible choices, and makes it almost impossible to choose the most efficient alternative from the huge range of those available. Of the thousand or more ways in which we can move an arm from down behind us to up in front of us, only a few are economical and effective – but which ones are they? We cannot possibly analyse and compare every one of those alternatives before making the movement. That would take too long and would overload and fatigue the brain, and movement would then be extremely fatiguing for the brain. So there must be a mechanism in the motor control system that eliminates the inefficient alternatives and selects the right one (Figure 3.1). According to Bernstein, the essence of motor control is more or less automatic elimination of superfluous alternatives or degrees of freedom.

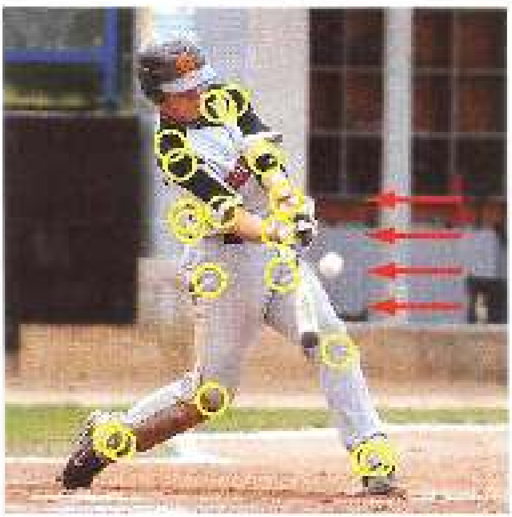


Figure 3.1
Batters can make adjustments in numerous joints in order to hit a ball heading for them at various heights and speeds. In addition, each joint often has several alternative muscles that can be used for this purpose. This can no longer be controlled.

Additional information

A ball-and-socket joint such as the shoulder has six degrees of freedom, and hence is difficult to control. In contrast, with only two degrees of freedom the elbow only flexes and extends, and hence is easier to control. When the movements of several joints are combined, as happens in all contextual arm movements, this creates very many alternative movement patterns, which makes the control of movement extremely complex.

Compare controlling the body with steering a car. If each wheel were controlled separately (four degrees of freedom), the car would be impossible to steer. Instead, the rear wheels are fixed and the ranges of motion of the front wheels are linked, so that there is only one degree of freedom left and the car can be steered.

The moving body also attempts to limit degrees of freedom in order to keep the movement under control. An example is the way in which the running cycle is structured. If the athlete starts by jogging slowly, gradually picks up speed and finally sprints, the transitions have to remain controllable. This means limiting the number of degrees of freedom in the transitions, and hence changing as few components of the running cycle as possible. One component that does not change as speed increases is the relative structure of the running cycle. The ratio between time on the ground and time in the air does not change. Nor does the ratio of time in the flight phase between the pendular motion of the lower leg and the scissor movement that place the legs in the right position for landing. In the stance phase the heel of the stance leg moves upwards at more or less the same moment. The potential degree of freedom of a differently structured running cycle is not used. The cycle thus remains the same, but is simply faster. This can be done by regulating stiffness, i.e. muscle stack. Muscle stack thus has not only an important performance-limiting function, but also an important function in keeping movement performance under control (Figure 3.2).

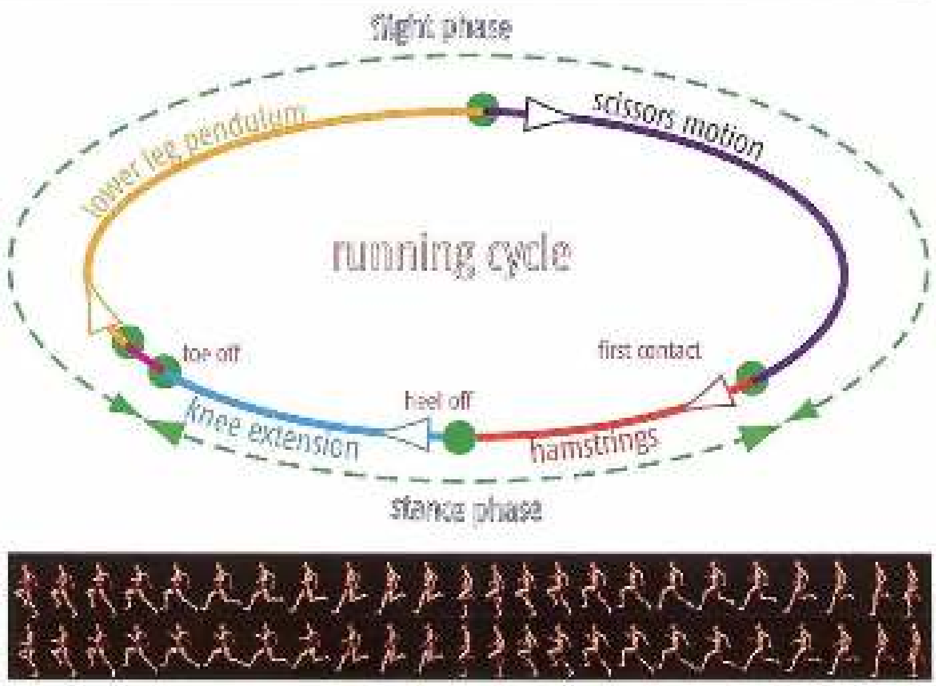


Figure 3.2 Top series of pictures: jogging; bottom series: high-speed running. The relative structure of the running cycle remains the same as speed changes. The absence of an extra degree of freedom owing to the changing structure of the cycle makes changes in running speed easier to control.

Additional information

Besides the problem of degrees of freedom, Bernstein described a second important problem regarding the control of movement: the problem of 'contextual variability'. This motor control problem arises because the forces acting on the moving body from the environment have a major influence on the eventual movement of the joints. In a changeable environment the forces will constantly be different, so the same command from the central nervous system to the muscles (a motor programme) will generate different movements in different environments. If performance of a movement (the total of joint angles) must always be the same, regardless of the environmental influences, the muscles will have to be controlled differently in each situation. This means that if the intended joint angle in a movement pattern must always be more or less the same (e.g. in a wrestling move), a different selection of muscles and muscle action may be required in each situation (the movements and the opponent's resistance) (Figure 3.4). This means in turn that a motor skill cannot be designed in a linear manner, not only because of the problem of degrees of freedom but also because of the influence of variable opposing forces.

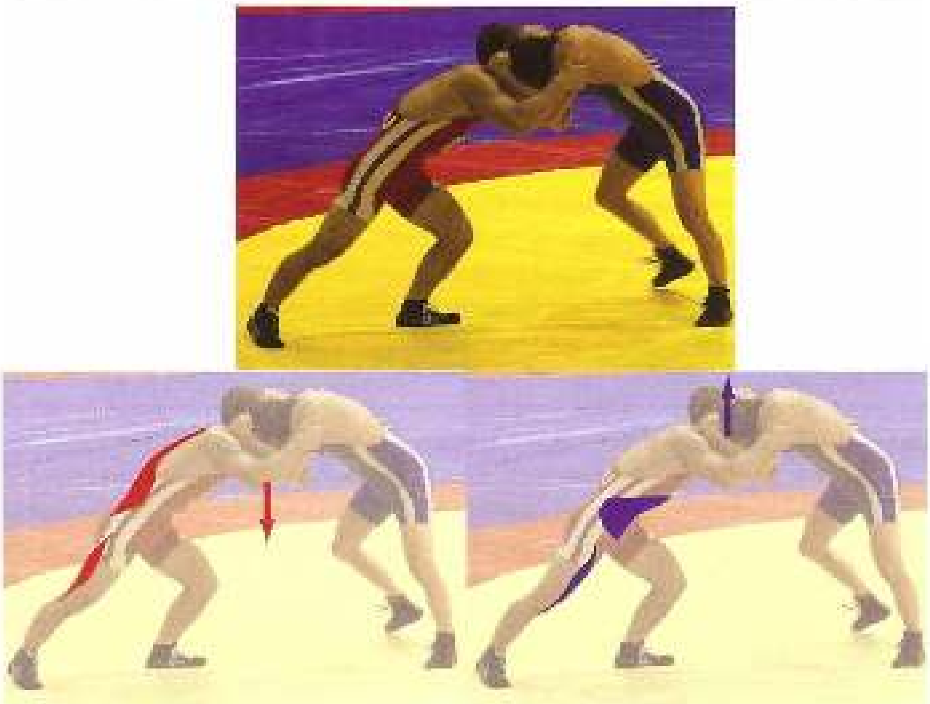


Figure 3.4 The wrestler in red is attempting to push his opponent back, and to do this he tries to keep his trunk at the same angle in relation to the ground.
Depending on what his opponent does, he will have to use different muscles. If the opponent tries to push him downwards, he will above all have to produce force in his back muscles and the hamstrings of his right leg. If the opponent tries to push him upwards, the abdominals and the rectus femoris of the right leg come into play to maintain the intended posture. The muscles will therefore be controlled differently in each situation.

3.2.2 *Non-linear motor skills*

A motor control system that rapidly selects from the various ways of performing a movement can be compared to finding a book in a library. Entering the first letter of the title in the search system eliminates most of the possible titles. Entering the first letter in fact means operating a highly abstract control mechanism that still has nothing to do with the subject matter of the book. Entering the first word of the title further reduces the number of possibilities. After a few words only one book is left. Much the same happens when selecting a motor skill. There are rules that rapidly eliminate numerous alternatives until the right one is found. The rules of motor control are remarkable, for they have to lead to the most efficient performance of the movement – in other words, the performance with the lowest energy costs, with sufficient stability to withstand perturbation of the basic pattern (for instance an opposing force that is greater than expected) and sufficient flexibility to adapt the basic pattern to the circumstances. Since efficiency and stability are so important to satisfactory movement, movement is not designed in the most linear manner possible. In linear motor skills, the muscles and their actions are selected directly when designing the movement. They perform the movement as a single incident. Such motor skill design can be compared to selecting a book from the library on the basis of subject matter, with all the attendant problems. In motor skills, however, the selection process is complex and non-linear. First a preliminary design is made, comprising generally applicable patterns of efficiency and firmness that do not yet include specific movement patterns – like entering the first letter or word of the title of the book. Only at the end is this preliminary design specifically converted into activation of muscles. This filter of abstract rules for efficiency and stability eliminates all sorts of alternatives early on in the design, so that movement can be designed with a limited number of efficient movement patterns. So, reasoning from the rules of non-linear control, running in ball sports will resemble running in athletics in essential ways, and will only differ in ways that are less crucial to the core organization of the running pattern. The trick, of course, is to find out which ones are which.

What is striking here is that people all over the world find more or less the same answers to movement problems. People all over the world run up a flight of stairs in the same way. In the mangrove forests of Borneo, where villages often have a small volleyball court but not a single television set, the best player hits a smash in precisely the same way as an indoor player in Italy, without ever having seen a smash on TV. The way in which the search system looks for efficient, robust movement patterns does not vary so much from individual to individual, and must therefore be linked to features that are much the same in all humans, such as the overall structure of the musculoskeletal system.

However, I am not arguing here for some rigid notion of a single ideal technique. Besides major similarities between individuals' musculoskeletal systems, there are always minor differences. In some movement patterns these may have a relatively large impact on the performance of a movement pattern (see Section 1.1.2, Figure 1.2), so that substantial differences may arise between individuals. This greatly complicates the debate about what are individual variants in movement patterns and what are errors in performance.

3.2.3 *Fixed principles of movement*

It seems that to a certain degree we all solve movement problems in the same way and easily manage to move efficiently. The overall structural plan of the musculoskeletal system is linked to a strategy for eliminating inefficient, unstable movement patterns that is deeply anchored in the system. This means we do not design movements as series of isolated movement incidents, but as coherent applications of fixed movement principles. The movement patterns that we choose are thus interconnected. The fixed principles are as universally valid as possible – in other words, the more movements a principle can be applied in, the more important it is to the system. The body has little interest in learning principles that only work in a limited number of cases. Rather than tricks, it wants to learn a flexibly applicable technique.

When we learn movements, we thus primarily learn to find and apply the 'mathematical' rules that filter out inefficient ways of performing the movement, in order to avoid rigid use of muscles and hence rigid movement patterns. A library-type retrieval system has to be set up. In walking, one such mathematical rule is the 'Froude number', a mathematical formula for the relationship between the leg length, walking speed and step frequency of all terrestrial animals (Alexander, 2003). There are also other rules in walking, such as rules for the transition from the stance phase to the swing phase (Prochazka, 1993). After these mathematical rules are selected, they are combined into larger systems, such as the principles of cocontraction. These larger systems are then translated into generic subsystems of contextuality, such as the organization of trunk control, cooperation between the lower leg and the foot at push-off, fixed cooperation between muscle groups that can move the pelvis, transmission of kinetic energy to distal parts of the body and so on. These fixed structural components of movement are further combined into larger contextual units, generic patterns of jumping, running, etc. Since the system seeks to maximize its use of generally applicable rules, movements become more similar. There is thus not all that much difference between running and single-leg jumping apart from forward speed. Running is basically a succession of jumps from leg to leg, in which the basic arrangement is not so different from long jumping or high jumping with a run-up. This efficiency strategy of emphasizing multi-purpose control is thus the basis for the 'specificity' and 'transfer' of training patterns.

3.2.4 *Attractors and fluctuations*

Movements are thus designed by eliminating degrees of freedom until a robust, efficient pattern is left. Here 'robust' means stable and hard to perturb, and 'efficient' means performing the movement with minimal energy costs. The theory of phase transitions is described in Section 1.1.2. It shows that a movement pattern seeks to be stable, and may suddenly shift to a different stable pattern if stability is lost. The theory states that such a phase transition may sometimes occur even in response to minor perturbations. Sudden phase transitions thus occur in a movement landscape that includes both stable and unstable patterns, and in which the moving body attempts to shift from one stable pattern to the next by skipping unstable patterns wherever possible (Figure 3.4).

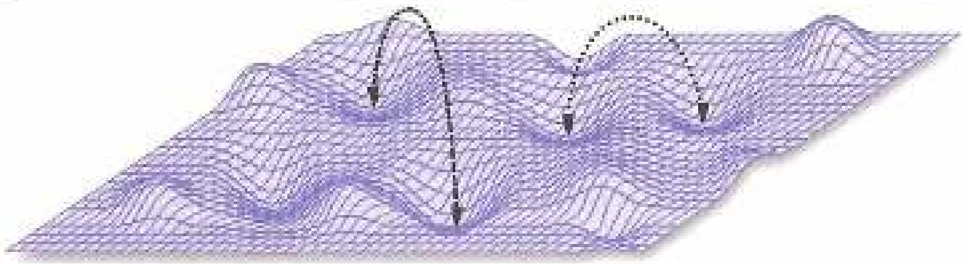


Figure 3.4 The movement landscape, consisting of wells with stable movement patterns and peaks with unstable ones. Movements preferably jump from one low-energy stable pattern to the next.

Stability and efficiency (economy) of movement play a role not only when choosing movement patterns. Even within a single movement, the various components of the movement are arranged into stable, low-energy components and unstable, high-energy ones. Stable, economical components of the movement are referred to in the literature as 'attractors' and unstable, high-energy ones as 'fluctuators' (also known in phase transition theory as 'order parameters' and 'control parameters' – Kelso, 1995). The fluctuators are needed in order to adapt the movement to the shifting demands of the ever-changing environment in which the athlete is moving. If a movement were to consist solely of stable factors, the movement would be performed rigidly, and influences from the environment could not be effectively incorporated into the movement pattern. A contextual movement therefore consists of a blend of attractors and fluctuators, which must satisfy two main criteria:

- The whole movement must be as stable (and hence economical) as possible.
- The number of fluctuators must be as small as possible, yet sufficient to meet all the demands of the environment.

The number of fluctuators must be as small as possible because the movement is only controllable if there are only a limited number of variables to be controlled. In the aforementioned comparison to a car, the steering wheel is a fluctuator (or, if you like, a degree of freedom). The fewer steering wheels a car has, the more easily it can be steered. The fewer variable ranges of motion there are, the easier the contextual movement can be controlled. In other words, learning to move is not just a question of learning the various components of the movement, but also of learning the ratio between stable and unstable components (Davids et al., 2008).

The learning process for contextual movements therefore takes longer than is generally thought, for it is not enough to learn the right components of the movement – it is also necessary to learn which components have to be used in a stable manner and which in a variable one (Figure 3.5).

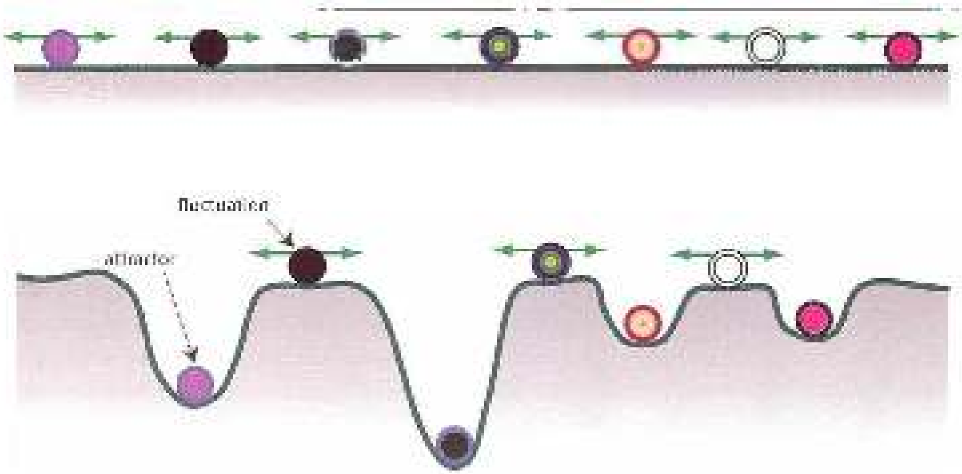


Figure 3.5 Top: a flat landscape in which all the mastered components of a movement may vary (roll from left to right). All these mastered components must be controlled at once – which is impossible.
Bottom: a mastered and applicable movement pattern in which the ratio between stable components (the wells) and variable ones (the peaks) is such that the overall movement can be controlled.

Additional information

The learning process only partly consists of learning to perform the various components of a movement. An important part of the motor learning process focuses on the correct division of components of the movement into attractors and fluctuators. At some stage in the initial learning process, especially in more complex movements, the attractors and fluctuators that develop may not meet all the criteria for optimal and efficient movement in the environment, and hence will be undesirable. The resulting attractors must be perturbed again in order to create a new, better arrangement. Perturbing the existing inadequate movement patterns is therefore a key step in learning new, better movements – and one that tends to be neglected in the design of the learning process. Instead, the focus is on learning the new pattern, which is impeded by stubborn old patterns. In some situations, the real problem in the learning process may in fact be the perturbation of old attractors, whereas learning a new pattern is a relatively simple matter. Indeed, there is plenty of evidence that this often happens in injury rehabilitation.

Once the correct ratio between stable and variable components has finally been found, the wells in which the stable components are located are further deepened. In other words, the stable factors become even more stable and low-energy (Figure 3.6; Thelen, 1995).

The red lines in Figure 3.6 represent five clearly distinguishable stages of the learning process:

1. When making the first attempts, use is made of fixed components (attractors) of movements from other, already known movement patterns. This is useful because it limits the number of degrees of freedom, the movement can be controlled and initial success can be achieved.

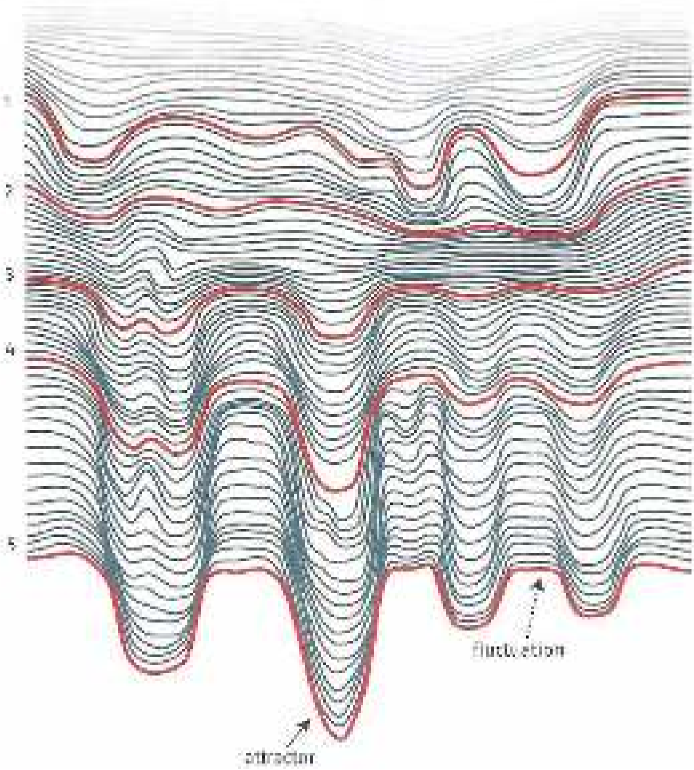


Figure 3.6 Attractors and fluctuators in the learning process (see Thelen, 1995). The landscape of attractors and fluctuators changes from the start (top) to the end (bottom) of the learning process.

2. The selected fixed components are not suitable for improving performance of the movement, and must be perturbed and detached. This makes the movement difficult to control. Performance skill is reduced, and as an emergency measure a number of joints are immobilized in order to make the movement controllable once more. This is known as 'freezing' degrees of freedom, and there are plenty of examples of it. Children who are learning to throw tend to put the wrong foot forward. This is in fact a strategy for freezing degrees of freedom and keeping the complex throwing movement under control. Similar strategies include keeping the knees extended on a slippery surface, immobilising the trunk in the first attempts to hit a golf ball, and making the backswing with an extended leg when shooting in soccer.
3. Better attractors and fluctuators are found, in a ratio that can be used in many different circumstances. The movement becomes more economical and can adapt to many circumstances with a limited number of fluctuators.
4. The attractor wells are further deepened in order to make the performance of the movement more suited to the body and efficient. Strength training can play a key role here (see Section 4.3).

5 The movement is so stable, and there are such effective ways (fluctuators) to adapt the movement to the demands of the environment, that control can become automatic. The movement is mastered and can, for instance, be combined with other tasks such as making tactical decisions.

This division into stable and unstable components cannot possibly develop from hierarchical, top-down organization of the central nervous system (the brain is the command centre for the design of the movement, and the muscles merely carry out those commands without influencing them – see Section 4.4). Differences between individuals in movement patterns would be much greater in high-intensity movements than what is actually observed in sporting movements, for the brain is extremely plastic, and extremely varied movement patterns could result. Moreover – and perhaps far more important – a hierarchical arrangement would inevitably lead to injuries. In high-intensity movements, large forces act on the body. This makes many structures vulnerable, with some at greater risk than others. When throwing, for instance, the shoulder joint is at greater risk than the lumbar spine. It is therefore important to keep the shoulder joint in a position such that the muscles can protect the joint as well as possible and to use spine movements in such a way that the arm reaches in the right position in space. Such an arrangement of joint angles can only be achieved if it is also based on bottom-up organization. This means that a non-linear arrangement of movements can also be justified in the interests of preventing injury. Higher parts of the system create the general, more abstract rules of the movement, while specific muscle actions and ranges of motion tend to develop from self-organization of the musculoskeletal system.

Additional information

As we have seen, it may be a useful strategy to distribute the movement over several joints and so maintain optimal stability in the joints that require this in order to prevent injury and perform the more unstable movements in joints that are less at risk. We may therefore question the common physiotherapeutic strategy of eliminating compensatory movements in rehabilitation exercises as much as possible, for example by forbidding patients who are recovering from a shoulder injury from bending the trunk sideways when abducting the arm during an exercise. Compensatory movements are a strategy by the body to achieve as healthy a ratio of ranges of motion as possible, and it may be wiser to use this as the starting point for exercises. In that case, compensation is allowed, and gradually adjusting and altering the exercises eventually restores the original level of contextuality. Exercises are thus based on the self-organizing capability of the body, which is not forced into the 'ideal' movement pattern. There is therefore a great need for more research into how the body uses self-organization to prevent at-risk joint positions. This will not only provide valuable information on self-organization in the body, but also a clearer perspective for assessing the value of various tests, such as the functional movement screen (FMS), that are designed to chart athletes' development. Such tests tend to assign great value to maximal ranges of motion and the amount of force that athletes can produce in extreme joint positions. How contextual these tests are, and to what extent the measured values are related to an athlete's potential for development and risk of injury, is therefore questionable.

Every athletic movement therefore includes components that are fixed and unchangeable, and components that can be adapted to the demands of the environment. A very precise description of technique, especially one that prescribes ideal joint angles of the movement, is therefore incomplete. Besides describing the required joint angles, it should also indicate which of the joint angles should be invariable (attractors) and which should be incidental (fluctuators) – otherwise this can lead to persistent misconceptions. One example is how high the knee should rise when running at high speed. In athletics this often leads to strict instructions about knee positions for sprinters. In the earlier comparison between running in athletics and running in other sports (such as soccer), the difference in knee position is often used to argue that running in athletics is little related to running in ball sports (Sayers, 2000). Since elite sprinters lift their knees higher than elite soccer players, it is argued that their technique is essentially different – but this is not true. Knee height is not so much an attractor as a fluctuator: the higher the running speed, the higher the knee, and sprinters have substantially higher top speeds than players of ball sports. This means that knee height is largely irrelevant to technique. Identifying such fluctuators as key features of technique leads to a series of incorrect conclusions about similarities in movement and hence transfer between different movement patterns. It is not enough to rely – as researchers into ideal performance of movement often do – on differences in range of motion.

Two examples

1. Throwing a ball creates a high-intensity load that puts the shoulder joint at risk. There is a remarkable similarity between individuals in the position of the shoulder joint when throwing. Not only experienced, trained throwers but also beginners abduct their arms to an angle of about 90° when throwing. This position hardly varies, and hence is an attractor. The idea that it arises in the body in a self-organizing manner is supported by the observation that beginners also place their arms at an abduction angle of 90°, even though they have not yet mastered the overall movement. However, there is much more variation in sideways bending of the trunk between beginners and advanced athletes, and this partly depends on the direction the ball is thrown in. The movement of the trunk is hence a fluctuator.
2. Baseball batters have to adapt their swing to the height and speed of the ball. As we have seen, the adaptation cannot be performed in all the joints at once, since the movement cannot then be controlled. Good batters therefore adapt their swing with only a few variables: bending their trunk according to the height of the ball, rotating their pelvis according to the position of the ball on the inside or outside of the home plate, and flexing their front knee according to the speed of the ball. These are the main fluctuators in batting. The batting swing movement of the arms is not adapted, and hence is an attractor in the movement pattern. Adaptation through variability of the knee and trunk is not explicitly taught, but occurs automatically with emphasis on not varying arm movements (Figure 3.7).

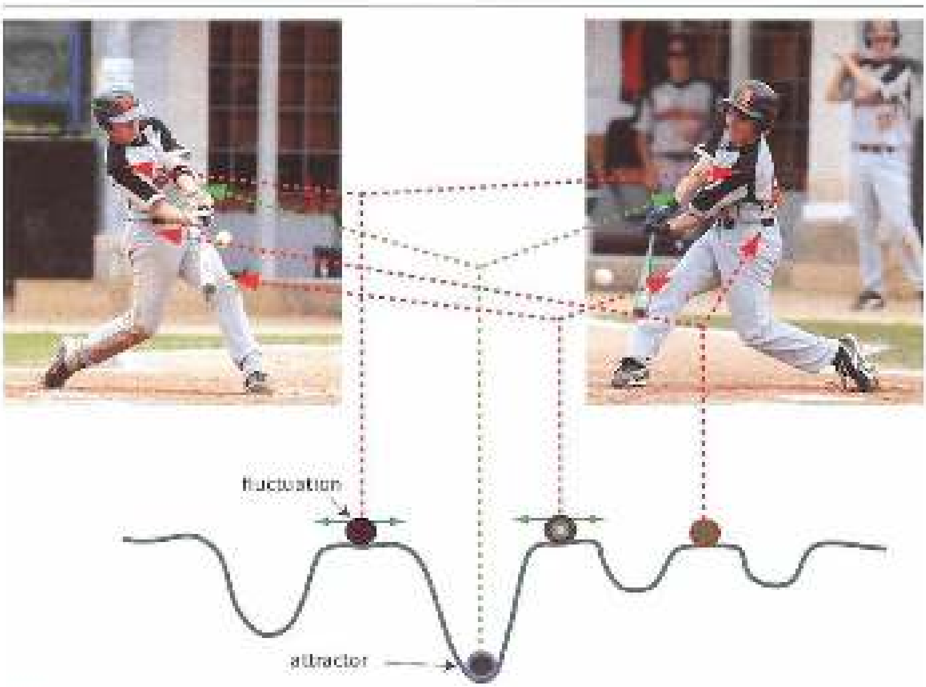


Figure 3.7 Attractors and fluctuations in baseball batting. The swing must be adapted to the variable height and speed of the ball. This is done by varying the angle of the front knee in response to the speed of the ball (the knee more extended if the ball is fast), opening the pelvis further if the ball is closer to the inside of the home plate and the trunk flexing sideways further if the ball is pitched lower. All three are fluctuations. The batting swing movement of the arms does not change, and hence is an attractor.

3.2.5 *Analyzing open and closed skills*

In conclusion, even when performing extremely open skills that require major adjustments to the environment and appear to be completely dictated by the environment, the excessive number of degrees of freedom must be controlled by building attractors into the movement pattern. An extremely open skill is thus a blend of closed- and open-skill components that must be chosen as fixed components that are efficient in all such situations.

Furthermore, movement in open skills becomes uncontrollable if the athlete has to keep shifting from one movement pattern to another. Constantly changing the organization of control makes considerable demands on the system, because new decisions constantly have to be made. Extremely open skills make numerous demands on a limited capacity of attention that is soon unable to cope. Soccer players who not only have to make tactical decisions and time their movements in relation to the environment but must also keep switching between very differently organized movement patterns while running will soon have too little control capacity to do all this successfully. It is therefore important to be able to perform key parts of the full range of actions, such as all running movements, using a single form of organization.

The demands made on the control of well performed movements – being able to control the various degrees of freedom, mastering the complexity of the open environment, keeping movements economical and protecting the athlete against injury – are such that even highly improvised movements must be built around solid, fixed basic components of movement. This means, for example, that sprinting on an athletics track, running in a marathon, running in a straight line and even improvised changes of direction during, say, a soccer match must all be based on the same fixed basic structures of starting/acceleration and running at speed – otherwise the motor skill no longer works.

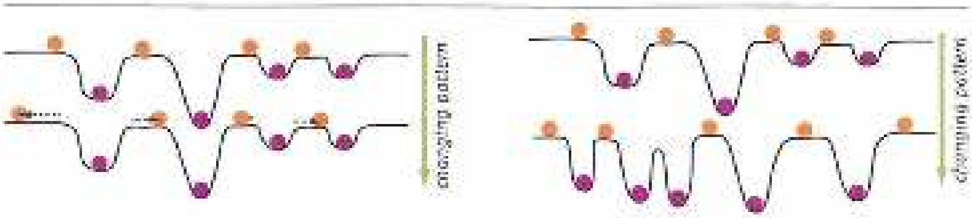


Figure 3.8 Changes in running movement, e.g. in ball sports.
Left: the attractor/fluctuator landscape does not essentially change (from top to bottom). The changes only occur because of shifts in fluctuators.
Right: the running pattern changes because of a substantial change in the whole landscape. Different attractors and fluctuators develop (from top to bottom).
The fewer changes that are needed to the attractor/fluctuator landscape, the more the quality of movement technique in agility improves.

The same applies – to a lesser extent – in closed skills. Every closed movement is a strategic blend of variable and non-variable components, because even in the most closed skill adjustments must be made to minor perturbations in the movement pattern. This means that analysis of both closed and open skills runs into the same problem: the attractor components must be distinguished from the fluctuator components, in order to determine what is permanent and what is incidental within the movement pattern. This may be much harder to determine in open skills than closed. In closed skills, however, the difference between winning and losing depends on much smaller technical differences than in open ones. 200-metre runners who are determined to keep their knees high while running at top speed may actually run more slowly because a high knee position may be far from optimal owing to athletic capabilities, decreasing running speed on the straight and increasing fatigue.

The distinction between basic structures such as attractors and variables such as fluctuators has far-reaching implications for training of movement patterns, and especially for the organization of strength training.

3.2.6 Attractors and agility in sport

So far there has been very little specific research into attractors within athletic movement, so there are no usable lists of attractors in throwing, running and so on. However, given what is known about motor control, we may conclude that such attractors develop mainly through self-organizing mechanisms in the body and are therefore closely linked to the architecture of the musculoskeletal system. Using anatomy, we can therefore draw up some basic rules for identifying stabilizing attractors within optimal movement technique. An important factor here is that the central nervous system is relatively slow and imprecise. The greater the demands made on control of movement, the less effective the control by the central nervous-system and the more self-organizing influences from the body will play a part in movement control and hence in the development of the attractor/fluctuator landscape. The following conclusions can be drawn in the light of this:

- In non-linear control it is essential that principles should be generally applicable (for instance Froude numbers). This means that attractors are seldom specific body postures. Movement attractors are relatively abstract principles of movement.
- The greater contextual variability, the greater the problem the central nervous system has to solve, and the more stability based on self-organization (attractors) will be required.
- The central nervous system is relatively slow. If the movement has to be controlled under time pressure, the role of peripheral influences will increase. These influences create attractors (see the operation of prefiexes in Section 4.3.3).
- Structures that are at risk, such as the shoulder and elbow when throwing or the hamstrings when running (see Section 5.6), should preferably be in an attractor state when under a heavy load – this protects them best.
- Movements have to be decelerated. At that point the body is particularly vulnerable, which means the movement should preferably be decelerated in a stable attractor state. This fits in well with the notion of intrinsic knowledge of result and end-point orientation (see Section 4.1.1)

Of course this list is incomplete, but it can be used hypothetically to describe a number of attractors for athletic movement.

We have already seen that agility in sport is best achieved by ensuring that the structure of the attractor/fluctuator landscape changes as little as possible when switching from one type of running to another. It is therefore important to formulate a number of attractors for running and agility, using the above 'search rules'. Even the following list of eight agility attractors is incomplete, has hardly been researched and is mainly based on theoretical principles and best practice.

1 Lock position of the hip

'Force closure' (a closed web of muscular forces enclosing passive anatomical structures) based on cocontractions round the hip has previously been described (see Section 1.3.3, groin injury) as a key mechanism for protecting the pelvic area against large, unpredictable opposing forces. In addition, this movement attractor (which develops from cocontractions) is a key reference point for the self-organization of toe-off (see Section 4.4.1, dominant intrinsic information). At toe-off, extension of the leg in the sagittal plane can be performed safely if it ends with a protective 'hip lock' (see fig. 4.29).

In good runners this dominant cocontraction round the hip is particularly noticeable in abrupt changes of direction such as a sidestep and a stop-and-go, and in the first phase of straight-line acceleration.

Figure 2.7 Abrupt change of direction and acceleration, both with an excellent hip lock.

2 Swing leg traction

In the last part of the swing phase the hip is extended just before the foot plant, then knee extension is added, and finally also ankle extension just before contact is made with the ground. If very carefully timed, this 'triple extension' ensures optimal pretensioning of the muscles just as the ground reaction force has to be absorbed during the stance phase. The biarticular muscles have a key part to play in this pattern of extension prior to contact with the ground. Through their rope function they pass on the energy of the hip extension from the hip to the knee (via *rectus femoris*) and from the knee to the ankle (via *gastrocnemius*) after the hamstrings have transported energy from the knee to the hip. There are good reasons to assume these muscles function almost isometrically in the running movement (see Section 5.6, function of hamstrings and specificity).

In all the varieties of the running movement, this triple extension should remain intact until contact is made with the ground. In addition, it should be reversed as little as possible on first contact with the ground, because stiffness provides resistance to the opposing ground reaction force.

When running in a straight line, being able to extend the hip until the foot plant does not at first seem a serious problem, even though the quality of the build-up of tension in the triple extension, and hence also the quality of the push-off, greatly depends on good hip extension just before the foot plant. The function of the hamstrings in directing the horizontal component of the rearward push-off is particularly sensitive to the quality of the hip extension.

However, during sharp changes of direction such as a sidestep or single-leg jumps with a run-up, hip extension does become a problem and often seems to stop just before contact is made with the ground. The foot then lands too far in front of the body and is not sufficiently retracted. Because hip extension stops, there are no satisfactory cocontractions round the hip and pelvis and not enough 'force closure', leading to loss of performance and increasing the risk of injury. 'Swing leg retraction' before contact is made with the ground should therefore be a major component of movement technique, and attention should above all be paid to it when performing a sharp change of direction (horizontally in a sidestep and a stop-and-go, and vertically in a single-leg jump and even while bracing the front leg when throwing or bowling in cricket.

Figure 3.30 Swing leg retraction in a sidestep and a vertical single-leg jump.

3 Foot plant from above

A key basic component of a technically satisfactory running movement is the 'foot plant from above' principle. This means that just before the foot plant the foot should move towards the ground as parallel as possible to the direction of the ground-reaction force that will occur in the subsequent ground contact. If the foot moves towards the ground at a large angle to the direction of the subsequent ground-reaction force, the foot is 'slapped in'. As a result, the ground-reaction forces cannot be properly absorbed and the stance phase proceeds less effectively. This principle is again essential not only when running in a straight line, but also for all sorts of agility and all vertical single-leg jumps with a run-up. To achieve a good foot plant from above, the foot must be brought to the ground from a slightly greater height. This conflicts with the traditional notion that changes of direction are best performed by keeping the feet as close as possible to the ground.

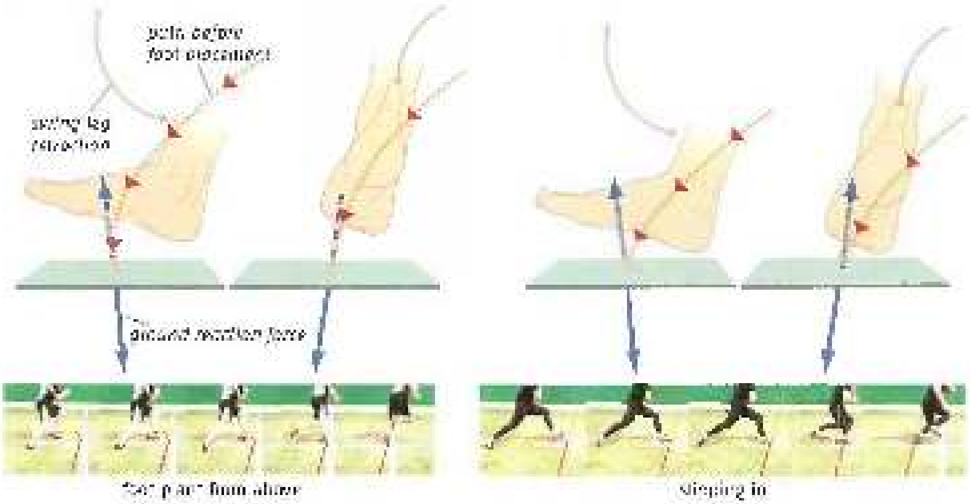


Figure [illegible] Left: the 'foot plant from above' principle, with the movement just before the foot plant as parallel as possible to the direction of the ground-reaction force. In the series of photographs this principle is applied satisfactorily in a change of direction. Right: the foot is 'slipped in' because it is too close to the ground just before the foot plant. The foot motion and the expected ground-reaction force are at an angle to each other when contact is made with the ground. In the series of photographs the direction of the ground-reaction force is unfavourable in the first part of the stance phase. The runner must therefore wait until his hips are above his feet before producing force and changing direction.
Hip lock, swing leg retraction and foot plant from above clearly have a major influence on each other and are parts of one basic movement pattern. The position of the pelvis at the start of the triple extension determines how the leg extends, and together they determine the foot's path towards the ground. The overall pattern ensures that in acceleration and changes of direction the place where the foot touches the ground in relation to the body matches the direction of the expected ground-reaction force.

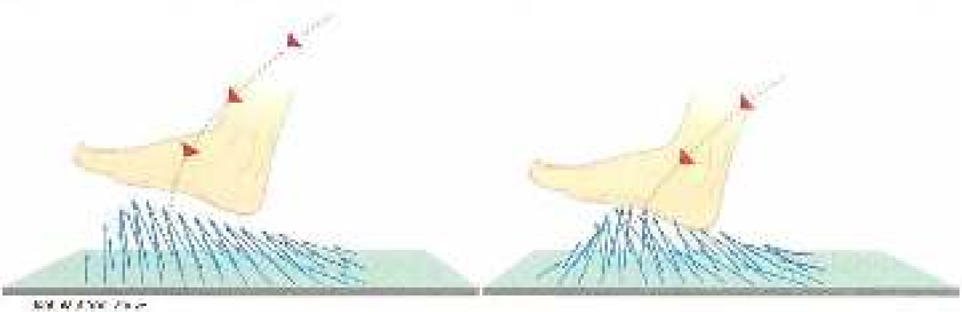


Figure [illegible] Foot plant from above and the 'landscape' within the placement and direction of the ground-reaction forces resulting from the right combination of hip lock and swing leg retraction. When the foot is 'slipped in', this landscape becomes much more untidy.

4 Positive running position

To perform an effective swing leg retraction, the swing leg must be brought into a correct starting position before the triple extension begins. However, this position of the leading leg (with sufficient hip flexion) must not be seen in isolation, as the potential for proper swing leg retraction very much depends on how fast the trailing leg can move forwards. If the trailing leg is a long way behind and has to be moved forwards from that position, the leading leg must 'wait for the trailing leg to catch up' – otherwise there will be anterior rotations that will have to be corrected by decelerating during the next step. In reality the leading leg always 'listens' to the trailing leg, and waits for it if necessary. For running actions that have to be performed at relatively high speed (such as swerving and a sidestep at speed) the body must therefore assume a 'positive running' posture as much as possible: the upper body is upright, with the hips forward and the knees pointing forward as much as possible. Such an extreme body posture is of course difficult to achieve in many situations, such as when changing direction during acceleration, but it should nevertheless be aimed for: as upright as possible, with forward pressure from the hips and the knees as far forward as possible. In this position the body is as well-balanced as possible, swing leg retraction is possible, and force can be produced as well as possible during toe-off.

Figure 3.27 Positive running at top speed and in a sidestep

5 Keeping the head still

Visual information is important when controlling movement. To absorb information from the environment as well as possible, vertical head movement should be kept to a minimum. In that case vision does not need to be calibrated so often and visual information can be processed more effectively.

In sports in which reading the opponent or intercepting the ball is a key aspect of performance, this movement attractor is therefore crucial to agility. A judoka moving across the mat with his opponent, a defender shadowing an opponent by moving sideways, a baseball player accelerating to intercept a ball or a pitcher who needs to observe where the ball will end up in the strike zone should keep vertical movement of the head to a minimum.

There is more vertical oscillation at low step frequency than at high step frequency. Athletes who need to gather visual information should therefore adapt (increase) their step frequency, preferably without harming other attractors such as swing leg retraction and foot plant from above.

In agility at high speed and high step frequency, vertical head movement can therefore always serve as one of the criteria for quality of performance. This means that technique may be judged less satisfactory if the athlete moves too deep into a squatting position when changing direction abruptly.

In some movement patterns, keeping the head still is so important that it may be the main source of feedback for an effective learning process. Examples include sideways movement in defence, a tennis player's movements during a rally, and hurdling (the best hurdlers should keep their heads as level as possible).

Figure 3.[illegible] Keeping the head at the same height in sideways movement.

6 Upper body first

Running at speed generates torsion in the upper body. One of the functions of this is to increase body tension, and it is, for example, one of the reasons why the Fosbury flop has become such a successful high-jumping technique (torsion is lacking in other techniques).

This increased body tension is also important in various types of change of direction such as the sidestep, the stop-and-go and the swerve, and can be summed up as the 'upper body first' principle: all changes of direction should be initiated from the shoulder girdle and upper trunk. Wherever possible, the upper trunk should start to turn in the direction the athlete wants to go in before the push-off. This leading role for the shoulders creates extra body tension, making it easier to cope with the large opposing forces of the change of direction. An important added benefit of the 'upper body first' principle is that the 'swing leg retraction' principle automatically improves if the movement is performed correctly.

Figure 3.45 The upper body first principle in a sidestep.

7. Extending the trunk while rotating

In many ball sport situations, athletes must be able to accelerate and run at speed with their upper bodies turned to one side: a soccer player about to receive a long pass, a rugby player who expects to be tackled and so on. They must be able to run with the upper body rotated so that their leg movements are impeded as little as possible by torsion. This means that torsion should not influence the position of the pelvis. The best way to achieve this is by combining rotation with trunk extension. When rotation and flexion are combined, the pelvis will inevitably change position in the running movement, and the 'launch platform' for leg movements will be at an angle. This is clearly visible in soccer players, when rotation of the trunk clearly leads to loss of speed. In sports (such as soccer) in which it is important that the upper body should move independently from the lower body when running, the rotation/extension technique may have a major impact on performance skill.

Figure 3.46 The principle of extending the trunk while rotating.

8 Distributing pressure when decelerating

Decelerating is a key aspect of agility. Since, biomechanically speaking, it has the least in common with the running movement, decelerating situations are more isolated.

The greatest pressure when decelerating is likely to be on the knee joint, so limitation of performance initially occurs in the muscles round the knee. To shift this limitation and minimize the load on the knee, the decelerating technique should ensure that the force acting on the body is distributed over the greatest possible area. Such distribution of peak forces is vital whenever decelerating. When decelerating after a throwing movement, for example, the force acting on the back of the shoulder should be distributed over as large an area as possible and hence be partly conducted away to the trunk by rotating the upper body round its longitudinal axis (rotating to the left when throwing with the right arm).

When decelerating during running, the peak force must likewise be distributed over a large part of the body, for instance to the hip and the back. This can be done by moving the upper body forwards when decelerating —not through flexion of the trunk, but by flexing the hip joint and at the same time rotating the pelvis forward. Opposing forces are then conducted away from the knee by using cocontractions of the abdominal and back muscles (ideally at their optimal length) to move the pelvis forwards, which then increases the traction on the hamstrings cocontracting with the *quadriceps* (again ideally at their optimal length), so that the flexing torque and the shearing force round the knee are reduced.

This rather abstract principle of movement must develop in a self-organizing manner and should be a key feature of exercises that focus on decelerating. When rehabilitating after forward cruciate ligament (ACL) reconstruction, for instance, we must consider whether anterior movement of the upper body should be built into several movement patterns at a much earlier stage than is customary.

3.2.7 *Implications for strength training*

Clearly, the process of deepening the right attractor wells and using fluctuators flexibly is largely self-organizing. Since no-one knows precisely how all this happens, no-one can directly teach the learning system how to organize. All a coach, physical therapist or movement expert can do is create conditions that optimize the self-organizing ability's chances of finding generally valid principles and satisfactory solutions. Among other things, this requires knowledge of how motor learning processes work. Such knowledge has so far had very little impact on sports practice – and this is probably the main blind spot among coaches, rehabilitation professionals and others involved in sport. Especially within strength training, little thought is given to how people actually learn. Yet thinking about motor learning processes is surely an essential part of sport-specific strength training. Strength training is very suitable for supporting learning processes with regard to basic components of the movement (attractors), but less suitable for learning the eventual, environmentally appropriate performance of the overall movement, and hence less suitable for learning overall patterns, including the function of fluctuators. Strength training can thus be used to improve important coordinative building blocks of movement, such as the aforementioned agility attractors. This changes the meaning of sport-specific strength training within the overall training plan.

Strength training is thus particularly suitable for teaching the general fixed principles of movement and studying these attractors in greater depth. Especially in open-skill sports, in which transfer from the strength exercise to the sporting movement is hard to identify, analysing athletic movement and identifying the stable components of the movement are key steps in devising a meaningful strength training programme. The methodology focusing on individual muscles, which is derived from bodybuilding, yields no improvement in essential movement attractors and hence is not so suitable for sport-specific strength training. In other words, strengthening individual muscles in the absence of any context merely focuses on the final stage of the movement design (individual muscle action), and hence serves little purpose.

'Strength is coordination training against resistance' is therefore a good definition of what sport-specific strength training ought to be. Since it is basically coordination training, strength training needs to take account of the laws of motor learning, as described in Chapters 5 (specificity) and 6 (overload). Chapter 7 identifies (where possible) various attractors in open and closed skills, and translates them into the practice of strength training.

3.2.8 *Over-reliance on measurement*

Focusing on strength training as a means of improving technique comes at a price. It is difficult, if not impossible, to measure adaptation through technically oriented strength training. What this means for coaches is there is little point in regular testing to measure the impact of strength training. Classic jump tests, power measurements and so on do not mean all that much if the focus is on coordinative transfer of strength training. This is not actually such a bad thing, for the predictive value of such classic measurements for performance in the sporting movement has always proved disappointing (Walsh *et al.*, 2004). It is unrealistic to rely on measurement. Focusing on test results should be replaced by a constant search for the link between the strength exercise and athletic movement. Of course, this makes for greater mental demands than merely focusing on measurement results. Athletes as well as coaches must try to imagine just how technique is transferred. Having a mental idea of how the strength exercise will be 'manifested' during athletic movement (visualization) will increase the amount of transfer. In other words, mental training works (Shackell & Standing, 2007).

3.3 Summary

Sport-specific strength training only makes sense if adaptations are transferred to athletic movement. In order to understand the capabilities of transfer, not only the strength exercise but also athletic movement must be analysed. In complex movements there will be many different alternatives for almost every contextual movement, given the many different joints and muscles in the body. Selecting from this vast number of alternatives, or degrees of freedom, is the basis for the design of the movement. The difference between open and closed skills is crucial here. Open skills are particularly hard to analyse, given the highly improvised nature of movement patterns.

Constant changes in movement patterns ensure the necessary adjustment to the environment. This adjustment must be controlled. If it could take place anywhere in the movement, the number of degrees of freedom would become too great and the movement would cease to be controllable. Every movement, including open-skill movement, therefore consists of both fixed, unchangeable components and changeable components that can be adjusted to the environment. The fixed components (attractors) are stable and economical. The changeable components (fluctuators) are unstable, and have high energy costs. It is extremely difficult to tell which components of movement are fixed and which are changeable. However, this must be done in order to design training schemes in which the fixed components are an intrinsic part of the movement pattern.

Strength training may be suitable for training many of these basic components of movement. The desired coordinative transfer of strength training cannot simply be identified by measurement. Training must therefore focus on more than just test results.

4
Fixed principles of training: contextual strength and coordination

Strength training leads to adaptations, which may be physiological and/or coordinative. Physiological adaptations within strength training include increasing the physiological cross section by synthesizing more proteins (hypertrophy), making the aerobic (FT IIa) fibres work more anaerobically, and changes in hormone balance. Coordinative adaptations include better intramuscular coordination (e.g. by better recruitment of muscle fibres or better linkage of force production to speed of muscle shortening) and better intermuscular coordination. Athletes can attempt to achieve adaptations through sport specific strength training in either a physiological or a coordinative manner or a combination of the two.

4.1 Physiological or coordinative adaptations in strength?

Coaches prefer to maximize whatever physiological and coordinative adaptations occur. However, this is by no means always possible or useful, and a choice must often be made between the two. Which approach is most effective will depend not only on how useful the adaptation is in improving athletic movement, but also on the extent to which adaptations due to strength training can be transferred to athletic movement. Especially in explosive sports the physiological approach is unlikely to be profitable, for excessive hypertrophy is undesirable (except perhaps in the proximal monoarticular muscles) and the hormonal response is too unpredictable to serve as a marker for organizing training. Furthermore, coordination at a high degree of mastery is the most performance-limiting factor in almost all explosive sports. It is therefore useful to take improvement of coordination as the aim of strength training for explosive sports.

4.1.1 *Speed skating*

In endurance sports the tendency is always to opt for the physiological approach. The metabolic effort in endurance sports has traditionally been seen as the most performance determining factor. Many endurance sport coaches therefore show little interest in the coordinative side of strength training. However, this is not always a good idea, for performance in some endurance sports is far more dependent on technique than is usually thought. Of course, it is safe to assume that movement technique is less difficult in endurance sports than when, say, sprinting. Naturally it is more difficult in long-track speed skating to take a bend at sprinting speed than at marathon speed. Yet it should not simply be concluded that technique is irrelevant to endurance sports and hence that the focus should be entirely on metabolic performance. The role of technique in speed-skating performance is well understood. Long-distance speed skaters also pay a great deal of attention to technical execution. Many speed-skating coaches therefore consider it extremely useful to approach strength training from a coordinative angle.

4.1.2 *Running*

Much less attention is paid to technique in long-distance running, and many marathon runners do no technique training at all – but this is a mistake. Even in running, endurance performance largely depends on technique. Elite runners may differ greatly in running efficiency. The main difference between the world champion marathon runner and the twentieth-placed runner is efficient technique. In endurance running the stance leg absorbs three to four times the runner's body weight on each landing. To keep this up for a long time, endurance runners must be able to recruit relatively large motor units with submaximal effort – in other words, long-distance runners have to be strong. Such high recruitment is trained by strength exercises, and combines well with the oxidative action of the FTIIa fibres. Such recruitment of oxidatively acting FTIIa fibres in addition to ST fibres (see Section 2.2.1) is essential to performance, for there is a close link between running economically and being able to perform isometric muscle action that produce high force. Major force production ensures that muscle fibres can remain at more or less the same length in the stance phase, without having to lengthen or shorten. The elastic parts of the muscle then change the length of the entire musculotendinous unit (see Section 2.1.4). Muscle fibres that stay at the same length produce force, but do no work: F (force) x s (distance covered) = W (work) when s = 0. Fibres that change in length produce force along a shortening path, and hence do work. Doing work generates higher metabolic costs than force production in isometric conditions. A technique in which a runner mainly produces force (only possible if enough muscle fibres are recruited) is therefore much more economical than one in which a lot of work has to be done (Figures 4.1 and 4.2). This influence of increasing energy costs in the transition from mainly producing force to doing more work is a far more plausible explanation than the exhaustion of energy supplies for why marathon runners 'hit the wall' at the 30-km point. However, the idea of increasing energy costs as the reason for the collapse in performance cannot simply be transferred to other sports. In sports such as cycling, in which [illegible] of energy through elasticity scarcely plays a role, other causes must be [illegible].

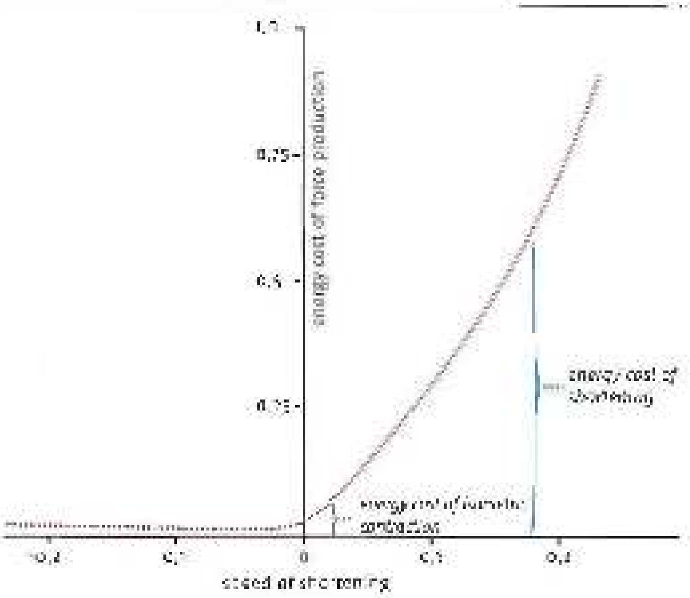


Figure 4.6 The energy costs of a muscle in force production in isometric conditions and when doing work. The energy costs in muscles that have to shorten are substantially higher.

Figure 4.7 Benefit in running: the gastrocnemius and the hamstrings hardly change in length. The lengthening at the knee is compensated for by shortening at the ankle and the hip, enabling these muscles to work very economically when running.

4.1.3 *Cycling*

Technique appears not to play any part in cycling – and yet it does. Two cyclists with the same body weight who can produce the same power per kilogram of body weight in aerobic endurance on a home trainer often turn out to perform very differently when cycling uphill. If technique played no part, there would be no such difference. However, muscles may cooperate in slightly different ways when cycling uphill. Since gravity then pulls in a somewhat different direction relative to the body, different intermuscular coordination may be required of energy-transporting muscles (hamstrings, *rectus femoris* and *gastrocnemius*), The difference in mastery of this technique may partly explain the difference in performance. Yet many experts claim that the influence of technique is only of marginal importance in cycling (putting a thick telephone book under the front of the home trainer may help during winter training). It must therefore be wondered whether cyclists can actually benefit from training their technique. Is it worth the investment, and should they not focus on the physiological aspect of strength training instead?

Between coordinatively complex sports and endurance sports in which technique barely plays a part, there are numerous sports in which the influence of coordination is less clear. In such sports the influence of technique on performance must be analysed in depth in order to decide how relevant technique can be built into strength training. For example, technique clearly plays a part in a sport such as rowing. A champion on a rowing ergometer is not necessarily a champion on the water. Which aspects that make the difference between the rowing ergometer and the boat can be usefully translated into strength training technique – only the building blocks of power production during extension, or is there more to it than that? Should the physiological approach to strength training be abandoned in favour of the coordinative approach – or vice versa? Much the same questions can be asked about motocross, cross-country skiing and so on.

4.1.4 Swimming

Besides the importance of technique in the sporting movement, there is, as we have seen, another factor that plays a part in deciding which approach to strength training to adopt – namely, to what extent can coordinative patterns be transferred from the strength exercise to athletic movement? The problem of transfer is particularly great in the case of strength training for swimming, for strength training is done on land and swimming, of course, in the water. This means that one of the most important constants when learning coordinative patterns – the emphatic 'presence' of gravity – is different in training on land and in the water. The main element of a movement pattern is sensorimotor orientation within the surroundings, reachable space in combination with being able to resist gravity. Owing to the upward pressure of the water, gravity is effectively absent. As a result of this difference in physical setting, a movement in the water must be organized differently than it is on land. This means there is very little transfer between land and water training in this primary element of movement. In addition to these motor differences, there are major sensory differences between moving in water and on land (known as 'feel for the water').

Such major differences may explain why it is so difficult to find an effective approach to land-based strength training for swimmers. One common strategy that is effective in all types of sports for guaranteeing transfer from strength training is choosing exercises that look like the sporting movement. In swimming, this strategy (strength training on a swimming bench) does not necessarily improve performance. When swimming in conventional swimsuits, using a swimming bench to increase muscle power had little effect. It was not until the development of superfast suits – which were in any case banned in 2010 – that greatly increased strength training suddenly proved highly effective for top performance over shorter distances.

Such elusive coordinative transfer suggests that transfer in swimming should instead be sought in physiological factors – although this may be just as hard to identify as coordinative transfer. In conclusion, then, it may be said that strength training for swimmers is extremely hard to design, because there is so little automatic transfer. Which type of strength training works in swimming may therefore have to be considered on a more individual basis than in other sports. A more aggressive approach to loading may work better with one swimmer, whereas another may benefit from as economical an approach as possible.

4.2 Adaptations in strength: the physiological approach

In endurance sports, strength training is always something of a problem. Many endurance athletes sooner or later include strength training in their programme, but eventually abandon it because they are not quite sure how it can contribute to their endurance performance. Endurance athletes rarely focus on technical details of movement and have little interest in, for example, barbell techniques. In addition, strength training for endurance sports requires a very different mental skill than is needed in endurance performance: within strength training, high intensity movement must be achieved in a few repetitions, and this requires an aggressive attitude. Such an attitude is not practised in endurance training. If anything, the opposite is true – aggressiveness must be curbed so that energy can be distributed over a long period of time. This is why endurance athletes do not usually feel at home in gyms.

In addition, there are few endurance coaches who are familiar with the theory and practice of strength training. Such coaches have quite a few prejudices about strength training, for instance that it always increases muscle mass – which would not be helpful in endurance sports. That is why endurance athletes seldom get a good education within strength training during their early years.

Besides the 'cultural gap' between endurance and strength, another factor is that the ways in which strength training can contribute to endurance performance are hard to identify. In explosive sports the link between the strength exercise and the sporting movement is easier to 'feel', whereas in endurance sports it is far less direct. Endurance athletes and coaches may even doubt whether such a link exists, especially if the technical aspects of performance are considered far less important than the physiological ones.

4.2.1 *Research*

A great deal of research has been done into the effect of strength training on endurance performance. If such research is to be sound, it of course requires a good instrument for measuring progress in performance. The performance must be measured in a sport-specific setting, which is not as easy as it seems. If we assume that strength training mainly makes its contribution through efficiency of movement, the measurement must come as close as possible to the sporting movement. With cyclists this is fairly simple – measurements on a cycling ergometer are considered representative of cycling performance. With runners, measurements are a lot more difficult. They are usually carried out on a treadmill. However, it is questionable whether running on a treadmill is sufficiently similar to running on a road or a track to be representative – especially if the measurement is made on a slightly sloping treadmill. Certain aspects of running technique that are important to running efficiency when running on a road may be far less relevant when running on a sloping treadmill (Figure 4.3). If, in addition, those very aspects of technique (converting the kinetic energy of the vertical motion into elasticity during the following stance phase) can be greatly improved by strength training, the result of the measurement on the sloping treadmill cannot simply be generalized to endurance performance on the road.

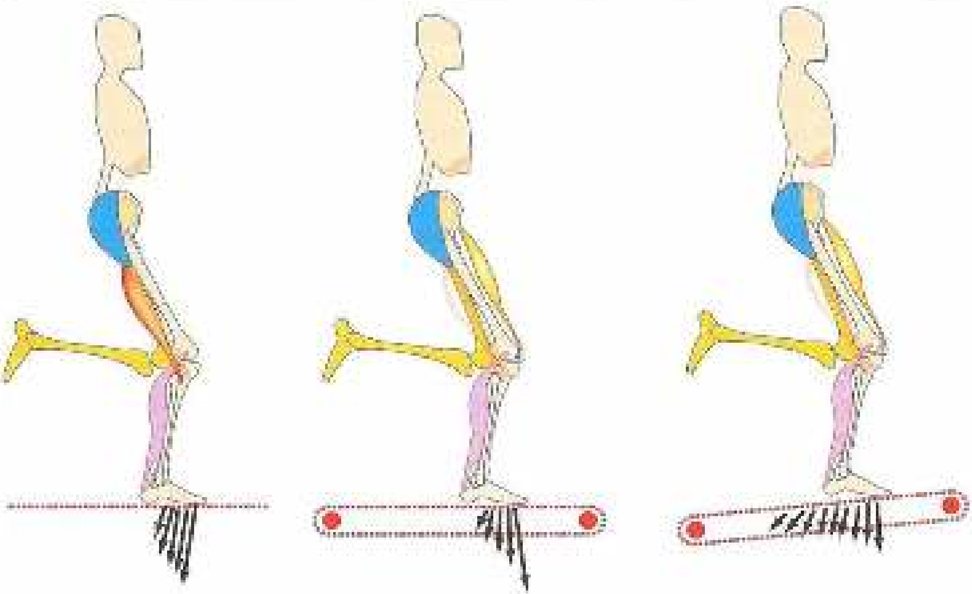


Figure 4.3 Running on the flat (left) requires a lot of activity from the hamstrings in directing force backwards. When running on a treadmill the force is directed more downwards, with more activity by rectus femoris. When running on a sloping treadmill there is less drop height to be processed elastically at stance (see also Figure 6.7).

In many scientific measurements, little thought has been given to whether the test setting is typical of the endurance sport concerned. However, this is not the greatest failing of research into strength and endurance. The main error is the failure to realize that transfer of training can only occur if strength training meets the needs of specificity (see Chapter 5). This means that, if generalizing statements are to be made about the impact of strength training on endurance performance, specificity must be very precisely analysed. The strength training carried out in the research must be described in very great detail – which adaptations occur, and in what way they are specific. Where adaptations include a coordinative aspect, the way in which technical performance is monitored during training must be described very precisely. If all these criteria are met, more or less generalizing statements can perhaps be made about the impact of strength training on endurance performance.

No studies have so far provided such a precise description of the research training process so that a generalizing statement can be made – if anything, the contrary. A lot of research is done without realizing what actually happens during training, and without understanding all the processes involved. This may be due to the widespread assumption that strength is an independent quantity that does not need to be carefully designed within research. The only conclusion that can be drawn from such inadequately described research is that the endurance performance has, or has not, improved during the research. General statements cannot be made about the impact of strength training on endurance performance.

Positive influence of strength training on endurance performance is most often found in runners, and much less often in cyclists, swimmers, speed skaters and so on. This makes sense, because major peak forces must be absorbed when running. Even less well designed strength training with large barbell weights will create a number of conditions for absorbing peak forces when running. This is like scoring in a large, empty soccer goal. In sports involving smaller peak forces, the goals in which players can score as a result of strength training are much smaller. In long-distance skating, for instance, force during the push-off does not exceed about 130–140% of the athlete's body weight, despite the high peak power (Houdijk *et al.*, 2000). Recruitment of fast-twitch fibres therefore has less direct impact on performance than in running. What matters in speed skating is how strength can be transferred to the ice in the right direction early in the push-off. Good skaters can produce force effectively at the start of the push-off, whereas technically poorer skaters can only do so later. Types of strength training that help athletes produce force earlier are much harder to find than ones that will improve maximal recruitment. It will therefore be easier to 'score a goal' in research involving runners, in which a number of fairly random types of strength exercises are trained, than in research involving skaters, cyclists and swimmers.

4.2.2 ST and FT fibres

Slow-twitch (ST) fibres are usually associated with endurance sports and fast-twitch (FT) fibres with explosive sports. For many coaches this is sufficient reason to think that strength training serves no purpose for endurance athletes. types of strength training that come close to maximal strength train FT fibres, which is supposedly 'useless in endurance sports', since these fibres work anaerobically and hence are only suitable for short-term effort. It is also reasoned that if you want to train ST fibres you must first exhaust the FT fibres (according to the size principle) in order to give the ST fibres the required overload. Such training leads to hypertrophy, which is very bad for endurance athletes.

However, a simple division into ST fibres for endurance sports and FT fibres for explosive sports is too simple. The distinction that is frequently made between ST and FT fibres mainly involves mechanical properties (the speed of muscle action and the speed with which muscle action can be initiated) rather than the physiological elements (which energy systems are used). Here it is assumed that the mechanical properties of muscle fibres determine ability for explosive sports and that sprinters have a higher percentage of FT fibres than endurance athletes. These mechanical differences between ST and FT fibres can scarcely be influenced by training, and so the performance level that can be achieved in an explosive sport largely depends on the relative distribution of ST and FT fibres ('it's hard to turn an endurance athlete into a sprinter'). We should also note here that the mechanical properties of the fibres are influenced by physiological properties (the contrast between the two types of property is therefore rather questionable) and that there are other important factors that determine ability for explosive sports, such as coordinative capability. Explosive athletes usually have a greater talent for movement than endurance athletes.

For endurance athletes the metabolic properties of muscles fibres are of particularly high importance. The metabolic differences between ST and FT fibres may be reduced as a result of training. FT fibres may start working more aerobically and become metabolically more like ST fibres ('a sprinter can be turned into a fairly good endurance athlete': Holloszy & Coyle, 1984). However, FT fibres will never achieve the mitochondrial and capillary density of trained ST fibres, and hence will never be able to work quite as aerobically as ST fibres. This means that, however well trained they are for endurance efforts, they will always produce a certain amount of waste (lactate, IMP, H+). Excessive accumulation of lactate in muscles is a problem for endurance performance. On the other hand, this problem is partly remedied by the 'lactate shuttle': the lactate produced by the FT fibres is absorbed not only by the liver but also by the ST fibres and serves as fuel for further aerobic energy conversion. So lactate is no longer the villain it used to be.

The benefit of training FT fibres for endurance athletes is that it increases the volume (aerobic power) of the body's aerobic engine. The drawback is that the engine does not process fuel quite so efficiently. So the situations in which the increased capacity of the engine makes up for its reduced efficiency need to be identified. In other words, in which endurance sports is it useful to recruit FT fibres (via maximal strength training), and in which ones is it not? It is always useful in running, for the muscles are required to produce large bursts of force. A second candidate for strength training in support of endurance performance is rowing, which also requires relatively large muscle action. In sports such as cycling, however, it is much harder to get the balance right. ST fibres work best at a given shortening speed, and hence a given pedalling frequency; if this is too high or too low, the ST fibres will work less efficiently. In some cases, such as cycling uphill, the pedalling frequency may fall too low so more FT fibres will have to be recruited to make the ST fibres work at their optimal shortening speed. Whether the total metabolic efficiency improves endurance performance will depend on the physiological properties of the additionally recruited FT fibres (greater efficiency through optimal shortening speed of the ST fibres versus the use of less efficient FT fibres). The value of strength training will therefore depend on the extent to which the recruited FT fibres can adapt physiologically to the demands of the endurance performance. All in all, striking the right balance between the benefits and drawbacks is no easy matter (Van Dieën & Bastiaans, 2006). The efficiency of metabolic processes and the improved efficiency of motion that can be achieved by strength training are two different factors, both of which must be considered when deciding whether strength training is useful for endurance athletes.

Additional information

Highly trained endurance athletes reach a ceiling in their oxygen uptake (VO_2max). Greater mileage will not improve it. However, athletes who wish to increase their VO_2max still further can consider maximal strength training as a means of doing so, for improved recruitment will bring more muscle fibres into play. This increased active muscle mass can then 'attract' more oxygen. Take the example of a cyclist whose VO_2max measurements are always the same. Since he has switched from cycling to triathlon, it is decided that his VO_2max will now be measured on a treadmill rather than a cycling ergometer. He will now be using far more muscles than when cycling, and his VO_2max will be significantly higher. At the same time, the extent to which VO_2max affects performance remains uncertain.

4.2.3 *Conflict*

There is an added complication when fitting strength training into training programmes for endurance sports: the 'interference effect'. It may be difficult to combine aerobic training with strength training geared to hypertrophy, for they can 'sabotage' each other. To put it briefly, aerobic processes are parasympathetic (it focuses on the organs and reduces muscle mass), whereas hypertrophy training is sympathetic (it increases muscle mass, producing a different hormonal response). So it is not helpful to do both during a given training period, for the conflicting training stimulus detracts from the intended adaptations. Aerobic training and maximal strength training can be combined much better, for maximal strength training mainly acts on the central nervous system and hence less on metabolic processes. However, the order in which they are provided and the intervals between them need to be carefully planned. To avoid such potential training conflicts, many training methods are organized into 'blocks', focusing on one type of adaptation during certain periods and on others during others – for instance one block focusing on strength training, followed by two blocks focusing on endurance capacity.

However, this arrangement is now less and less common in training for explosive sports, for large amounts of strength training are no longer considered so important, and intensity of training is increasingly reflected in high-quality performance of strength exercises. Reducing the amount, and increasing the quality, of strength training means that athletes suffer less fatigue and that strength training does not interfere so much with other types of training. As a result, strength training can fit into the overall training schedule all year round. This is also important within strength training for endurance athletes. Besides the fact that strength is relatively irreversible (it decreases only slowly if the athlete stops strength training, whereas aerobic capacity is extremely reversible), the 'block' arrangement must also take account of the fact that the physiological stress of strength training reduces the scope for endurance training. Less extensive, more efficient strength training somewhat reduces the stress and hence the need to train in blocks. This means that athletes can continue types of strength training for longer – a particularly important factor for endurance athletes, in whom strength is somewhat more reversible. Reversibility of strength varies greatly from individual to individual, and is usually greater in women than in men. Strength training is particularly hard to fit into training programmes for endurance athletes, whose strength levels rapidly decrease in the absence of training; their strength training must be organized as economically as possible in order to meet all the other training requirements (Figure 4.4).

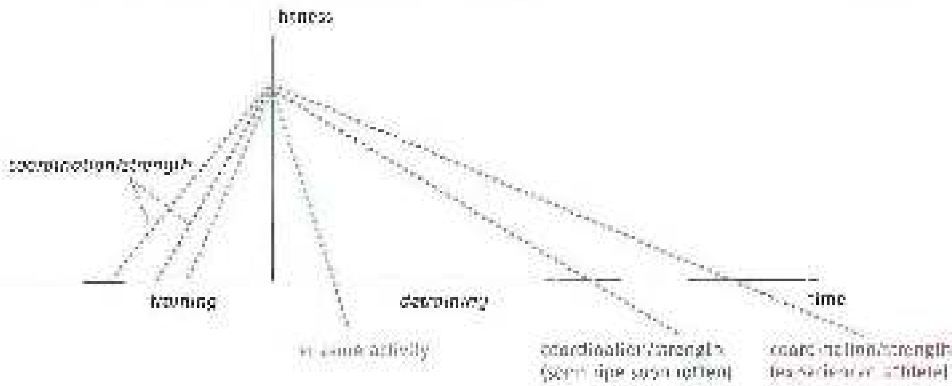


Figure 4.[illegible] Strength and coordination are far less reversible than enzymatic actions within aerobic processes. The shorter an athlete's training history, the more reversible his strength and coordination become ('soon ripe, soon rotten').

As we have seen, there is a price to be paid for including strength exercises in training programmes for endurance athletes. It is not a good idea to add strength training onto a complete endurance programme, for instance by planning long-distance running or some other low-intensity endurance activity right after strength training. This is very tempting, because strength training does not seem too fatiguing for the athlete and recovery is rapid. So opting for strength training also means opting for less endurance training. Many coaches are reluctant to do this, for fear of providing too little training. Others, however, are not, and claim that many endurance athletes do too much endurance training anyway, so a bit less will hardly be missed. At the same time there are other factors that are influenced by strength training, such as testosterone levels, which may have a positive impact on the athlete's load capacity.

In conclusion, strength training is useful for endurance athletes if it helps them move more efficiently. Perhaps the most important factors here are the neural aspects of improved force production (Häkkinen & Keskinen, 1989). This is always true of runners (thanks to improved coordination, even with less sophisticated types of training). In the case of cyclists and rowers, positive transfer may mainly occur when they have to produce a great deal of power (efficient use of FT and ST fibres). Furthermore, strength training must be built into the training programme in the right way, so as to avoid physiological or other conflicts due to the concurrent training stimulus. It would be beyond the scope of this book to discuss physiological aspects of strength training in more detail, but readers are referred to the large amount of literature on the subject.

4.3 Motor control and limiting influences on force production

Explosive sports usually make far greater demands on technique than endurance sports, and so technique is more important to performance. In sports in which technique has a decisive impact on performance, it may be useful to gear strength training to coordination as far as possible. Another factor here is that most physiological parameters in explosive sports respond very haphazardly to training, and so the resulting adaptations will be hard to predict. This means that physiological parameters can hardly serve as a reliable guide to the design of training.

If strength training is to focus on coordination, two key questions must first be answered:

1 Is the quality of the explosive sporting movement limited by the demands that motor control makes on its performance, and, if so, how does this limitation occur? Here we have to consider the problem of degrees of freedom (Section 3.2.1), and how degrees of freedom will be controlled in a landscape of attractors and fluctuators. The need to control degrees of freedom may be one of the factors with the greatest impact on performance.
2 Under what conditions can strength training help shift this limit?

Only after both questions have been answered can sport-specific strength training geared to coordination be usefully developed. The first question will be answered in this chapter, and the second in Chapters 5 and 6.

4.3.1 *Load capacity of the locomotor system as a limit*

In endurance sports, technique serves a twofold purpose: preventing injury, and minimizing the energy costs of movement. Minimizing energy costs is not all that relevant in explosive sports – the aim here is to maximize output, regardless of how much energy it takes. Furthermore, maximal output in explosive movement is not usually determined by the amount of energy produced or the speed with which energy becomes available. However, this is not to say that motor control is the only possible performance-limiting factor left. Another potential candidate is the load capacity of the locomotor system.

If muscles are called on to produce their maximal strength or power, or if tendons and other passive tissues reach the limit of what they can absorb in terms of tensile forces, this will limit performance. In that case it is useful to try and shift this load capacity limit by means of, among other things, strength training. If muscles become stronger and passive tissues can absorb more tensile forces, performance will automatically improve. Coordination does not then play a limiting role. If strength is seen as an independent, more or less isolated quantity among the range of performance-determining factors, the value of strength training should logically be sought in such limitation of performance.

In sports practice, however, we see again and again that in complex movements the limit of performance is not determined by the maximum that can be obtained from individual muscles and passive tissues. Strength is not an independent phenomenon. The strongest athletes are by no means always the fastest sprinters, and evaluation of training always shows that, in technically somewhat complex sports, increased force production does not automatically lead to improved performance. Apart from energy production and load capacity of the locomotor system, it seems there are other factors that may limit performance and may even do so before such factors as maximal strength and power production can become the limiting factors.

7.3.2 Motor control as a limit

In explosive sports, performance is largely limited by the requirement that the movement must be controllable. As movement becomes more intensive, the central nervous system will be less and less able to control it, and the movement pattern may then become unstable. The central nervous system avoids such high-intensity unstable movements because they become dangerous, and limits performance before the muscles and tendons reach their limit of tensile load capacity. If the athletes reached the limit of what their muscles could absorb in terms of opposing forces, they would of course regularly go just beyond that limit and hence would sustain injuries. In a 100-metre sprint, some runners would lose their balance and fall (for instance with a tail wind). The hamstrings, which are never exposed to a greater load than when running at high speed, would often be loaded beyond their limit and tear. In javelin-throwing, extreme bracing of the front leg at the point of release would produce such forces on the muscles that shoulder girdle and arm muscles would sustain damage, and so on. In practice, however, most athletes are rarely injured. The obvious explanation is that, in a healthy body, muscles probably never reach the limit of their load capacity when movement is technically satisfactory, and hence that there is a built-in safety margin that limits performance. Factors such as the size principle and the limit of recruitment of muscle fibres have already been discussed (see Section 2.4). Factors that keep the movement stable and controllable can be added here.

A movement is only controllable if it can withstand external and internal perturbations. External perturbations include the influence of an uneven surface when running, a ball that is heavier or lighter than the catcher expects, different bounce times at different points on a trampoline, unexpected movements by the horse during an equestrian event, opponents when fighting for a ball, and so on. The main internal perturbation is fatigue. Control of movements includes built-in mechanisms to ensure that such perturbations have only a limited impact on the performance of the movement. Thus there are features of the motor control system that ensure the movement is performed robustly. One of the most important mechanisms for controlling movements and making them robust is the influence of cocontractions in what is known as the 'speed/accuracy trade-off'.

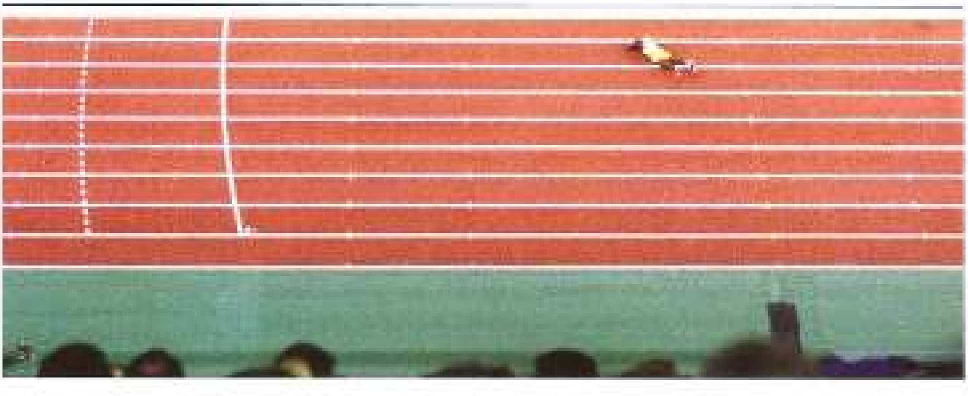


Figure [illegible] Injury in an international race event

4.3.3 *Cocontractions and preflexes in explosive movements*

The faster a movement is performed and the more force produced in the process, the more errors ('noise') there will be in the signals transmitted to the muscles by the central nervous system. This increasing noise will lead to errors in how the movement is performed. In order to achieve a usable, accurate movement pattern, the 'noise' must somehow be dampened. This is done by activating not only the agonists (the muscles that ensure the intended joint motion), but also antagonists. Agonists and antagonists then perform what are known as cocontractions (Figure 4.6; Van Galen, 2006; Kelso, 1998; Turvey & Carello, 1996).

When agonists and antagonists contract at the same time, they keep each other more or less balanced. This dampens any errors in the signals from the central nervous system. The right balance is thus struck by a number of muscle properties that are not subject to neural control, such as the force/length and force/velocity characteristics of muscles and the elastic properties of tendons. These properties affect how muscles respond to signals from the central nervous system. The series elastic components of the musculo-tendinous unit thus affect the change in muscle length, and the muscle length affects the force of the muscle action.

The effect of these mechanical properties is known as 'preflexes' (mechanical muscle properties that influence the eventual performance of the movement without involving the central nervous system). The action of preflexes within cocontractions forms the basis for muscles' self-organizing ability and makes movements respond robustly to both noise within central nervous system signals and unexpected environmental influences (external forces). The effect of preflexes can be compared to the action of shock absorbers in a car's suspension system.

Three examples:

- When running on an uneven surface or walking down steps, we want the body's centre of gravity to move as linearly as possible or follow a fixed, slightly undulating downward curve, rather than follow the irregularities of the surface. To do this, we adapt our ankle movements – not so much by altering the muscle-fibre length of the muscle (*soleus*) as by using the Achilles tendon to absorb external forces, much like a car's shock absorbers (Grimmer *et al.*, 2008).
- In an explosive start or when running at speed, the direction of the ground-reaction force changes so quickly (along the front and back of the knee) that the athlete cannot respond to it. The changes in the ground-reaction force are corrected for by cocontractions in the flexor and extensor muscles. If the ground-reaction force moves along the front of the knee, this acts eccentrically on the hamstrings, whose force/length characteristics shift to a greater length, generating more force. At the same time the length of *quadriceps* is reduced, so that less force can be produced by that muscle group. As a result, the flexing torque is corrected. If the ground-reaction force moves towards the back of the knee, the effect on the muscles is the opposite. This dampens errors in the direction of the ground-reaction force (Figure 4.6). Of course, other properties such as tissue elasticity and muscles' force/length characteristics also play a part in the corrective action of preflexes when running.

Figure [illegible] Joint coordination in the sprinting start. Muscles that are trying to flex the knee (the hamstrings) are active at the same time as muscles that are trying to extend it (quadriceps femoris).
Left: the ground reaction force moves towards the front of the knee, which may therefore be extended [illegible] early [illegible] [illegible] extension. The hamstrings (a) may then lengthen and the quadriceps (b) may shorten. Both muscles may lengthen (in the direction of the dotted line), in such a way that the hamstrings will produce more force. This will correct the 'error' in the direction of the ground reaction force.
Bottom: the ground reaction force moves towards the back of the knee. This time it is quadriceps that supplies more force to correct the error.

- When throwing, the shoulder joint needs to be abducted 90° – the best position for the joint to perform its task, with the least risk of injury. This position is achieved not only by the central nervous system, but also by muscles and muscle properties. During the throw there is a high-intensity action of the abductor muscles together with the adductor muscles (Minetti, 2006). If the arm is too low, the abductor muscles will have a better length for producing force than the adductor muscles, and the arm will move upwards. Conversely, the adductor muscles will predominate in the same cocontraction if the arm is too high. The muscles will compensate for the errors. The correct shoulder angle needs to be only partly learned. The central nervous system emits the signal for the powerful cocontractions, and the correct shoulder angle is then largely 'organized' by the muscles themselves (Figures 4.7 and 4.8).

The compensating effect of preflexes improves with increasing speed of movement and force production. Movement at average speed with submaximal force production is therefore less accurate than full-speed movement, even though signal noise is greatest at full speed. This is illustrated, for example, by the fact that pupils learning a tennis stroke are above all encouraged to hit the ball *hard* once they have mastered the basic pattern. Preflexes then compensate for minor errors in the performance of the movement, and mastery of the stroke improves.

One major benefit of preflexes is that response time is nil (0 milliseconds), so they can make corrections when the central nervous system has insufficient time to intervene. Preflexes are an increasingly important concept in our understanding of how motor control works (see also Section 2.1.5).

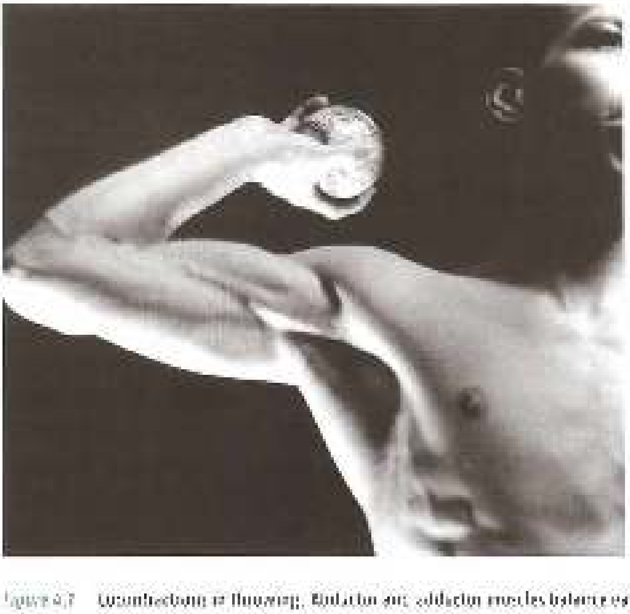


Figure 4.7 Cocontractions in throwing. Abductor and adductor muscles balance each other, and the shoulder is abducted 90°.

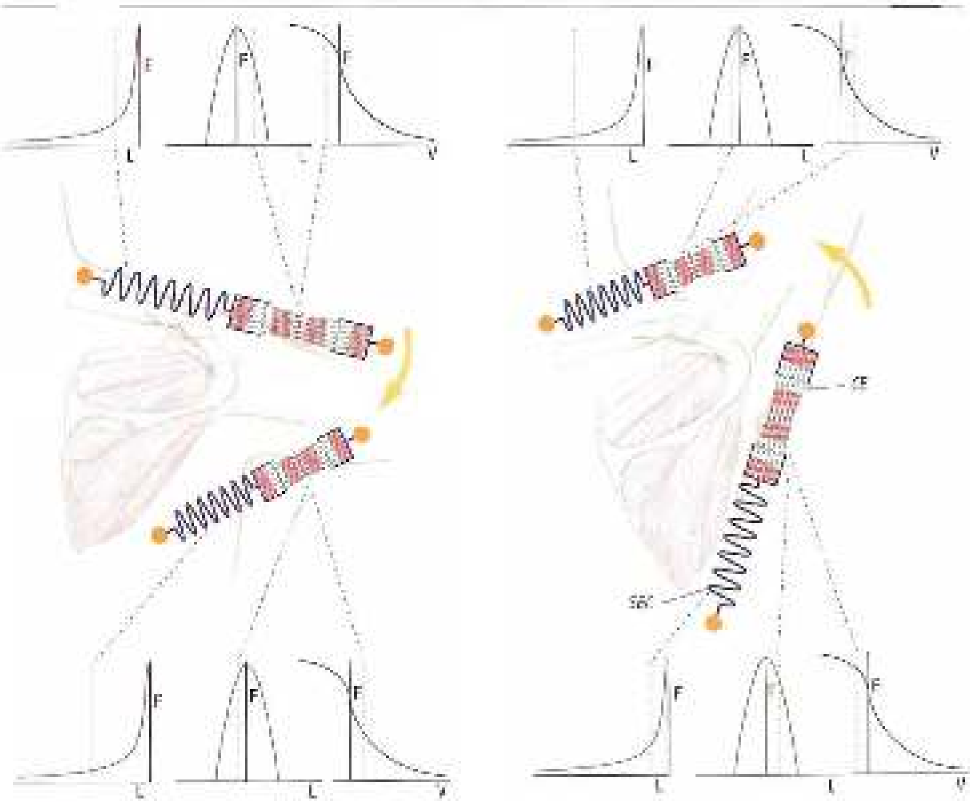


Figure 4.8 FL characteristics, FV characteristics and elastic properties when bringing the arm into the right abduction position. Left: the arm moves down from the ideal position. The FL characteristics, the FV characteristics and elastic properties shift to less favourable positions for producing force in the adductor muscles, but to more favourable positions for producing force in the abductor muscles (shown by the position of the dotted lines).
Right: when the arm moves up from the ideal position, the opposite happens.

4.5.4 *Speed/accuracy trade-off in explosive movement: a stand-off*

Cocontractions thus correct errors in the movement. At the same time, however, the antagonists inhibit the speed of muscle action of the agonists, and the movement will be performed more slowly. In other words, the more speed, the more noise, the more cocontractions, the more the speed of movement will be inhibited. The movement will thus be limited by mechanisms that make the movement respond robustly to external perturbations (such as ground-reaction forces or opponents) and internal errors (control errors) before the limit of load capacity is reached. If an athlete (e.g. a baseball pitcher) throws a ball at full speed, the shoulder joint will be under great pressure. The strong external rotation and the huge forces acting on the shoulder joint put it at risk. The structures in and round the joint can only be properly protected if the upper arm is abducted to approximately 90°. As we have seen, this position is guaranteed by cocontractions. The joint is protected by the mechanical properties of the muscles, but at the expense of speed of movement. In fact, it is remarkable what a key role cocontractions play in throwing, even in young children, who have scarcely mastered the throwing technique, the arm adopts the 90° abducted position almost automatically. This position may be one of the most important, deepest attractors in throwing (see Chapter 3).

The speed/accuracy trade-off due to cocontractions is found in many types of movement. It has been found that the faster cyclists turn the pedals, the more cocontractions there are. Simultaneous agonist/antagonist muscle action provides considerable stiffness round runners' knees during the stance phase, and during vigorous movements such as jumping and throwing, the trunk is made rigid by the activity of large, mutually antagonistic muscle groups, and so on.

Cocontractions do not just inhibit the intensity (speed and strength) with which a movement can be performed – they also have a positive impact. This is because they reduce muscle slack. Muscle slack limits intensity of movement. Reducing it by creating pretension with the help of cocontractions reduces this limiting effect on the potential intensity of the movement, and so allows the athlete to move faster and with more strength.

There is thus a stand-off between maximizing the intensity (speed and strength) of movement, controlling movement through cocontractions, and reducing muscle slack through cocontractions (Figure 4.9). Since this process involves several mechanisms at once, the impact on potential performance is considerable. This intrinsically stable overall mechanism may explain why eight rather differently built athletes may line up for an Olympic 100-metre sprint final and why there are seldom differences of more than a few hundredths of seconds between the medal winners. Performance may be determined not by a sum total of athletic properties but by the aforementioned stand-off, which results in only small differences between athletes.

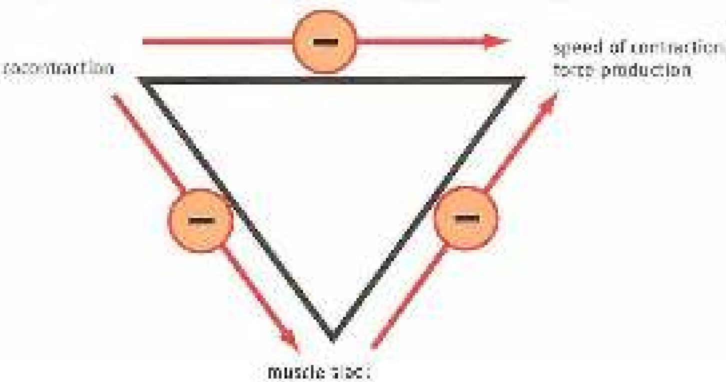


Figure 4.9 Muscle slack, cocontractions and intensity of movement are interrelated. Cocontractions inhibit speed, but at the same time they reduce muscle slack, which in turn inhibits intensity of movement. Intensity of movement is thus influenced, and probably limited, by various basic principles of movement.

In conclusion, the limit on performance in explosive movements is probably determined by the demands that motor control makes on intensive movements. This limit, which occurs before the limits of the healthy locomotor system are reached, serves two purposes: (1) keeping the movement controllable in an environment in which several unforeseeable perturbing forces will act on the mover (keeping the movement robust), and (2) protecting the athlete from injury by limiting the load on the locomotor system.

4.3.5 *Implications for strength training*

The requirement that a movement pattern should not be maximal and at the same time fragile (i.e. injury-prone), but should simply be 'good enough' (submaximal) and also robust, means that high-intensity explosive movement should be designed to provide a reserve of load capacity. Performance of intensive athletic movement is thus limited by what the load capacity of the body will allow as well as by minimal demands on stability. Stability of movement is a key component of technique. This means that movement technique has more influence on potential performance than may at first be thought. Two strategies for training in general, and strength training in particular, can be derived from this idea:

1. One which uses strength training to raise force production by muscles as far as possible, in the hope that the submaximal ('good enough') level, as well as the robustness of the movement, will increase together with the maximal level. The maximal level raises the submaximal level along with it, as it were.
2. One which primarily seeks to increase the robustness of the movement, so that the 'good enough' level will shift towards greater force production during athletic movement. The submaximal level then shifts towards maximal level without the maximal level needing to rise.

The second strategy approaches strength training much more in terms of movement technique than the first. The great advantage of this is that the load within strength training is significantly lower than when the robustness of the movement plays only a secondary part. In practice, of course, there will almost always be a blend of the two strategies, for instance by alternating simple movements with a large barbell weight with coordinatively complex ones with a light weight. This may be the most efficient solution. Yet this new knowledge of motor control, which shows that the need for robustness restricts the potential for movement, makes clear why in many cases there is no reason for explosive athletes to try and lift heavier and heavier weights within strength training. Above a certain limit, strength exercises that are performed with heavy weights, and hence have to be coordinatively simple, no longer help to make movement patterns robust.

Coaches who sense this usually say 'strong enough' rather than 'good enough'. 'Strong enough' is something most athletes can easily achieve, and it is pointless to invest in anything more. Making movement patterns more robust through strength exercises, and hence technically challenging strength exercises, is the most useful strategy for elite athletes to develop their skills. However, developing this strategy depends on a good knowledge of how the sporting movement is structured.

4.4 The laws of motor learning and training

The previous chapter stated that, when learning movements, we primarily learn to find and apply the rules that filter out inefficient execution. Clearly, if we opt for an approach to strength training that follows the coordinative track, knowledge of how people learn movements is essential. The mechanisms and features of this learning process are among the main topics of the science of motor learning and control. This field of research is rapidly expanding, and around the 1980s and 1990s it received a huge boost from new ideas such as Gibson's direct perception theory and further elaboration of Bernstein's insights into dynamic systems theory. This new understanding of how movement can be controlled has also led to new insights into the mechanisms of motor learning. However, these insights have so far had little impact on sports, and none whatsoever on thinking about strength training. Sports methodologies are still largely dominated by the older cognitive schema theory. This states that all the information about how to execute a movement is basically generated by the central nervous system. The system is also more or less hierarchically structured, with a clear command centre in the brain (Schmidt & Lee, 2008). Methodologies for learning movements therefore rely greatly on the cognitive aspects of learning (cognitive learning of the explicit rules for ideal execution of the movement pattern). Ideas about cognitive learning are by no means out of date, and there is little point here in discussing the battle between supporters of hierarchical theory and decentralized-control theory – this is a matter for real specialists in that area. What is useful, however, is to look at a number of principles of motor learning and their implications for an approach to strength training that takes greater account of more recent ideas on self-organization of motor patterns, such as the aforementioned mechanisms for making movements robust. Two aspects of motor learning that will play a key part in Chapters 5 and 6 respectively and are introduced in this chapter, are (1) the importance of knowledge-of-result (KR) feedback in terms of the intention-action model and (2) the importance of variable learning.

Being able to apply both KR feedback and variable learning is essential when supervising learning processes that provide room for self-organization of motor patterns.

The role of KR feedback will be examined in terms of the intention-action model for designing a movement pattern. Intention and result are of course interconnected, and are driven by the external focus of attention. The simplest forms of result-driven movement design are those in which the result is located outside the mover's body. The focus of attention is then also outside the body. For example, the result of a perfect forehand stroke is that the ball lands on the opponent's baseline, and the focus is then on the intended point of landing. Cause-and-effect relationships are somewhat harder to grasp if the result of the movement is not so clearly located outside the body and the relationships between the various sub-movements have to be sought within the movement pattern (such as improving trunk control by focusing on the position of the arms). The underlying mechanism that makes this so useful is somewhat harder to identify.

Besides result-oriented learning, learning through variable practice plays an important part in the learning process. The value of variation in practice is generally acknowledged. However, if we can say why such variation is so useful, we may be able to devise a better, more systematic approach to it. That is why it is important to understand the relationship between variation and monotony and the relationship between variation and self-organization.

4.4.1 *External focus and the result of the movement*

The intention–action model

The human brain is so complex that its workings can only be understood by using greatly simplified models, of which there are many: the stimulus-response model, the chaos model, the hierarchical model, the stimulus-perception model for sensory function and the intention-action model for motor function (known jointly as the perception-action cycle) and so on (Kok 2004). 'Evidence' can be found for each, depending on how the experiments are designed.

The intention-action model is a well-researched idea about how motor patterns arise, with major implications for the design of motor learning processes.

The intention of the movement to be made is first constructed in the higher parts of the brain (near the cortex): what should the situation be after the action is completed? Get the ball into the basket, put the shot as far as possible, make a stable landing, come out of the bend at the right point and so on. Then the appropriate actions are constructed in a number of stages. Abstract, more or less mathematical principles of movement are first selected, then the fixed principles of how muscles cooperate are determined deeper and deeper inside the central nervous system, and finally the specific muscle action is selected at the deepest level. Because of this order, the way in which the muscles are used is flexible. This is especially important with open skills, for the movements must always be initiated from a different starting position. If the way in which muscles are used for a judo throw or a rugby tackle is strictly determined in advance, serious problems will arise if the opponent makes an unexpected movement. A small step to the side will substantially change the starting position, and the muscles to be used will no longer have been correctly selected to deal with the new situation. The later and more flexibly the muscles are selected, the more effectively the execution of the movement can cope with perturbing movements by the opponent (see also Bernstein's degrees of freedom problem and the problem of contextual variability).

Of course, the intention to action process is far more complex than this – but what is essential is that, when we generate movements, the muscles we are going to use are not selected in advance, but only at the last moment.

The intention-action model is endorsed by both central-command and decentralized-control theory (Figure 4.10). The differences in interpretation lie in the assessment of how accurate the output from the central nervous system to the muscles should be. The command-centre theory sees it as complete (it contains all the necessary information); the decentralized control theory sees it as incomplete, and important 'decisions' in the body are made by peripheral self-organization of movements (for example through reflexes).

The intention-action model allows two interesting conclusions to be drawn as regards the design of strength training:

1. Because movements start with the intention, movements without a clear intention will not fit as well into the organizational structure of the system as movements with a clearly defined goal. Non-contextual movements are therefore somewhat undirected; they float round in the brain and do not have a logical structure. The problem with many strength exercises is that they lack a clear intention. If we move a ball upwards, it is quite clear whether the movement is executed well or badly if the intention of the movement is that the ball should land in the basket. But if we move a dumbbell upwards, the vague intention of the movement means that it is far less clear whether the movement is executed well or badly. This will be of far less use to the learning system – and so less will be learned (Figures 4.11 and 4.12).

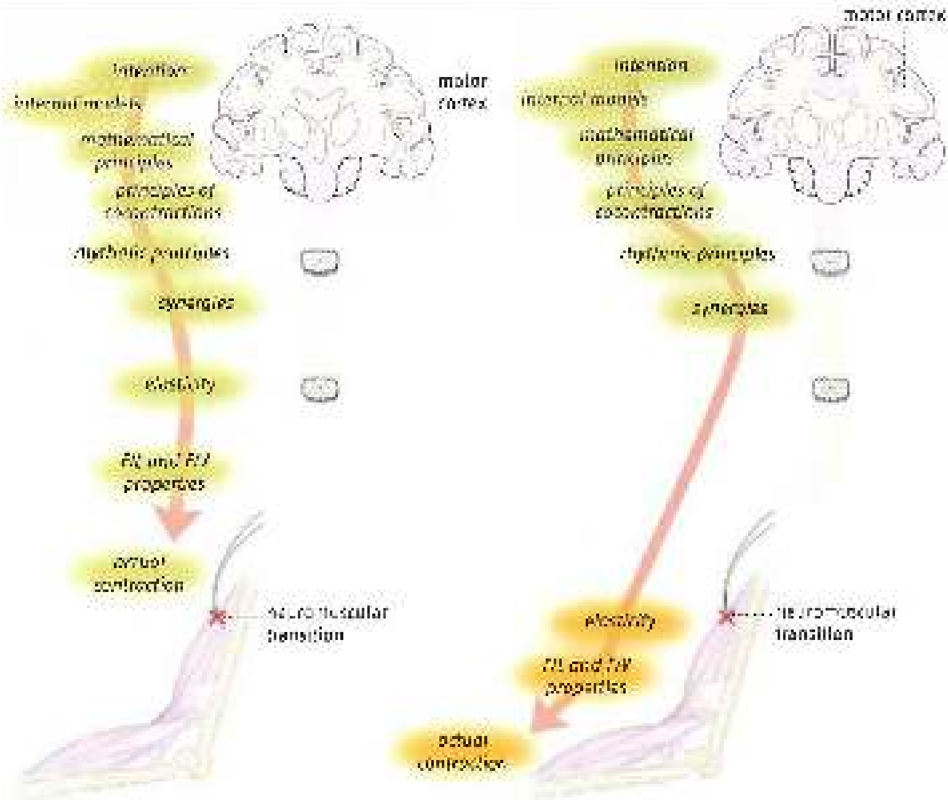


Figure 4.5 Intention–action model. The cross marks the neuromuscular transition: on the left, structured according to the central-command model; on the right, according to dynamic systems theory. In the central-command model, the signal from the central nervous system contains all the information that is needed in order to execute the movement correctly. In the dynamic systems theory model, the signal from the central nervous system is incomplete, and part of the movement is designed without the help of the nervous system. The steps, and the order of the steps, between intention and action may differ from those illustrated here.

Figure 4.6 An exercise without a clear intention.

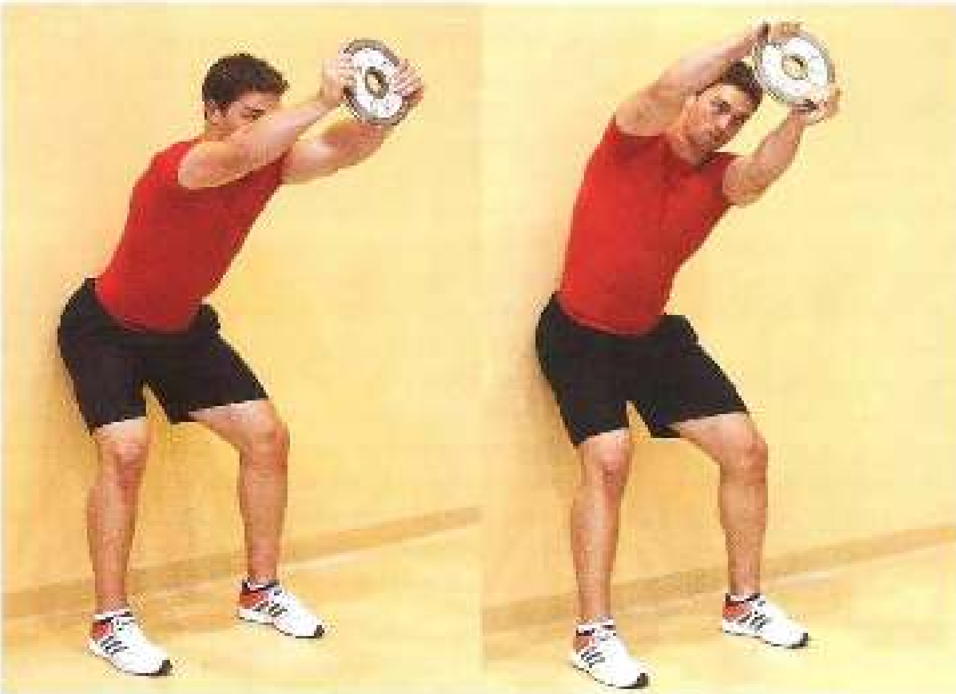


Figure 4.12 An exercise with a clear intention. Trunk control must be created by a coordination of all the muscles that influence the spinal column. The intention of the exercise is to push the weight away as far as possible. This will produce the cocontraction of the trunk and abdominal musculature (even without any instructions). The exercise can also be executed in a rotating manner (the shoulders turn with the plate; look under your arm). The rotation increases the load, especially on the abdominal muscles.

2 Strength training based on training of specific muscle groups, which is customary in the fitness, bodybuilding and sports injury rehabilitation world, intervenes in the final stage of the movement design; as a result, it does not train the flexibility of muscle use, and hence is less effective. This is particularly true when applied to open skills (Figure 4.13).

A contextual sporting movement should not be an incident, and practising a sport should certainly not mean carrying out a series of incidents. Movements should be part of a coherent matrix, and there should be relationships between categories of related movements. Bernstein used the term 'motor equivalent' to describe these relationships (Bernstein, 1996). If two or more movements share the same intention, the system marks them as related. The system is designed to execute the movements with great variation in the muscles used. A classic example involves drawing circles in space with your hand: first large circles at the side of your body and then suddenly in front of your body in the frontal plane, then small circles close to your hip, then as far behind you as possible, and so on. The transition from one type of circle to the next is never a problem, for the intention remains the same, namely drawing a circle. But muscle use varies considerably from one type of circle to the next. This appears not to cause any difficulties (Figure 4.14).

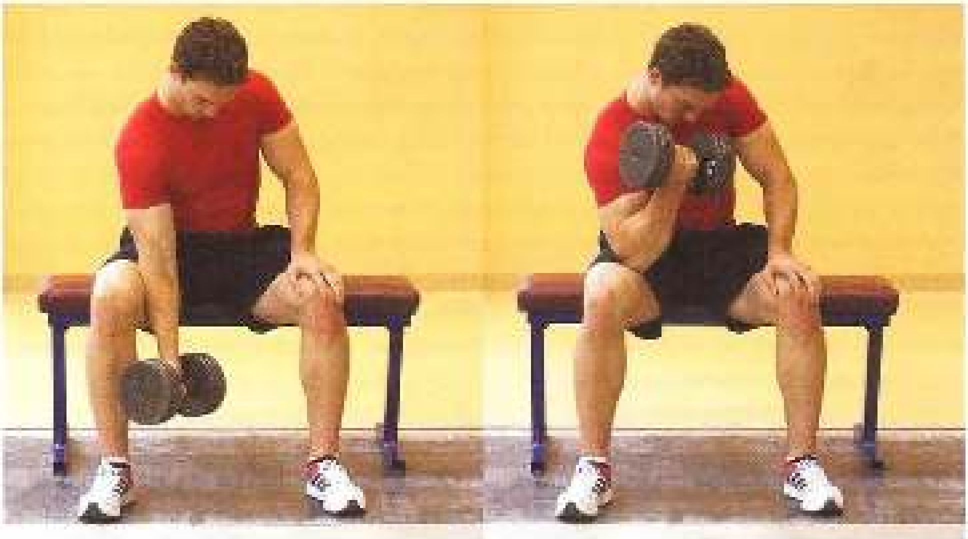


Figure 6.17 A biceps curl, in which the muscle is trained in isolation. There are hardly any contextual movements in which the biceps works in this way, i.e. concentrically with movement in the elbow but not in the shoulder.

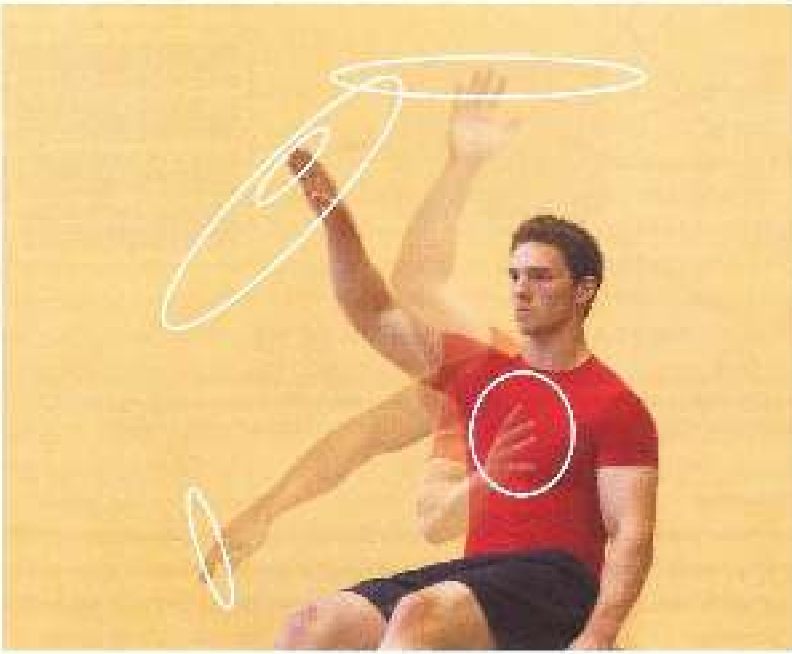


Figure 6.18 No one has any difficulty in drawing circles and varying them anywhere in space. Different muscles are used each time, but this is never a problem. We switch effort easily from using muscles round the shoulder to muscles in the forearm and wrist. The organism does not think in terms of muscles, but in terms of outward movement patterns.

Nor, once we have mastered a movement pattern, do we find it difficult to execute it with a selection of active muscles we have never used in this way before. Few people will ever have written their signatures large enough to fill an entire school blackboard. Yet this proves to be no problem, even if it has to be done while gripping a book under one's arm. We seem to be able to do this at once, using muscles we have never previously used for this purpose. Movement is extremely flexible in performance.

So we organize our movement solutions in clusters of similar intentions, rather than clusters of similar muscle activity. And the system tries to reason from the movement solutions to the specific action – from the conclusion to the arguments, as it were (figure 4.15). When teaching a movement, coaches should therefore make sure not to reason the other way round – for this would mean 'driving the wrong way' in the learning system, against the grain of natural movement processes.

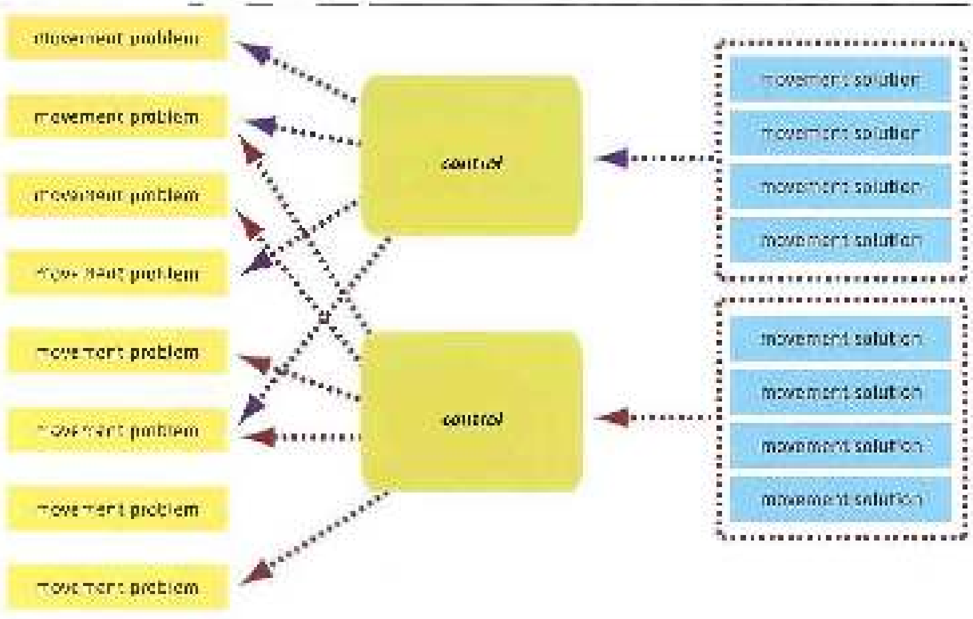


Figure 4.15 Movement problem, control and movement solution. The learning system clusters movement solutions that are related (the result achieved) and can be achieved using the same control. The system is not interested in how movement problems are related (how the movement should be executed), so movement problems are not grouped by similarity.

Additional information

The fact that our movements are above all intention-oriented and we are hardly concerned with muscle activity is apparent from small children's ability to imitate people. When an adult shows how a tennis ball should be placed in a tennis ball can, a small child can imitate this easily. But if we look carefully at this imitation, we discover that the child only copies the adult's intention. The child's actions are very different from the adult's. The adult opens the lid with a thumb movement, picks up the ball with two fingers, moves it by rotating his upper arm and forearm to the right position and pushes the ball into the can. A small child needs all its fingers to remove the lid, then has to pick up the ball with both hands and has to make large shoulder movements

> to get the ball into the right place. But the child evidently makes this conversion with ease. Muscles don't matter – all that counts is the result of the action.

So the body does not think in terms of processes, but in terms of the results of the movement. There is a closer relationship between the intentions of various movements than between the muscle movements involved in making them. The specificity of the movement pattern thus mainly focuses on the outside of the execution of the movement – how it looks in the space around us – rather than which 'motors' are actively involved in producing the movement. Similarity in the intentions of the movement is therefore a key aspect of the transfer that occurs between movement patterns when learning (see Chapter 5). In coaching, including sport-specific strength coaching, this transfer must be effected as well as possible. That is why it is a good idea, wherever possible, to add an intention to a strength exercise that has no clear intention of its own. For instance, the aforementioned strength exercise in which a dumbbell is lifted from the ground to above one's head becomes more meaningful if the dumbbell is supposed to reach a target, such as a tennis ball hanging from the ceiling by a thread. Especially in sports injury rehabilitation, exercises often lack a specific goal, whereas an intentional aspect could very easily be added.

Finally, it may be a useful idea that, in a movement pattern that has not yet been fully mastered, there may be somewhat more transfer at the lower level of the motors of the movement, and at higher performance levels above all at the high level of the intention of the movement. So far, however, there is no scientific evidence in support of this idea.

The relationship between external focus and intention

Intention is a key component of contextual movement. It is therefore also likely that there is a mechanism that supports control of the movement on the basis of the intention. A great deal of research has shown that intentional movement is strongly directed by attention.

Attention is a remarkable psychological phenomenon: everyone knows what it is, yet it is hard to define. Attention is the searchlight we use to explore the circumstances that we find ourselves in and that we are aware of. It may be either passive or active. Passive attention is drawn by a sudden sound or an unexpected movement in our environment. It is directed by the sensory system, and is a bottom-up system that is already present in infants. Active attention is directed by an internal process (the brain) and is organized top-down, towards the senses, as it were. For instance, a baseball pitcher pays active attention when he focuses on the catcher's glove. To some extent, then, we control how our attention is directed.

> **Additional information**
> Since the eye is our most important sensory organ, focused attention and focused gaze usually coincide. When vision coincides with focused active attention, central vision – the part of the visual system that we use to focus our gaze and consciously observe objects – is our attentional searchlight (Fak, 2004). Besides central vision, which is mainly responsible for registering objects (object information), the eye also gathers information by means of peripheral vision, which is mainly sensitive to movement (spatial information). It is generally assumed that this information, which is very important for controlling movement in the environment, is processed unconsciously. This means that attention, which usually coincides with central vision, does not select

which information we use but is above all a searchlight we train on one point so that we can also observe and process other (including peripheral) visual information more effectively and link it to automated movements (Vickers, 2007). The importance of both visual observation systems is highlighted by the fact that they are not only located more or less separately in the eye (central vision in the fovea, peripheral vision outside it) but the information is transported along separate routes (the 'what' information via the parvo pathway and the 'where' information via the magno pathway) to the visual cortex, where it is processed separately (the 'what' information via the ventral route and the 'where' information via the dorsal route) (Figure 4.16; Carey, 2010).

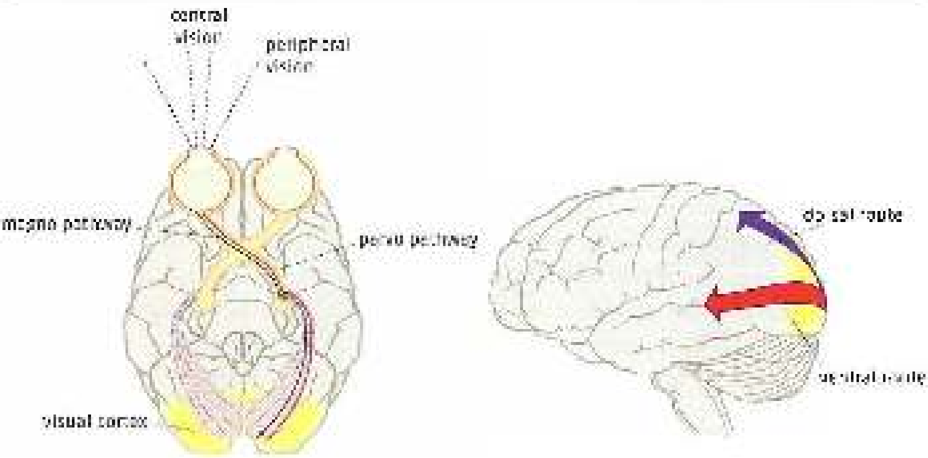


Figure 4.16 Two routes for processing visual information: the central vision ('what')-parvo pathway-ventral route and the peripheral vision ('where')-magno pathway-dorsal route.

The information from the 'where' route is processed unconsciously and is strongly linked to automated movement control (the dorsal processing route, see Figure 4.16). The 'what' information is processed via the ventral route and is linked to conscious control.

'What' information, ventral route:

- observing colour and shape
- cognitive processes
- giving meaning to objects and events
- anticipation
- planning
- new movements
- slow processing

'Where' information, dorsal route:

- observing movement
- processing information under time pressure (quick processing)
- automated movement

Research has shown that there is an important difference, when moving, between the attention paid to processes within the body (how do I move?) and the attention paid to processes outside the body (what happens around me as I move?). The distinction between internal and external attention, or focus, is often linked here to the distinction between broad-focus and narrow-focus attention. Athletes are therefore divided into four groups as regards their focus:

1 Narrow internal focus: attention to small features of the movement that occur within the body (e.g. is my stance leg knee sufficiently flexed when I take a free kick?)
2 Broad internal focus: attention to large features of the movement that occur within the body (e.g. full-body tension needs to unload in one go in the kicking movement)
3 Narrow external focus: attention to small features of the movement that occur outside the body (e.g. aim twenty centimetres to the right of the man on the right of the 'wall').
4 Broad external focus: attention to large features of the movement that occur outside the body (e.g. should I shoot straight at goal, or is a teammate suddenly in a good position?) (Nideffer, 1993)

If attention is focused outside the body on features related to the movement, the movement and motor learning processes will be controlled more effectively. Controlling movements effectively is thus a matter of focusing attention externally and hence using vision effectively (making optimal use of central and peripheral vision). We all know this from personal experience. It is easiest to walk along a balance beam if you focus your gaze on the far end of it. It is easiest to take a bend on a motorcycle if your gaze is 'parked' a long way round the bend. If your gaze and attention while taking a left-hand bend are focused on moving your right hand upwards and your left hand downwards, you are likely to end up in the ditch.

The reason why external focus works better than internal focus is quite simply that external focus concerns the result of the movement (the end of the bend and 'how to come out of it' when skating, the basket the ball has to go into, a stable landing in gymnastics), whereas internal focus concerns how a movement is executed (getting the hip position right in relation to the skate when taking the bend, keeping the elbow close to the body when shooting for the basket, extending in time for landing). Directing attention at the result and hence the intention of the movement provides room for the intended organization of the movement, from the intention to flexible use of the muscles. This principle – that movements can best be controlled on the basis of the intention – is known as the 'action effect hypothesis' (Wulf & Prinz, 2001; Wulf et al., 2002). Coaching in which attention is directed internally, i.e. on the process of the movement, goes against the grain of the natural route of design for controlled movement patterns – 'driving the wrong way' on the motor control route.

This means that a person's own preferred focus is not always the most appropriate focus for learning a particular movement or correcting a particular error. Internal focus – especially narrow internal focus – is less suitable for learning a movement and performing well. Now, if we look at how instructions affect the way in which attention is focused, we have to conclude that the great majority of instructions given by coaches and rehabilitation specialists direct attention internally; internal focus rates of up to 94% have been measured in education, 84% in academic coaching and 69% in mid-competition coaching (Fishman & Tobey, 1978; Porter *et al.*, 2010). In other words, most instructions are of the 'driving the wrong way' type and are not only less effective, but actually counterproductive. The notion that giving plenty of internal-focus instructions can never do any harm is mistaken. Research into easy tasks has often shown that giving no instructions at all and simply encouraging the learner usually has a better impact on the learning process than giving internal-focus instructions on how to execute the movement (Wulf & Weigelt, 1997). Especially in practice situations involving practice of a non-complex movement, as in rehabilitation and strength training, this is an important – and possibly disconcerting – discovery. Precise movement corrections by strength coaches and physical therapists, for instance during trunk control exercises, are thus not so much a sign of professional expertise as a sign of ignorance about how movements are controlled. Well-intentioned but misapplied expertise can often be highly damaging (Wulf, 2008).

Augmented KP and KR

The way in which feedback is given greatly affects the learning result one wishes to achieve. It is therefore useful to distinguish between the various types of feedback. First we need to distinguish between:

- Intrinsic feedback: feedback that the learner receives from executing the movement itself (including proprioceptive feedback);
- Augmented feedback: external feedback (from coach's instructions, video pictures and so on).

Feedback can also be divided into:

- Knowledge-of-performance (KP) information;
- Knowledge-of-result (KR) information (i.e. knowledge of the state achieved following the movement).

In the scientific literature (which is always very cautious) the distinction between KP and KR is usually only made in the case of augmented feedback. It is not so explicitly mentioned in the case of intrinsic feedback – possibly because the two categories are hard to describe in relation to intrinsic feedback, and hence hard to study. Research into the effect of KP and KR feedback is therefore almost always about augmented KP and KR.

Since the difference in the effect of augmented KP and augmented KR is so great, it is useful to emphasize this difference in the coaching methodology. It is also very useful to look for KR feedback in practical work, so that it can replace over-dominant KP feedback (Figure 4.17).

So what is the difference in results between learning through KP and KR information? There is a famous and much-replicated study of discus coaching, involving two groups. One was given instructions about correct technical performance by a leading coach (KP), the other was simply given a tape measure to record how far the discus was thrown (KR) (one German study provided information about the angle of release). The results were measured in terms of 'retention under stress', i.e. some time after the practice sessions in competition settings with no opportunity to practise at length beforehand. The improvement in performance in the second group was at least as good as in the group with a leading coach. Were instructions from a leading coach no better than a tape recording?

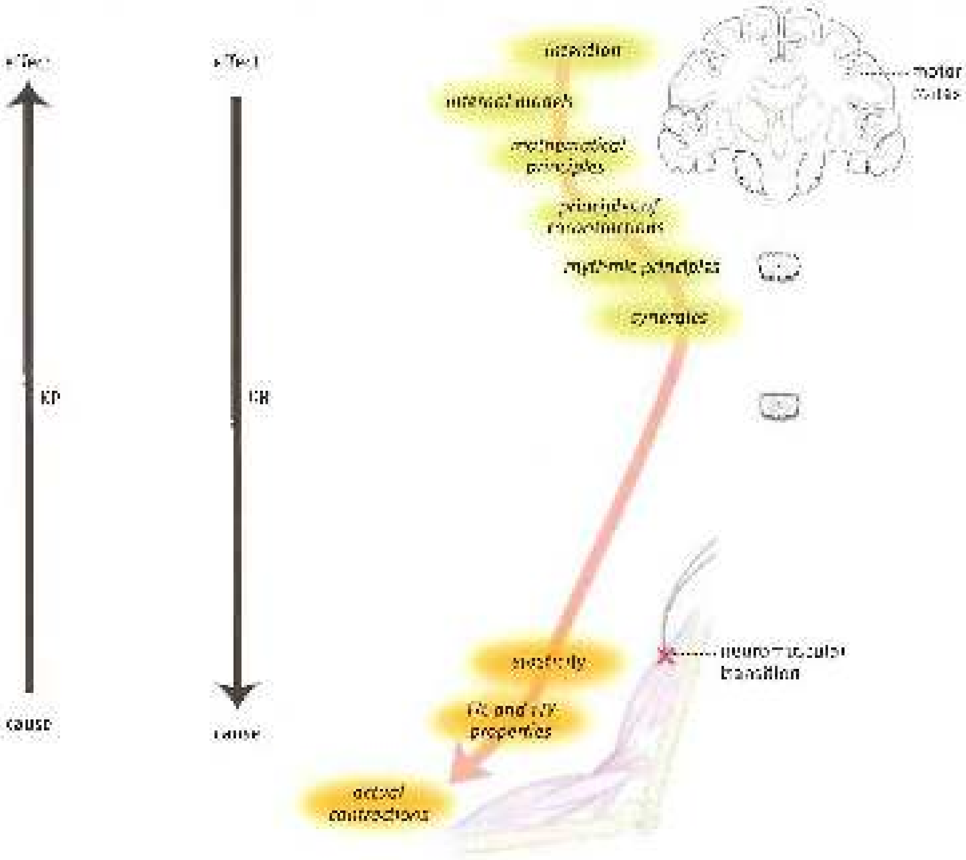


Figure 4.7 Movement is organized from intention to action. External focus and KR information follow the route. Internal focus and KP information 'drive the wrong way'.

The learning system evidently tries to focus on the result to be achieved (the intention) – especially if the measured results are objective and accurate (unlike precise feedback from a coach, which always includes subjective and partly incorrect components) – and is not particularly interested in *how* the result is achieved (Ballreich & Press, 2000; Farfel, 1977). Compare this to learning a language. People are interested in communication (the result) rather than the underlying grammar (the process). You learn a language by making yourself understood – achieving results. The same is true of motor patterns. The system does not learn well if it is fed the successive stages of the movement (process) without knowing what the result is supposed to be, or without the stages being directly linked to a result.

The 'knowledge' referred to here is not explicit cognitive knowledge (being able to name the result of the movement), but intrinsic knowledge whose point of reference is located within the body.

Many studies have shown that learning via the result of the movement (KR) can be extremely efficient (Wulf & Shea, 2001) – in contrast to somewhat older views that some learners, given their personal preferences, would benefit more from process information and others more from result information (Pijning, 1978). The question now is how such a result-oriented approach should be seen in practice. This brings us back to insight into non-linear learning. Learning is being able to fit into underlying abstract measuring and regulating mechanisms (the speed/accuracy trade-off, and so on). When the learning system knows what the result of executing the movement is, it can adjust the regulating mechanisms appropriately and calculate the movement. Learning then moves from the result to how you can get there – a non-linear process. One of the consequences is that the system tries to achieve results as soon as possible in the learning process, even with a very imperfect execution of the movement, but one that will do as a temporary solution. This can be observed when striking a ball with a golf club. The body is primarily interested in striking the ball, and only later in such things as using the hip. To achieve a result, the beginner's body opts for an uneconomical solution by not using and immobilizing the joints of the upper body ('freezing degrees of freedom'). Not initially using the joints allows the ball to be struck.

If coaching is process-oriented and the various components of the movement are offered without the context of the result, the movement is not seen from the perspective of the underlying regulating mechanisms and there is no intrinsic reason for the body to decide whether the movement is right or wrong. Examples include practising the glide phase of shot putting in isolation, moving the javelin past your ear, learning just the right grip for the baseball bat, and all types of exercises in the gym. The fact that there is a coach on hand to say whether the movement has been executed rightly or wrongly may well be of interest to the athlete's obedient consciously cognitive brain – but the body couldn't care less. As far as the body is concerned, the coach's instructions are nothing but chitchat. It will briefly attempt to do what has been asked of it – but it certainly won't store it away in a memory system. Forget it as soon as you can, says the body, otherwise your motor memory will get cluttered up. The main thing the body learns is how it itself perceives the execution of a movement pattern. The body is interested in the result, and uses this to guide its control of the learning process by applying underlying abstract principles. Result-oriented coaching can even be applied in movement types such as dance, even though this does not appear to be so result-oriented. Even dancers may well learn movements better on the basis of the result. The intended expression – what the dance communicates – can be used as an effective guidance mechanism in the learning process.

From this we can definitely conclude that there is a strong link between the effect of directing attention (internally or externally) and the application of augmented KP or KR information. KP information leads to internal focus and KR information to external focus – with all that that entails. So does this mean that coaches no longer have any part to play and should only provide augmented KR information – i.e. simply read the tape measure? Not at all! But what it does mean is that they should be gardeners rather than conductors. Rather than indicate which component of the movement should be learned at which point in the learning process, they should above all create conditions that are in keeping with the character of intrinsic learning, and so optimize the learning process. Gardeners do not decide when or how fast plants should grow – when the next step should be taken in the learning process – but simply hoe and fertilize. The coach's task shifts to creating ecologically valid practice situations – in which the environment is organized so as to help find the right movement solution and the learner implicitly learns to recognise a biomechanically optimal solution (Davis, 2007).

Here are two practical examples of how KR information can be applied in the learning process:

1 Learning to somersault on the mini-trampoline. The idea is that the athlete should make a perfectly balanced landing. In 'KP thinking' this is a question of extending in good time, perhaps keeping your chin on your chest (or not), and so forth. The athlete can attempt to do this; but in the initial learning stages it is far from certain that this will lead to a correct result, for the body does not know what to focus on, or what other factors need to be properly coordinated besides extending in good time (such as what else you might be able to do when you land). This is like swimming in the ocean without knowing where the shore is. KR is more effective. The task is now to somersault, land and instantly obey a command that is given right after landing. The command is to perform either a forward or a backward roll. The athlete must therefore be able to perform either movement, and this can only be done if the landing is stable. The body itself seeks a landing position from which either task can be executed, and the landing quickly becomes stable. The body may discover that the moment of extension is not so very crucial, but that what matters is how you fight to achieve a stable position as you land.
2 A youth soccer goalkeeper always lands on his elbow after diving, and so tends to lose the ball. KP ('get your elbow into this or that position as you dive') does not help. KR does: if you don't land on your elbow, you not only keep hold of the ball but can also dispose of it quickly after landing (e.g. throwing the ball away with both arms while lying on your side). First the keeper practises throwing the ball away while lying on his side, and then prior movements are added, e.g. first go from a crouch to a side-lying position, and then throw the ball or keep hold of it on command. This is practised without any further instructions. Finally a dive is added. And it really does work – the problem is usually solved in a single coaching session.

The result of the movement is thus a key regulating mechanism in learning to move. That is why it is so important (when preparing and evaluating coaching) to distinguish between KP and KR feedback, and to replace at least part of the KP feedback with KR feedback. This means that KP information is certainly not useless (the group of discus throwers who were given instructions by a coach also made progress), and that alternation between KP and KR information may be the optimal solution for the learning process. Besides providing information for learning, KP information also increases motivation, which may well be the most important driving force in learning. So coaches need be able to identify both intrinsic and augmented KP and KR feedback and apply them in the learning process. Their task is not reduced – just different.

Intrinsic KP and KR

As already mentioned, researchers mainly focus on augmented KP and KR when describing feedback. The fact that the difference between intrinsic KP and KR feedback is so hard to pin down does not, however, mean that the dynamics of intrinsic feedback are unimportant – quite the contrary, as practitioners are well aware. The role of intrinsic feedback in the learning process is probably greater than that of augmented KR.

Just like augmented feedback, intrinsic KP and KR feedback differ greatly in terms of their impact on learning. The difference between intrinsic KP and KR is not as easy to state as with augmented feedback, but it *is* important for understanding how learning processes work, especially when – as in the case of strength training – the learning process focuses on self-organization of the 'building blocks' of movement. The differences between intrinsic KP and KR do not lie at the boundary between 'internal' and external focus, but are determined by more abstract concepts.

Intrinsic KP information is the constant flow of sensory information that is released when we move. This constant flow gives us information about the process: how far the knees are flexed in a volleyball pass, how tensed the muscles are during push-and-pull movements in judo, how the muscles change in length when pushing off in skating and so on. This information guides the execution of the movement.

As skill increases, the sensory system becomes more sensitive to certain information ('sensitization') and less sensitive to less relevant information ('habituation'). Signals from the sole of the foot and the speed at which tension increases in the muscles tell a good sprinter just how hard the track he is running on is. Milliseconds of difference in rebound time between a hard and soft track are faultlessly registered. In contrast, after a few seconds the athlete no longer notices the pressure of the well-tightened laces of his spikes. The change of length in the muscles during the clean is perfectly registered, so that the weightlifter can immediately tell whether or not the attempt is going to succeed. The temperature of the barbell bar is scarcely registered at all.

Intrinsic KP feedback provides an uninterrupted flow of information. The process of sensitization and habituation means that this permanent flow is used optimally. Intrinsic KP information has a positive impact on motor control and motor learning. As we have seen, this positive impact is not necessarily present in the case of augmented KP or even (though to a lesser extent) augmented KR. In fact, too much augmented KP information is bad for the learning process, and can easily interfere with learning. The optimal frequency of augmented KP feedback thus turns out to be surprisingly low. Giving feedback once in every five practice attempts has a better impact on the learning process than once every attempt. Furthermore, very precise augmented feedback is usually less effective than feedback that provides an overall assessment of the quality of the attempts. All things considered, the learning system does not really seem designed to deal with augmented feedback. There is no such problem with intrinsic feedback (Beekers et al., 1994; Chambers & Vickers, 2006; Vickers, 2007; Winstein & Schmidt, 1990).

Dominant intrinsic information

The question now is whether the motor system focuses completely on the permanent flow of intrinsic process information that is released, or whether a more result-oriented strategy is adopted when processing intrinsic information. Although research has not yet revealed much about this, practice also points emphatically to result-oriented (KR) control of movement in the case of intrinsic feedback.

When an athlete executes a movement pattern, there is a more or less clear result. The result may be located outside the body but also within it. Results within the body are not registered via the augmented feedback route, but are registered via the intrinsic route. Results that occur during movement both within the body and in the environment are, as it were, beacons that the moving organism focuses on. It attempts to move from beacon to beacon, and focuses on the position of the beacon. It reasons back, as it were, to the information it ought to receive if it were to manoeuvre efficiently towards the beacon. Once a beacon has been reached, the next one becomes the point to aim for. This enables the organism to click its way through the movement landscape. Control of movement thus does not involve permanently processing permanently released sensory information (like water from a tap), but working towards results of the movement that are planned in chunks (Figure 4.18).

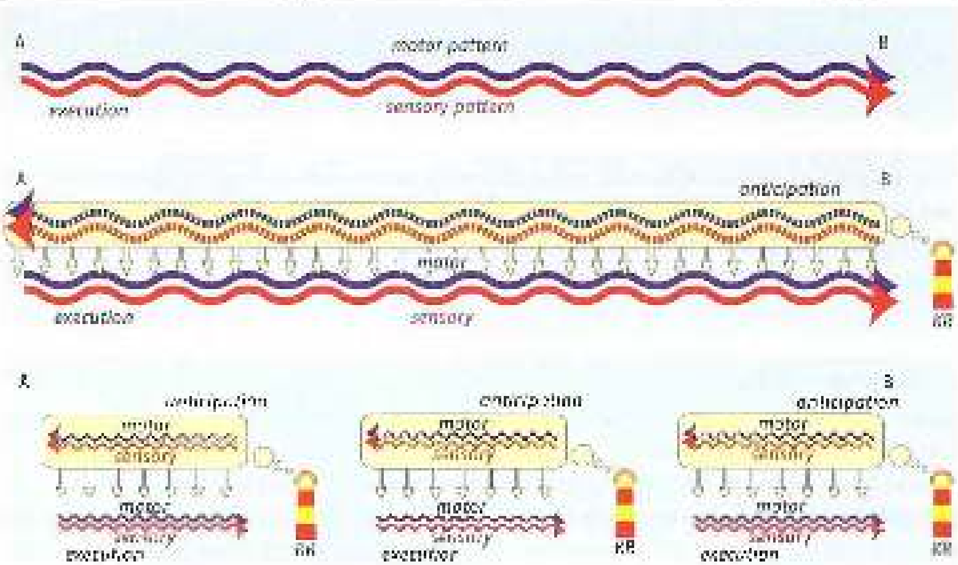


Figure 4.18 Top: execution of a movement from A to B without intrinsic KR. Since the intention of the movement is lacking, the motor and sensory patterns cannot be anticipated.
Centre: execution of a simple movement from A to B with only KR information at the end of the movement. During execution of the movement, the motor and sensory patterns can be compared with the motor and sensory patterns to be expected in the light of the intention. The movement is thus automatically adjusted.
Bottom: in complex movements there are more and successive KR beacons information.

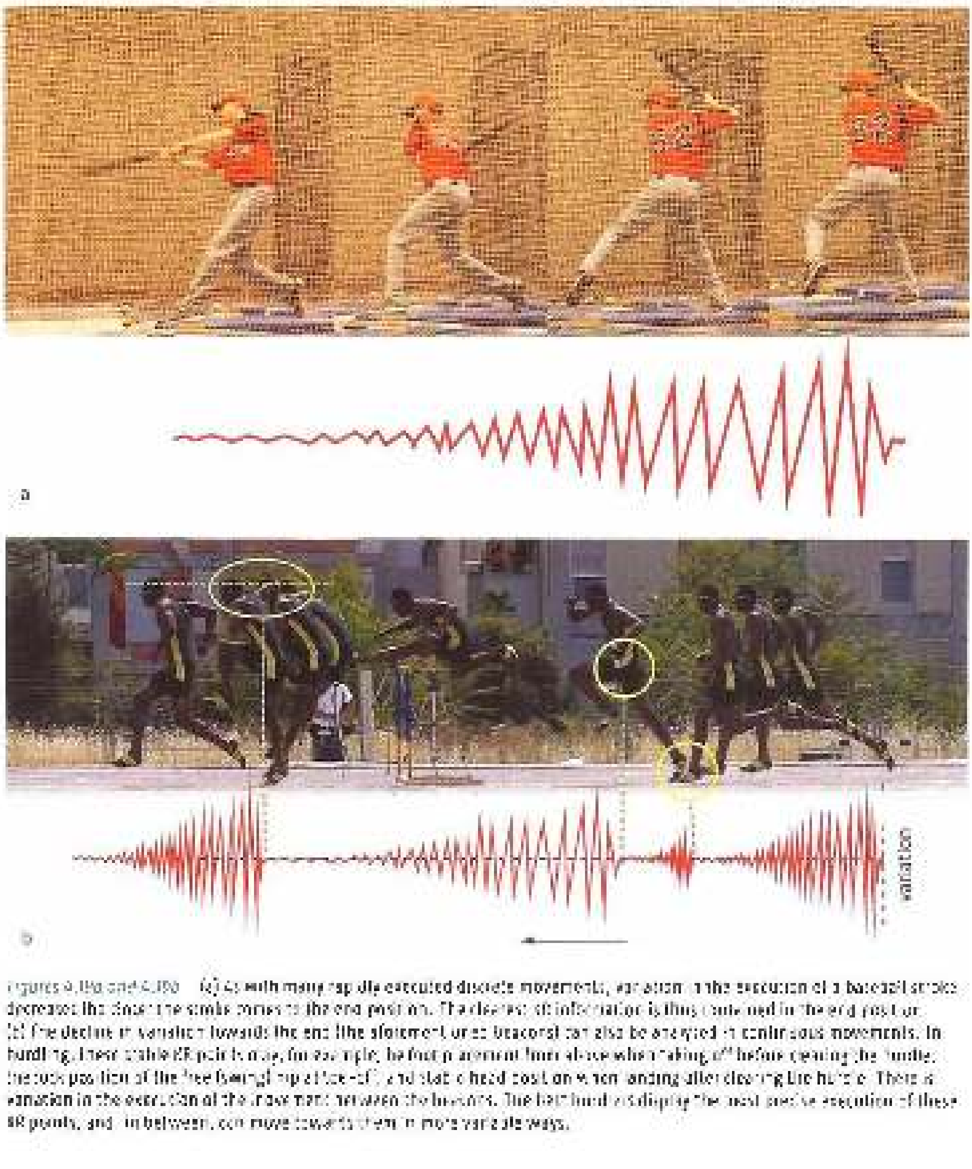
Terminology

Since the concept of intrinsic points of focus has been a topic of little interest to researchers, there is still no standard terminology to describe it. The term 'intrinsic KR' and the 'beacon' metaphor are therefore very provisional, and may eventually be replaced by more appropriate terms. In sports practice, the concept of 'result beacons' is occasionally used in an intuitive way, especially in closed skills such as gymnastics, in which the importance of not performing exercises hastily, and hence missing out 'beacons', is acknowledged. When performing the exercise, gymnasts must insert 'rest points' in order to 'reset', which basically means using those moments to focus on the next beacons in the movement.

Researchers will object to the term 'intrinsic KR' and claim that the information released in the examples described here is strictly speaking KP information. However, if the stable position of the head when tumbling (as described below) is termed KP information, this fails to take account of the specific function of this information – which is that the keeping the head still is the result to be achieved through organization elsewhere in the movement pattern. This means it is KR information. This (KR) function of the beacons in the movement design requires these components of the movement – keeping the head still – to be executed very precisely, otherwise they cannot be planned in advance. The clearer the goal is, the better the athlete can anticipate and the more effectively he or she can work towards the goal. In contrast, components of the movement that are not beacons are variable. This distinction between precise beacons and variable components of movement is crucial. In this book we have therefore given the distinction between a precise effect of the movement and the variable process of the movement priority over the distinction between information within and outside the body. A term that covers the concept of intrinsic beacons, is not confusing and comes to be accepted in the literature may one day be formulated.

The existence of these KR calibration points in movement is obvious. Controlling the movement by means of KR calibration points is a useful strategy that allows the moving organism to link higher cognitive and even conscious intentions to highly automated movements. Such a link must be created somewhere if movements are to be learned in a completely economical and meaningful way and within an acceptable period of time. Furthermore, the beacons must be meaningfully aligned. For example, if the final beacons in the movement pattern are unclear or missing, the earlier beacons in the chain of movements will lose much of their effect. That is why it is often a good idea to teach a movement pattern backwards, from the end to the beginning. First teach the last beacon, then the second-last, and so on. This principle, known as 'backward chaining' or 'end point focus', is regularly used in sports such as baseball when teaching pitching and batting. In pitching it means first learning the wrist movement, for example pitching the ball into the ground to isolate the wrist movement. Once this has been sufficiently mastered, prior components of the pitching movement are then added by pitching more horizontally and finally with a curve.

To clarify this concept of result information that is more guided by 'will' and guides the adaptation of deeper automated control, a number of examples from fields other than strength training will now be discussed. The examples show how the organism reasons



Figures 4.19a and 4.19b (a) As with many rapidly executed discrete movements, variation in the execution of a baseball stroke decreases the closer the stroke comes to the end position. The clearest KR information is thus contained in the end position. (b) The decline in variation towards the end (the aforementioned beacons) can also be analysed in continuous movements. In hurdling, these stable KR points may, for example, be foot placement from above when taking off before clearing the hurdle, the lock position of the free (swing) hip at take-off, and static head position when landing after clearing the hurdle. There is variation in the execution of the movement between the beacons. The best hurdlers display the most precise execution of these KR points, and in between, can move towards them in more variable ways.

backwards when controlling and learning movements. Some examples of KR-oriented strength coaching are discussed in 5.3.5.

1. High jumpers end their run-up to the Fosbury flop technique with a curve, which is crucial to their performance. They must therefore have a good sense of how speed and change of direction can best be combined within the curve. How do they learn this? The purpose of the curve is to help them perform the rotations that are needed during takeoff. The rotations occur if the end position at takeoff is correct. High jumpers who have not mastered a clear, precise end position never achieve fully satisfactory control of the curve.

There is no beacon. 'Power jumpers' have a less precise end position than 'speed jumpers', and so run less precise curves. Their curves are far more hit-or-miss. This contrast in jumping styles is very significant even among top-level athletes (Figure 4.20).

Figure 4.20
Left: power jumper with a variable end position at takeoff. Right: speed jumper with a more stable end position at takeoff. The left-hand jumper's curve is therefore less controlled than the right-hand jumper's.

2 A gymnast takes off from a mini-trampoline (Figure 4.21). This requires considerable body tension. Body tension is a highly automated technique. It can be controlled by keeping the arms high during takeoff. If the arms are high, body tension will automatically increase. The fact that the arms can be kept high is considered the result of keeping body tension high; the arms cannot be kept high if body tension is too low. This principle of controlling body tension (the process) by being able to keep the arms high (the result) is universal. As well as in gymnastics it can be found in exercises for running technique, basketball layups and so on.

Figure 4.21 Taking off from a mini-trampoline with the arms held high. All good gymnasts control body tension in this way.

3 Baseball pitchers often tap the ball with their glove just before pitching. This releases result information that tells them their shoulders are properly turned – otherwise they would be unable to touch the ball with their glove. Pitchers have no idea why they do this (maybe it feels good). This goes to show that looking for result information in movement is a fundamental strategy of the organism. This trick of bringing both hands together after picking up the ball can be used to teach players of other fielding and throwing sports such as cricket, the most effective starting position of a throw.

4 The wrist movement in pitching is the final beacon in the movement. That is why the wrist movement is learned first when learning 'backwards'. When first throwing a javelin, someone who can pitch a baseball fairly well executes the movement remarkably badly. The reason for this is that the wrist movement used in throwing a javelin is quite different from the one used in pitching a baseball. This means that the final and most important beacon in the movement is different and so the orientation that is used when pitching a baseball suddenly no longer works. Contrary to expectations, good baseball pitchers are not automatically good javelin throwers (though there are exceptions, such as former baseball player and javelin world-record holder Tom Petranoff).

Although the intrinsic KR feedback described here is focused on a process within the body and the objections to internal focus could also apply, this system of feedback does turn out to be very effective in practice (Figures 4.22 and 4.23). This may be because the KR feedback is processed entirely within the organism. Augmented feedback first has to leave the organism and must then be translated (put into words) several times before it reaches the perception system. This may create 'noise' that makes the feedback less effective. With intrinsic KR feedback, which occurs in the body, nothing is 'lost in translation', and this may mean that the information is processed entirely unconsciously, without any adverse impact from internal focus.

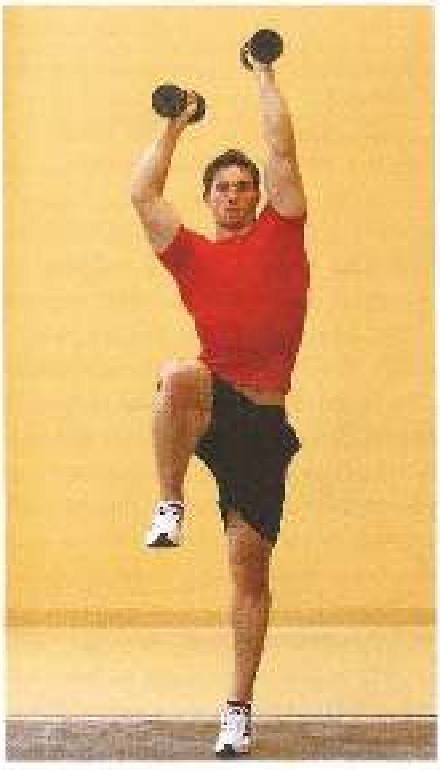


Figure 4.22
Technique exercise for running. With many runners the free (swing) side of the pelvis does not elevate sufficiently during the stance phase. This impairs running efficiency, and must be corrected.
KP correction: elevating the free (swing) side of the pelvis during the stance phase.
KR correction: being able to keep the arm on the stance-leg side extended and high is the result of keeping the free (swing) side of the pelvis high, and so can be used as result information for the correct position of the free (swing) side of the pelvis. The dumbbell on the stance-leg side must be pushed as far up as possible.
Using this intrinsic KR strategy can also be effective in exercise programmes for the treatment of Trendelenburg gait.

Figure 4.[illegible] Technique exercise for running. Many runners rotate excessively around the longitudinal axis at the end of the stance phase, at toe-off. Rotation must be compensated for, and this impairs running efficiency. Good technique eliminates this rotation.
KP correction: prevent rotation by adapting the body position during the stance phase.
KR correction: the stick must be kept as still as possible when running. This can only be done in the absence of rotation, and so can be used as result information to change the body position without explicit instructions.
Running with a whirling skipping rope has a similar effect and may provide even more useful intrinsic result information.

Since intrinsic KR is focused on a result within the body, it may be more useful to distinguish between the cause and the effect of the movement than between internal and external focus, for the effect of the movement may be located within the body. A possible rule here is that the further the effect of the movement is located from the cause of the movement, the better the movement can be controlled. Especially if it is important to improve the underlying mechanisms of movement, for instance when the technique is to be applied in open-skill situations, optimizing intrinsic KR information is an effective way to enhance learning. The main focus of sport-specific strength training should be on improving these underlying mechanisms. This means it is well worth examining how not only augmented KR, but also intrinsic KR information can be used within strength training.

An example

When hitting a top-spin forehand in tennis, attention can be focused strategically:

- Remote and external: where the ball lands after it is hit
- Less remote and external: the curve of the ball as it crosses the net
- Close and external: the place in space where the ball is hit

- External and process-oriented: the turning of the racket when the ball is hit
- External and result-oriented: where the racket ends up in space after the stroke
- Narrowly internal and process-oriented: the wrist movement when the ball is hit
- Broadly internal: extension of the body when hitting 'through' the ball
- and so on.

Some attention-focusing strategies work better than others. And there is a simple rule that indicates what works best: the further away from the process, the better.

An underlying connection

Chapter 5 discussed dynamic systems theory, with its division of components of movement into attractors and fluctuators. An interesting question that then arises is whether there is a connection between attractors and fluctuators in (a) movement and (b) the intrinsic process and result information yielded by that movement. Researchers have not yet asked, let alone answered, the question about this connection, yet it is very evident (and hence at the same time very speculative). The quality of intrinsic result information largely depends on how clear the information is, and that of course depends on how stable it is. This means that good intrinsic result information can only be found in attractor components of movement. Result information in fluctuators would be unpredictable and hence impossible to anticipate. Anticipating result information is the core of motor control, and the clearer the information, the better the control.

Reinvestment

Internal focus is increased by over-use of augmented KP feedback and does not enhance the learning result, because augmented KP feedback is processed 'the wrong way round', against the natural direction in which controlled movement patterns are designed. Apart from this drawback, which has already been mentioned, frequent use of augmented KP feedback has a second adverse effect on performance. Movements learned with a great deal of augmented KP feedback are less stable and less reliable especially in stress situations – for instance during competition. The famous 'choking' phenomenon that occurs during competition may well be due to the way in which the movements have been taught.

Movements can be designed and controlled in two ways: through processes based on declarative or explicit memory (the working memory) or through processes based on procedural or implicit memory (the hard disk) (Edwards, 2010). We have access to declarative memory through conscious processes, whereas access to procedural memory is unconscious and automated. The above analogy with a computer (the working memory and the hard disk) is useful up to a point. When we use a computer and when we control movements, new structures are designed on an accessible working memory, which yields incidental, temporary results. If we want to store these results permanently, we must copy them from the working memory to the hard disk. Computers are designed to make this easy – and that is where the similarity between a computer and the motor control system ends. Things designed on the working motor memory cannot simply be copied to the hard disk. The hard disk has a specific structure, and any information that we want to make permanent has to fall within that matrix. If not, the information will be deleted (forgotten). We have already seen that the matrix consists of abstract rules of movement, such as rhythms, coordination principles and so on. We have also seen that internal focus and KP feedback yield results that do not fit into the matrix ('driving the wrong way') and so take a long time to make the movement automatic.

If a lot of augmented KP feedback has been used in the learning process and the movement has eventually been copied to the hard disk, the learner has learned not only the automated movement but also how to reconstruct such a movement quickly in the working memory. In stress situations the athlete no longer relies on the automated movement and switches instead to incidental control through the working memory. However, this leads to a less efficient movement pattern than tried-and-test control by the permanent memory. This phenomenon, which is the reason for underperformance under pressure, is known as 'reinvestment'. A clear example of reinvestment can be seen when someone who has no trouble walking along a 10-cm-wide beam that is on the ground has to do the same on a beam placed a metre above the ground. Suddenly the person finds it much harder to walk and keep his balance, having reinvested in the ad hoc reconstruction of walking and keeping his balance in the working memory. This incidental arrangement dampens automated control. Fear of failure, and reinvestment, are what make footballers miss penalties and tennis players hit the ball into the net at match point (Gray, 2004; Masters & Maxwell, 2004).

If more KR feedback is used rather than frequent KP feedback, athletes cannot learn how to use the working memory for incidentally designing a movement pattern. This makes reinvestment difficult, for it is never explicitly learned; and this in turn reduces the likelihood of failing under pressure.

In fact, it is striking how much the quality of the working memory can differ in practice from that of the hard disk. Top athletes in open-skill sports often turn out to perform remarkably poorly when they have to try out a new skill for the first time. The working memory performs rather poorly, whereas control via the hard disk yields brilliant results – one more reason to make a clear distinction between conscious and automated control of movements.

4.4.2 *Variability and monotony*

The role of motivation

We learn by practising. The more we practise a skill, the better we learn it. This is an obvious and undisputed observation. It has even been suggested that athletes need 10,000 hours of deliberate practice to reach the very top of their sport (Ericsson *et al.*, 1993).

However, the effectiveness of practice varies throughout the learning process from beginner to expert. At first, practice leads to great progress, but later the same amount of time spent practising will have far less effect. At expert level, so little progress is derived from practice that athletes feel they have reached a 'performance ceiling' – they are no longer able to make any progress. Of course, the decrease in efficiency is not the same for all skills in all settings, and moments of increased learning effect may occur in the midst of a general decline (Figure 4.24). Yet the 'power law of learning' (Fitts, 1964), which captures this decline in the effectiveness of practice in a mathematical formula, is generally accepted as one of the most robust pieces of data in the field of motor learning. In other words, the decline in effectiveness of coaching is fairly compelling and inescapable (Newell, 1991; Schmidt & Wrisberg, 2008). So athletes who want to reach the top will simply have to put in the hours. Especially as Fitts's power law seems so universal, it is tempting to think that this mechanism of reduced additional yield cannot be influenced. To imprint a movement, athletes must be patient and frequently repeat the movement they want to impress.

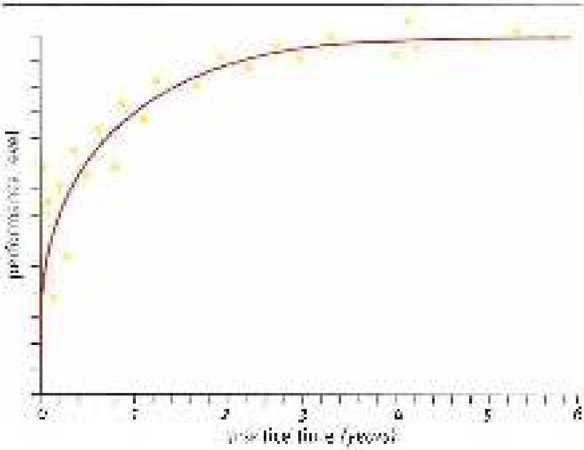


Figure 4.24
The power law of learning: the better the movement is mastered, the more practice time will be needed to keep making progress

Yet there are good reasons to look at more than just the repetition strategy for achieving a high level. Repetition has not only the advantage of imprinting a movement, but also the disadvantage of reducing motivation. Motivation is a basic condition that must be met if the organism is to learn. In this context, motivation should not only be seen as a conscious, cognitive, perhaps even moral factor to be worked from (top-down). The underlying motivation in motor learning is also strongly connected to 'lower' mental aspects such as arousal, fear and activation, and is fuelled from the body bottom-up. It could be said that the organism needs to become interested in the movement patterns to be learned, and that this interest must be present at both cognitive and more animal levels, such as the emotional desire for improved performance, the basic state of the central nervous system (alertness) and signals from the body that it is worth while making adaptations. Motivation is thus a state of the entire organism, and can be seen as the thermostat of the learning process.

The question now is how to trigger motivation during training. Some of the main factors that fuel the learning process are of course top-down. Identification with the goal to be achieved, moral reasons to pursue the goal, a social environment that fuels the will to pursue the goal and so on have an enhancing impact on learning. This has been shown by, among other things, research into the effect of the content of performance feedback (KP). If a group of basketball players are given good KP feedback when learning to shoot, there will be a learning effect. The temptation is then to ascribe the entire learning effect to the quality of the information that has been given. But this is wrong – for if a similar group is given no substantive instructions but simply encouragement, there will also be a learning effect (although it will be less than the effect of substantive feedback). From this we may conclude that the learning effect is only partly due to the content of the feedback. A considerable part of the effect comes from the additional motivation provided by the encouragement that is contained in the substantive feedback (Wallace & Hagler, 1979).

Motivation is not only fuelled 'top-down' but also 'bottom-up'. Under certain conditions the signals given by the body will increase arousal, activation and motivation, and in others they will reduce them. Of course it is important to know which sensory (reafferent) information triggers learning, and which dampens it. This knowledge can then be used to organize coaching more efficiently. A more or less general rule here is that the learning process will not be greatly stimulated if the sensory information is well-known and is released in a well-known blend of sensory information and executed motor patterns. However, if the sensory information is new, or if the composition of the sensorimotor information released while practising is different from what the athlete is used to, the learning system will be activated. In other words, if the movements during coaching are repeated again and again in an unchanging environment, the learning effect will be less than if the performance and the practice environment keep changing. The link between sensory and motor patterns must be shaken up in order to generate motivation to learn. Sensorimotor chaos is, if you like, the basis for learning (Schöllhorn et al., 2009). Repeating the ideal execution of the movement in a standard setting does not lead to chaos – variation in the execution of the movement in unfamiliar settings does (Figure 4.25).

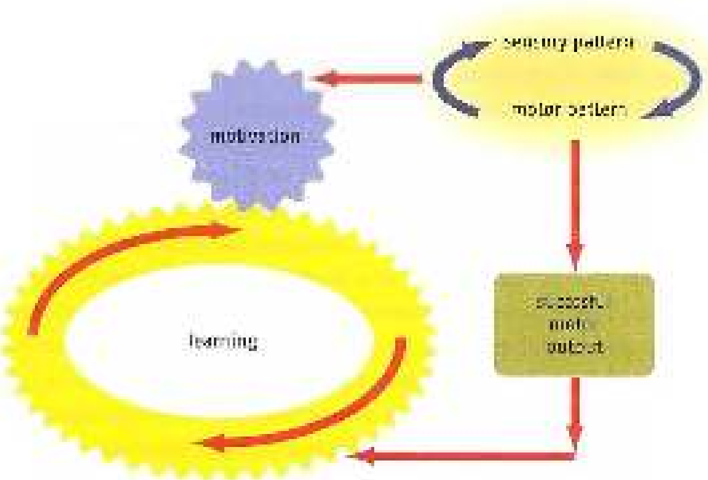


Figure 4.25 Motivation is the motor of the learning process. Unfamiliar links between sensory and motor patterns activate motivation, which triggers the learning process when the movement is successfully executed.

To achieve an optimal learning effect, movements should therefore not constantly be repeated in the same way. Variation is the key to efficient coaching. This does not only apply to optimizing motor learning processes. When planning physiological adaptations during training, variation should again be the main feature of the training. It is so important that the reason why periodization models appear to work so well is not the perfect planning of the components in relation to one another, but above all the simple fact that periodization leads to variation in training. This variation is merely a by-product of periodized planning, so it is unlikely that its effects are fully exploited (see Section 1.1.1). Variation is therefore the first and most important training principle, along with individualization. The realization that variation is the key to good training planning can be seen in the way many outstanding coaches approach training planning in the course of their careers. After organizing training in an increasingly complex way for many years, in order to capture more and more aspects of training in their planning, many experienced coaches eventually turn out to go back to a far simpler approach. They then opt for simple planning in which variation in training stimuli becomes the key to the planning of training.

The idea that the system should become interested in the movement to be learned before learning processes can commence can be seen as part of thinking within the central governor and probabilistic prognosis theories (see Section 2.4). These theories emphasize mechanisms that limit performance before the physical limits are reached. In fact, like the regulating mechanism motivation, they are conservative systems that allow adaptation only in exceptional cases.

Monotony in coaching

So the value of training variation lies in avoiding monotony. Monotony in coaching stimuli has several adverse effects on performance improvement.

1 *Monotony and physiological adaptation*

In training, a balance needs to be struck between effort and recovery in order to prevent eventual overtraining. The amount of training an athlete does is an important factor here, but not the only one. Another key factor is the amount of variation in training sessions. With the same amount of training, the balance between effort and recovery may be perturbed because there is too little difference between the various training units over a given period. If the sessions are similar in intensity, volume and performance, the athlete's ability to recover is reduced. Especially in endurance sports this is an ever-present risk. Not only does training then become mind-numbing, but the athlete's ability to adapt is diminished. Monotony in training, combined with high intensity of training, turns out to be related to overtraining. Overtraining, reduced resistance to illness and increased risk of injury can thus develop even if the amount of training is not increased (Anderson *et al.*, 2003). Of course, monotony in training is a difficult parameter to calculate. The most commonly used way to measure monotony is to identify the athlete's subjective perception. There are a number of protocols that measure 'perceived exertion' and convert it to tables in which threshold values can be indicated (Delattre *et al.*, 2006).

Of course, it is training for endurance sports that is at most risk of monotony, owing to the emphasis on the physiological side of the training and because such sports tend to involve endless repetition of the same movement pattern. However, if the athletes include strength training in their total programmes, variation increases. That is why strength training for endurance athletes yields not only coordinative and perhaps physiological benefits, but also the important benefit of reducing monotony in coaching.

2 Monotony and the environment

The training environment has a strong (top-down) influence on variation in training. A physical environment that is always identical, and a social environment that never changes and always generates the same perception of the sport, will also impair performance. Top-class athletes who regularly attend training camps do not do so to pamper themselves, but from necessity. Especially athletes who are somewhat older and have been performing at a high level for a long time will need to find environmental factors that can give them a new reason to keep on performing.

An anecdote

Grzegorz Sposób (b. 1976) was a top-class Polish high jumper. He started high jumping relatively late in life, after a career as a volleyball player. Most people around him could not understand his decision to switch. 'Grzegorz, why do this? What good will high jumping do you? You can't make a living out of it. And at your age....' Yet he remained an elite athlete until well beyond the age of thirty. Asked how his coaching had changed over the years, he replied: 'Nowadays the focus of my coaching is on finding a reason to keep on jumping all-out, to keep on going for that takeoff even though it could really hurt. This is no longer about training my body – it's about training my willpower.'

3 Monotony in practising and coordinative adaptation

The need for variation when learning movements was described by Bernstein as 'repetition without repetition'. We do not learn by constantly repeating the same solution to a movement problem, but by constantly solving a new movement problem. Learning and motivation are stimulated by the constant emergence of unfamiliar totals of sensory and motor patterns that do not fit into existing, familiar sensorimotor relationships. We learn through confrontation with something new, rather than imprinting something familiar. Because of their versatility in performance-related aspects, many types of sport suffer little from insidious monotony. However, this does not apply to endurance sports, or to strength training; within strength training, only a limited number of movements are executed in comparison with other athletic movements. These movement patterns are usually not complex or diverse. In sensory terms, moreover, strength training is frankly 'meagre' in comparison with athletic movements. In particular, sensory information from the environment is minimal and has little controlling influence. In contextual sporting movements, the visual system nearly always has to do a lot of work in estimating and calibrating central and peripheral vision. For instance, central vision is used to judge time to contact – something that plays almost no part within strength training. Peripheral vision is important if controlling information is released through optical flow when moving in space, whereas within strength training the athlete does not usually move in space. That is why the learning system usually finds strength training monotonous and boring; the only difference between movements is the variation in load on the barbell. Such monotonous training leads to a decrease in corticospinal activity, which impairs when learning new skills (Jensen *et al.*, 2005). This monotony within strength training has an adverse impact on the training effect. It also impairs coordinative transfer, and so it will not be easy to organize strength training in a way that creates the conditions for such transfer. Variation and avoiding monotony must therefore be major cornerstones of strength training. This demands a great deal of coaches. However, it is extremely useful to increase variation within strength training, for, as we have already seen, strength training is a very suitable way to teach basic components of the movement. Variation within strength training will be discussed in more detail in Section 6.4.

4.4.5 *Finding generalized rules through variability*

The organism does not seek to store motor patterns as a catalogue of isolated incidents, but where possible as a coherent whole. A principle of movement will be learned and automated more quickly if it is generally valid and hence can be applied in many different movement patterns. It is absolutely essential to seek generally applicable principles of movement. In the absence of such generalized rules, movements cannot be controlled. If a catalogue of isolated movements were to be drawn up, it would be so big (the 'storage problem') that it would be impossible to find a given movement pattern. And finding the right movement under extreme time pressure – which is necessary in sport – would be quite out of the question. Using generalized principles when designing movements keeps the total catalogue limited, and hence manageable.

What all motor control theories have in common is the need to reduce the size of the catalogue of movements. In schema theory this is done by applying a recognition schema for sensory patterns and a retrieval schema for motor patterns. Following on from schema theory, generalized motor programme theory goes a step further by describing variant and invariant features of a motor programme, so that a principle of movement becomes even more generalized (Schmidt & Wrisberg, 2008). Proponents of direct perception theory (Gibson) and dynamic systems theory (Bernstein's legacy) consider this reduction in the size of the catalogue still far too small, and they propose even more mechanisms to further reduce it.

Whether a movement principle is generally applicable only becomes apparent when it is tested and executed in all circumstances – in other words, in variable training. If, for instance, we take a set of ten related movement patterns, several control mechanisms will appear to work if only the first variant is executed. If we leave things at that, we will rely completely on that set of mechanisms. However, if we go on to practise a second variant of the movement, some of the control mechanisms will turn out to be no longer reliable. Only a small number will work for both variants. If we go further and also practise variants three to ten, more and more mechanisms will drop out. Eventually only those that are reliable for all ten variants of the movement will be left. The number of effective control mechanisms will have been greatly reduced, and only the remaining principles will need to be stored in the long-term memory (wherever that is). Motor learning is far more a matter of eliminating movement patterns that work only incidentally than linear design of a single solution that is perfect for a given situation (Figures 4.26 and 4.27).

An interesting question here is why learning reaches a ceiling. As we have seen, one of the basic mechanisms that causes this is monotony. It may be wondered whether it is possible to keep raising the limit even at a high level of performance by varying things differently each time. The problem for athletes who have more or less reached their learning ceiling is that there is also a limit to the value of variation. Because in top-level sport any improvements in movement patterns are smaller and smaller and the most generally applicable control mechanisms have already been identified, more and more time will have to be spent finding that one additional effective rule. The costs of training stress will soon exceed the benefits, and additional training will then be counterproductive. That is why many older athletes who have reached their learning ceiling deliberately train less in order to achieve the required standard through more rest and recovery. However, it should be mentioned here that variable training is in practice stopped much too soon. Practising non-standard movement patterns appears to have an impact even with athletes who have reached their ceiling (Davids et al., 2008).

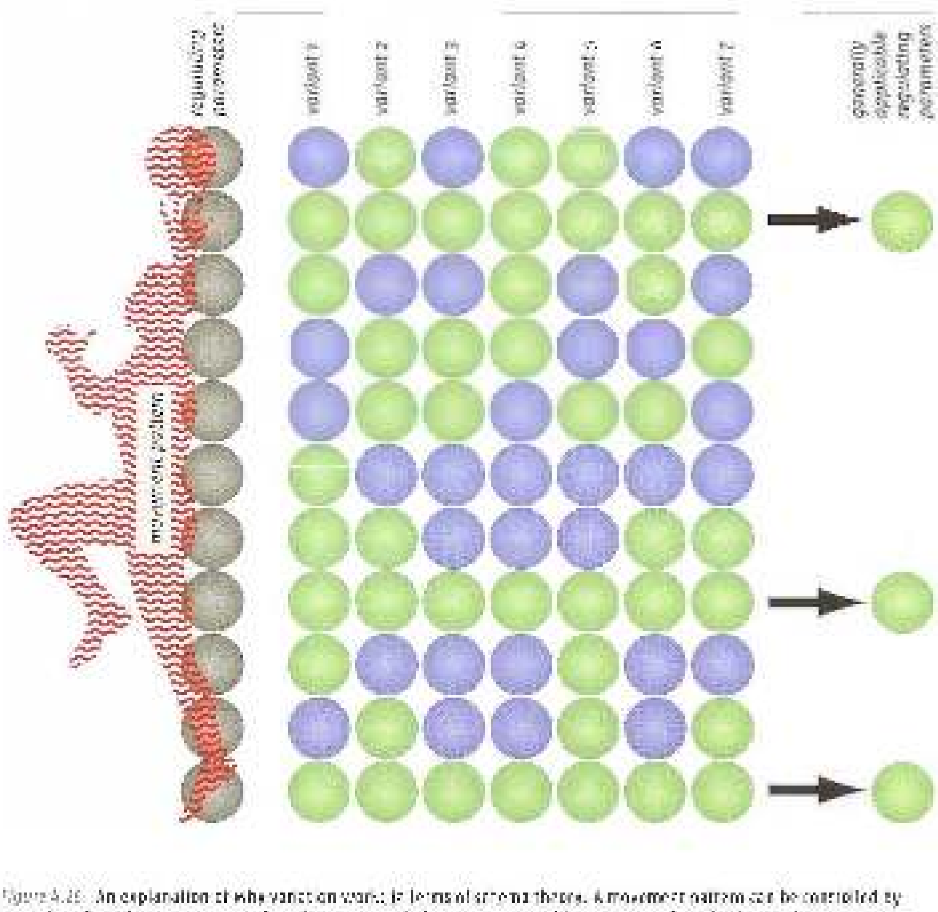


Figure 4.26 An explanation of why variation works in terms of schema theory. A movement pattern can be controlled by a number of regulating parameters (in this case eleven). Some parameters (shown in green) work with particular variants, others (shown in blue) do not. By practising more variants of a movement pattern, the organism can discover which regulating parameters always work (in this case three, shown on the right in green). The organism tends to opt for the parameters that can be used with as many variants as possible.

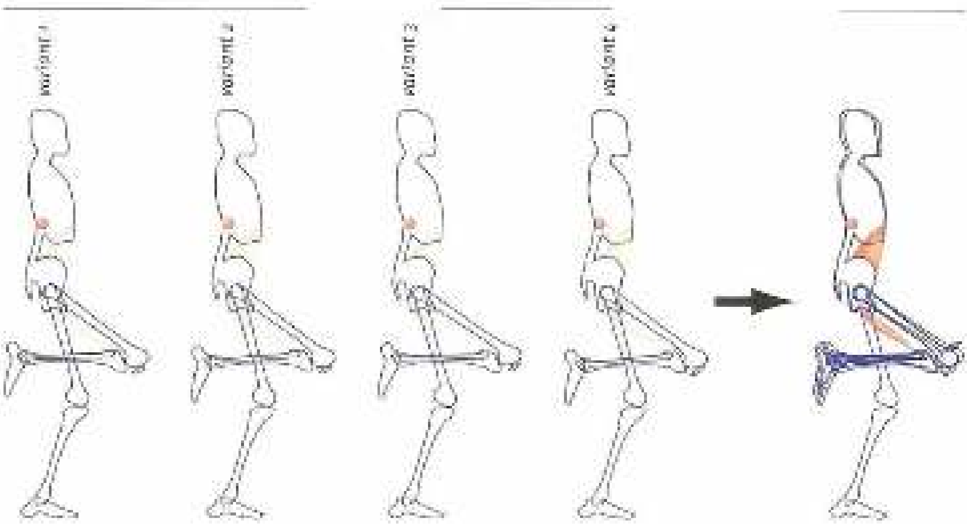


Figure 4.27 Another way of showing the principle in Figure 4.26. The four figures on the left represent four performances of the same exercise. The differences between the performances – what the coach can see and the athlete can consciously discern – may not be determined visually. Yet the unconscious regulating level is able to superimpose the performances and compare the differences transparently (on the right).

Variation and self-organization

The importance of dynamic systems theory in controlling movements was discussed in Chapter 3. When learning movements, there is a division into components of movement that are stable (attractors) and variable (fluctuators). A meaningful division of the movement into attractors and fluctuators makes the movement effective in the environment where it is taking place. This division into attractors and fluctuators cannot be consciously controlled. Not only does the learner have little or no cognitive access to this process, but it is almost impossible for the coach to control or guide the process. The process develops through 'self organization' of the organism. The coach can only contribute to it by arranging conditions for learning so that the learning process can take place satisfactorily. The ideal way to do this is variation in training.

If an athlete still has to learn the sporting movement or has not yet mastered it, there is also a division into stable and variable components of movement. However, this is not an optimal division. The movement is therefore uneconomical, and this impairs performance. The athlete attempts to improve this by practising. In order for new and better stable patterns to be incorporated into a movement, the existing stable patterns must first be 'detached' from the movement pattern. This is an important and often misunderstood step in the learning process. The sooner the old patterns are made unstable, the more effectively new meaningful patterns can be established (Vereijken & Thelen, 1997). Of course, this process of detachment is easier if the athlete is only just learning the movement than if it has become routine ('you can't teach an old dog new tricks'). It can be achieved by greatly varying the movement patterns and introducing new, unfamiliar environmental factors during practice. That is why variation is so necessary in the initial stages of learning movement (Figure 4.28).

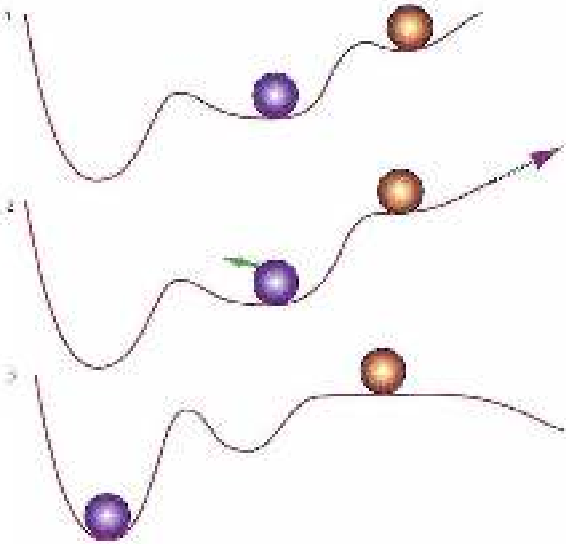


Figure 4.32
An explanation of why variation works in terms of dynamic systems theory, based on phase transitions and perturbation of an attractor. Stable components of movement are not always efficiently chosen (1). When increasing an error in the execution of the movement the existing attractor must be perturbed (2) before a new, better attractor can be built into the movement pattern (3). Variation ensures an unfamiliar blend of sensory and motor patterns and so perturbs stability.

Learning by varying should also play an important part in the teaching of strength training techniques, but in practice it seldom does. Training has to be safe and free of injury risk, is the reasoning, and so the execution of the strength exercise must remain within strict technical limits. But what makes training unsafe is the use of heavy loads. There is far too great a tendency to use high resistance. This makes it impossible to learn through variable training, and an essential step in an athlete's development is omitted. If the load is kept low, strength training can be safe and variable. Variation helps athletes develop the building blocks of movement control, and hence is an indispensable part of that development. With beginner athletes, coaches would be advised to spend a few years building up a good catalogue of basic components through variable training. Only then does it make sense to start using larger barbell weights. This does not mean that they should wait a long time to introduce strength training for young athletes. Because the training loads are kept low, resistance training can be started at a much younger age. There is no reason not to train 14-year-olds in a resistance training setting, provided the loads are strictly limited.

For example, when performing step-ups with a heavy barbell weight (a key exercise for jumpers), no-one will consider making the barbell heavier on one side than the other. Nor will the step-up height be varied too much. A safe margin (no more than about 70% of the length of the lower leg plus the foot) will be observed. This limits the variation in step-ups. For beginner athletes, however, it may be very useful to come into contact with many different variants of the step-ups. For instance, movements in the sagittal plane (extending the hip, knee and ankle) can usefully be combined in a number of ways with stabilization and use of opportunities for frontal plane movement (adduction and abduction of the stance-leg hip). This could be done by using a low barbell weight (e.g. an unloaded bar) that protrudes to one side, so that the load distribution changes. If the load protrudes to the stance leg side, there will be more pressure on extension in the sagittal plane than on frontal plane movements. If the bar protrudes to the other (free/swing) side, there will be more pressure on abduction and adduction in the frontal plane. A second alternative is to vary the step-up height, even to the point where the step-up is difficult to perform without tilting the pelvis backwards. This puts pressure on cooperation between muscles that can tilt the pelvis. A third alternative is to vary the movement of the free (swing) leg during the step-up, for example by pulling it in extremely fast after leaving the ground. A fourth alternative is to vary horizontal movement during the step-up, in order to place more load on the hamstrings. A fifth alternative is to combine the step-up action with torsion in the upper body (something that is usually strictly forbidden when carrying a heavy barbell weight because of the risk of injury). All these variants can be combined, and executed at various speeds. The organism thus learns to coordinate the various alternative movements and find the right overall control, which is essential for such things as running and jumping. For beginner athletes this is an investment that will later pay dividends. For example, many ball sport (soccer, rugby, basketball etc.) players are restricted in their skill performance because they cannot optimally combine extension of the leg in the sagittal plane with frontal plane movement control when running and changing direction. Their acceleration, top speed and changes of direction therefore remain at a lower standard than they might otherwise be (Figure 4.29).

We have already seen that the learning organism is very interested in general rules that can be applied in many situations. These generally applicable basic components of the movement have attractor characteristics. They are stable and economical. Variable training can separate the stable (i.e. generally applicable) components of the movement from the changeable, fluctuating (i.e. incidental) ones.

A movement must be truly mastered if it is to be executed stably as well as in a changing environment. Performing well in competition depends on this combination of stability and adaptability. This means that top performance depends not only on perfect execution of the movement (the attractor part of the movement) but also on its flexibility. It may therefore not be such a good idea to focus entirely on the perfect sporting movement as an important competition approaches. What may be very useful in the run-up to the performance is to retain some variability in training and so make the difference between the stable basic components and the variable incidental components of the movement even clearer to the organism (Figure 4.30). This means continuing to practise not just the sporting movement but also all types of other forms, provided these help to separate attractors and fluctuators. On the other hand, the training should not be so fatiguing that recovery becomes a problem.

In this sense there is also room for continuing strength training in the run-up to top-level performance. In Verkhoshansky's blocked periodization, strength training only takes effect after a while, and it is not a good idea to continue strength training right up

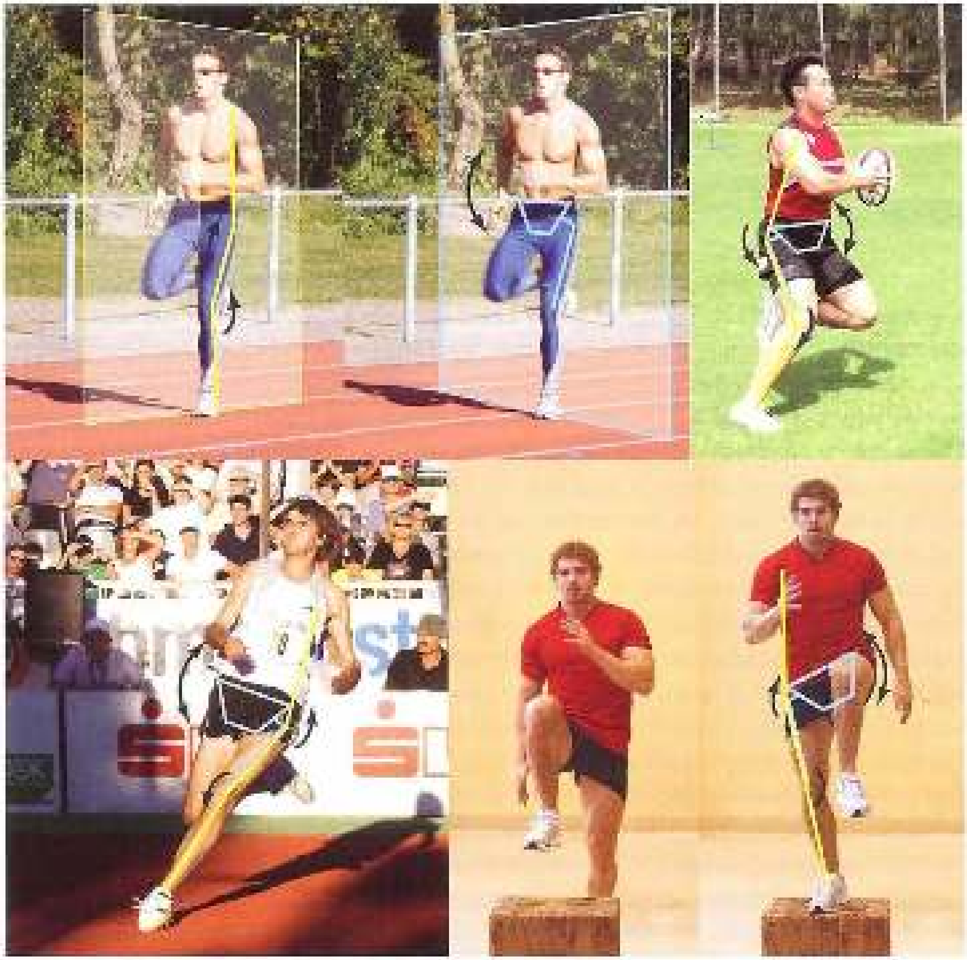


Figure 6.29 Good running technique is marked by a correct relationship between movements in the frontal and sagittal planes. The same applies to side steps and single-leg takeoffs. Variants of step-ups can train this relationship.

to the performance (Verkhoshansky, 1984). It has already been observed that this delay is most likely to occur if strength training is seen as synonymous with heavy loads. The accumulating fatigue then requires long recovery, delaying the effects of the training. But if strength training is seen as 'coordination training against resistance', this fatigue can be avoided and variable training with resistance becomes a useful way to explore stable patterns and make incidentally intended patterns more flexible. In particular, women that practise explosive sports often need to continue strength training until they are close to the top-level performance. It is useful for them to base their training on variation rather than intensity and volume. Account should also be taken here of the differences in load and recovery time between the various types of strength training (see, for example, Section 7.2.5).

Figure 4.26 In a single-leg jump the frontal plane movements act as fluctuators. Movements in the sagittal plane are kept as stable as possible. The same thing happens in a step-up action on an unstable surface.

Variable training can thus play an important part at every stage of the athlete's development: beginner athletes can use it to find a meaningful distribution between attractors and fluctuators, and elite athletes to further increase the difference between attractors and fluctuators in the contextual movement.

Forms of variable learning

There are, roughly speaking, two categories in the methodology of variable learning: differential learning, and random learning. Both strategies are in accordance with Bernstein's adage 'repetition without repetition'.

- Differential learning: learning by frequently alternating many variants of one movement in a single session.
- Random learning: learning by frequently alternating many different movement patterns in a single session.

1 *Differential learning versus learning the ideal technique*

When learning the ideal technique, athletes usually try to approach the right movement pattern (or components of it) as closely as possible when executing the exercise. They also try to make the environmental factors as similar as possible to the competition setting. The learning process is thus linear: athletes try to deviate as little as possible from the right path to the ideal sporting movement. In differential learning, on the other hand, they deliberately deviate from it. Execution of the exercise differs each time, and differs from the sporting movement. One reason for the differences is that the environment in which the movements are taking place is changeable. For instance, the movements may be executed on various surfaces, with varying resistance (e.g. throwing medicine balls with different weights). Gymnasts can practise a movement on different types of apparatus, swimmers can use different hand paddles, baseball players can use different balls and bats, and so on. In each case the different environmental factors lead to different execution of the movement. Performance can also be varied by making different demands in each repetition. Tennis players can move as quickly as possible to another part of the court after hitting the ball, gymnasts can combine each exercise with a different practice component, and so on.

The effects of learning the ideal technique and differential learning are different. Learning the ideal technique will yield faster results – but this is deceptive, for the effect is usually temporary. Practising the ideal technique leads to solutions that are not suitable for inclusion in the coherent catalogue of multi-purpose movement solutions. This is rather like a temporary dental filling: quick, but not lasting. Not only is the solution quickly forgotten, but it cannot easily be transferred to other movement patterns. So it lingers in the learning system for a while and may then disappear.

In differential learning the immediate results (the 'practice results') are not so good, but the eventual impact on the sporting movement turns out to be better and more lasting (the 'learning result'); it can also be transferred more easily to related movement patterns and is more stress-resistant. The underlying mechanism has previously been described in connection with how the organism tries to find generally applicable rules for performance by comparing movements (Schöllhorn *et al.*, 2010).

If the learning process relies heavily on differential learning, two possible disadvantages of this must be taken into account. First, the learning result is invisible. This makes it difficult to assess the effect of the training, and thus how fast the exercises can be covered. So it is not entirely clear when to step up the difficulty of the exercises. This is a problem not only for coaches who design the programmes, but also for athletes, who generally prefer to work from visible progress to visible progress. Second, heavy reliance on differential learning when learning complex movements (such as high jumping or pole vaulting, the butterfly stroke in swimming or a combination in gymnastics) creates the problem that athletes have little insight into the structure of the movement and ultimately not fully understand it. Differential learning can therefore be frustrating and demotivating, for there is no obvious goal to aim for.

With complex movements, a more linear approach may be unavoidable. If a more linear approach to training is chosen to make the structure of the movement clearer to the athlete, it is important to be aware of the serious risk of reinvestment in stressful situations.

Additional information

An example of the pursuit of pointless ideal technique is the rule, often applied in sports injury rehabilitation, that the stance-leg knee should never be in front of the toes in squatting movements. The reason is that the roll-and-glide movement will then be unsatisfactory (too much glide), which can eventually cause injury. This rule is sometimes applied very strictly, forcing execution of the movement into a straitjacket. But the rules ignores an important aspect of flexing one or both knees. In correct technique there is an interplay between moment arms and

joint torque, with the moment arms distributed at the joints in such a way that performance-limiting factors ideally do not occur in a single isolated joint. In some cases the ideal posture may be with the knee in front of the toes. There is no problem with this as long as the pelvis does not move towards the knee – but that usually happens if the barbell weight is really too great to be technically controlled. So the 'rule' that the knee should not be in front of the toes does not necessarily lead to correct execution of the movement; it is really only meaningful if the barbell weight is already excessive (in other words, if the exercise is already unsuitable for the athlete). This calls for variable training with lower barbell weights, so that self-organization of joint torque is given room to find a contextual solution to the movement problem (Figure 4.37).
The most marked 'straitjacket exercise' in the knee-flexion family is the forward lunge on a flat surface. Even with a low barbell weight (50% of body weight), self-organization into meaningful contextual movement is almost impossible in this exercise. We can then introduce strict rules such as 'knee in front of the toes' – or we can drop the exercise altogether, for it may safely be assumed that any exercise with such a barbell weight, which leaves little room for contextual self-organization, will also provide little transfer to contextual sporting movements. Forms of the lunge, which must provide transfer to a sporting movement (for instance tennis or squash), should therefore be performed with a barbell weight so low (e.g. just an empty bar) as to allow self-organization, such as the self-organization of deceleration described in Section 3.2.6 (Attractor 8: distributing pressure when decelerating).

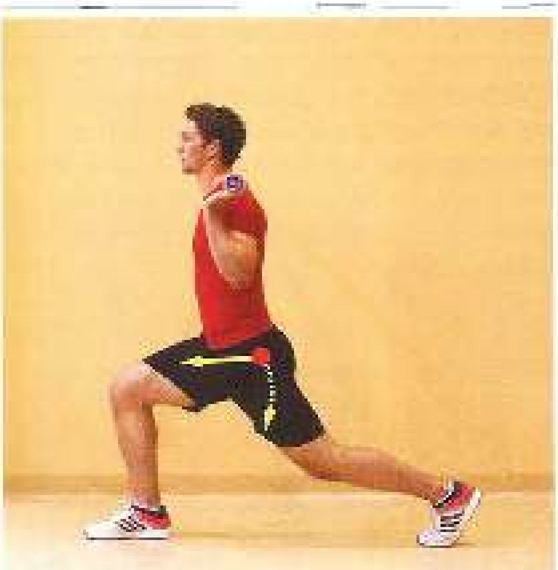


Figure 4.37
In a variant of the knee flexion, there must be room for self-organization of the movements in the leg joints (i.e. hip rotating round, or moving towards, the knee). In a forward lunge this is only possible with very low barbell weights.

2 *Blocked versus random organization of the exercises*
If four different, unrelated movements are practised in a training session, we can start by performing all the repetitions of movement 1, and then move on to movements 2, 3 and 4. This is known as 'blocked practice'. But we can also structure the same number of repetitions differently, alternating the movements frequently and in a random order. This is

known as 'random practice'. Total practice time and the number of repetitions are the same in both cases, but the effect is clearly different. The difference is similar to the difference between ideal-technique exercises and differential learning. In many motor-learning settings, blocked organization yields a better practice result and random organization a better learning result; the earlier comments on the reversible learning process also apply here.

Additional Information

Research has frequently revealed a clear difference between the practice result and the learning result. The practice result is the performance level achieved at the end of the practice session or sessions. The learning result is the level that can eventually be reproduced permanently. Research repeatedly shows that the better the practice result, the worse the learning result tends to be. Solutions that yield quick results are apparently only suitable for temporary use, whereas solutions that can become permanent do not yield such quick results in the very short term.

The solutions that are designed in the practice session are stored in the working memory. Whether they are copied to the permanent memory after the practice sessions (real learning) depends on whether they fit into the way the permanent memory is organized. If not, the information is simply deleted. Nothing has been learned. Coaches that let themselves be guided by the performance at the end of the practice session or sessions do not usually get the learning results they want (Figure 4.32). So it is not such a good idea to make great use of tools that enhance the practice result, such as cognitive transfer of precise KP information.

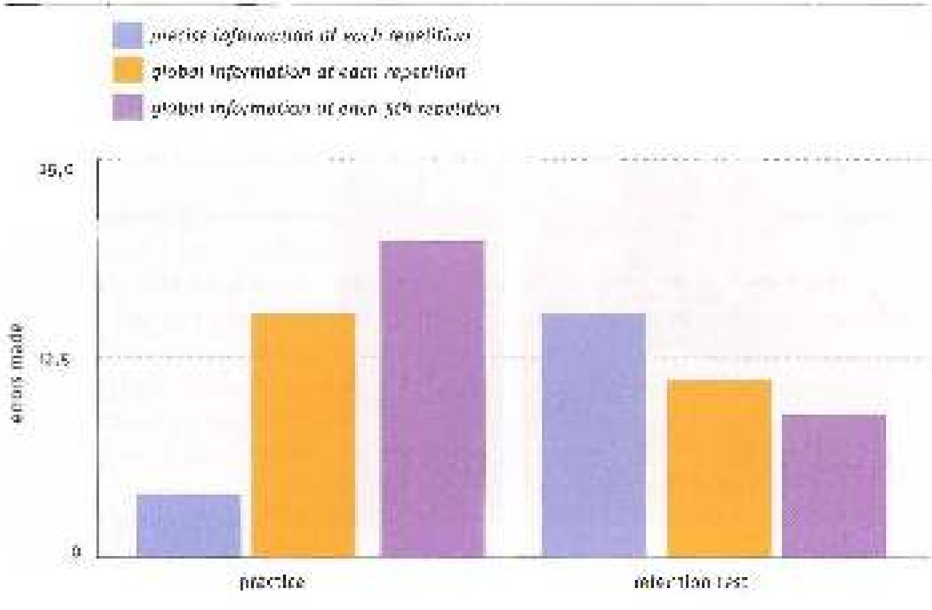


Figure 4.32 An overall picture of the difference between practice result and learning result. In the practice sessions the first group is given precise feedback after each attempt. The second group is given overall evaluation feedback after each attempt. The third group is given overall evaluation feedback after five attempts on the average of the five attempts. The first group makes the fewest mistakes, but learns the worst. The third group makes the most mistakes, but learns the best. This difference between practice result and learning result is revealed by a retention test (Winstein et al., 1990).

Research has not yet provided a clear answer to the question of why random practice has such a different impact on learning than blocked practice. The issues regarding the impact of random practice are focused on 'contextual interference' mechanisms (Lee & Simon, 2004). One explanation seeks the difference in the fact that retrieving a motor pattern may be an additional difficulty when executing movements. We can understand this by looking at sport-specific warming-up before a competition. When preparing for a competition, athletes usually execute the sporting movement: practice starts, practice jumps, shooting at goal, serving and so on. Everyone considers this meaningful. A high jumper who shadow-boxes during warming-up seems ridiculous. Warming-up has no learning effect, for there is simply not enough time between practice and consolidation (the competition). In that sense, sport-specific warming up is pointless. The point of it lies in retrieving the right motor pattern for the competition. This is prepared during warming-up and it seems that retrieving a complex motor programme is difficult.

If training is largely organized on a blocked-practice basis, the motor programme only has to be retrieved once, and so retrieval is not practised or learned. Performance at the end of the session is good, because in each case the motor programme is all ready. In random practice, a different motor programme is practised each time, and the athlete constantly practises retrieving the motor programme. When applying motor programmes in competition, the effect of learning to retrieve motor programmes is important. It should be noted here that this is not just a matter of retrieving relevant motor patterns, but also relevant sensory patterns. A skier who visualizes series of bends before a downhill competition without actually executing the accompanying motor patterns is also attempting to get relevant patterns 'all ready' for the movement.

This explanation is easy to 'grasp', but given the powerful impact of random-practice training there must also be a number of 'ungraspable' mechanisms at work that invisibly organize learning effects deep within the organism. It may take quite some time for researchers to identify these mechanisms. All we can do for the time being is speculate. Until then, practical experience must be our main guide to how random-practice training can be used.

Random practice usually produces a better learning effect, and especially better transfer to related skills, than blocked practice. Young people turn out to be remarkably sensitive to the effects of random practice. This is therefore highly suitable for use in the initial years of learning basic strength training techniques, during which heavy loads are avoided and the emphasis is on establishing basic movement patterns. With older athletes, the use of random practice is interesting because a different, and possibly better, training effect can be achieved without having to increase loads.

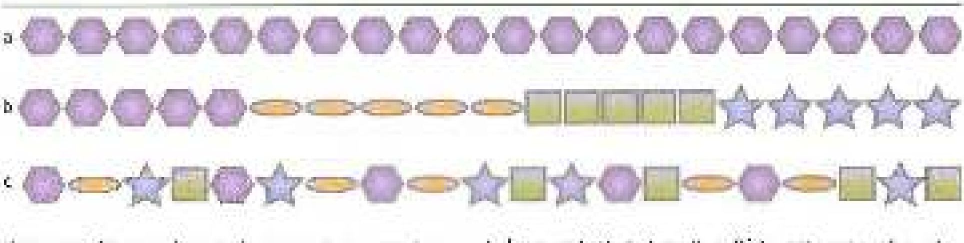


Figure 4.[illegible] (a) A practice situation is repeated over and over again (completely blocked practice). (b) A small number of exercises are alternated after several repetitions (between blocked and random practice). (c) There is no structure in the order of the exercises. All types of practice situations are practised at random (completely random practice).

There is a considerable debate on how different exercises should be in order to optimize the learning effect. Does the optimum lie in alternating related movements (differential learning) or in largely or completely unrelated movements (in the direction of random learning)? Some studies indicate that a mixture of very different movements yields a good learning effect, and may be more important when learning athletic movement than using exercises that differ less from the pattern to be learned. In that case, exercises could perhaps be a lot more varied than may at first seem advisable (Figure 4.33) – but this is not yet entirely clear. What is known, for example, is that shot putters who have mastered their sporting movement to a very high level and appear to reached their ceiling can still benefit from learning movements that are meaningless in terms of direct performance (such as a double rotational technique), and even from learning to perform incorrect movements on demand. The organism seems to want to learn from much greater contrasts than has hitherto been assumed especially in cognitive-oriented theories.

The answer to the question of how much variation is most effective when learning may lie in a simple observation. This chapter has discussed two major theories of motor learning: result-oriented learning, and learning through variation. The most effective learning setting may involve taking account of both theories. Exercises should be variable, but at the same time they should have a clear result that can be defined in advance (KR, intrinsic or augmented). Moreover, the result must be clearly linked to the intended result of the sporting movement to be improved. Variability meets the need for generic ways of keeping the intention-to-action process flexible, and result (successful or unsuccessful performance of the movement) meets the need for intentional markers for the learning system's self-organizing ability, and hence the stability of the intention-to-action process.

If there is little variation in the exercises, general principles will be hard to find, even if the exercises yield clear result information. If there is more variation, generic principles will be easier to find and they can be linked to the still available relevant result information. However, if there is too much variation, it will provide plenty of information that can be used to formulate generic principles, but the result information will be less relevant (the exercises will not yield the same result as the sporting movement to be improved). With even more variation, in the direction of random movement, anticipated and achieved results can no longer be compared, and the learning process will stop. This will lead to a typical minimal-optimal-maximal bell curve in which variation and focus on results are in conflict.

From all this we can derive a simple rule of thumb for varying exercises: choose exercises with maximal variation, in which the result/intention of the exercise remains close to the result/intention of the sporting movement to be improved.

4.5 Summary

Strength training based on the physiological adaptations that occur seems most suitable for endurance sports. Coordinative adaptations as a result of strength training seem less relevant. However, the two cannot easily be separated, especially in a sport such as speed skating. In sports that seem technically less complex, the main value of strength training lies in oxidative recruitment of FTIIx fibres. This makes sense in endurance running, since major external forces must be absorbed in the stance phase. Being able to absorb these isometrically is a very important factor in optimizing running efficiency. In cycling, on the other hand, the value of strength training is much harder to determine, and recruitment of larger muscle fibres should be seen as a way of optimizing the speed of muscle action of ST fibres, which is particularly important when cycling uphill. One problem with strength training for endurance athletes is that the adaptations sought appear to conflict with the required endurance adaptations.

When opting for strength training based on coordination, the way in which motor control limits performance must first be identified. If performance is only limited by limits to the physical load capacity of the musculoskeletal system, there is little point in basing strength training on coordination. However, performance is limited not only by limits to recruitment but also by the way in which movement control is designed. Control by reflexes based on cocontractions dampens and limits performance in high-intensity movement, and hence it is useful to take account of the consequences of motor control within strength training.

The laws of motor learning are therefore of relevance to strength training based on coordination. The consequences of the intention to action organization of motor function, and variability of training, are key factors here.

Motor function is intention-oriented. The purpose of the movement is therefore an essential part of movement control. First the purpose is designed, then the basic abstract rules of the movement are formulated, and finally the specific muscle action is determined. External focus and knowledge of result feedback facilitate this way of organizing motor function. Internal focus and knowledge of performance can tempt the athlete to 'drive the wrong way', activate the working memory and, in stress situations, lead to reinvestment.

The motor system only tries to learn if it is challenged. Exercises that generate a new and unexpected blend of sensorimotor information are essential in developing the necessary motivation. Monotony stops the learning process. This means that exercise variation is crucial to learning: it ensures that generally applicable rules for motor control can be found by comparing the usable control mechanisms in the various ways of performing the movement. These mechanisms, which are generally applicable in various conditions, are then stored in the memory for automatic control. There are two strategies for variable learning: differential learning, in which several versions of a given movement pattern are practised, and random learning, in which a number of unrelated movements are practised in succession. It is not yet clear which strategy is the most effective.

Since the purpose of strength training is to improve the building blocks of movement patterns, it needs to be intention-oriented and varied.

5
Specificity within strength training

5.1 Specificity and transfer of training

The first and most important feature of any type of training is that it should help to improve athletic performance. The contribution that practising a particular movement pattern makes to improving another movement pattern is known as transfer of training. Everyone knows intuitively that transfer of training can only occur if the exercise in some way resembles the sporting movement. Tennis players are unlikely to start hurdling in order to improve their footwork. Marathon speed skaters sense that there is some resemblance between muscle use in their sport and cycling, and hence do some of their training on bicycles. However, it is not enough to approach similarity of movement between exercises and sporting movements simply on the basis of intuition. That will only work with exercises in which the similarity is very obvious – but often it is not. In the past, for example, some long-track speed skaters – especially sprinters – included 100-metre sprints on athletics tracks in their training programmes because they felt it would improve their speed; but others queried the value of running training for skaters, if only because the skating and running postures are so different. Transfer from the running movement to the skating movement is evidently rather questionable.

Intuition is also usually inadequate in approaches to strength training, for it is hard to identify similarities of movement between strength exercises and sporting movements. Strength exercises seldom outwardly resemble athletic movement, and the way in which the exercises contribute to the competition is therefore also fairly 'invisible'. To assess the value of strength exercises, we must analyse the movement patterns between which transfer is supposed to occur. Such analysis of the type of exercise and the sporting movement will identify both the visible and the invisible components of the movements. We can then determine which components are similar enough to allow transfer. Similarity of movement between types of exercise is known as specificity. Specificity of the type of exercise is the main guarantee for transfer of training to athletic movement, and a systematic approach to specificity in exercises is therefore critical to the effectiveness of sport-specific training.

Strength exercises for improving sporting movements are almost always part-practice exercises, in which a small number of aspects of the sporting movement are trained in isolation. Part practice allows greater emphasis on those isolated aspects, which can be trained without the possible disruptive influences of other components of the sporting movement. This allows overload to be created in the exercises. Such isolated training greatly improves individual aspects, which can have a positive impact on the overall sporting movement. However, such transfer of improved individual aspects is not self-evident, for the sensorimotor links (the way in which sensory and motor information forms a contextual whole in the movement) will be different in the part-practice exercise and in the sporting movement. This is often the case with strength training. The common practice of choosing strength exercises that are solely based on similarity in joint angles and angle changes cannot therefore guarantee the intended transfer. In some cases there may be so little transfer from part-practice exercises that the value of investing time in exercises solely based on similarities in the external structure of the movement must be questioned. This is a vital issue for strength coaches and sports injury rehabilitation professionals, and an important reason to seek a better justification for choosing exercises than the purely intuitive, 'obvious' ones. Choice of exercises must be based on deeper insight into the possibilities and limits of transfer.

Additional information

Another key term besides 'transfer' is 'generalization'. 'Transfer of learning' means the way in which two more or less related movement patterns may influence the performance of the other and may lead to accurate, stable performance of the movement (through fixation of movements in closed skills: Gentile). 'Generalization' means the way in which a movement pattern may be made flexible via a number of adaptations so that it is appropriate in a changing movement environment (through diversification of movement in open skills: Gentile). Sporting movements must meet both criteria: fixation of stable patterns, and flexibility in a changeable environment.

5.2 Limited transfer of strength and power

5.2.1 Maximal strength and transfer

When drawing up a training plan it is best to approach strength not as an isolated quantity, but as integrated aspect of the eventual performance – something that interacts strongly with other performance-determining factors. The eventual performance is not just the sum total of the various components to be trained, but is also determined by interaction between them. This also applies to maximal strength (the greatest muscle action force that a muscle can produce).

Maximal strength is contextual, and not universally applicable during movement. This limited applicability of maximal strength is very useful. The limit of possible recruitment of muscle fibres for movements that are not mastered at a high level (e.g. a 100-metre sprinter who suddenly tries to throw a javelin) has traditionally been assessed at about 75% of the total number of muscle fibres in a muscle. This percentage can be raised by training, and so the maximal achievable strength will increase. However, the higher percentage will mainly be achievable in movements related to the movement being trained. The fact that strength can only be transferred to related movement patterns is a key mechanism for protecting the body. Without it, athletes would often suffer injuries. Suppose that elite high jumpers decide one day to start throwing a javelin. If, from a high run-up speed, they could produce the same maximal force (up to ten times their body weight) when bracing the front leg during the throwing action as when taking off for a high jump, they would be at high risk of serious injury. This is because they would lack the skill to plant their front foot from a cross-over step so as to keep the external forces completely under control. The risk of serious injury would likewise be great if good shot putters who tried throwing a javelin for the first time were able to produce the maximal force of their shoulder girdle muscles. They would lack the skill to protect the joints in their throwing arm by means of muscle action against the extreme ranges of motion in the throw (external rotation in the late cocking phase). The muscles themselves would also be at risk, as a shot putter is not used to absorbing large tensile forces acting on the muscles over a very short time. To prevent such situations from arising and to protect the body against excessive, uncontrollable interplay of forces, trained properties such as force production are therefore mainly transferable to related movement patterns. This creates a link between how many muscle fibres an athlete can recruit and the athlete's skill. A high percentage of recruitment in a movement that has been mastered will only be transferred to related movement patterns, i.e. those that have also been mastered. This again highlights the fact that strength is in fact dependent on coordination.

This contextuality of transfer of maximal strength applies not only between different sporting movements, but also between strength exercises and sporting movements. An important question that a coach should ask – in addition to the question of specificity – in order determine the effectiveness of the exercises is how the amount of forced produced in the strength exercise relates to the amount of forced produced during the sporting movement. This is because coaches are far too likely to assume that the maximal amount of force that a muscle can produce is always achieved in the gym. We have already seen that a number of sporting movements (running) display an extreme pattern of excitation and inhibition at spinal-cord level owing to reflex support (the stumble reflex, the crossed extensor reflex, the foot-sole reflex and so on). As a result, larger peaks of force production occur in such movements than those that can be achieved through maximal voluntary contraction (MVC – see Section 2.3.2).

Movement patterns in heavy strength training are seldom if ever reflex-based. Determining the value of heavy strength training therefore involves looking at more than just the maximum that can be achieved during the exercise, and a comparison must be made with force production during athletic movement. If the maximal force that can be produced during an exercise is less than during the sporting movement, the exercise serves no purpose when it comes to providing overload. This can be seen in practice, for example, from the fact that in sports requiring very high maximal force production (such as high jumping) quite a few of the very best athletes do little or no strength training (Figure 5.1). Simply jumping evidently makes you very strong without any need for strength training. For some muscle groups, in fact, training with heavy barbell loads is completely pointless in terms of overload, because barbell training simply cannot provide overload compared with the load in the sporting movement. The barbell load needed in order to create such overload compared with the sporting movement is much too great to perform the strength exercise (see Section 6.2.1: Overload within strength training of calf muscles).

Since force production evidently can be increased by more than just work in the gym, it is important to seek the value of strength training not only in the amount of overload, but also in interaction between force production and other aspects of performance.

This means that the overload aspect in maximal strength training cannot be approached one-dimensionally (purely quantitatively) as often happens in practice, i.e. by interpreting the maximal load an athlete can lift during a barbell exercise as the maximal force a muscle can produce; within strength training, overload has to mean more than just 'more and heavier' in purely Newtonian terms (see Chapter 6).

5.2.2 *Power and transfer*

In many sporting movements, maximal strength is not a performance-determining property. The intention in many sports is to accelerate either the athlete's own body or an object that is already moving at some speed. To do so, muscles must produce force while shortening rapidly. This link between force and speed is known as power. A rower making a stroke at speed will have to apply force to water that is already moving at a given speed in relation to the boat; force will therefore not be produced statically but dynamically, and the muscles must have a given speed of shortening in order to apply force to the moving water. The same applies to a cyclist who wants to turn an already turning pedal even faster, or a shot putter who increases the speed of the shot halfway through the put (Figure 5.3). Expressed in a formula:

$$F \text{ (strength)} \times v \text{ (speed of shortening)} = P \text{ (power W)}$$

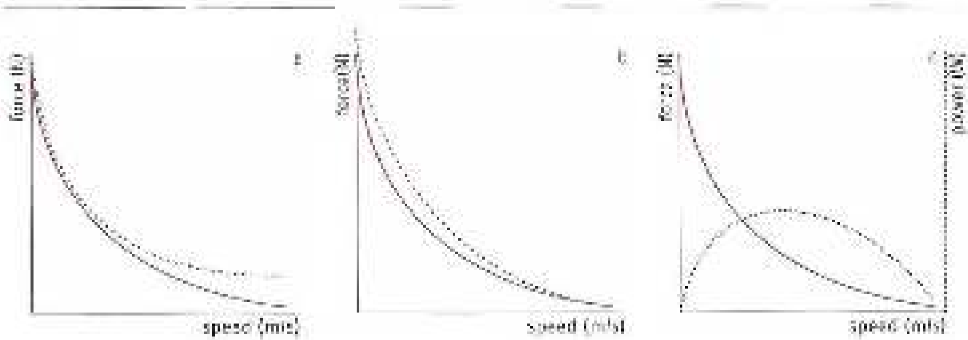


Figure 5.2 a A force/velocity relationship for muscles. The dotted line indicates the shift in the graph owing to training at high muscle action speeds.
b The same, but with a shift in the graph owing to training with heavy barbell loads.
c A force/velocity relationship for muscles and the corresponding power curve (dotted line).

Linking strength to speed of muscle action is a problem in view of the force/velocity relationship. The greater the speed of muscle action, the less force the muscle fibre can produce. With extremely rapid muscle action (e.g. a table tennis forehand), very little force production will be possible. This difficult link between force and velocity can be influenced by training, and is task specific. When training by slowly moving large weights, it is above all the link between great force and low muscle action speed that will improve. The link between force production and rapid movements will barely change, if at all. Conversely, training by quickly moving small weights will have no effect on maximal strength. Clearly, athletes will benefit from opting for power training to create a link between force production and velocity such that optimal adaptation occurs in the part of the force/velocity curve that is relevant to the sporting movement.

Just like the concept of maximal strength, the concept of power has acquired a rather independent status in the training world – see, for example, the increased popularity of power measurements (in isokinetic tests). Considerable importance is therefore still attached to the results of power measurements, which are sometimes even treated as crucial when adapting training plans. However, this is questionable. Are power measurements an accurate reflection of an athlete's performance? Can the results of the measurements simply be transferred to the sporting movement if this is based on the same link between force and velocity as the measurement, or is the ability to produce power movement-specific, just like maximal strength? And are there sports with a fixed speed of movement? In many sports there is a change in speed, and hence a change in the amount of force produced. A baseball player stealing a base or an athlete pushing a bobsleigh will achieve a higher speed of movement during his acceleration and so be able to produce less and less force. Obviously, measuring the maximal possible force at a single speed will not tell us anything about a performance in which speed keeps increasing.

Even if there seems to be no problem of increasing speed and the sport in question seems to have a more or less constant speed of movement (rowing, speed skating and so on), the speed of muscle action does vary within a cycle. Power measurements may therefore be rather questionable if they are carried out in a movement pattern that differs from the sporting movement.

Since not only adaptations in maximal strength but also adaptations in power production are very situation-specific, it is questionable whether a training structure based on 'general' and 'specific' blocks of strength training is a good idea. General strength presupposes a type of strength that is generic and is not transferred in a situation-specific way, and hence has an impact on all types of training to be carried out later. Yet in fact there is no such thing (Baker *et al.*, 1994). The possible value of a general block of training must therefore be sought in an area other than applicability within the sporting movement, such as increasing the robustness of the musculoskeletal system.

There are two important considerations regarding power production in sporting movements:

1. The total amount of power that an athlete produces very much depends on how fast force can be built up in the muscles. This build-up takes place in a number of steps:

- from the stimulus at the neuromuscular junctions of the motor neuron to the start of muscle-fibre action (electrochemical delay);
- from the start of muscle-fibre action until force is applied to passive series elastic components (SECs);
- from the moment that force is applied to the passive SECs until stiffening of the SECs passes on all the force of the muscle fibres to the muscle attachment points;
- the unloading of the SECs together with power production in the muscle fibres (Roberts & Konow 2013).

There are several commonly used terms for the build-up of force, such as electromechanical delay (EMD), muscle slack and rate of force development (RFD).

- EMD means the time between the start of the stimulus received by the muscle in the neuromuscular junctions and the moment when the muscle fibres begin to apply force to the muscle attachment points.
- Muscle slack means the time between the start of muscle-fibre action and the end of the build-up of force in the muscle (until peak force/power is reached at the attachment points), i.e. without the (electrical) delay between the stimulus at the neuromuscular junctions and the start of muscle-fibre action.
- The point at which RFD starts is less clearly defined. It is usually assumed to be start of force production at the muscle attachment points. However, an earlier point after the stimulus to the muscle fibres is sometimes chosen instead. The end of RFD is the same as the end of muscle slack.

In terms of sporting contextuality it is mainly important to know how force is built up in the whole muscle from the point when the muscle fibres become active to the point when the build-up to peak force is completed, as well as how this can be influenced by training. The delay between the stimulus and the start of muscle-fibre action is thus ignored here. That is why the terms 'muscle slack' and 'rate of force development (RFD)' will be used indiscriminately in this text, both referring to the time between the start of muscle-fibre action and the end of force build-up at the muscle attachment points.

Muscle slack is one of the most performance-determining factors and is highly movement-specific. It arises because muscles hang between their attachment points like slack ropes; they must first be tensioned and enough stiffness must then develop in the SECs before they can produce their peak force (see Section 2.1.5). The amount of muscle slack to be overcome greatly depends on the situation in which the muscle is working. If, starting from a squat position, the athlete extends without a barbell load by producing as much power as possible, there will be plenty of muscle slack. When extending with a barbell load this will be much less of a problem, because in the squat position the load acts eccentrically on the muscles, thus providing the pretensioning needed to reduce the muscle slack (Figure 5.3).

RFD measurements are nearly always carried out against resistance. Sometimes a pull against a fixed bar is measured, allowing the build-up of isometric force to be quantified. Sometimes a barbell is accelerated from a standstill, allowing the build-up of power to be quantified. A precise justification is seldom if ever provided for the size of the weight to be accelerated. The way in which strength is built up in such situations may be fundamentally different (through the external force of the resistance and/or allowance of a countermovement) than in RFD without much external resistance (using pretensioning from cocontractions). This fundamental difference is so great that measuring RFD with resistance cannot simply be assumed to tell us anything about the quality of RFD without resistance.

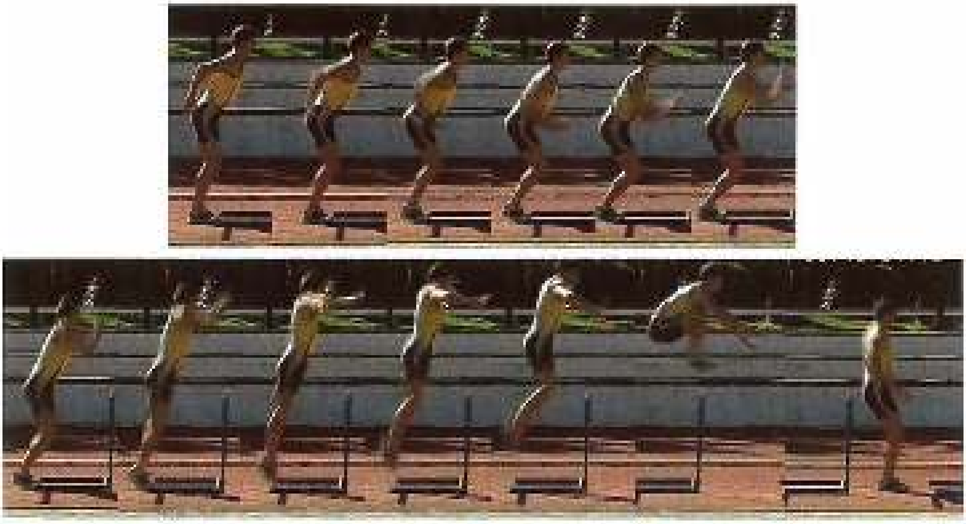
The difference between moving with and without a barbell load is similar to the difference between rowing in a competition boat and on a rowing ergometer. Indoor rowing champions are seldom champions on water. This is because in a boat the resistance of the water can hardly be used to reduce muscle slack at the start of the stroke, and hence good technique with cocontractions is needed for effective build-up of force on the oars. Things are much simpler on an ergometer – it is easier to use external opposing forces to solve the problem of muscle slack at the start of the stroke (using the resistance of the machine by jerking the handle at the start of the stroke). Because of this, there are lightweight rowers whose technical skill allows them to achieve relatively better performances on water than on an ergometer.

As well as under the influence of a barbell load, major differences in muscle slack may also occur because of differences in the postures in which power has to be built up. A weightlifter may be able to do a higher vertical squat jump than a 100-metre sprinter, but will be unable to accelerate out of the starting blocks as fast. A single-leg and a double-leg push-off are relatively dissimilar. Differences in body posture between the measurement and the athletic movement may therefore further reduce the relevance of the measurements.

To this can be added the fact that there is little correlation between the first milliseconds of RFD against resistance and force production after 150–250 milliseconds (Andersen & Aagaard, 2006).

All this makes it very difficult to determine whether, in sports in which the first tenth of a second of the action is crucial (i.e. virtually all explosive sports) and no major external resistance has to be overcome, training of RFD against resistance and performance of strength exercises lasting longer than about 150 milliseconds (heavy strength exercises) yield positive, neutral or negative transfer to the quality of RFD without resistance (Blazevich, 2012; Gruber & Gollhofer, 2004; Marques et al., 2011). This conclusion challenges many of the ideas from classic training theory about supposedly automatic transfer of RFD and reveals a flaw in the traditional mechanistic way of thinking. If there is no demonstrable link between RFD with and without resistance, the idea of automatic positive transfer of heavy strength training to 'athleticism' is largely nullified. Strength training can then only contribute to better performance in explosive sports under certain conditions that are hard to identify: there will then be substantial differences between and within individuals, and in some conditions, such as excessive strength training, transfer may even have an adverse impact. In other words, what works for one individual need not work for another, and what works for one individual today need not work in the same way for the same individual a year from now. This is particularly true of experienced athletes (Marshall, McEwen & Robbins, 2011; Häkkinen et al., 1987). In beginners, strength training usually does have a positive impact on some aspects of RFD. The RFD transfer problem may well be the biggest problem in sport-specific strength training. This means that a can't-do-any-harm approach to sport-specific strength training should be avoided at all costs.

2 Besides the limitation of muscle slack at the start of the movement, power production is further limited because the movement has to be decelerated at the end of the acceleration path. In a squat jump, for instance, the knee extends. Muscle force must be produced to decelerate this knee extension, otherwise the joint will be damaged. The same applies to extension of the elbow in shot putting, boxing and so on. Although this limitation on the ability to produce full power up to the end of the acceleration path tends to be overlooked, it has a major performing-limiting influence on sporting movements. This anatomical influence is movement-specific (Zatsiorsky & Kraemer, 2006). A power measurement can therefore only predict competition performance if movements are decelerated in the same way as in the competitive movement. The first principle and rule of thumb for determining specificity is that deceleration is substantially different in a double-leg and a single-leg push-off. The same applies to deceleration of a double-arm or single-arm thrusting movement. When extending during a single-leg explosive jump (e.g. the first step out of the starting blocks), the knee extension may be decelerated at the same time as the free (swing) side of the pelvis (on the same side as the swing leg) elevates. This may produce very different muscle deceleration characteristics and a different influence on total power production at push-off than when measuring power in a double-leg squat jump. The same will apply when decelerating elbow extension in a thrusting movement (shot putting or boxing). In a sporting movement such extension is always accompanied by rotation of the trunk, i.e. movement in the shoulder from the flexion position to the abduction position. This means that the total deceleration characteristics will be different than in a power measurement when a weight is thrust as explosively as possible with two hands (and hence without rotation of the trunk).



a

b

Figure 5.7 An athlete attempting a maximal-height squat jump over a hurdle from an almost upright position without a countermovement cannot help making a slight one: see (a), pictures 1–6. Even with excellent technique, there will always be too much muscle slack to reach the required height. In a clean from an almost fully extended position (b) the countermovement can be omitted, for the barbell load will provide the necessary pretensioning. This means that training with barbells only provides limited transfer in terms of pretensioning. In other ways, too, a clean and a jump are far from similar.

Like muscle slack, the barbell weight may facilitate such deceleration, especially when power is produced with high resistance. The high resistance reduces the speed of movement as well as the problem of deceleration at the end of the power production, so that muscles can keep producing their power for longer. This means that high-resistance power exercises are less specific than movements with low external resistance.

In power training, high resistance therefore facilitates both the beginning (RFD) and the end (deceleration) of the movement, thus making the movement easier to coordinate. Especially at a high level of mastery, such 'simplification' of coordination of course does little to help athletes learn the complexity of explosive athletic movement.

The power produced during a contextual movement is almost always the result of cooperation between muscles. This does not mean that each individual muscle has to produce as much power as possible in order to achieve optimal performance. Some muscles (the biarticular muscles) perform an energy-transporting function and do not themselves do any work (work is producing force along a given shortening path). They remain more or less isometric throughout power production. Simulation models have shown that potential performance may be highly dependent on the quality of intermuscular cooperation. If an important muscle group such as the hamstrings contracts a fraction of a second sooner or later in a squat jump than it should in the ideal movement pattern, there will be a sharp deterioration in performance (Bobbert & Van Soest, 1994; Van Ingen Schenau & Van Soest, 1996). This means that the total power produced (as measured in, say, a push-off) may be limited by the maximal power that muscles can produce, by the limits of their energy-transporting capacity and/or by poorer intermuscular cooperation. The limiting factor may therefore differ between two different movement patterns. If, for instance, the energy-transporting capability of gastrocnemius were the limiting factor in a clean from above the knee, but not in a push-off in skating or a start in swimming, the performance-enhancing effect of the strength exercise would be reduced for these sporting movements. However, it is very hard to determine if this is really so.

Besides the aforementioned limitations on power production, there is also a limitation due to the required robustness of the movement, further reducing the predictive value of power measurements for movement patterns that differ from the measurement (Wilson *et al.*, 2007).

What this means is that, just like maximal strength, power is not an isolated, universally valid phenomenon. Like strength, it depends on coordination and on the context in which the movement is performed.

5.3 Categories of specificity

Two movement patterns are usually considered 'specific' in relation to each other if their features are more or less similar. These similarities can be divided into five categories (Figure 5.4).

1 Similarity in the inner structure of the movement:
   * intramuscular similarity: similarity in coordination within a muscle (e.g. similar muscle action in the lower limb during a trampoline jump or a jump to block in volleyball);
   * intermuscular similarity: similarity in cooperation between different muscles (e.g. similar cooperation between the erector spinae muscle and the gluteus maximus muscle when jumping to block in volleyball and in a swimming start).
2 Similarity in the outer structure of the movement (the similar extension of the joints, e.g. similarity in the movement of the shoulder girdle in a baseball pitch and when serving in tennis).

3 Similarity in energy production.
4 Similarity in sensory patterns:
   - similar sensory patterns when monitoring the environment;
   - similar sensory patterns when monitoring the body (proprioception).
5 Similarity in the intention of the movement.

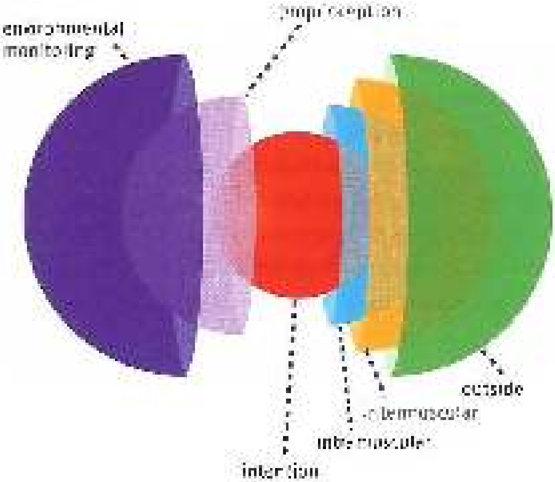


Figure 5.4
The three-layer model of specificity. In the centre, the intention of the movement, and around it to the right the spheres of the inner and outer motor characteristics. The spheres on the left represent sensory patterns: proprioception (perception of the body) and environmental monitoring.

Similarity of movement, and hence the value of a strength exercise, can be determined by analysing the five categories of specificity. Obviously, most types of exercise are not similar to the sporting movement in every category, and hence are only partly specific. This is particularly true of strength exercises. In most types of strength exercise, transfer should therefore not be sought in transfer of the entire movement pattern, but in one or more aspects of it.

5.3.1 Similarity of movement through similarity of inner structure of the movement

This aspect of movement similarity can be divided into two groups: similarity of movement in how action takes place within a single muscle (intramuscular coordination) and similarity of movement in how muscles cooperate (intermuscular coordination). Even researchers find intermuscular and intramuscular coordination very hard to measure, and research has so far revealed too little to make conclusive statements about what is going on 'under the skin'. Features of intermuscular and intramuscular movement patterns are so far mainly determined from anatomical models of movement. In combination with practical training experience and such research findings as do exist, it is possible to develop an idea of what happens at the intramuscular and intermuscular levels. The similarity between the two types of exercise can then be identified.

Intramuscular coordination

Strength training is highly suited to optimizing coordination within a single muscle. Co-operation between the muscle fibres within one muscle during action is very complex and will vary greatly depending on the setting in which the muscle has to work. In other words, intramuscular coordination is also very complex. A well-known principle of control is the 'size principle' (see Section 2.2.1); however, this describes only a small part of the organization of control. There is much more to it than this. For example, cooperation between the muscle fibres and the passive tissues that transmit the force of the muscle action has not yet been thoroughly studied. There is a one-dimensional model of this (Hill's model), but a three-dimensional model is needed in order to describe reality.

Muscles can do their work in more than one way. The various types of muscle action (concentric, eccentric, isometric and elastic) differ considerably, and when a movement is executed correctly there is no gradual transition between them. In practice they turn out to be compartmentalized, and so the specificity of an exercise at intramuscular level very much depends on the type of muscle action taking place.

The first step in making strength training specific therefore lies in similarity in the type of muscle action. But this alone does not optimize specificity. Even when the type of muscle action is the same, the intramuscular pattern of cooperation can still be variable. If a muscle has to act concentrically when cycling, the system will organize cooperation between the muscle fibres differently than when the muscle acts concentrically in another activity, such as rowing or accelerating when playing soccer.

The question is whether it is always wise to maximize specificity within strength training. In some cases it is, but in others making the strength exercise even more specific makes the exercise too difficult or even dangerous. For instance, running at speed places an elastic load on the hamstrings, especially during the pendular motion of the leading lower leg at the end of the flight phase. Loading of the hamstrings can be properly trained within strength training. However, since the exercise cannot be made specific by imitating the pendular motion of the lower leg, that part of specificity is disregarded within strength training.

Intermuscular coordination

Strength training is also very suitable for optimizing cooperation between muscles. In many sports, intermuscular coordination is the factor on which performance most depends. This is particularly true at high levels of mastery. Intermuscular coordination is so complex that at least two requirements must be met when executing sporting movements.

1. The movement must be executed efficiently and economically. In contextual total movement patterns this means attempting to make optimal use of the elastic capabilities of muscles together with isometric muscle action, so as to reduce the amount of concentric and other work and the accompanying high energy costs (see Section 4.1.2). Because the muscle architecture differs considerably, some muscles are better than others at dealing with elasticity in a complete pattern. The same applies to force production. Good intermuscular cooperation should take account of the specialization of muscles.
2. The movement must be controllable. This is only possible if the movement patterns are constructed on fixed principles that are flexibly integrated into a complete pattern. Among other things, this requires cocontractions and synergies that make execution of the movement resistant to breakdowns and errors in control.

In order to meet both requirements, the body has an abundance of different muscles. For instance, there are a whole series of muscles that can produce hip extension in the hip joint (e.g. *gluteus maximus*, hamstrings, *adductor magnus*). It seems they are needed in order to act flexibly in various settings. In order to meet both the efficiency and the flexibility requirements, specific cooperation between the muscles in a contextual movement is based on fixed building blocks of intermuscular cooperation. In terms of contextual transfer to non-linear control of the sporting movement, strength training is particularly suitable for exercising and improving these building blocks. The increased resistance in the exercises allows numerous factors of relevance to intermuscular interplay (such as the influence of reflexes) to play a part in the design of the building blocks.

Take, for example, cooperation between the back muscles and the hamstrings, both of which are attached to the pelvis. The back muscles can rotate the pelvis anteriorly. The hamstrings span both the hip joint and the knee joint and can rotate the pelvis posteriorly. They play an important part in many movement patterns because they are loaded with elastic energy in an open chain before the pendular motion is reversed (e.g. in running) and then convert that energy into an opposite movement. In a closed chain, the hamstrings produce hip extension torque in relation to the hip. This can result not only in posterior movement of the leg but also in posterior rotation of the pelvis. Rotation of the pelvis is usually undesirable in a closed chain, so it is important that the back muscles act to prevent posterior rotation of the pelvis. They can only act if the back is positioned in sufficient lumbar extension (lordosis). If the back is too rounded, the back muscles may not be properly tensed and the pelvis can easily rotate posteriorly. Tension on the hamstrings, which is crucial when running, will then be reduced. In closed-chain movement patterns, during which hamstring action is important, the back muscles must therefore be active. That is why all elite sprinters run with their backs well extended. Strength exercises such as step-ups and variations of clean are particularly good for standardizing and improving this fundamental cooperation between hamstrings and back muscles (Figures 5.5 and 5.6).

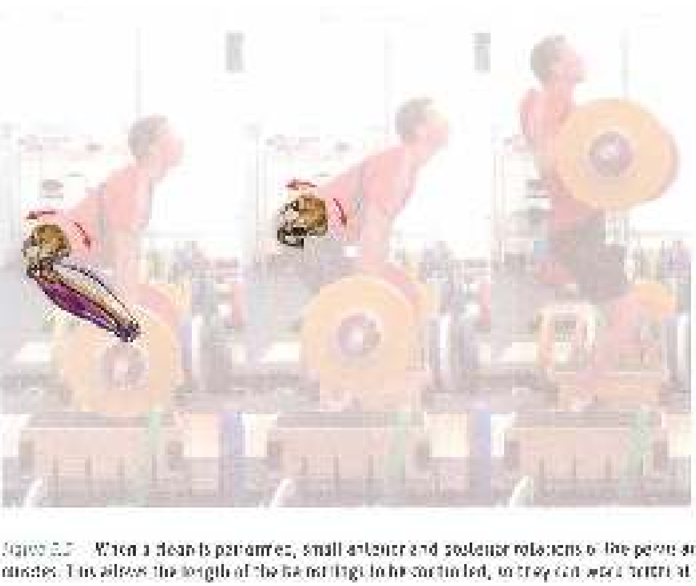


Figure 5.5 When a clean is performed, small anterior and posterior rotations of the pelvis are controlled by cooperating muscles. This allows the length of the hamstrings to be controlled, so they can work better at optimal length.

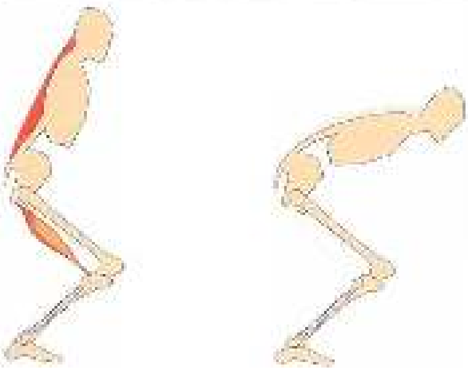


Figure 5.6
Comparison between body position in a squat jump (left) and in a push-off in skating (right). In a squat jump, the back muscles are well tensed, so that the pelvis is fixed and the action of the hamstrings contributes to hip extension. In skating, the back is rounded and the action of the hamstrings will partly result in posterior rotation of the pelvis. That is why the action of the hamstrings is of only limited importance to skaters. During the push-off the thigh does not extend, but moves mainly sideways. Because of the limited load, skaters seldom suffer from hamstring injuries.

A key basic component of barbell training is the position of the spinal column when carrying the barbell load. The back must always be extended, with the back muscles tensed, so that the movements can be properly executed and injuries prevented. In good technique, cooperation between back muscles and hamstrings is specific for high-speed running in all exercises in which a barbell is carried on the shoulders. That is why it is so important to pay attention to cooperation between back muscles and hamstrings in barbell exercises. It may also be very useful to do exercises that put pressure on this cooperation, for example in a variant of a single-leg squat in which the body's centre of gravity moves forwards after the downward movement, increasing the moment arm of the mass above the hip. The back muscles must remain at their optimal length, and at the same time the hamstrings must generate a great deal of force to fix the hip (Figures 5.7 and 5.8).

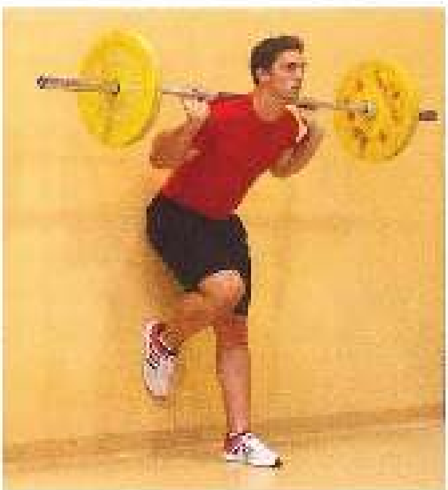


Figure 5.7
Single-leg good morning against the wall. The back muscles must be well tensed, creating a rigid unit of spinal column and pelvis that is kept in position by activity of the hamstrings in cooperation with other muscles.

The position of the pelvis is of course controlled by activity of even more muscles, such as the *iliopsoas* and the abdominal muscles. These muscles also cooperate on the basis of underlying principles. This underlying cooperation helps in rotating the pelvis. The abdominal

Figure 5.8 Single-leg squat in lunge position: moving the trunk forwards at the lowest point (right) puts great pressure on the fixing action of the back muscles and the hamstrings.

muscles can posteriorly rotate the pelvis. The *iliopsoas* ensures flexion of the hip joint, while the *psoas* portion also acts on the lumbar spinal column. Because a muscle always acts in two directions, the *iliopsoas* can also anteriorly rotate the pelvis. The two muscle groups therefore cooperate during contextual movements. Abdominal muscles are also able to act elastically after stretch. The *iliopsoas* is thereby able to generate power in hip flexion. In movement patterns in which one or both legs must be moved forwards after hip extension, cooperation between the two muscle groups is very important, for instance when initiating the anterior movement of the swing leg after leaving the ground when running. In this movement there must be a proper balance between the elastic posterior rotating action of the Abdominal muscles and the concentric anterior rotating action of the *iliopsoas* (Figure 5.9)

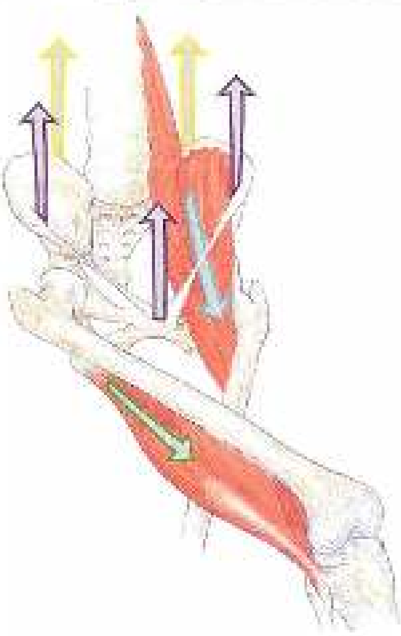


Figure 5.9
Muscles influence pelvic rotation: the iliopsoas muscle (anterior, blue arrow), the abdominal muscles (posterior, purple arrow), the hamstrings (posterior, green arrow), the back muscles (anterior, yellow arrow).

Additional information

Fixed cooperation between quadratus lumborum and erector spinae

Erector spinae acts on the spinal column. The muscle group can contract with very great force. It is important that the forces acting on the spinal column (and hence on the intervertebral discs) are well controlled by muscle action. Information on the position of the spinal column is then of course crucial. *Quadratus lumborum* is a muscle group that contains a large number of proprioceptors (muscle spindles and Golgi tendon organs). These sensors provide information that is important for the overall coordination and control of position of the spinal column. In a sense, *quadratus lumborum* is the eyes and ears of *erector spinae*.

Systems in which certain muscles provide important sensory information for the control of major movement systems can be found in various parts of the body. The abdominal muscles are the eyes and ears of *iliopsoas*, and so on. Proprioception does not always work very well, and needs to be trained. During rehabilitation of sporting injuries (which nearly always involves a serious deterioration in proprioception), a certain amount of time is needed to restore proprioception. But it must always be remembered that the role of proprioception may change in a phase transition to high-intensity movement (see section 1.1.2).

Similar fixed patterns of cooperation as building blocks in complete patterns can of course also be found during athletic movements of the shoulder girdle. 'Scapulohumeral rhythm' is a fixed rhythm involving lateral scapular rotation in relation to abduction in the shoulder joint. In this rhythm there are fixed principles of cooperation between muscles, e.g. the trapezius muscle (upper and lower fibres) and deltoid muscle. Another example of a fixed principle is the combination of protraction of the shoulder blade by the serratus anterior muscle and flexion by the *pectoralis major* muscle. In a complete pattern such as a throw or a tennis smash, the use of these building blocks (attractors) results in a highly contextual universal movement pattern. The pattern is efficient because internal rotation of the shoulder joint (combined with pronation in the forearm) occurs in all types of throwing and smashing. The pattern is flexible because it is applicable in tennis, volleyball, handball, water polo and so on. By using free weights during strength training (e.g. a single-arm overhead press with a dumbbell) these building blocks of the movement of the upper limbs are practised and improved. The arrangement of the building blocks must be self-organizing. It is therefore not a good idea – as often happens – to force the execution of a movement pattern within strength training into too tight a straitjacket by greatly reducing the number of degrees of freedom of movement in the exercises. Isolating muscle groups and range of motion of a joint by eliminating other joints (such as doing shoulder stability exercises in a sitting position, or practising trunk control in a prone plank position with four supporting points) prevents such self-organization from taking place.

5.3.2 *Similarity of movement through similarities in external structure of the movement*

If the movement results of the various movement patterns are externally similar, there is a degree of specificity. Here we can consider similarities in joint angles, speed of movement or angular velocity in the joints, and the direction in which force is applied.

Similarity in the external form of movement is traditionally the most function guide to choosing strength exercises, especially when it is not possible to maintain similarity of movement in the inner movement structure during strength training. For example, it is almost impossible to imitate the throwing movement with a large resistance (weight). Even if the angles and changes of angle in the joints can remain similar to the sporting movement, similarity in type of muscle action can in any case not be maintained. When practising throwing and striking movements, besides training with very low resistance in which specificity in the inner structure can also be aimed for, it is therefore above all the similarity in outer structure that guarantees specificity when training with heavy weights.

Similarities in outer movement structure is an important starting point in the pursuit of high specificity and efficient transfer of the exercises, among other reasons because the form of the movement is important for the vertical representation of a movement pattern that is summoned by the athlete. The form thus helps integrate the intention, the sensory patterns and the appearance of the movement.

5.3.3 *Similarity of movement through similarity in energy production*

The feature of specificity that is least applicable to strength training is similarity of energy production. Strength training can seldom if ever meet this criterion for transfer of training. In explosive sports it is not useful to link force production to the necessary energy-supply processes, for energy production is rarely a limiting factor in performance; the limitation lies far more in neural factors. In endurance sports, linking force production to relevant energy systems means that deployment of force must be very low and hence that there will be no adaptation in the ability of the muscles to produce more force (see also Section 7.1.3).

5.3.4 *Similarity of movement through similarity in sensory patterns*

Besides the three aforementioned 'classic' aspects of specificity, it is useful to add sensory similarity as a feature. For purposes of training practice a useful distinction can be made between sensory organs that register the environment (the eyes, the ears, the vestibular system, touch and so on) and those that register the state of the body, a process known as 'proprioception' (muscle spindles, tendon sensors, joint sensors). This is because sensory patterns have a major influence on motor patterns. They belong together. The brain does not simply design a movement (motor programme) that is sent to the muscles and carried out unchanged. Instead, the movement is constantly appraised and adjusted in the light of sensory information ('closed loop' control). As a result, the actual movement is always carried out differently than in the original design. Adjustment of motor patterns in the light of sensory information largely determines the quality of execution of the movement. It is therefore safe to assume that similarity in sensory information has a major impact on transfer of training and specificity. This is clearly reflected in the fact that 'dry' exercises – in which the movement is made outside the context (sensory and otherwise) of the real sports situation – have little effect. The most obvious example is swimming, in which the specific environment (water) interferes with the transfer of movements practised on land (see Section 4.1.4). A similar problem arises in the transfer of rowing on a rowing ergometer to rowing in a competition boat.

important sensory information from the environment is rarely specific between the strength exercise and the sporting movement. The main specificity of sensory information within strength training therefore lies in proprioception. This plays a particularly important part within strength training as regards transfer of training during complicated movement patterns. Of course, these are above all the movements involving only slightly increased resistance (dumbbells, medicine balls, hand paddles in swimming). It therefore seems useful to pay great attention to sensory information while performing these types of exercises. Feeling resistance and body movements may well yield quick results with this type of training. However, the problem that then arises is that proprioceptive information is mainly processed by unconscious processes and that awareness of information from the body (internal focus) does little to improve motor patterns. This means that focusing on maintaining muscle tension and changing joint angles, as frequently happens in rehabilitative exercises ('arch your back and rotate your pelvis while keeping your abdominal muscles tense'), contributes far less to the learning result than is generally thought. Conscious focus on the movement improves the practice result (the level achieved at the end of the practice session). However, it will not be easy to convert the practice result into a learning result (the level of skill that becomes permanent).

In conclusion, sensory information is very important to the learning process, but it is very difficult and often impossible to use it as a direct starting point for teaching.

5.3.5 *Similarity of movement through similarity in the intention of the movement*

The intention-action model was described in Chapter 4. The learning system tries to reason from the intention of the movement to the process (the muscle action), and mainly uses intrinsic knowledge of results to do so. In this sense it is clear that an exercise will above all result in transfer to a sporting movement if the intention is the same in both cases. Similarity in the intention of movements is therefore a feature of specificity.

In the practice of strength training it is therefore difficult, or even impossible, to cover all the categories of specificity in a single exercise. Reduction or absence of specificity in one or more categories will therefore just have to be accepted (see also Section 6.1.3). Just like similarity in the outer structure (form) of the movement, similarity in the intention of the movement may be difficult or impossible to achieve. However, in order to understand the transfer that occurs, it is useful to analyse the intention of the movement closely. There are two main questions here: what is the quality of the intention, and how is intention linked to other categories of specificity?

Strength exercises and intention

In movements that have a strong contextual relationship to the context in which they are executed, the intention of the movement will be fairly obvious. When throwing at a target, the obvious intention is to hit it. When taking a bend in long track speed skating, the intention is to end up at a particular point on the track. When striking a golf ball, the first intention is to hit it and the next intention is for it to land in the intended place. When running, the first intention is to keep upright, the second intention is to move from A to B and a possible third intention is to do so in a given time. If running on the flat, the first intention will be easier to achieve. If running on ice, it will be much harder. The intention to keep upright will be a greater constraint when running on ice, so that the process (the running technique) will self organize to adjust to the need to keep one's balance and hence the horizontal component of the push off will be reduced. The more sharply the intention of the movement is defined for the learning system, the more automated and faster self-organization towards that intention will be. Especially if the intention is intrinsic to the movement, this mechanism will be enhanced. If there is no intention, or if it is vague, the learning system will find it hard to organize the execution of the movement properly. The body will not know where it is supposed to end up, and hence will not know how to find the way there.

In many strength exercises the intrinsic intention is unclear, or completely lacking. A weight or resistance is moved from A to B, but strength exercises seldom have a very clear and obvious (contextual) end point. This lack of an obvious end point (which is not lacking when throwing at a target) is at the expense of self-organization of effective movement patterns towards the target, and hence learning of useful basic components of movement patterns. In particular, learning intermuscular (chain) patterns with considerable force production will have only a limited effect if the result of the movement is unclear. This problem arises, for example, when training with kettlebells. These weights, which are lifted and swung with one hand, are used to load and train muscle chains in a number of ways. The swinging and lifting movements can be executed with optimal freedom of movement in three-dimensional space. There are therefore high demands as regards control of the movement. Yet the learning effect in terms of intermuscular coordination is less than might be expected, for it is very difficult to include a precise and above all targeted end point within kettlebell exercises.

If a coach thinks in terms of isolated training of muscle groups, it will scarcely be possible to achieve meaningful intentionality within the exercises. The athlete will become 'strong', but will be unable to apply this strength effectively during athletic movement. This notion is becoming increasingly important in many sports, and there is an observable general tendency towards less, but more specifically designed, strength training in which the intention of the exercise plays an important part. For example, many leading athletics coaches now say that two one-hour sessions of strength training a week may generally be sufficient for a jumper or sprinter. Seeking specificity through similarity in the movement intention is therefore useful within strength training when carrying out complex patterns involving chained muscle action and movement patterns in a similar fashion to sporting movements. The focus here is on intrinsic KR information. This KR information works best if it is clear, and a clear end point helps here. The learning system can fairly easily use this clear sensory KR information to find an efficient process towards the end point.

Examples of KR focus within strength training

1 A step-up action onto a 25-cm box with a heavy barbell weight (e.g. equal to the athlete's body weight). The aim is to extend the stance leg as explosively as possible in the step up action. The associated end point is an extended body position with a fully

extended stance leg, while the swing leg is flexed sixty to eighty degrees in the hip, the knee is at an acute (fully flexed) angle and the ankle is in dorsiflexion. This position of the swing leg (Figure 5.10) is the end position that guides self-organization in performing the step-up effectively. Pelvic movement (elevation of the free (swing) side of the pelvis during the step-up) is a key aspect of powerful reflexive extension of the step-up leg. This is emphatically controlled by the aforementioned end position of the swing leg. To make the end position in a step-up an even greater constraint, the movement can be executed with a light barbell weight held above the head with extended arms. The constraint in the end position of the swing leg is now supplied by the requirement that the arms be kept extended and fully elevated, for this can only be done if there is sufficient body tension and the free (swing) side of the pelvis is elevated early in the step-up (Figure 5.10).

Figure 5.10 A step-up, performed here in series. The movement of the swing leg to the end position increases the extension of the stance leg.

2. A classic clean with a heavy barbell weight (e.g. equal to the athlete's body weight). A clean can be used to train the correct timing of proximal–distal extension of the joints in the lower limbs and the accompanying intermuscular coordination. First the hip is extended, then extension of the knee is added, and finally ankle plantar flexion, with the biarticular leg muscles working to transport energy. In the absence of clear intrinsic KR information, this timing is very hard to learn. That is why every weightlifter tries to build in clear result information when performing a clean. This can be done by looking at the ceiling during the clean. If the ceiling can be seen throughout the clean, this means that extension has been correctly timed. If the timing is wrong, with the knees extended too

soon and too fast, the trunk becomes more horizontal and the athlete cannot look at the ceiling when the barbell is accelerating. Looking at the ceiling during the clean thus provides KR information that assists self-organization of extension in the lower limbs.

Additional sensory input can also be added to the KR information when performing a clean. The clean ends in a position with the barbell resting in front of the body on the chest and fingers. The athlete must be in a well-balanced position in which the weight can be carried comfortably. This position releases information that indicates whether the clean has been performed correctly. More stringent requirements for this end position will make the end point even clearer. For instance, the feet must briefly leave the ground at the end of the clean, then the athlete must land on the balls of the feet, and finally the body must remain completely motionless for three seconds. Even small errors in technique are then instantly revealed (Figure 5.11).

Figure 5.11 When performing a clean, there is a risk that the knees will be extended too rapidly during the first part of the movement. Looking upwards prevents this excessively fast extension. The end position also includes extensive sensory information about the way the clean is performed. A well-balanced stance and comfortable flexion in the hips and knees, with the barbell stably positioned on the chest, indicate that the exercise technique has been executed correctly.

3 Running is a cyclical or continuous movement. It does not seem to have any intention or end result, since there is no clear beginning or end to it. And yet it does have such an end result. In correct technique there is no anterior rotation or rotation round the longitudinal axis at the point when the foot leaves the ground (toe-off). A well-balanced position thus provides intrinsic KR information that the risks of rotation at toe-off have been correctly compensated for. A balance exercise becomes more specific if there is no residual rotation at the end of the toe-off. By placing the swing-leg foot on a box with a slight delay after the balance movement, the athlete shows his or her ability to execute the movement without residual rotation. Residual rotation results in loss of balance, which leads to premature landing (Figure 5.12).

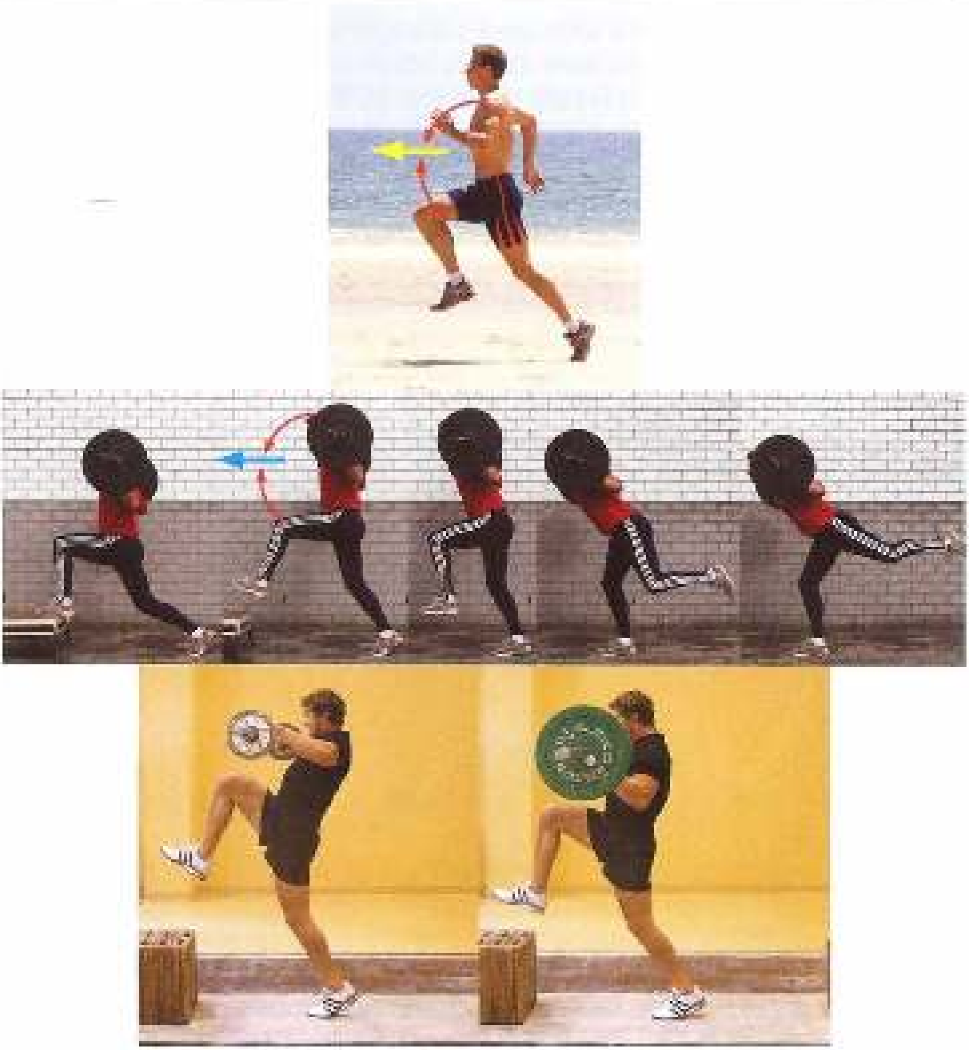


Figure 6.12 When running, the various rotations must cancel each other out at toe-off. Some barbell exercises such as the 'balance' are suitable for imitating this. Since toe-off is a key moment in the running cycle, there is more transfer than might be first thought. The 'balance' exercise can also be executed with a snatch (balance-to-snatch) or a clean (balance-to-clean) as well as with the barbell on the shoulders.

Within strength training, coaches must decide whether the transfer benefits outweigh against the investment in increased specificity. This is often very difficult, especially during strength training for upper limb contractility. For example, throwing is dominated at the intramuscular level by elastic muscle action, and at the intermuscular level by the transport of energy from proximal to distal. In outward appearance it is marked by a movement pattern from external to internal rotation in the shoulder, and from neutral or supination to pronation in the forearm. The intention of the movement is of course defined by the trajectory of the ball. When training with heavy weights there is little point in trying to cover all these categories of specificity, and it is usually wiser to be satisfied with specificity at the recruitment level in more or less isometric conditions, as this is a key component of elastic muscle action. Specificity in terms of intention (the wrist movement that is the end position of the movement, and the trajectory of the ball) cannot be guaranteed for long when training with increasing resistance. Attention is therefore only paid to this specificity requirement in types of training in which resistance is close to the resistance faced during sporting movements, such as throwing weighted balls. The debate on the value of the 'long toss' when training baseball pitchers should be seen in this light (Fleisig et al., 2011).

However, within strength training of the upper limbs for other sporting movements, such as thrust and push movements with a clear beginning and end, specificity can be increased simply by moving towards a clear end point in the exercise (e.g. a ball hanging from a string) and touching that target at the end of the movement. Especially in a sports injury rehabilitation setting in which basic movement patterns are to be restored, such a simple addition to the exercise can be useful.

5.4 Barbells versus strength machines

Accelerations and decelerations constantly occur during athletic movement. The interplay of external forces acting on the body and the force generated in the body (the torque round the joints) keep changing and interacting. Sporting movements are not just about getting an object – a ball, a javelin, a bobsleigh, the athlete's body – to move, but also about making an already moving object change direction, speed up or slow down. This constantly changing interaction between external forces and forces in the body calls for accurate timing. This interaction of forces, joint torque, accelerations and inertia within sporting movement is like the interaction of the instruments in a symphony orchestra. Music can only be made if all the instruments play their notes coherently.

To meet the requirements of specificity, strength exercises must likewise allow the aforementioned natural interplay of forces. Training with Olympic barbells automatically fulfils these physical criteria, because there are enough degrees of freedom during the exercises to create a self-organizing interplay of joint torque. When an athlete performs a clean, the body posture and hence the torque the muscles can produce round the joints will be such that proximal–distal action can be optimized. At the same time, the reflex action of the muscles can correct minor perturbations in coordination (noise), allowing a well-balanced final posture (Van Ingen Schenau & Van Soest, 1998). Hence barbell training is naturally specific.

Strength-training machines do not necessarily allow the natural interplay of forces when an object is accelerated, for they reduce the number of degrees of freedom and force the movement in a direction that is often not natural. As a result, there is seldom an opportunity for accelerations to occur naturally when undertaking machine-based resistance/strength training. In order to be specific, machines must imitate these natural characteristics, and they rarely can. This is because the margin between a technically correct or incorrect acceleration, linked to a given body posture, is so small that it can hardly ever be imitated by a machine.

Sport-specific training on machines is therefore hardly feasible, especially with movement patterns that involve working in chains with many different muscle groups (a large orchestra with lots of instruments). How difficult it is to perform a forced movement naturally is apparent from the problems that athletes accustomed to squat exercises with free barbells encounter when training on a squat rack with a guide rail (Smith machine). The barbell can then only be moved up and down in a precise vertical movement. In a squat with a free barbell the movement can be varied horizontally to some extent, so that the upward path will differ slightly from the exact vertical line. Although the difference is scarcely visible to the naked eye, most athletes accustomed to free weights will avoid a Smith machine, because it forces the movement too much and feels unnatural. This is particularly true of rapidly performed squats. The same applies to other apparatus that fixes the movement path, such as resistance machines that imitate the throwing movement.

Transfer through similarity in the intention of the exercise is also important when comparing free barbells and guided equipment. A great drawback when training on guided equipment is that the self-organizing effect of end-point orientation is lacking. Although the end point of the movement is clear, there can be no meaningful relationship between self-organizing mechanisms and the end point, and hence no meaningful transferable patterns. This is simply because incorrect organization of intermuscular patterns leads to the same end point as correct organization. The learning system is unable to distinguish between efficient and inefficient movement patterns.

5.5 Limitations on specificity of strength training

5.5.1 Overload versus specificity

In order to be useful, training types must meet not only the specificity requirement but also the overload requirement. Within strength training, overload is usually easy to arrange by ensuring a higher resistance than the athlete is used to overcoming. However, the apparent ease with which overload can be achieved within strength training entails a risk that athletes will not think beyond heavy loads. Not only may overload turn out, for instance, to be very disappointing in terms of maximal force production, but an exercise that only involves overload is not enough to achieve the intended adaptations. Training with overload but without specificity usually has very little positive impact on athletic movement. This is particularly true of athletes who are already trained to a high standard. Useful sport-specific strength training therefore involves keeping more than one ball in the air, and this turns strength training into a difficult juggling act.

Types of training that easily meet the overload requirement are far less likely to meet the specificity requirement. Overload and specificity are in conflict. Identifying and guaranteeing specificity with sufficiently intensive loading is therefore one of the most difficult aspects of developing a strength training programme (see Section 6.1.3).

5.5.2 *Specificity in rapid and slow sporting movements*

The best approach to the problem of specificity within strength training varies from sport to sport. It is therefore useful to classify sporting movements according to the speed at which joints flex and extend. Speed skating is a sport with high forward speed but slowly performed movements, such as extension of the push-off leg. In running, forward speed is far lower, but the angular velocities achieved in the joints are far faster. The movements in skating are therefore performed more slowly than in running.

In sports with high-speed motion (sprinting, jumping and so on), specificity within strength training must above all be sought in the inner structure of the movement. Intramuscular and intermuscular coordination is thus the starting point for coordination. There are two reasons for this. First, in such sports it is nearly always impossible to imitate the overall movement with high resistance and hence low speed of movement (the external form). In addition, high-speed sporting movements make very high demands on intramuscular coordination. Muscles must work in a specialized manner (see the discussion of the centrifuge model of muscle action in the box). For instance, they are subject to extreme elastic loading or have to be able to shorten quickly with high force. It is therefore useful to train this specialized intramuscular coordination specifically during strength training. On the basis of this specialized muscle action, intermuscular coordination can also be improved by means of strength exercises. When performing the relevant intermuscular patterns, similarity of outward form is a potential bonus. The movement thus proceeds outwards. In sports with low speed movement (swimming, speed skating, rowing, cycling) specificity within strength training can best be approached from the external structure of the movement. In that case the movement proceeds inwards. Since the sporting movement is slow, it can be imitated fairly well with additional resistance. Intramuscular and intermuscular aspects of the movement can then improve within relevant postures. In slow sporting movements, the intramuscular demands on muscles are not as extreme as during high-speed movements. In particular, elastic action due to counteracting peak forces is less relevant, or irrelevant. The problem with slow sporting movements is far more the way in which intermuscular coordination is linked to body posture during the movement. For speed skaters the problem is not how much power can be generated within individual muscles, but how early muscle power can be transferred to the push-off on the ice in the uncomfortable skating posture. Strength exercises in the skating posture help here.

For rowers, a clean is an excellent exercise for approximating the external structure of the sporting movement. With some adjustment to the exercise technique, the body posture in the clean will more or less match the sporting movement, and in performing the movement the athlete can focus on its external structure, e.g. which joint requires acceleration at which point of extension (for instance, when to flex the arms). Of course, coaches should attempt to train this without immediately giving internal instructions.

Additional information

The centrifuge of muscle action as a model for rapid movement

It is difficult to determine how muscles cooperate during a movement. It is hard to measure from the outside how exactly changes in length occur when muscles act. It is therefore a good idea to look at differences in the structure of individual muscles. Owing to anatomical differences, they are not all suitable for the same type of muscle action. The fact that a muscle is suitable for a particular type of muscle action does not automatically mean it will always act in that way. Gymnasts who slowly raise their legs to horizontal on the rings will use their abdominals concentrically rather than elastically, even though the structure of the abdominals is less suited to slow shortening. We then have to consider when a muscle group will act according to its specialization within the sporting movement, and when it will work differently.

The centrifuge model may be used to answer this question. When the muscles are in the middle of the centrifuge, they may be used for more types of muscle action and even perform intermediate types of muscle action. For instance, they may act at different lengths, and at the same time partly do positive work and partly transport energy, absorb limited external forces and act concentrically, and so on – in the Hill model, partly display CE behaviour (the contractile element of the muscle) and partly SEC behaviour (the serial elastic component of the movement). When the muscles are close to the mantle of the centrifuge, they can only really do the work they are structurally specialized in: either they can only function well at one length and behave in an elastic or isometric manner, or they can behave in a positive concentric manner, and so on.

If the centrifuge revolves slowly (i.e. during slow sporting movements), the muscles may be in the middle of the centrifuge. If it starts to revolve faster, the muscles are pushed outwards, and

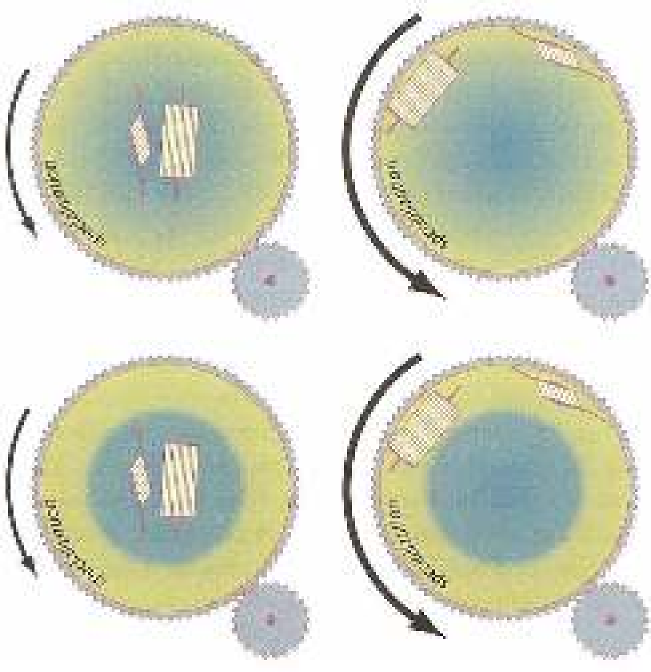


Figure 5.17 When the centrifuge is turning slowly (left: low-intensity movement), the muscles can work outside their speciality. In high-intensity movements (right) the muscles are forced into their specialty and can then only contribute to performance in that capacity. The transition from broad action to specialization may be gradual (top: transition from the blue to the green area) or sudden (bottom).

at a high revolution (i.e. maximal-speed sporting movements such as high-speed running, a full-speed javelin throw, a full-force volleyball smash and so on) the muscles are pressed against the mantle and can only make a useful contribution to the overall movement in their specialized function. This movement towards the wall as the centrifuge revolves faster need not be gradual, but may be very sudden – the phase transition in dynamic patterns theory (Figure 5.13).

Speed of movement when running is high in comparison with most other sports, and muscles therefore behave in a specialized manner when running. If this centrifuge model is further extended to running, a sprinter will use his muscles in a more specialized manner than someone running at a calm endurance pace; and elite sprinters may well be able to recruit their muscles in a more specialized manner than more average sprinters. What is interesting here is how to estimate the speed of rotation when running at, say, an 800-metre pace. If this can be done so that the muscles have already moved a long way towards the wall (consider the already very short ground contact times), training of muscle specialization during strength and technique training may be considered useful. Such an analysis is also useful for long-distance running coaches.

Muscle action is therefore less marked at low speeds of movement than at high ones. This means that the specificity of a type of strength training at intramuscular and intermuscular level is easier to determine for rapid sporting movements than for slow ones.

5.5.3 *Open-skill sporting movements and specificity*

Sporting movements can be roughly divided into closed-skill movements (athletics, gymnastics, speed skating, baseball pitching, tennis and volleyball services and free throws in basketball) and open skills (judo, field play in basketball and soccer, tennis and volleyball rallies and so on). Specificity is easy to identify in closed skills, for the movement is fixed. In open-skill movements this is much harder. In order to design effective specific strength training for open-skill sporting movements, they must first be broken down into their closed-skill and open-skill basic components. The closed-skill components provide a guide for selecting strength exercises. The selection of exercises for these closed-skill components of open-skill sporting movements is based on the same specificity and overload principles as with closed-skill movements.

Given the great variety of open-skill sporting movements, it may be wondered how useful it is to try and imitate this variety within strength training. There is no simple answer to this, for the result of the cost-benefit analysis that must precede it very much depends on the situation.

Knowing how to apply closed-skill components of training within open-skill settings is part of what makes a coach excel. If the closed-skill components are too isolated, there will be no optimal transfer to the open-skill setting; and if too many components are treated as open skills, and hence athletes are not instructed how to perform them, the technique on which improvisation during a sporting movement is based will not be well developed (see also Section 5.2.5).

Examples from an open-skill sport such as rugby will make this clear:

1. An essential part of rugby is tackling. It requires great trunk and upper-limb strength. The strength of the arms can be trained during closed-skill strength exercises, including

bench pressing and pull-ups. In the sports setting, however, pushing and pulling movements are always improvised. Consideration may then be given to doing part of the strength training with dumbbells, so that the athlete can train more three-dimensionally and the shoulder girdle becomes more stable. To take another step towards open skills, use can be made of unpredictable external resistance, such as partially water-filled tubes, medicine balls or physio balls. Finally, the most open-skill setting can be chosen by devoting time to game-like application of strength in wrestling and judo exercises. The most effective solution is not easy to determine, for it greatly depends on the athlete's individual standard and requirements.

2 Rugby is a running sport as well as a wrestling sport. When running, players improvise a great deal by accelerating and changing direction. This is not to say that there are no rules for movement when running in rugby (or other ball sports). Just as in upper limb movement patterns, only part of running involves improvisation, involving variation on a small number of fixed generic principles (see Section 3.2.6). When designing strength training for the lower limbs, this should mean the same as for the upper limbs. However, carrying a barbell load limits scope for training in an open skill setting, because of the need to ensure safety and avoid trunk rotation, lateral flexion and even flexion. It may therefore be better to perform the fixed patterns for strength training of the lower limbs in a closed skill setting, and to train the improvised forms without barbells (a broomstick, for instance, is often sufficient).

3 Training rugby players' abdominals and back muscles confronts coaches with a dilemma: Should strength training mainly focus on the direction of the variability that is required of muscles when tackling and in the ruck, or is it more effective to focus mainly on the demands that the closed-skill components of running make on the abdominals and back muscles? Which combination is best? The answer to this question greatly depends on the specific requirements of the sport and the individual player's athletic abilities.

5.3.4 *Negative transfer*

Exercises that are coordinatively unrelated to the sporting movement will do nothing to improve it. At best, they will have no effect on the movement; at worst, they will impair its quality. It is simply not true to say such exercises 'can't do any harm'. They may create 'negative transfer' by reducing the quality of sporting movement.

A badly timed amount of strength training – too much, or stepped up too rapidly – may place such a load on the system that it cannot recover sufficiently, at the expense of the sporting movement. Endurance athletes are especially wary of this effect, but even in sports that make many different demands on the athlete and hence require highly complex training programmes, such as pole vaulting, it is vital to estimate carefully the optimal proportion of strength training exercises within the overall range of training. Many athletes do too little strength training – but some do too much.

Besides the negative effect of too much strength training, the wrong exercises can easily impair the sporting movement. There are two reasons for this: unfavourable coordinative changes transferred to the sporting movement, and anatomical changes that make the body less capable of performing the movement optimally.

Direct negative effects on coordination

Learning processes are very slow. They not only involve mastering the various components of the movement, but also identifying its stable and changeable components and its main sensory features (see Section 4.4.1). In particular, the wrong type of training may make it more difficult to distinguish the fixed and the changeable components of the movement (attractors and fluctuators). Errors in the design of strength training may adversely affect both components.

The better the athlete masters his sporting movement, the more the types of muscle use (e.g. producing power and working elastically) becomes a fixed, unchangeable component, for there are no longer any inefficient combinations of explosive and elastic muscle use at intramuscular level. Muscles will be employed within the movement in the way they work best. If athletes are frequently made to use their muscles in a way that is not the most effective, their muscle use will shift from fixed to combined, and the sporting movement will be performed less effectively. Speed skaters or swimmers who use elastic loading within strength training will disrupt the system of fixed components, just like javelin throwers who do a lot of high-resistance power training for the upper limbs.

A crucial changeable component is the amount of muscle slack within the movement pattern. A jogging runner has less stiffness (i.e. less pretensioning and hence more muscle slack) than one who is sprinting. The movement pattern can be adapted to the required speed of movement by regulating muscle slack. The usual problem in explosive sports is excessive muscle slack, so types of training must be found that will reduce it. Strength exercises that include explicit countermovements (a downward movement before explosive extension from a squat, or a large backswing before throwing a medicine ball) will reduce stiffness, so it is a good idea to minimize such countermovements. It should be noted here that barbell training always has a negative impact on the control of muscle slack (see Figure 5.3), because the barbell weight facilitates muscle pretensioning.

Negative coordinative effects of strength-training-related morphological change

Strength training may cause changes in the body that are detrimental to performance. They may even have a negative impact on the way the movement technique is performed. Hypertrophy training is particularly likely to produce this effect. Intermuscular coordination is the decisive factor in mastering a skill at a high level. Intermuscular cooperation must be very accurate when it comes to producing strength and/or power and timing muscle activity. The central nervous system has a key function here, but the reflex effects between the various muscles are also important. Especially if hypertrophy training is based on bodybuilding principles (training of isolated muscles), the interplay of muscles will deteriorate. Control by the muscles will then suddenly lead to a completely new interplay, because some instruments (muscles) have rapidly become much larger and "louder". Hypertrophy training is therefore bad for coordination, and it is clearly going out of fashion within sport-specific strength training, even in sports that require great force production, such as throwing events in athletics.

Hypertrophy is not the only change that strength training may cause in a muscle. Researchers are finding more and more evidence that training causes many types of muscle changes, and that all these changes are interrelated in contextual movements. Besides well-known adaptations such as hypertrophy and changes in muscle length, hypertrophy training may, for example, adapt the muscle fibre pennation angle to the line of action (Aagaard et al., 2001). All these adaptations are specific, they depend on the type of training load, and they help the athlete to cope with that load better and more efficiently. This 'pennate angle' affects both intramuscular and intermuscular coordination. The right training will alter the structure of the muscle so that it is optimally adapted to the demands of the particular sport. The demands made on an ice-hockey player's hamstrings will be different from those made on those of a sprinter. This means they may have to load their hamstrings differently when training. A runner who often does concentric exercises for his hamstrings, or lets large forces act on a greatly shortened hamstring, runs the risk that the structure of his hamstrings will change in a way that is detrimental to fast running.

It is not only muscle fibres that may adapt unfavourably as a result of strength training – the same may happen to passive tissues such as fascia, the connective tissue entwined with and connecting muscle. Opinions about the role of connective tissue in the locomotor system are changing fast. It is now increasingly assumed that, for instance, fascia transmits signals and hence information when stretched, and that other parts of the locomotor system, such as muscles, respond to this (Myers, 2009). Besides the influence of the central nervous system, muscle activity is thus also adapted when passive tissue is stretched. A classic example is the effect of frequent deep squats on the fascia anterior to the spinal column. When making deep squats with large barbell loads – which remains a common feature of training programmes for elite rugby players and others – it is difficult to keep the spinal column under control; this leads to extreme lordosis in the back and hence stretching of the fascia anterior to the spinal column. These tissues change in length and no longer supply any information to, for example, *iliopsoas* and the abdominals. These become passive, and what you then get is rugby forwards with the typical posture of hyperlordosis, and permanent anterior pelvic rotation. The same thing can often be seen in female 100 metre sprinters. Whereas this used to be attributed to an overactive *iliopsoas*, it may also be due to the same mechanism as in rugby players. Very deep squats may not in fact be the cause here, but constantly rotating the pelvis a long way forwards at the end of the stance phase while running, owing to lack of strength and activity in the abdominals and *iliopsoas*. What we then have here is underactive rather than overactive abdominal and *iliopsoas* muscle complex – not a favourable posture for anyone who wants to run fast.

So the can't-do-any-harm mentality should be avoided in sport-specific strength training. We need to consider not only which usable adaptations we want to achieve, but also which training effects we want to avoid.

5.6 An example: hamstring action and specificity

Hamstring injuries are common, and hamstrings are very important during the running movement. They are therefore an excellent example of how to analyse the problem of specific training in detail.

5.6.1 *Specificity*

Given that an exercise is only useful if there are certain similarities to the associated sporting movement (specificity), proper design of training depends on a sound understanding of how the hamstrings work within contextual movements. In the literature on hamstring injuries this tends to be described in astonishingly imprecise terms. Even serious studies (Thelen *et al.*, 2005) often provide vague, undetailed descriptions. Hamstring action is frequently described very generally as 'eccentric–concentric', even though there is no way to measure how an eccentric–concentric movement can develop within the muscle. As a result of this imprecise approach, many hypotheses have been drawn up about hamstring mechanisms that may not in fact exist. Not surprisingly, the studies draw hardly any conclusions about what causes injuries that could be of use within prevention and rehabilitation programmes.

The whole issue is approached from too narrow an angle, usually setting out from simple biomechanical models. Neurophysiological influences – particularly the way in which motor function is controlled – are ignored. For a fuller understanding of specific training for the hamstrings, it is therefore useful to analyse hamstring action in terms of every aspect of specificity, as well as the specificity requirements resulting from analysis of the intention of hamstring action in contextual movements such as running. We also need to analyse the links between the various aspects of specificity. Only then can we obtain a useful picture of how to specifically train the hamstrings.

5.6.2 *Degrees of freedom in the hamstrings*

Several anatomical features of the hamstrings lead to the conclusion that in principle the muscle group has a large number of degrees of freedom – i.e. it can behave in a very differentiated manner when moving (Figure 5.14).

1 Apart from the short head of *biceps femoris*, the hamstrings pass over two joints: the hip and the knee. This means that the length of the muscle is not directly related to the position of either joint. With a large degree of hip flexion and an extended knee, the hamstring may be extremely lengthened. However, if the knee is flexed, e.g. in a squat, the hamstrings will be close to their optimal length. There is thus a dynamic relationship between hip and knee movement and changes in hamstring length.
2 The muscle group extends the hip joint and flexes the knee (or, as in a push off, is active during knee extension). The medial hamstrings (the semi-group) cause internal femoral rotation, and *biceps femoris* external rotation. This means that different activity is required from the various parts of the muscle when running bends or swerving.
3 The hamstring muscles have a pennate structure, with a complicated architecture of active and passive parts. As a result of this complicated structure, various parts of the muscle can, as it were, perform scissor movements in relation to each other (Gerritsen & Heerkens, 2008).
4 The short head of *biceps femoris* is innervated by a different nerve than the long head. This ensures that the parts can contract independently and at the same time perform very different muscle actions. Activation of the entire muscle group must therefore involve effective interplay between both sources of control. In high-speed running, timing of the relative activity of various parts of the hamstrings may be of great importance (Higashihara *et al.*, 2010).

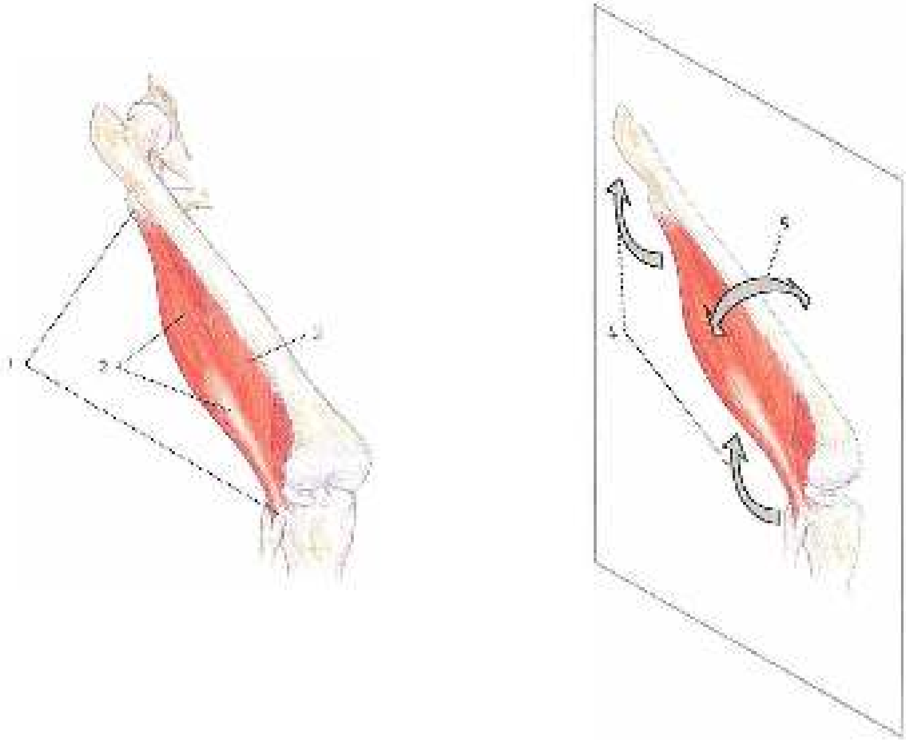


Figure 5.16 Hamstrings pass over two joints (1), with a complex architecture (2), one monoarticular head (3), movement in the sagittal plane when acting together (4) and also internal and external rotation when acting in a differentiated manner (5).

There are many different ways for the hamstrings to act within a single movement pattern. During rehabilitation after hamstring injury, it is therefore often assumed that all these various ways should be retrained. This means that the muscle group should be loaded at every possible length, fast and slow, and the load should be combined with internal and external rotation of the hip, and so on. However, this may not be the right conclusion. If the hamstrings are not used in such a varied manner during movements that put great pressure on them – movements in which large forces act on the hamstrings, e.g. high-speed running – but their action is limited to just one type of muscle action, it makes more sense to train them in that one characteristic action type.

Knowledge of motor control thus suggests that it may be better to train the hamstrings in a specialized manner for high-speed running.

5.6.3 *Eccentric–concentric, or not?*

During high-speed running, the largest external forces act on the hamstrings when the lower leg swings forwards with great force (Chumanov *et al.*, 2011; Schache *et al.*, 2012). This causes the whole muscle to lengthen. Is this due to lengthening of the muscle fibres, or lengthening of the elastic components? Perhaps an even more important question is there an essential difference between the two eccentric forms? If this difference is only marginal, both types of muscle action are specific for running, and so they need scarcely be taken in account when choosing exercises. If, on the other hand, the difference is crucial, it makes sense to focus closely on correct hamstring action within training.

5.6.4 Motor control and specificity

If we look at the ways in which a muscle can act 'eccentrically-concentrically' in terms of the last feature of specificity (the result of the movement), we can see a fundamental difference between the two types of muscle action. When the muscle fibres lengthen during the eccentric phase, the energy of the opposing force is absorbed and is mainly converted into heat; but if the lengthening in the eccentric phase occurs in the elastic components, the energy will be converted into muscle recoil. For example, if an athlete lands from a height of thirty centimetres, eccentric action in the muscle fibres will bring him to a standstill, while stretch in the elastic components will make him bounce back up again.

The same applies to hamstring action during the pendulum phase of running. If the eccentric action takes place in the muscle fibres, the pendular motion will only be decelerated. But if the elastic components are stretched, the leg will move backwards during the subsequent discharge of elastic energy. The result of both types of muscle lengthening will be very different when landing from a height and when the lower leg makes a pendular motion during running. We may thus assume that the two possible types of lengthening (in the muscle fibres or the elastic components) are relatively unspecific in relation to each other. It is therefore useful to identify this difference clearly, and to examine what really happens in the hamstrings during high-speed running.

It seems very likely that the hamstrings work elastically when running, for elastic loading and unloading is the ideal way to reuse the kinetic and other energy that is stored in the scissor movement during the flight phase in order to start reversing the scissor movement

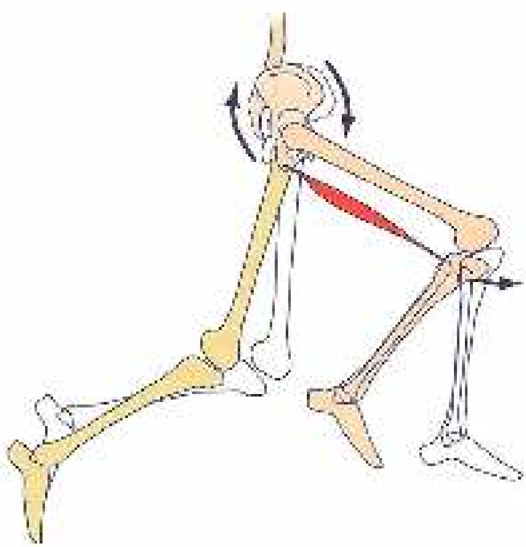


Figure 5.18
The pendular motion of the lower leg may load the hamstrings with an opposing force. Anterior rotation of the pelvis may also stretch the hamstrings (Chumanov et al., 2007).

after the pendular motion of the lower leg. Extending this idea further, soccer and rugby players may be at greater risk of hamstring injuries since they use both types of eccentric muscle action in turn: lengthening of the muscle fibres after shooting the ball, and isometric action of the fibres when running at speed. This may occasionally 'confuse' the body, and cause injury. However, this is still mere speculation.

5.6.5 *Soccer and athletics*

Converting the kinetic energy in the scissor movement into storage of elastic energy in the hamstrings in order to reverse the movement economically is based on an ideal model of movement. This model may be applicable to 100-metre sprint athletics, which can be seen as a closed skill (see Section 3.1). Soccer, rugby and other ball sports, on the other hand, are open skills. The environment (especially the opponent) is so changeable that the way the movement is performed is not predetermined, and constant adjustments must be made to the demands of the moment. Running in the woods, on an uneven surface, can also be seen as a more or less open skill.

This raises the question of whether the hamstrings function the same way in a closed-skill and an open-skill setting. In a high-speed rush during a rugby match, will hamstring action have to adapt so greatly to opponents and conditions on the pitch that it can no longer be purely elastic, and will there also be major eccentric action of the muscle fibres?

5.6.6 *Attractors and fluctuators*

Dynamic patterns theory shows that an open skill consists not only of variable components of movement (fluctuators) but also of fixed, invariable ones (attractors). This means that in movements requiring the hamstrings to absorb large forces, such as running at speed, not every component of the movement will be varied, but only a limited number of components will be used to adapt the movement to changing environmental factors. Within optimal technique, it is the components most suitable for adapting the movement to the environment that will be fluctuators (Figure 5.16).

Is hamstring action when running at speed a fluctuator or an attractor? It seems likely that the hamstrings act as attractors in well-performed open skills, for two reasons:

1. Hamstring action is a fundamental part of the running cycle. It is almost impossible to control or alter the timing of hamstring activity while running. This can only be deliberately influenced by keeping the knees lower at the end of the swing phase. Other components, such as movements of the ankles or trunk, are far easier to control. It makes sense that components less fundamental to the running cycle will be more suitable as fluctuators (see Section 2.2 on swing leg retraction).
2. The external forces acting on the hamstrings during high-speed running may well be the greatest forces they encounter within any movement. Since the hamstrings are at the limit of their load capacity when running at maximal speed, there is little scope for variation. Other loads, such as external forces acting on the calf muscles, do not bring the muscles as close to their limits, so there is more scope for variation (see Section 5.5.2: the centrifuge model of muscle action).

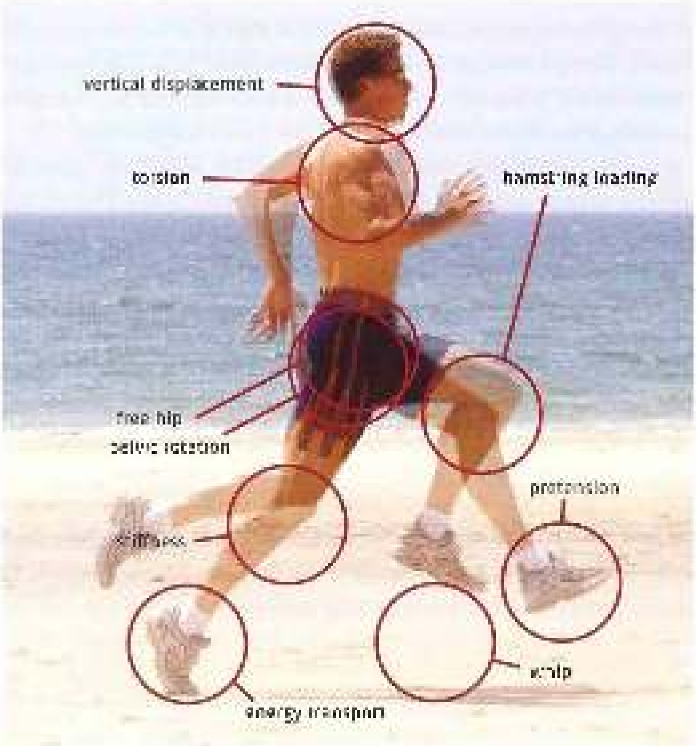


Figure 5.9 The running cycle is made up of many different components, which cannot all be adapted to the demands of the environment. Which ones will be fixed as attractors?

5.6.7 *Training*

The above ideas about specificity and degrees of freedom lead to a model in which hamstring action is a fixed, invariable attractor in open-skill running. Elsewhere in the body, fluctuators ensure adaptation to the demands of the environment. The hamstrings work purely elastically as attractors in order to optimally return kinetic energy. Conversely, if technique is such that the hamstrings work non-isometrically during high speed running and there is eccentric-concentric action in the muscle fibres, hamstring action will become a fluctuator. This is detrimental to running performance, and increases the risk of injury.

All this is still just a hypothetical model, for science is not yet capable of measuring it. The measuring errors are still usually too great and muscle slack also makes it difficult to interpret what happens in a muscle when its attachment points move apart. However, the model is an interesting one, because a number of conclusions can be drawn from it and applied to rehabilitation and training (Thelen, Chumanov, Best *et al.*, 2005; Thelen *et al.*, 2006; Chumanov *et al.*, 2012; Orchard, 2012).

One interesting idea that arises from the structuring of movements into fluctuators and attractors is that this helps explain why injuries often do not occur singly but seem to move through the body in sequence. If, for instance, the Achilles tendon is injured, ankle movements become less effective dissipators. The required adaptation to the environment may then deteriorate. As a result, the hamstring may cease to act purely as an attractor and may start to become more of a fluctuator, making it more vulnerable. This idea has various implications for rehabilitation after hamstring injuries, for instance that an extensive recovery programme for coordination of the overall movement pattern is needed.

The model also has implications for performance training. The hamstring must be trained specifically for isometric muscle action and for recoiling elastic energy. This means that hamstring training should focus on maximal strength training at optimal length. Range of motion training, for instance on a leg curl machine, should therefore be avoided at all costs. Maximal strength training should then be combined with training of elastic hamstring action using elastic loading (again at optimal length).

To ensure that loading takes place at optimal length, the exercises must meet two criteria (Figure 5.17):

- The hamstring must try to extend the hip and resist knee extension. Exercises in which the hamstring flexes the knee (leg curls) are counterproductive in developing the right coordinative pattern.
- The pelvis must be able to rotate anteriorly and posteriorly. This will enable the athlete to find the optimal hamstring length in a self-organizing manner.

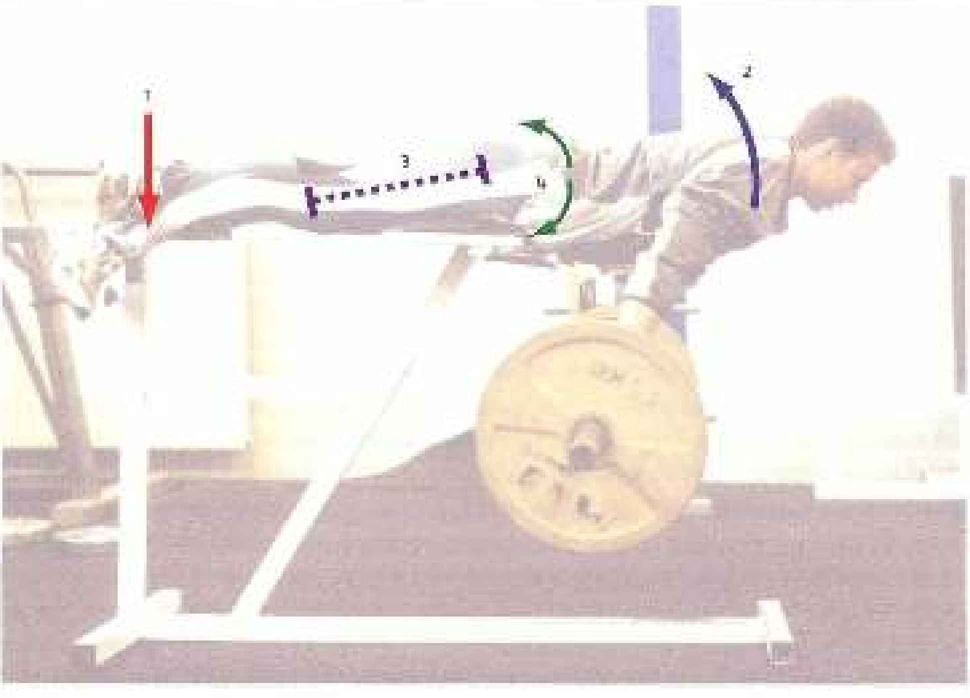


Figure 5.17 Criteria for hamstring training: sufficient overload (one leg fixed) (1), the hamstring tries to extend the hip and prevent knee extension (2) at optimal length (3), with the pelvis able to rotate in order to find the optimal length (4).

Besides these intramuscular patterns, the hamstring should also be trained in complex intermuscular patterns. In addition to hamstring action, the focus here should be on mastery of pelvic rotation (i.e. trunk control) and cooperation between the hamstrings and calf muscle during knee extension. This can be done using complex types of strength training as well as running training. Running on varying and undulating surfaces, for example, can help to further differentiate attractor and fluctuator functions.

5.6.8 *Summary of hamstring action*

- Hamstrings have a complex structure and are crucial to complex coordinative patterns.
- In order for rehabilitation and training to prevent the occurrence of hamstring injuries, we first need a precise description of how hamstrings act within the running movement.
- Scientific knowledge about motor control must be included in conceptual models when developing training theory.
- During high-speed running the hamstrings probably work at their optimal length, elastically and with isometric action of the muscle fibres.
- It is a good idea to training the hamstrings with the same type of loads as when running, using maximal strength training and elastic loading.
- The quality of fluctuating components within the running cycle must also be constantly monitored.

A rehabilitation protocol for hamstring injuries in athletes

A protocol for rehabilitation after hamstring injuries can be designed using the aforementioned model of hamstring action within contextual movement. The protocol is based on the principle that only one new type of stressor at a time is added to the training load during the rehabilitation process. These stressors are determined by identifying the biomechanical components factors that play a part in the overall loading of the hamstrings during high-speed running. This is because phase transitions may make hamstring action at a low-intensity level aspecific for hamstring action at a high-intensity level. High-intensity movement is thus the starting point, and every aspect of this is introduced step by step into the protocol. As a result, the structure of the rehabilitation process is similar to that for a calf strain injury in Section 4.5.3. Since rehabilitation should resemble training, the exercises are mostly also suitable for training uninjured athletes.

The aim of the protocol is to retrain the hamstrings as quickly as possible in absorbing external forces within isometric conditions. Even well-trained athletes remarkably often turn out to have weak hamstrings. The hamstring is then more or less ignored in the movement pattern, is no longer sufficiently loaded and is no longer capable of high-force isometric muscle action – possibly a common cause of injury, and particularly re-injury, especially with team ball games. The hamstring can be ignored during the running movement because its function can be taken over by other muscles: the adductors and gluteus maximus as hip extenders, and gastrocnemius for inhibiting knee extension. Especially in players of ball sports whose pelvises quickly rotate a long way anteriorly just before toe-off, it may eventually be more and more difficult to recruit the hamstrings during the subsequent pendular motion of the lower leg, without this being readily apparent from the outward appearance of the running movement. Other muscles take over this function, albeit inefficiently, thus concealing the problem.

The single-leg isometric exercise in Figure 5.17 can serve as a guide to the right level of recruitment within a properly functioning hamstring. In this exercise, a well-trained athlete who is also exposed to large loads during competition should be able to lift an additional barbell load equal to about 60% of his own body weight and fixate it with his body extended for about two seconds in order to remain in the safe zone for maximal strength. Good sprinters can often perform three to four repetition of a lift of a weight close to their own body weight.

Strong isometric muscle action is a key feature of the various stages of a rehabilitation programme, and is therefore trained first. The external force can either be constant or take the form of elastic loading. Loading of the semi group can also be differentiated from loading of the biceps group by rotating the trunk (with or without a load) during the trunk extension movement on the Roman chair (see Figure 5.17).

After the isometric stage, basic patterns of intermuscular coordination are trained, without major external forces. Vertical and horizontal explosive jumps train the role of the hamstrings in contextual patterns.

Larger external forces are then added to the load. This is done by means of running exercises – quite simply, the higher the speed, the greater the external forces. That is why the first exercises involve running up steps.

The last stressor to be added is repetition. Coordination within the hamstrings is complex, and performing this complex movement in an endurance form is a coordinative load that has to be trained separately.

Some features of the various stages of rehabilitation for a grade 2 hamstring strain:

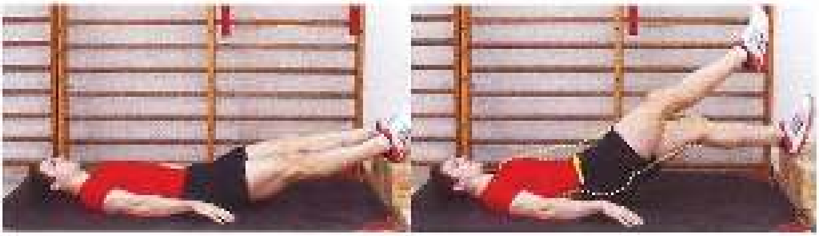
- Stage 1, day 1–3. Acute protection phase with regular physical therapy.
- Stage 2, starting on day 3–5. Supine bridge exercises with a two-second isometric hold (only shoulders and heel on the ground, and supporting-leg knee flexed at thirty degrees). First double-leg, then single-leg. This stage is a preparation for stage 3, and is meant to test whether the hamstring can cope with the load. Progression to Stage 3 is allowed as soon as the athlete can do three sets of six single-leg bridges.
- Stage 3, day 4–8. Single-leg Roman chair (see Figure 5.17) in order to build up isometric muscle action for transport of energy from the knee to the ankle, provided there is no pain:
  - First of all, just the athlete's own body weight. The additional weight is increased slowly.
  - Small-amplitude bouncing movements with the body extended, letting the upper body fall ten centimetres at a time and then bounce back. Elastic loading under isometric conditions, with six to eight repetitions at a time.
  - Increasing the isometric load by adding to the barbell weight. The athlete must eventually be able to lift about 60% or more of his own body weight.
  - Rotating the upper body in the extended position on the Roman chair, possibly with a weight (e.g. a 10-kg plate) that is held with arms extended and moves from left to right with the body.

- Local flexibility is also introduced during this stage by stretching the tensed muscle at a variety of hip and knee angles (Figure 7).
- Stage 4, day [illegible]. Intermuscular patterns during vertical jumps onto boxes:
  - Countermovement first onto low, then high boxes.
  - Squat jump with pretensioning onto high boxes.
  - Pretension jump with extended knees (with only hip flexion) onto high boxes.
  - Jump with extended knees and horizontal movement.
  - Single-leg vertical squat jump, repeated with one foot on a small step and the other on the ground.
- Stage 5, day [illegible] and beyond. Running:
  - Running repetitions up fifteen to twenty steps. Single steps initially. Hamstring load can be increased by taking two or three steps with each stride.
  - Submaximal overground running (60–70% of top speed), accelerating twenty to forty metres.
  - Accelerating on uneven surfaces and small hurdles spaced at varying intervals.
  - Changing direction and moving sideways on uneven surfaces (for players of ball sports).
  - Sport-specific types of running (for players of ball sports).
  - Repeated sprint (endurance) running.

Grade 2 Hamstring strain	Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
Phase	Date																												
1 Acute Protection		P	P	P																									
RICE / POLICE		P	P	P																									
Partial Weight Bearing & Crutches?		P	P																										
Stationary Bike				P	P/T	P/T	T	T	T	T	T																		
2 Bridging																													
Double Leg				P																									
Single Leg (vary knee flexion)				P	P																								
3 Single Leg Roman Chair																													
Body Weight					P	P	P																						
Manual Resistance					P	P	P	P																					
Elastic Bands								P	P	T	T																		
Fit ball (Check >80% BW)									P	T	T																		
Plane Rotations										P	T	T																	
TRX Flexibility							P	P	T	T	T	T	T	T	T	T	T	T	T	T									
4 Box Jumps																													
Counter Movement									P	P	T	T	T	T	T	T	T	T											
Pretension										P	P	T	T	T	T	T	T	T	T										
Bent Leg										P	P	T	T	T	T	T	T	T	T										
Straight leg											P	P	T	T	T	T	T	T	T	T									
For Distance												P	P	T	T	T	T	T	T	T	T								
Single leg step jumps												P	P	T	T	T	T	T	T	T	T								
5 Running Progression																													
Stair Running								P	T	T	T	T	T	T	T	T													
Sub-Max Accelerations													P	P	T	T	T	T	T	T	T	T	T	T	T	T	T	T	
Uneven Surface Accelerations														P	P	T	T	T	T	T	T	T	T	T	T	T	T	T	T
Lateral runs & side stepping																P	P	T	T	T	T	T	T	T	T				
Non-Contact Rugby drills																		P	P	T	T								
Endurance																						P	P	T	T	T	T	T	T
Full Rugby Training																													T
Return to Play																													

Figure 5.10 Checklist for the rehabilitation protocol for a hamstring injury (developed for the Wales National Rugby Union Team in partnership with physical therapist Craig Ranson).
P = physical therapy
P/T = both physical therapy and fitness aspects
T = training and fitness aspects (note, running, day and off feet day strategy is not in this table)
RICE = Rest, Ice, Compression, Elevation; POLICE = Protection, Optimal Loading, Ice, Compression, Elevation

A number of exercises from the protocol:



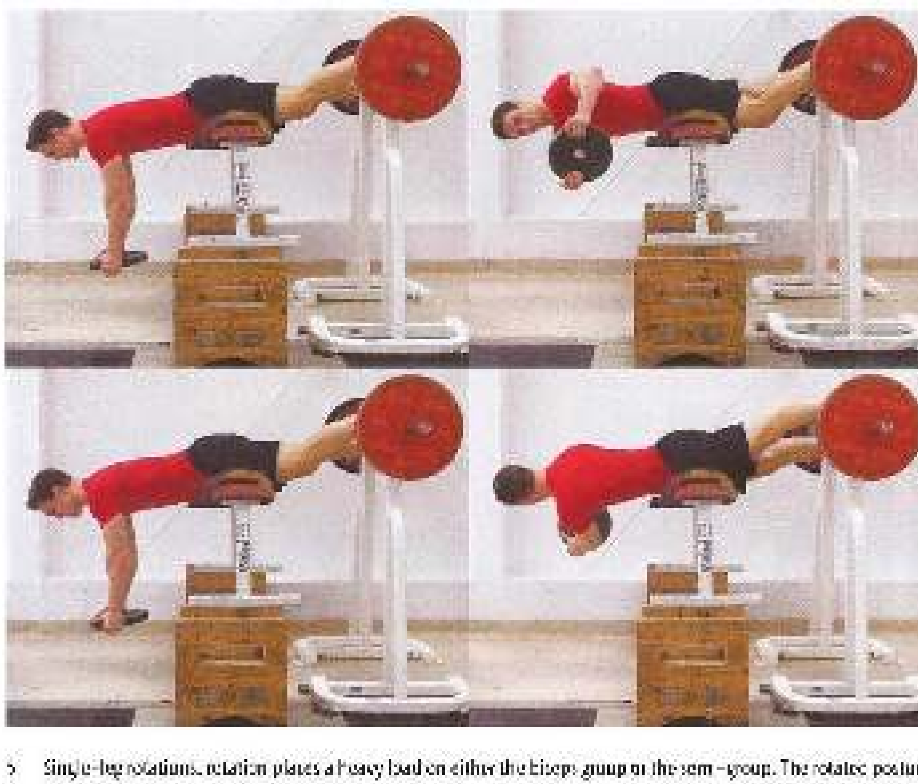
1 The bridge exercise is performed as the first test to see whether isometric muscle action can be performed. Lying on his back with the heel on a 45-cm step or chair, the athlete first lifts his trunk from the ground, holding in neutral hip extension for two seconds, before lowering the trunk back to the ground.

2 Single leg isometric hamstring on glute-ham machine. Lifting the trunk without extra resistance, and with one leg fixed.

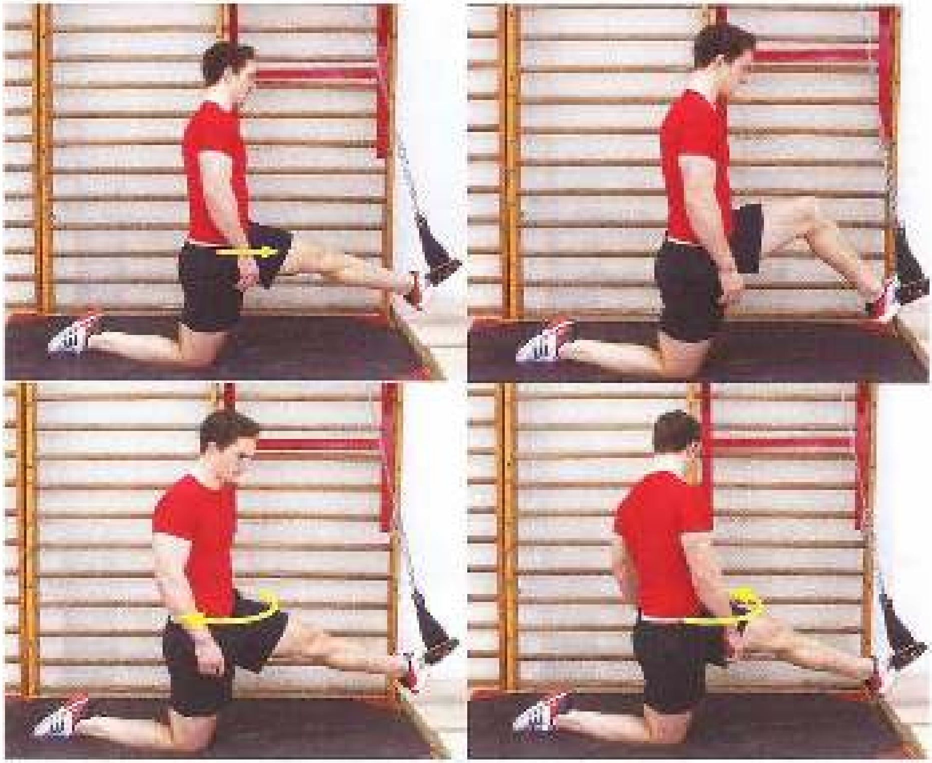
3 Lifting the trunk (single-leg); extra resistance is then provided manually.

4 Single-leg elastic loading: from an extended position, the upper body falls five to ten centimetres at a time, and immediately bounces back. A barbell plate can be held in front of the chest for additional load. The range of motion must remain small.

Single-leg: maximal strength in isometric conditions and at optimal hamstring length.



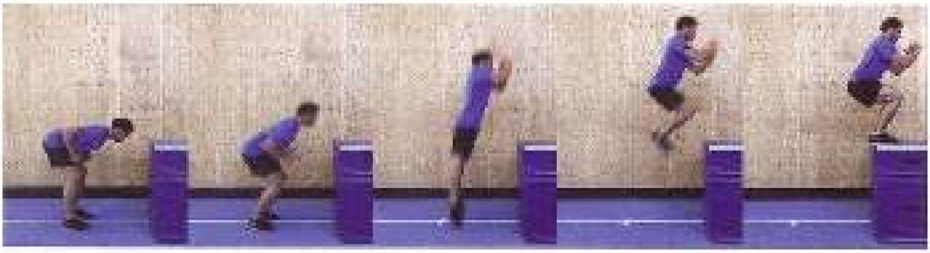
Single-leg rotations: rotation places a heavy load on either the biceps group or the semi-group. The rotated posture is briefly fixed.



7 Flexibility under tension. Total muscle tension in the hamstrings is increased by moving the hips anteriorly. Stretch also increases. When the knee is extended, stretch will occur low down in the hamstrings. When the knee is flexed, the proximal section of the hamstrings will be stretched.
Bottom: the biceps group is stretched by moving the free (swing) hip backwards. The semi-group is stretched by moving the free (swing) hip anteriorly.
By combining all these movements (moving the hips anteriorly for tension, extending and flexing the knee, and moving the free (swing) hip anteriorly and posteriorly to determine where the stretch will be), the athlete can stretch each specific section of the hamstrings without assistance.

8 Throwing a 4-kilogram shot backwards

9 Pretension squat jump with flexed legs.



10 Pretension squat jump with extended legs.

11 Pretension squat jump with horizontal movement.

12 Single-leg squat jump from a step.

18 Running up steps, focusing on short ground contacts and taking, progressively, one, two and three steps per stride.

5.7 Summary

Apparently the various motor programmes are not separate in the brain, but – if they are related – are linked together in some way. This linkage is known as 'specificity'. Transfer of training – the way in which movement patterns influence one another – is guided by specificity.

Transfer between movement patterns is limited. This limitation is an important mechanism for protecting the body against injury. Maximal strength and power production are only possible in movement patterns that are sufficiently mastered, so that the movement can be performed safely. Common classifications of types of strength are therefore not as generic as they may seem. An athlete's maximal level of strength varies from situation to situation, and is never absolute. Factors such the build-up of power production and deceleration at the end of the movement play a key role in total power production during a movement.

Specificity can be divided into five categories:

1. Similarity of movement due to similarities in internal structure of the movement; this can in turn be divided into similarity in intramuscular coordination and similarity in intermuscular coordination.
2. Similarity of movement due to similarities in external structure (form) of the movement.
3. Similarity of movement due to similarities in energy production.
4. Similarity of movement due to similarities in sensory response.
5. Similarity of movement due to similarities in intention of the movement.

Similarity in energy production and sensory response can hardly be achieved by strength training. Similarity in internal structure and similarity in external structure have traditionally been seen as the main features of specificity, while intention of the movement is a useful new addition borrowed from motor learning theory.

Specificity cannot be guaranteed by using strength training machines, but can best be achieved by using free barbells. Barbell training is more in accordance with basic physical principles (interaction with gravity).

Specificity between movement patterns is limited not only by the limits of similarity between movements, but also by several other factors.

- The need for overload in order to produce adaptations limits the scope for specificity.
- If a sporting movement has no marked intramuscular structure, as may be the case in slow sporting movements, it can best be approached on the basis of external structure. Rapid sport movements, on the other hand, can best be approached on the basis of intramuscular and intermuscular structure, because the external structure of the rapid movement is hard to imitate against high resistance.
- In open-skill sports, only the closed-skill components should normally be selected for strength training. Given their sensory focus, the variable components are far less suitable for strength training.
- Types of strength training may produce positive transfer to one aspect of the movement to be improved, but negative transfer to another.

Approaching specificity in terms of these five categories may result in a training programme that is substantially different from what is customary. For example, the addition of the 'similarity in intention' category greatly alters the nature of hamstring training and rehabilitation. Training maximal strength at optimal length to facilitate the elastic action of the muscle group then becomes an extremely important aspect of strength training.

6
Overload within strength training

6.1 Overload

Strength training nearly always represents part practice, in which it is always very difficult to think of exercises that will produce an interplay of specific sensory and motor information. That makes it difficult for strength exercises to achieve a high degree of specificity. There is only limited transfer from part practice to the sporting movement, and so transfer from strength exercise tends to be overestimated.

This makes it impossible to use only very specific strength-training exercises. The repertoire of exercises will be extremely limited if such a strategy is adopted, and constant repetition of the same material will inevitably lead to monotony, and hence reduced adaptation. Monotony will inhibit both metabolic and neural adaptation associated with strength training.

To prevent monotony and tempt the organism to adapt, less specific movements must also be practised. Reducing the specificity of the exercises will allow training stimuli to be given that the organism is unfamiliar with and not yet equipped for. Training stimuli that the organism is not yet equipped for and that call for adaptation will produce what is known as overload. Such exercises with less specificity but more overload are needed to ensure improved athletic performance.

6.1.1 *Definition in physiological terms*

Overload is classically defined as a stimulus that produces a greater stressor than the current stress resilience of the organism (Roovers, 1999). The term 'greater' points to a quantitative measure, and more or less implies that overload only arises if the stress in training is more intensive and/or lasts longer than the athlete is accustomed to. Thus greater load is necessary if adaptation is to occur.

The reason for this quantitative approach to overload is that adaptation is usually very much viewed in terms of physiology. More muscle proteins, more energy substrates and more enzymatic actions within the muscles, more neurotransmitters in the neuromuscular synapses and so on are seen as the basis for better performance. Physiological adaptation can occur by first exhausting the systems and subsystems. For example, to increase the stock of glycogen in the muscles, the existing stock will first have to be reduced to below a certain limit, which will depend on how well the athlete is trained. The training stimulus will therefore have to be stronger for a well trained athlete than for a beginner. In the rest period that follows an effort, the glycogen stock is replenished, to a higher level than the original one. This increases stress resilience, and hence ability to perform.

If the definition of overload is translated into strength training, this means that external forces on the muscle at a given speed of shortening must be greater, or that the muscles must do more work than the athlete is used to. In the period after training there will then be recovery to a higher level. This mechanism of exhaustion and supramaximal recovery mainly occurs in training designed to increase the amount of contractile proteins in the muscle ('hypertrophy training'). Muscle proteins are broken down to below a critical limit, and then more nutrients are used to create new proteins during the recovery period, so that the muscle increases in size.

In training theory, this process of breakdown (catabolism) followed by buildup (anabolism) is known as 'supercompensation'. The supercompensation model is also referred to as the 'single-factor model'. It is solely based on the immediate training effect of exhausting certain biochemical parameters on stress resilience. One of these is protein breakdown/protein resynthesis (Olbrecht, 2000; Zatsiorsky, 1995).

Apart from this mechanism, there will be several other breakdown and buildup mechanisms within strength training. However, these are less readily identifiable as supercompensation processes than protein resynthesis. For instance, increased recruitment as a result of maximal strength training cannot simply be cast in a model in which the amount of transmitter substances in the neuromuscular synapses is reduced to below a given limit through training, followed by supercompensation and increased maximal strength. The reality is far more complex, and in general it may be said that the supercompensation model is too simple to be true – as is apparent from the difficulty of pinning down the issues associated with overreaching and overtraining (*Morgan et al.*, 1987). A more complex reality is shown in the 'two-factor model'. This includes the physical fitness and fatigue factors in the modelling of changes in performance levels over time. Physical fitness as a component of ability to perform is seen as a slowly-changing parameter, whereas the fatigue component can change quickly. In this model, training is seen as a type of input that elicits two physiological responses: (1) increased physical fitness and (2) increased fatigue. The sum total of these responses is the output: the current ability to perform. Immediately after a unit of training, physical fitness has improved, but ability to perform is adversely affected by the fatigue component. In the subsequent recovery phase, fatigue disappears rapidly, yet the improvement in physical fitness decreases slowly.

Such a two-factor model is more in keeping with an adaptation model for strength training in which improvement of coordination plays a key part (Figure 6.1).

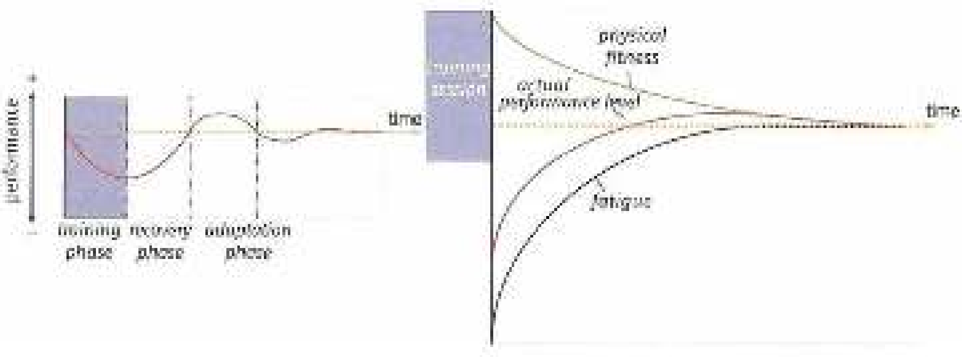


Figure 6.1 Left: the single-factor model; right: the two-factor model. In the two-factor model, ability to perform is divided into physical fitness and fatigue.

6.1.2 *Definition in coordinative terms*

If overload is not seen purely in terms of exercise physiology (exhaustion and supercompensation) but also, above all, in the far more complex terms of coordinative adaptation (especially if adaptation is ascribed to changed neural commands), it is useful to define *overload* as follows: a training stimulus that creates a *different* load than the organism can process without adaptation. The approach is then no longer purely quantitative, but can also be qualitative. This means that, in order to achieve adaptation, overload within strength training should not always be sought via increased force and power production. The intended adaptation can also be achieved by moving in different patterns and contexts within strength training, without muscles having to produce more force and/or power. This approach creates countless new opportunities to provide useful strength training stimuli. In particular, the basic mechanisms of intermuscular coordination – which may be seen as the most important performance factor in advanced athletes – can thus be improved through well targeted resistance.

Thus sport-specific strength training should not just focus on physiological aspects such as protein synthesis, but also on learning and improving motor patterns: not just improving the engine, but also its control. When designing strength exercises it is therefore useful to take account of not only physiological mechanisms, but also the requirements of motor learning, in particular:

- the need to see movements in relation to each other (specificity as a precondition for transfer: see Chapter 3)
- the need to stimulate learning by presenting new movement patterns as interesting and attractive.

When strength exercises are performed, just as in technical training, sensorimotor links or 'packages of sensory and motor information' are created. If the exercise has already been performed repeatedly and the sensorimotor packages are familiar, the packages will be uninteresting and the organism will have no stimulus to learn. Motivation to learn is triggered by exercises that produce sensorimotor links or 'packages' that are unfamiliar to the organism and are perceived as more or less chaotic. Processing this chaotic information means making changes to the neural system, i.e. learning. The disorder that is created teaches the organism to find control mechanisms that are equally valid for the different sensorimotor packages it confronts. In other words, variation introduces the system to universal control mechanisms and teaches it to tackle movement problems generically rather than ad hoc.

This means that the law of variation described in Section 4.4 also applies to sport-specific strength training. within strength training, too, the term 'overload' could more or less be replaced by the term 'variation'. Both quantitative overload (e.g. bench pressing with a heavier barbell load than the athlete is used to) and qualitative overload (e.g. switching from bench pressing using the standard Olympic barbell to one using dumbbells without increasing the weight) can be seen as forms of variation, and hence of overload.

This approach puts the role of increased resistance within strength training in a different light. Resistance becomes a way to produce unfamiliar packages of sensorimotor information, and variation in resistance and ways of performing the exercise is useful for the learning process. In the more linear, quantitative approach to overload, variation is much less useful.

For example, if a quick step-up is made onto a 30-cm box with a 100-kg barbell weight, it will make little difference in purely quantitative terms to vary the step-up by raising the weight to 110 kg and reducing the box height to 25 cm – the amount of power generated will be much the same. Yet, in terms of insight into the non-linear character of learning processes, such variation is useful. The power generated may be the same, but the sensorimotor packages will differ – and hence will give the organism a reason to learn/adapt.

Additional information

Many publications on sport-specific strength training contain calculations on the ideal barbell weight for a given strength exercise. For example, on the basis of laboratory measurements, the calculations can be very detailed, e.g. when determining the power characteristics (force x velocity) within which, for instance, a cycling track sprinter should ideally train. Again, the ideal relationship between stroke frequency and power production in rowing can thus be translated into the external load that should be used within strength training. However, it should be realized that only one aspect of specificity is then covered, and hence that transfer to the sporting movement is not automatically guaranteed. A proper training effect also requires variation (deviation from the very specific). Perhaps it can be said that the more important efficiency of movement is to performance, the more variation (i.e. intrinsic motor learning) should be built into strength training; and also, the more complex the sporting movement, the more important efficient performance of the movement (and hence learning) becomes. This would mean that using carefully chosen barbell weights for a very specific link of speed and force is only important in sports that are simple in motor terms, such as cycling. In coordinative terms, rowing may already be so complicated that speed-specific power training is less useful (Bell *et al.*, 1999). An approach based on exercise physiology (overload) is then less useful than one based on motor learning (variation). That a very specific link between speed and force within strength training for technically complex sporting movements may not be so very important is evident from strength training practice: good coaches of complex sporting movements such as athletics field events often seem to care little about which barbell weights they choose. Some even leave the choice to the athletes, provided there is variation.

6.1.3 *Overload and the central/peripheral model*

Describing overload as variation also gives the relationship between overload and specificity a far more interesting dynamic. A purely quantitative approach to overload involves little mutual influence between specificity and overload; in a qualitative approach (variation) the two are far more closely related.

The specific character of the movement (i.e. its relationship to other movements) is important, because movement control has to be structured: movements need to be stored in a matrix that creates links between numerous movement patterns. Movements that do not fit into the matrix are then simply 'tricks' – isolated skills that are difficult to sustain.

Movement control needs to be universal because motor memory – whatever that may be – has only limited storage capacity, and control mechanisms that can only be used for a few movement problems soon result in an overcrowded 'library' of control mechanisms.

Overload – in the sense of variation – teaches the system which control mechanisms are generally valid and hence can make the motor memory easier to handle.

The two features of workable control – fitting into the specificity matrix, and universal validity – ensure that control is rapid and adaptive. If the specificity matrix is inadequate and the control 'library' is too big, the right type of control cannot be found quickly enough. This underlying arrangement of movement patterns is therefore of great importance to sporting movement.

Each of these features of motor memory calls for a different learning process approach. Finding transfers through specificity means keeping close to the sporting movement; finding universal validity through variation means getting away from it. So learning basically depends on two more or less opposing stimuli that vie for control like partners in a bad marriage. The 'marriage' gets worse the longer the partners are stuck with each other – in other words, specificity and variation are more compatible in training programmes for inexperienced athletes than for those who are experienced.

This negative relationship between specificity and overload can be shown in the 'central/peripheral model' (Figure 6.2). Exercises that are close and very similar to the associated sporting movement (central) are very specific, but it is difficult to perform such exercises with overload. Exercises that are remote from the sporting movement (peripheral) are unspecific, but can easily be overloaded. There is a gradual decrease in specificity and a gradual increase in possible overload from central to peripheral. One key aspect of the model is that the link between specificity and overload is inescapable; according to this model, there are no exercises that can combine high specificity with significant overload. Coaches should therefore avoid trying to find that one set of exercises that contains the secret to success. There are no holy grails in training. Another key aspect of the model is that exercises that provide overload but are completely unspecific, or are very specific but provide no overload, are pointless. The risk of overlooking specificity or overload is greatest at either end of the model. So it is essential to keep a close eye on the exercises at both ends of the continuum (overload on the central side, and specificity on the peripheral side).

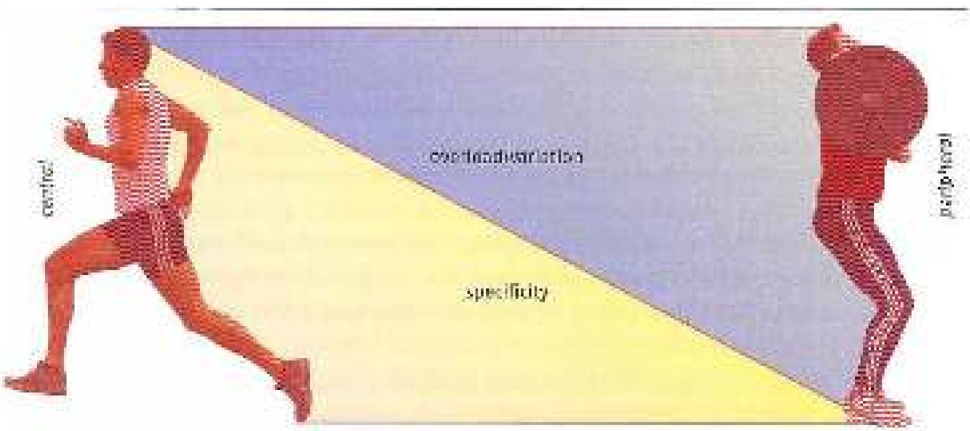


Figure 6.2 The central/peripheral model. Left: central (in this case the running motion). Right: peripheral (in this case the clean). Specificity and overload conflict.

On the central side there do seem to be exercises that break the golden rules of the model. They clearly produce very great overload, and also seem very specific. But if we look at them more closely, they turn out to differ from the sporting movement in essential ways. So they are less specific than they appear.

One good example is uphill and downhill running. For a runner, running on an athletics track is the sporting movement. Practising the running movement on a slight uphill slope is close to the sporting movement (central). Even on a slope so slight that it seems almost flat, the running movement may seem much the same, but overload increases very fast. The energy costs of running are suddenly much higher. The steeper the slope gets, the more the overload increases. This seems to violate the model. Yet if we look more closely at the uphill running action, it is essentially different from running on the flat, even if only on a slight slope. The key to running on a flat surface is elastic processing of kinetic energy upon landing. In uphill running the drop height per step is reduced, and therefore so is the amount of kinetic energy that can be converted. This means that additional (muscular) energy must be produced to generate the required vertical movement (Figure 6.3). So even on a slight slope there is an essential change in the running movement: force production must be replaced by work, and a substantially different muscle action is called for.

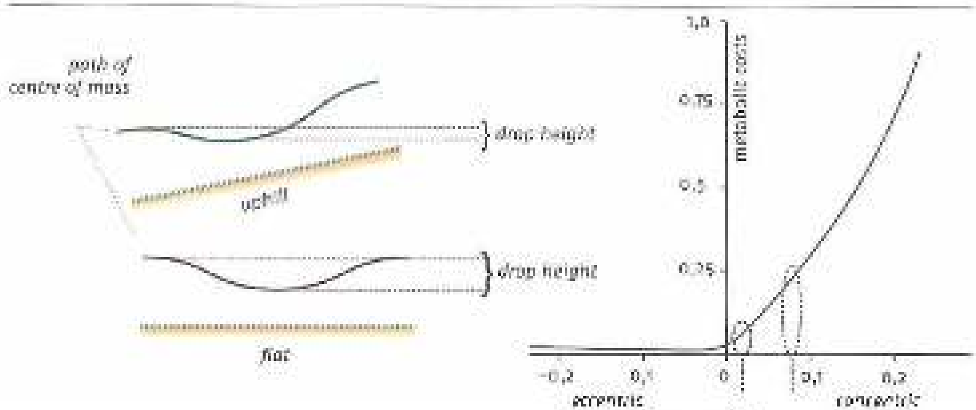


Figure 6.3 In uphill running the drop height per step is smaller than when running on the flat. Muscle action will therefore shift from isometric (elastic) to concentric, and energy costs will greatly increase (see also Figure 6.1).

When running downhill at high speed on a slope of not more than 4%, specificity may again seem very high, while the supramaximal speed that can be achieved guarantees great overload. Yet here again, there turn out to be essential differences from the flat-surface running movement. The ground reaction force is now directed anteriorly rather than posteriorly so as to use the additional drop height in order to maintain supramaximal speed (downhill running can be compared to running on a treadmill). As a result, intermuscular coordination is substantially different than when running on a flat surface. On a slope of just a few degrees, running becomes so unspecific that the exercise is pointless.

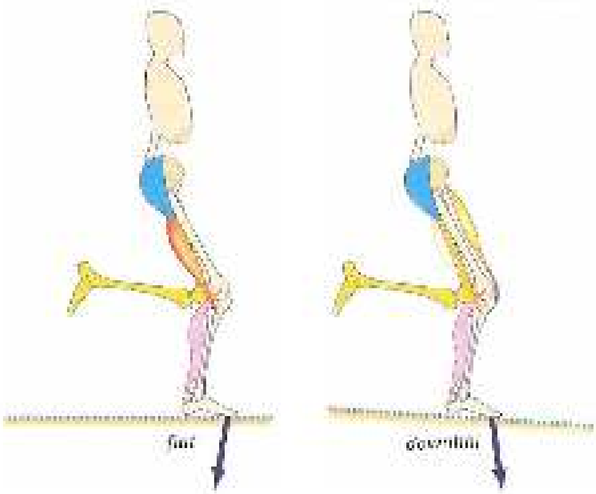


Figure [illegible]
Left: running on the flat: the hamstrings direct the force backwards. Right: downhill running: the rectus femoris muscle directs the force anteriorly.

Many types of exercise that are close to the most central part of the model thus seem more specific than they really are. Throwing with weighted objects (heavy balls, over-weight javelins) loses specificity because the most essential component of the movement, the transport of kinetic energy from proximal to distal, has to be timed differently. Training with hand paddles or flippers alters essential components of swimming, the balance between thrust and resistance and so on. The value of all these variants that are very similar to the sporting movement therefore mainly lies in the variation in the sensorimotor packages that are produced. So we need to look not just at the motor differences between exercises. These may be relatively small, so that the exercises may easily be considered very specific; yet the sensorimotor package may differ greatly from exercise to exercise, so that the exercises provide variety that may be essential to the learning process.

At the other end of the central/peripheral model is sport-specific strength training. However limited specificity may be, it must be carefully monitored in this area of the model. Intramuscular and intermuscular specificity is the easiest to monitor. Other aspects of specificity are much harder to guarantee, or cannot be guaranteed at all. It is also advisable to be somewhat suspicious of the amount of overload the various types of exercise appear to produce. Many explosive sports involve great force production. The force produced within strength training is often not much higher. On closer inspection the amount of overload proves rather disappointing. Overload and specificity in strength exercises must therefore be defined precisely if we are to gain insight into possible transfer. If the value of load within strength training is judged too uncritically, there is a serious risk that transfer will be overlooked and the quality of training will only be judged by how hard it is perceived to be.

6.1.4 *Strategies for the specificity/overload relationship*

Where can the greatest benefits be obtained in the central/peripheral model – at the ends, or in the centre? There is no simple answer to this question, for the effect of the type of exercise depends on more than just the specificity/overload link. Other factors that need to be considered include planning over the training cycle, motivation and training facilities. Many athletes find it hard to combine a technical approach to strength training (i.e. a shift towards the specificity side of the model) with the high activation or arousal that is required in many strength exercises. They need training in which they can 'switch their minds off'. This is easiest in exercises that are on the model's periphery. But other athletes find the technical approach to strength training the key to good training. They are motivated by the links they perceive between the strength exercise and the sporting movement. Some coaches and athletes need a very clear, unambiguous structure, with progress that can be measured via tests. Even if this structure creates a false reality it can boost the athlete's motivation – the trigger for adaptation – sufficiently to produce a great improvement in performance. Other coaches and athletes do not need this, and derive their motivation from creative approaches to exercises.

So there are no fixed rules about how to use the central/peripheral model; coaches must make cost-benefit analyses when choosing exercises. But the following factors should be borne in mind:

- For inexperienced coaches, the two ends of the model are the 'safe zones'. The questions that arise there are fairly easy to answer. These are good starting points for expanding the repertoire by adding types of training that are closer to the centre of the model.
- Transfer from an exercise to the sporting movement is direct. If an exercise does not itself contribute to the sporting movement, transfer will not occur through exercises located elsewhere on the central/peripheral scale. A classic example is the function of high speed power exercises, which are meant to convert slow strength exercises into speed. This is a highly questionable effect. The idea first arose from a highly reductionist view of adaptation mechanisms (see also Section 7.1.2).
- What works today may not work tomorrow. The interplay between the many components that determine how adaptation occurs is constantly changing. This means that strategies should always be treated as temporary.
- The big drawback within strength training is the lack of sensory specificity in the types of exercise. This can partly be accommodated by mental training. If the athlete can form an idea of how to apply the exercise within the sporting movement, there will be more transfer.
- Even if a form of strength training is chosen in which technical aspects are hardly present, it is still important to provide a good argument for it. Strength training for its own sake soon becomes counterproductive.

6.2 Force production in the sporting movement and overload within strength training

6.2.1 Overload within strength training of calf muscles

In many sporting movements the calf muscles are heavily loaded. Particularly great demands are made on them when running and jumping after a run-up, for instance in long jumping or a basketball lay-up. It then makes sense to consider strength training for these muscle groups. If the strength training is to be useful, not only the specificity requirement but also the overload requirement must be met.

During strength training, the quantitative overload requirement creates a problem for the calf muscles. When running at speed and jumping after a run-up, the calf muscles must produce a substantial amount of force. This is because two mechanisms play a part in the contextual loading of *triceps surae* (calf muscles): external forces, and rapid knee extension. The external forces arise at the foot plant. When running, and especially when jumping, this is a hard impact that attempts to dorsiflex the ankle and must be counteracted by the calf muscles at the rear of the lower leg (primarily *soleus*). The rapid knee extension at the end of the push-off must also be decelerated. This is done by the upper part of the calf muscle (*gastrocnemius*, Figures 6.5 and 6.6). The force then acting on the Achilles tendon when running may easily be three to six times the athlete's body weight (depending on running speed), and eight to ten times his body weight in maximal single-leg jumping from a run-up. The muscles and tendons must be able to absorb this tensile force. They are automatically trained to do so by practising the sporting movement.

If strength training is to give the calf muscles quantitative overload, resistance will have to be higher than an athlete is used to during his or her sporting movement. Assuming six times the athlete's body weight on one leg in sprinting, the muscles should be loaded with at least six times the body weight. For a body weight of 75 kilograms, this would mean a barbell weight of 5 x 75 = 375 kilograms on the shoulders (the athlete's body must be added onto this for the total load) – which is quite impossible. The barbell weight required for a double-leg load would be almost double this.

Since quantitative overload of the calf muscles is scarcely feasible via strength training, high jumpers – whose calf muscles are exposed to the greatest forces in all sports – usually avoid strength training for their calf muscles and focus instead on good running and jumping technique. When training the calf muscles, athletes should therefore stay in the central area of the central/peripheral model.

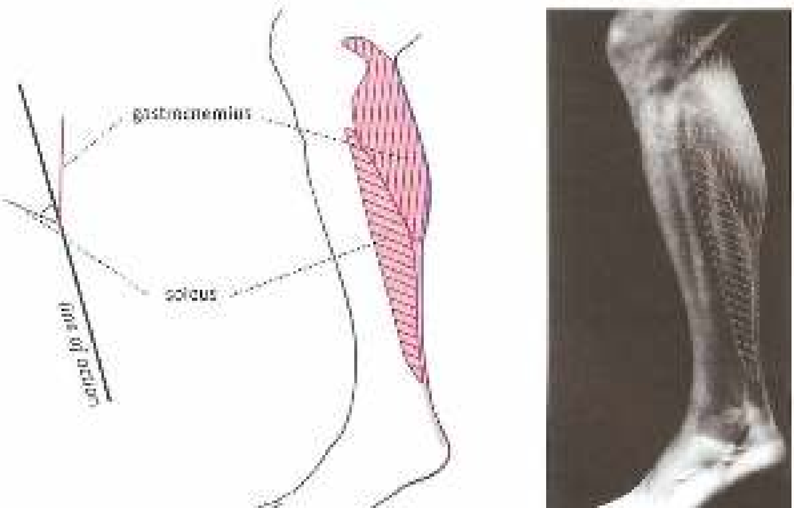


Figure 6.5 Both calf muscles (soleus and gastrocnemius) have a pennate structure. The angle of the soleus fibres to the line of action is greater than that of the gastrocnemius fibres.

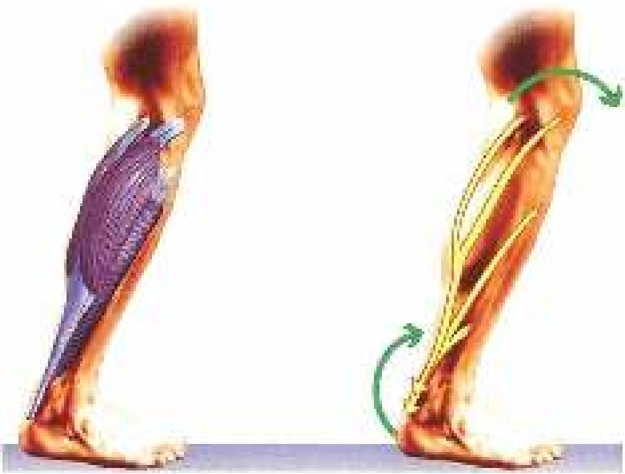


Figure 6.6
Soleus absorbs the external forces of landing, decelerates ankle dorsiflexion and stores the landing energy elastically. *Gastrocnemius* transports knee extension energy to the ankle.

6.2.2 *Overload within strength training of abdominals*

It is also difficult to create overload within strength training of the abdominals. In sporting movements in which the abdominals are under great pressure (throwing, baseball batting, running at maximal speed, single-leg jumping and so on), the peak loads acting on the abdominals are great. In such contextual movements the muscles transport a great deal of energy from one part of the body to the other, during which they act in more or less isometric conditions close to the optimal length of the muscle, and absorb large external forces. Their structure also makes them suitable for this, giving them a narrow force/length curve, and they lose much of their contractile force when they have to work concentrically. This means that during concentric muscle actions (e.g. sit-ups) they can only produce a limited amount of force, and hence no quantitative overload. Creating quantitative overload within strength training compared to the competition load in such movements as throwing, jumping and running is therefore only possible via imitation of isometric conditions at the optimal length of the muscle, with large opposing forces having to be absorbed.

In open-skill sports such as wrestling, rugby or boxing the abdominals are loaded more evenly. In wrestling, the abdominals often have to work outside their optimal length in order to absorb opposing forces (Figure 6.7). In such an open setting, there will also be concentric abdominals muscle actions. In such sports it is also a good idea to do strength exercises during which external forces must be absorbed outside the optimal length. Absorbing such forces is almost always the performance-limiting factor rather than concentric power production, for when absorbing external forces the abdominals must not only be able to produce large peak force and hence come under pressure, but must also be able to produce it very fast. Many athletes can produce enough force, but are often unable to build it up fast enough in competition settings.

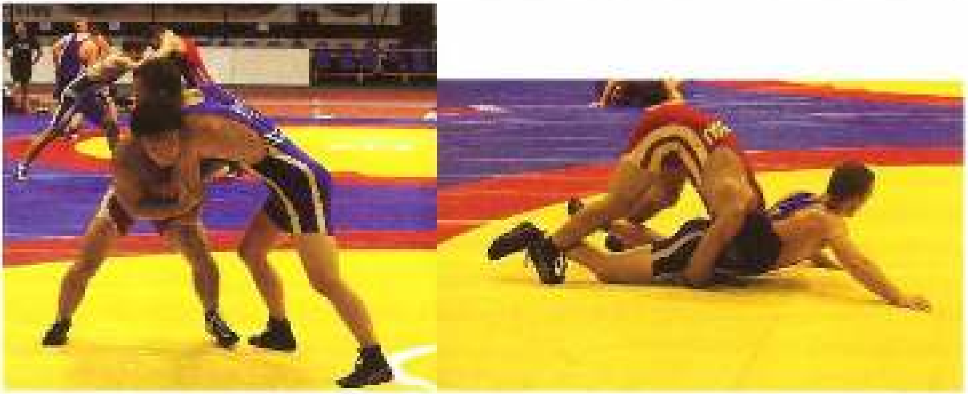


Figure 6.7 Activity in the abdominals outside their optimal length while wrestling. The muscles supply substantially less force outside their optimal length than at optimal length.

In Section 1.3 we saw that proprioception exercises do not do much to improve high-intensity movements, since these are probably controlled very differently than through proprioception. The specificity of these standard physical therapy exercises for high-intensity movement is thus limited. This means it is safe to assume that training the abdominals in overall contextual patterns, including technically correct absorption of large opposing forces in strength exercises, will be sufficient to optimize trunk control (Figures 6.8 and 6.9). It may then also be wondered how many standard physical therapy stability exercises are useful for healthy, well-functioning athletes. A limited amount of low-load training per week, in order to improve proprioception, as well as intensive training of preflex action and high-load exercises, may be enough both for athletes who load their abdominals heavily in the sporting movement and for those who load them so little in the sporting movement that they may eventually encounter posture control problems (such as cyclists, speed skaters, rowers and swimmers).

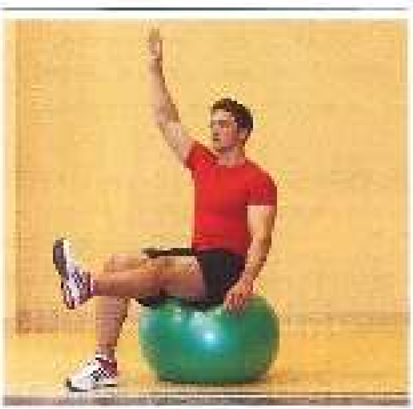


Figure 6.8
Physio ball exercise for trunk strength and control. Not only is specificity limited, but there is no quantitative overload. How useful are such exercises for healthy athletes?

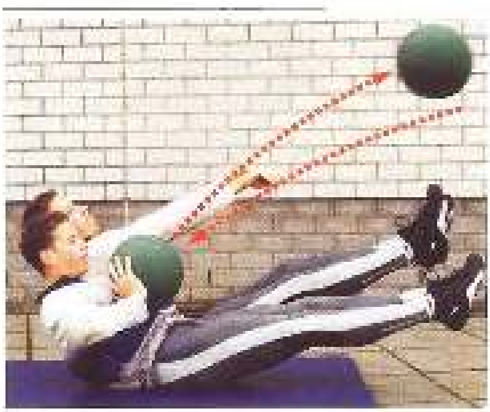


Figure 6.9
An exercise for the abdominals in which large opposing forces are absorbed and there may be quantitative overload.

6.2.3 *Overload within strength training of muscles round the shoulder*

Throwing is a sporting movement that puts the greatest pressure on the shoulder, which is then at risk. Immense forces act on it when decelerating the movement at the end of the throw. Among other things, this leads to numerous injuries at the rear of the shoulder, especially in *supraspinatus* and *infraspinatus* (rotator cuff muscles; Figure 6.10). The question is then what causes these injuries – poor timing of the activity or lack of maximal muscle force? Timing can really only be improved by choosing exercises on the central side of the central/peripheral model (practising the throwing movement). A problem here is that timing must be extremely accurate, and hence can only develop in a self-organizing manner. Focus on the final posture plays an important part in self-organization (Figure 6.11, 1st photograph). Variation and control based on the result of the movement (i.e. the final posture) are therefore key components in training the function of the rotator cuff muscles when decelerating the movement. Overload is then highly qualitative.

In deceleration muscles work eccentrically, with larger forces than may be producible within a strength exercise. Purely quantitative overload on the peripheral side of the central/peripheral model to strengthen the dorsal rotator cuff muscles is thus very hard to achieve. Only doing exercises with a heavy load will therefore have a limited effect. So it is a good idea to include rotator cuff exercises from the whole of the central/peripheral continuum in training, in order to meet the need for both accurate timing and great force production. It will take a long time to strike the best balance between a qualitative and quantitative approach for each individual athlete.

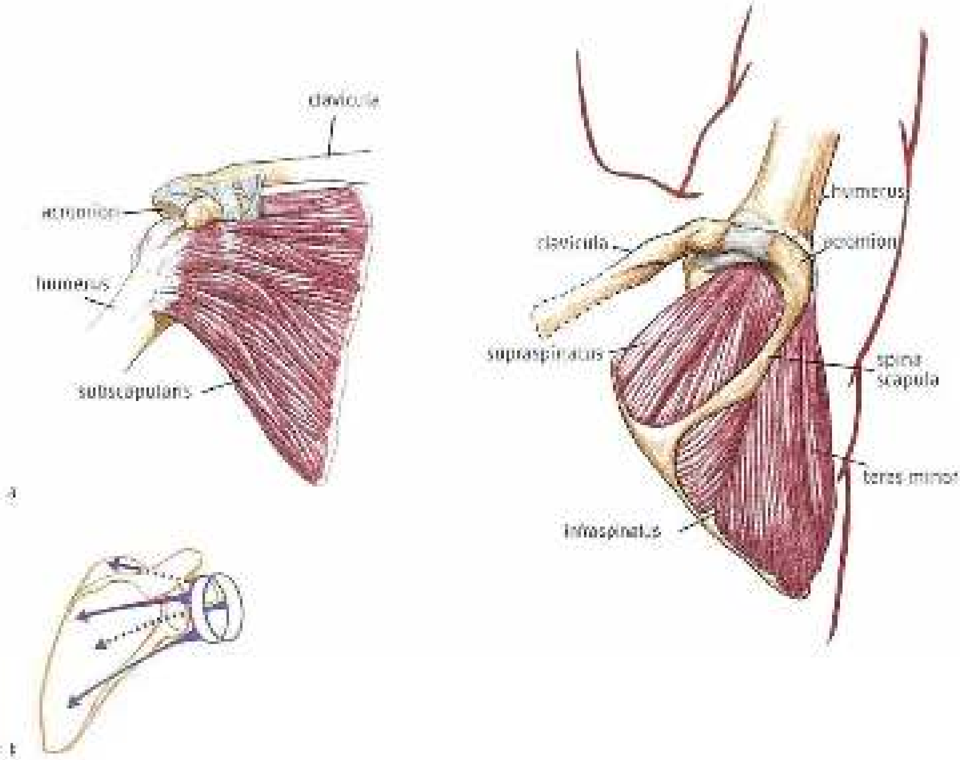


Figure 6.10 The rotator cuff muscles (a) and their stabilizing interaction/function in relation to the shoulder (b).

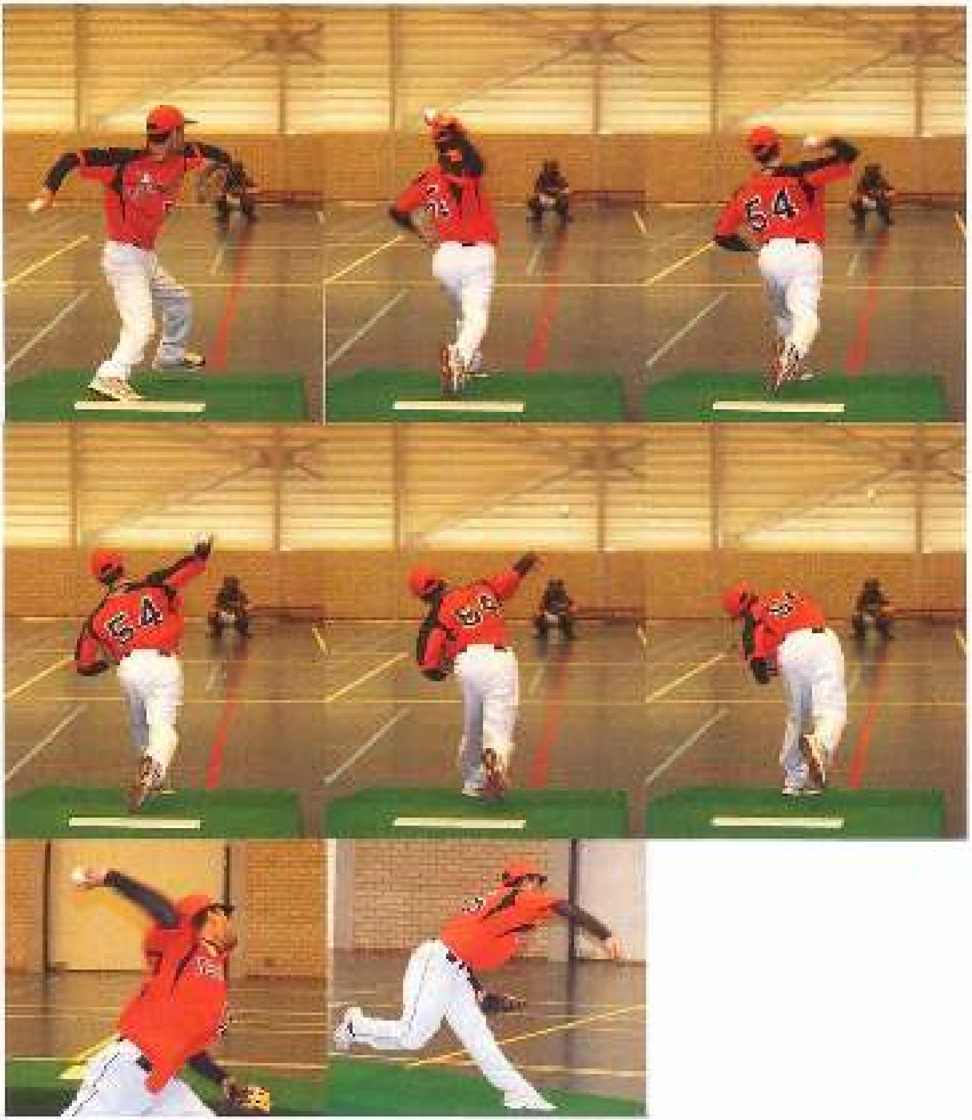


Figure 6.7 The load on the shoulder is enormous owing to the extreme external rotation that may occur when throwing. The eccentric load when decelerating is also very high. This means that the final posture is very important, with the trunk rotated round the longitudinal axis to the left in the case of a right-handed thrower, so that the shoulder ends in an abducted posture, with internal rotation of the shoulder and pronation of the forearm (see the last photograph in the series).

6.2.4 *Muscle pain and inappropriate muscle loading*

Why is training calf muscles by means of calf raises with a limited barbell load so hard? After all, training should be easy when the muscles are loaded so far below their capacity. Yet anyone who has done calf raises with a barbell load roughly equivalent to his own body weight knows that the exercise cannot be kept up for long, because the calves soon become so painful they seem about to explode.

Such pain is far too readily seen as proof of effective training. But good overload may not be the reason for the pain and the sensation of blown-up muscle bellies. Calf muscles are designed for very brief efforts, so that the muscle performs a pumping function for blood circulation. In long duration muscle actions such as calf raises, this function cannot be performed properly. Perhaps even more important is the fact that calf muscles are not very good at changing length with great force. If they have to absorb large opposing forces (several times the athlete's body weight) during sporting movement, this always happens at one length (the optimal length) of the muscle, without any substantial shortening and lengthening of the muscle fibres. Because of its structure, with pennate muscle fibres that are about 30 millimetres long (Agur *et al.*, 2003), *soleus* is not suitable for generating a large range of motion in the ankle, as is required in calf raises. In contextual movement, *gastrocnemius* acts when the knee extends while the ankle moves to plantar flexion. That is why this muscle also remains more or less at the same length (isometric) during contextual movement, especially in talented runners (Sano *et al.*, 2012). When performing calf raises there is no knee extension, and *gastrocnemius* must then work concentrically, just like *soleus*. The 'inappropriate' use of muscles in the calf raise exercise, with the ankle alternately flexing and extending and the muscle fibres changing length, is what causes the pain. Apart from the fact that the exercise does not provide overload in terms of force, there is too little specificity to make it efficient.

Seeing pain as proof of effective training is a surprisingly common error. The same thing happens when training the abdominals. Long series of sit-ups cause the same sort of pain as calf raises. The exercise appears to require a great deal of muscle power. 'Yes, I can feel it – it must be working!' is a misinterpretation of what pain means. If strength training is to improve the sporting movement, it should not just cause pain. In fact, transient severe muscle pain during a strength exercise is a sign that the exercise is not very useful – especially if no such pain occurs during the associated sporting movement.

For the same reason we can question the value of leg curl exercises for the hamstrings, biceps curls for *biceps brachii* and so on (Figure 6.12). In contextual movement, both muscle groups act as energy transporters and act within isometric conditions. The concentric loads in curl exercises are therefore aspecific, and muscle pain should be interpreted as a sign of inappropriate loading.

Rules of thumb for contextual training of muscles with a very narrow force/length relationship, such as the groups mentioned above:

- Muscles should be loaded by resisting external forces under more or less isometric conditions.
- Muscle loading should be typically neuromuscular. This means that fatigue should occur very suddenly after a small number of repetitions (six to twelve) and should resolve just as quickly. Nor should the fatigue be easy to locate – the athlete should not really be able to describe how it feels or where it is felt.
- There should be no local pain sensation.

This means that exercises such as calf raises only make sense in the physical therapy setting, in which the eccentric part of the exercise is claimed to help athletes recover from, say, Achilles tendon injuries. The accuracy of claims about eccentric exercises is beyond the scope of this book.

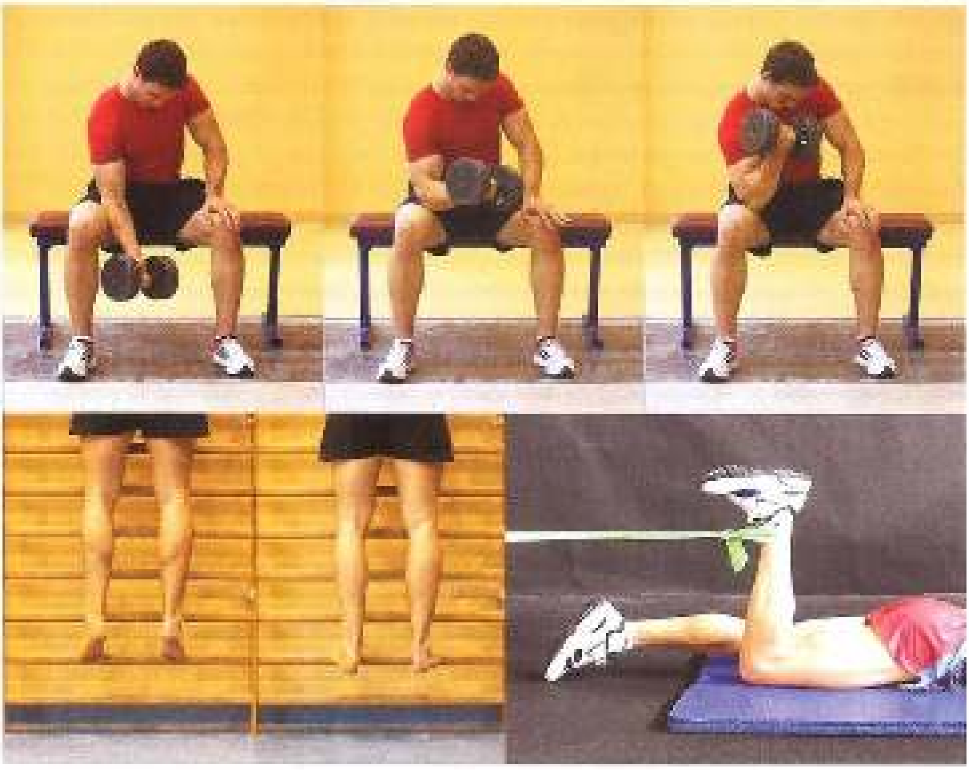


Figure 6.[illegible] Biceps curls, calf raises and hamstring curls. The sharp pain that occurs in these exercises is a sign that the muscles are being forced to work well outside their optimal range.

6.3 Newton's laws or the laws of motor learning?

Designing good sport-specific strength training programmes means understanding the various facets of the moving organism. Biomechanical, physiological, psychological, nutritional and other factors all have an impact on the adaptations that will occur as a result of strength training. An integrated approach is needed in order to control all these different influences. Besides listing the ingredients of training, recipes must be provided so that the ingredients can be turned into 'tasty' training, and such recipes – in the form training protocols – are provided in numerous books on strength training. So far no protocols have been provided in this book, and any remaining protocols will not be so specific that coaches can just follow them blindly. There will always be an important, and perhaps difficult, adjustment to the specific reality of training – one that cannot be described in a generally applicable text.

The language used in practically every book on strength training is the language of classical physics, based on Newton's laws. The main focus is on mechanical features of training, such as load, levers, torque, acceleration through power production, angular velocity, rotation, translation – the standard terminology used to capture the essence of an exercise. We can see this, for instance, in the way strength training is described in training logbooks. It is simply taken for granted that the essence of the training process can be recorded by describing the training in terms of physics (Baechle & Earle, 2008).

The weight of the barbell load is taken as the starting point when drawing up training protocols. The maximal weight that can be lifted in one repetition (1RM) often forms the basis for system design. Of course, 1RM is a key indicator of an athlete's level of strength. However, except in powerlifting, contextual sporting movements never just involve a single production of maximal force. To meet the demands of a given sport, several repetitions are required, so we must be able to determine which weights should be used in order to perform the chosen number of repetitions of an exercise. There are numerous tables showing which percentage of the maximal barbell load corresponds to a given number of repetitions for each main strength exercise (e.g. Table 6.13), and what the load should be for each exercise when comparing various exercises. For example, when performing a standardized squat, the maximal 1RM weight can be taken as the starting point. The maximal load an athlete should be able to lift at 1RM in a lunge, a step-up, a clean and so on will then be a given percentage of the 1RM squat for each of these strength exercises. The interrelated 1RM starting values for the various strength exercises are then used to determine which percentage of the 1RM load should be chosen for the required number of repetitions. The guidelines can also be used to determine whether there are any shortcomings in the training. If the 1RM value of a particular type of strength training is too low, the focus can be shifted towards it so that there is eventually a well-developed all-round level of strength.

The number of repetitions per set can be determined by the required adaptations. One to six repetitions will improve strength and power in the maximal range, eight to fifteen will produce hypertrophy in the muscles when power is supplied, and more than fifteen will train strength endurance. The intervals between the sets are designed to bring recovery into line with the required adaptation: three to five minutes to ensure sufficient recovery between exercises in the maximal strength and power ranges, one to three minutes to optimize the effects of hypertrophy training, and thirty seconds to one minute to reinforce the stimulus for improving strength endurance (these are very approximate indications). This determines the intensity of training.

To determine how much time should be devoted to the various aspects of training (volume of training), the number and sequence of sets can be fixed, e.g. three to five sets for maximal strength and power, three to six sets for hypertrophy training, and two to three sets for strength endurance. Finally, the way in which the sets interact can be translated into the organizational form that is being used: a pyramid, or inverted pyramid, a contrasting organization and so on.

upper body exercise (pull)	% of the maximal resistance to be used for 1RM of the starting exercise	3RM (= 95% of 1RM)	5RM (= 87% of 1RM)	9RM (= 77% of 1RM)	15RM (= 65% of 1RM)
supinated pull-up (starting exercise)	100	95	87	77	65
pronated pull-up	95	90	82.75	73.25	61.75
lat pull-down (narrow grip)	95	90	82.75	73.25	61.75
lat pull-down (wide grip)	90	85.5	78.25	69.25	58.5
lat pull-down on shoulders (wide grip)	75	71	65.25	57.75	48.75
seating rowing	75	71	65.25	57.75	48.75
bench pulls	65	61.75	56.5	50	42.25
upright rowing	50	47.5	43.5	38.5	32.5
single-arm dumbbell rowing	33 per dumbbell	31.25	28.75	25.5	21.5
Biceps curls	40	38	34.75	30.75	26

lower body exercise	% of the maximal resistance to be used for 1RM of the starting exercise	3RM	5RM	9RM	15RM
full squat (starting exercise)	100	95	87	77	65
front squat	80	76	69.5	61.5	52
lunge	40	38	34.75	30.75	26
step-up	40	38	34.75	30.75	26
single-leg squat	40	38	34.75	30.75	26
lateral lunge	25	23.75	21.75	19.25	16.25
Romanian deadlift	75	71	65.25	57.75	48.75
clean pull	80	76	69.5	61.5	52
clean from thigh (hang clean)	65	61.75	56.5	50	42.25

Figure 5.[illegible] Table showing resistance to be used during various exercises. The starting value is the maximal resistance for one repetition (1RM) in the starting exercise. For the other exercises in the same group (in this case two groups, for the upper body (pull) and the lower body), a percentage of this maximal weight at which 1RM should be achievable has been empirically determined. The percentages for 3, 5, 9 and 15RM can be derived from this 1RM value (tables used include those of the strength coach Bob Takano)

An additional way to use principles of physics in training is the power measurement described in Section 5.2.2. The power produced at a given speed is then an important guide to the design of training. Of course, power produced at a given speed of movement, the number of repetitions per set at a given barbell load, and the number of sets punctuated by appropriate rest periods can be further integrated with even more physics parameters to produce more and more complex training systems. The purpose of this is to design carefully controlled, highly efficient training that fits as neatly as possible into an overall training programme.

Applying the laws of physics is common practice, and quite definitely has its uses:

- It provides patterns and guidelines that can be used if the athlete's knowledge and mastery of the complex biomechanical, physiological and neural processes that take place during strength training are limited.
- There is a link with the outer layer of the specificity model described in Section 5.3 (similarity in externally observable movements).
- Protocols can be designed in a simple, standardized manner.

However, describing strength training in terms of principles of physics also has its drawbacks. One major restriction is that although the system may indicate the progress an athlete makes within strength training, it fails to describe how transfer to sporting movement occurs. There is little or no scientific evidence for the transfer mechanisms to the sporting movement that are taken for granted in these protocols based on the laws of physics. This is particularly true of many advanced protocols, which claim substantial transfer owing to the highly accurate design and timing of the training. The use of Newtonian terminology is therefore a clearly limited vehicle for registering transfer. Many of the factors involved in adaptation of the sporting movement as a result of a given overload within strength training cannot be described in this mechanistic terminology.

Improving coordination is a key component of strength training. The learning organism does not think in terms of kilograms and muscles, but in terms of sensorimotor links and control based on the result of the movement. To the learning organism, a hundred kilograms does not mean the same thing as two times fifty kilograms. That is why it is useful to use not only physical data (the quantitative approach) when describing overload within strength training, but also terminology based on knowledge of motor learning. This provides a better link with transfer to the sporting movement. Approaching overload not just as a quantitative variation but also as a qualitative one allows us to describe a number of new strength training arrangements that are useful for the sport-specific setting.

6.4 The law of variability as a guideline

6.4.1 *The constraints-led approach*

The most important adaptations in sport-specific strength training are coordinative. It is therefore useful to take not only physics-based mechanisms as a frame of reference for the systematic division of 'overload' within strength training, but above all learning mechanisms. This means a shift in overload based on the amount of force produced towards production of a new package of sensorimotor sets. Both quantitative (physics) and qualitative (the new sensorimotor package) overload thus depend on variation.

Performance of a movement depends on three factors:

- the environment in which the movement is made;
- the movement (task) being trained;
- the organism performing the movement.

Each of these factors forces performance of the movement to shift in a particular direction. The environment makes some ways of performing the movement more effective than others, and even rules some of them out. The same applies to the demands made by the task, and the properties of the organism. The interplay of the environment, the task and the organism, each with their own constraints on how the movement can and cannot be performed, creates a conditional framework within which movement can be usefully and contextually performed. Movement control thus mainly involves omitting movement patterns that do not meet the demands of the environment, the task and the organism. This idea is summed up in constraints-led approach theory. Constraints on ways of performing the movement can create sensorimotor interaction (the sets) that leads to motor control (Davids et al., 2008).

Additional information

The constraints-led approach is an attempt to link up the various motor control theories:

- Schema theory focuses on design of the intention and the movement plan (task). This theory claims that incoming sensory information is essentially meaningless, and only acquires meaning – i.e. becomes perception – in the brain (Schmidt & Lee, 2008; Schmidt & Wrisberg, 2008); it also claims that the central nervous system generates all the information that is needed in order to perform the movement appropriately, and that motor control has a hierarchical structure.
- Ecological theory (direct perception theory; Gibson, 1986) claims that sensory information from the environment is of such high quality that ways of controlling movement are already part of it, in the form of 'affordances'. Movement is therefore largely controlled by the environment.
- Dynamic systems theory claims that movement is designed by an organism's self-organizing dynamics, that body properties are decisive factors in what the movement will be like, and that motor control has a decentralized structure (Kelso, 1995).

The ecological approach in fact attacks the claim of brain-central (cognitive) theory that sensory input is interpreted entirely internally, whereas dynamic systems theory attacks the claim that movement is entirely designed by the brain. The constraints-led approach attempts to link up the three components – the brain (task), affordances (the environment) and dynamic self-organization (the organism) – and hence the three theories (Figure 6.16).

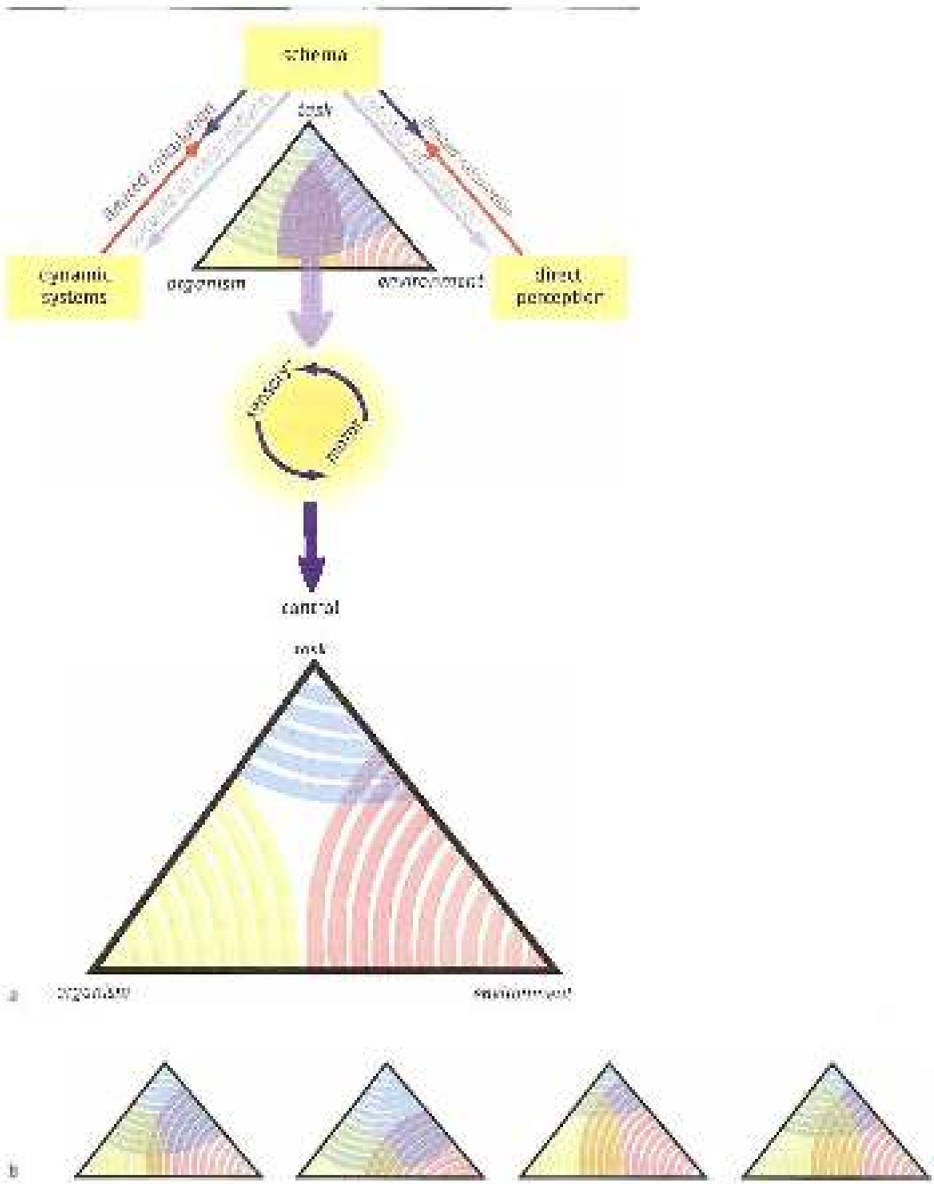


Figure [illegible] Top: the constraints-led approach attempts to link up the three main theories of motor control. Constraints from the environment, the task and the organism (representing the three theories) are placed in a triangle, each corner of which allows certain movement patterns. This is indicated by the range of the concentric circles that cover part of the triangle from each of the corners. Where the possible movement patterns overlap (the central area of the triangle), there may be a useful blend of motor and sensory input that allows the movement to be controlled.
Middle: if the possibilities provided from the three corners of the triangle fail to overlap, there is no effective movement pattern, and the movement cannot be performed and learned successfully.
Bottom: varying the exercises will produce different overlaps between the environment, the task and the organism. This will provide valuable information for finding universal principles of control.

This theory can be illustrated by a step-up with a barbell load. The way the movement will be performed is controlled from three angles:

1 The environment: the height of the box determines the required angle of the hip joint when the athlete begins to produce force. The size of the barbell load determines how fast the stance leg can be extended and how fast the free (swing) side of the pelvis can be elevated during the step-up.
2 The task: demands can be made on the performance of the movement, e.g. how firmly the trailing leg can be pushed off, which movement should be made by the swing leg after leaving the ground, how much horizontal movement is required, and so on.
3 The organism: sense of balance determines how fast the step-up movement should be made, and whether the athlete can end in a single-leg posture. Properties of the organism will also determine when the free (swing) hip can be moved upwards during the step-up – e.g. very late in the movement if the abductor muscles are relatively weak in relation to the hip and knee extensors.

What the eventual exercise will be like is determined by constraints from all three corners of the triangle (Figure 6.15). If the task is to step up as fast as possible without pushing off with the trailing leg and to elevate the free (swing) hip as early as possible, ending in a single-leg posture, the eventual performance of the movement will depend on the environment (the box and the weight) and the organism (strength and balance). The constraints from the three corners of the triangle may well be such that the movement cannot be performed properly. In that case, the exercise is too difficult.

Figure 6.15
The final posture in a step-up, with determining factors from the environment (= the barbell load and the height of the box), the task (= foot in dorsiflexion, acutely flexed knee and the elevated free (swing) side of the pelvis) and the organism (= balance and strength in the push-off leg).

When designing a strength exercise, coaches should therefore take account of influences from all three corners of the triangle. Especially if the type of exercise is rather more complex, difficult decisions may have to be made. Suppose, for instance, that a strength coach wants to use a clean to improve a basketball player's force production when jumping. Which type of clean is the most efficient, and at what weight, for an athlete who is two metres tall and has to overcome large levers? Given the physical structure (e.g. lack of flexibility) and components of the movement (single-leg push-off) that need to be improved, should the barbell be on the ground or on a box, or is the best starting position perhaps upright, with the barbell just above the knees? What is the optimal weight – a heavy weight, so that the initial acceleration of the weight is very important and the weight is caught in a squatting position, or a lighter one, so that the end of the acceleration is important and the weight is caught in a more or less upright position? How does strength in the shoulder girdle affect the choice of weights? And so on. These choices ultimately have a major impact on the intended transfer from the clean to the sporting movement.

Training can be varied by varying each of the three constraints. The resulting changes in each will affect the way in which the movement is controlled, and hence the sensorimotor sets that are produced. Producing variable sets will activate the learning process.

In strength training, the environment, the task and the organism can thus be varied. Little or no account is taken of the last alternative (varying the organism) within strength training. Yet motor learning theory suggests it can be very useful to vary the dynamics of the organism. The main factor that can be varied here is fatigue.

6.4.2 *Variability in environmental factors*

Strength exercises can be varied by varying the environmental factors, although the rules for performing the exercise do not essentially change. Varying the environment is a very useful way to apply differential learning. The size of the resistance can be varied and the resistance can be made more or less stable. The direction and stability of the surface on which the athlete is lying or standing can also be adapted. All these variations can be used to create differing sensorimotor sets.

The amount of variation in environmental factors is determined by three criteria:

- The movement should be contextual and intentional – i.e. there should still be an efficient way to perform the movement so that its purpose is achieved. For instance, if a complex exercise such as a clean with a submaximal weight is performed on an extremely unstable surface, the desired triple-extension movement can no longer be performed properly because of insurmountable balance problems. The sole purpose of the movement will then be to perform it without the athlete losing balance, and maximal acceleration of the barbell by timing the extension correctly will become a marginal factor. Obviously no-one is going to perform a clean on a 30-cm thick tumbling mat, for it is immediately clear that the movement will be disrupted by the soft surface. In other cases, however, the adverse effect of the surface will not be so obvious. If a clean is performed with a maximal or almost maximal weight, a much smaller instability in the surface may be enough to interfere with performance. To ensure good balance, the floor is always hard and flat, but the shoes athletes wear often have soft springy soles, and this may seriously impair performance when using heavy weights. Athletes who do high-level sport-specific strength training should therefore consider investing in special hard-soled weightlifting shoes.

2 Variation should not be so great that it results in a substantially different movement. For instance, an increased barbell load may require the muscles to act differently, in which case the learning mechanism based on comparing closely related movements will no longer work. Training will then to some extent become random practice. Examples of such a limit to variability are bounds with a load. If a barbell load is not used, jumping may be purely elastic, and the energy on landing is then used for a new vertical impulse. If the exercise is performed with a 7 kg barbell, the contact time will be slightly longer, but the push-off will be just as elastic. If the barbell load is then further increased, the contact time will be even longer. At this point the muscles can then no longer be used elastically; eccentric muscle action will be needed to absorb the landing, followed by concentric action in the next push-off. Suddenly the movement is no longer a version of elastic bounding. The same thing happens in exercises that require throwers to face the wall from close up and throw a medicine ball at it several times as hard and as fast as they can, with their arms extended above their heads. If the ball is not too heavy, the throws can be performed with a brief interval between loading and unloading of muscle tension, and the elastic properties of the muscles can be used to accelerate the ball. However, if the ball is so heavy that the contact time while throwing becomes too long, the muscles will have to act eccentrically and then concentrically.

3 The movement should be safe. The greater the resistance, the better the joints should be protected by muscle actions, and the fewer external perturbations should be allowed. When training with heavy weights, the focus must be on health and safety. Safety is only guaranteed if perturbations to balance are always sufficiently controlled by muscle power.

So the lower the percentage of maximal resistance that is trained, the greater the scope for variation. Stability of the surface can be reduced by making it more inclined, narrower and/or less stable. Resistance can be varied by making it less stable, more asymmetric and so on (Figures 6.16 and 6.17).

The differing sensorimotor sets develop because different proprioceptive information is released. This body-position information must be processed quickly and appropriately in order for the movement to be performed correctly.

Strength training with relatively large resistance will generally be chosen, in order to achieve the greatest and fastest progress. However, if the level of training is already high and the athlete already has vast experience of high-resistance training, there will be much less

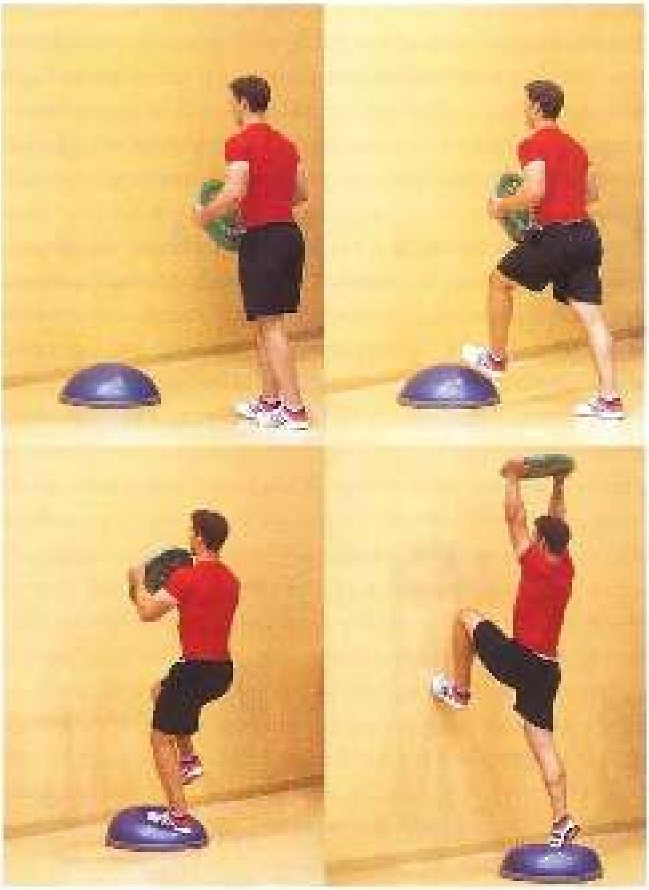


Figure [illegible] Variation in surface in a step-up. The unstable surface makes great demands on ankle control. Replacing the barbell plate with a water-filled pipe makes great demands on trunk control.

Figure [illegible] Examples of variation in surface and resistance. Flies on a physio ball with two dumbbells of different weights. To keep his balance during the fly movement, the athlete has greater elbow extension on the side with the lighter dumbbell.

progress, or indeed none at all. In that case it is very useful to look more closely at ways of varying the environment.

6.4.3 *Variability in the task*

Differential learning can be applied by varying the tasks within the limits of the features of the movement. Of course, this works best if combined with the aforementioned adaptations to the environment. This will create numerous opportunities for variable strength training, always focusing on the need for safety.

One serious problem always associated with stating how the task should be performed is that the focus risks becoming internal. This can to some extent be prevented by stating how the movement should be performed with reference to external features of the movement, for instance by stating how the barbell load should move, rather than how the movement should be performed in the body. For example, if versions of the clean are performed, possibly in combination with variations in the surface, it may be stated that the barbell should remain stationary after it is caught. The environment may also be designed so that the required form of the movement develops automatically. It is always a good idea to check which information can be transmitted by explaining things, and which information by changing the environment.

Varying the task is of course also suitable for creating a random practice setting. The strength exercise to be trained is then alternated with other differing movement patterns. If the patterns are very different or even contrasting, it will be difficult to switch from one exercise to another (see Section 4.4.3). This may create a new impulse to learn. Such use of major contrasts in exercises is very useful with athletes who are close to their adaptation ceiling.

Examples of high-contrast random-practice strength training:

1 Lower limbs:
   - Exercise 1: a deep (ninety-degree) single-leg squat with a high barbell load, performed slowly and with a two-second rest period at the lowest point, twice to the left and twice to the right – a maximal strength exercise in which *gluteus maximus* and *quadriceps femoris* are loaded at a relatively long (outer-range) length.
   - Exercise 2: a clean from an almost upright starting posture, with the barbell halfway up the thigh, with 85% of body weight resistance and three repetitions. This starting posture facilitates transport of energy from the knee to the ankle.
   - Exercise 3: a 'good morning' with the same weight as in the clean and four repetitions. Emphasis on the isometric strength of the back muscles.
   - Exercise 4: eight bounces with an empty 7-kg barbell on the shoulders. Emphasis on elastic muscle action.

Each of the exercises puts pressure on a different movement mechanism. It is important that the very first repetition of the next exercise be performed correctly. The great contrast between the exercises places great demands on motor control when switching exercises.

2. Trunk:
   * Exercise 1: hanging on the wall rack, with the athlete's back to the rack and with the feet close to the hands. The legs swing down to an almost fully extended hang, then the feet move back up to the starting posture. The feet must not touch the rack at the bottom of the movement. The exercise makes great demands on the rectus part of the abdominals, which has to cooperate with the iliopsoas. Since the range of motion is so large, the movement can only be performed if they cooperate efficiently. Ten rapid repetitions.
   * Exercise 2: the starting posture is a prone plank position, with the hands and feet almost as far apart as possible. One hand or one leg is then lifted from the ground in succession. The abdominals should act isometrically and absorb external forces in a different direction each time. Lift a hand or a foot from the ground for two seconds, six times in total.
   * Exercise 3: jumping onto as high a box as possible from a squat. At the finish, the abdominals must absorb large opposing forces to counteract anterior rotation of the pelvis, so that the legs (together with *iliopsoas*) can be rapidly tucked in after push-off. Ten jumps in succession.
   * Exercise 4: sit with the torso off the ground inclined at 45°, with the feet fixed and the hips and knees flexed. A 5-kg barbell plate is held in front of the chest, then thrust away as firmly and as far back as possible, and finally returned to the starting position. The abdominals are loaded elastically. Eight to ten repetitions.

The abdominals have to absorb opposing forces in a different overall movement pattern each time; these are built up slowly or fast and are absorbed in different cooperation with the hip flexors each time. Greatly contrasting patterns are sought. Once again, the very first repetition of the next exercise should be performed correctly.

Of course there are many different ways of organizing random practice. Exercises that call for complex coordination are particularly suitable for this. Strength exercises can also be alternated with sporting movements that make high coordinative demands.

Even though it is not quite clear how random practice works, it is still useful to distinguish between random and differential learning in training, for they may be based on very different mechanisms. If exercises are organized too arbitrarily and fail to distinguish between the two, variation may become too random, and the information – especially KR information – that serves as the basis for learning too disorganized.

6.4.4 *Variability in the organism*

When looking for ways of variation, the training organism tends to be overlooked. Body properties are considered rather invariable, for training adaptations take a very long time and hence are not very suitable for use in varying training over a short period. However, the two-factor model shows that there is another factor besides fitness that can vary, namely fatigue. Just like adaptations, fatigue alters the state of the body but fatigue and recovery are much faster processes. Strength training is the area of training in which fatigue occurs most rapidly. With maximal effort and a heavy load, it can reach extreme levels in a matter of seconds. In a strength exercise involving heavy neuromuscular loading, five or six repetitions in quick succession may be enough to prevent the athlete from completing the exercise. However, recovery after such neuromuscular training is just as rapid. The result is continual and extreme alternation of fitness and fatigue during training.

This rapidly increasing fatigue in strength exercises may be a key motor learning mechanism. The following explanation for this is still very speculative, and has not yet been researched. The idea is based purely on practical experience, but surely deserves further study. The rapid increase in fatigue means that the body responds differently to central nervous system commands during the last two repetitions in a set than during the first two. The resulting output may then be very different from what was expected, and the exercise may be a failure. In very simple movements such as bench pressing, this problem can usually be solved by a stronger stimulus from the central nervous system that causes more muscle fibres to be recruited; but more complex movements such as the clean and snatch in weightlifting make very great demands on intermuscular coordination, and merely strengthening the signal will not be so effective. This is because fatigue tends not to increase at the same rate in all the muscles, so the signal must be adapted for each muscle. This means the central nervous system would first have to measure how fatigue is increasing in the individual muscles, and then completely restructure intermuscular coordination. It is therefore extremely complicated and difficult to control complex movements when fatigue occurs, and the exercise is much more likely to fail. In addition, the movement can no longer be performed economically. If the last two repetitions are to meet the same criteria as the first two, the neural system will therefore have to find commands that are more or less unaffected by fatigue – commands that can override the current state of the muscles and are formulated so as to produce more or less the same movement in a rested body as in a fatigued one. The system will therefore have to find rules that are more generally applicable than the rules used to control a movement when the organism's output is constant and predictable. To put this another way, as the body's output is different from what was expected, the resulting sensorimotor package will be confusing. Such confusion will encourage the sensorimotor system to learn, for the learning system is fond of generally applicable rules that allow movements to be performed regardless of perturbations.

So there are good reasons to assume that fatigue, if well used, may be a motor learning tool. Coping with fatigue is a basic survival skill in nature. Animals that have trouble controlling their motor systems as a result of increasing fatigue, and whose performance therefore declines, are more likely to be caught and eaten by predators. Learning to make control resistant to fatigue is therefore crucial to survival.

The intention-action model also explains why fatigue and learning are related. The main feature of this model is that the result of the movement must be as independent as possible from the muscles that have to perform it. A movement is only mastered if the intention can be achieved through all the ways in which it can be performed. It therefore makes sense that control also seeks to be independent from muscle fatigue. Learning to cope with fatigue is therefore in line with a key rule of motor control. Of course, the intention-action model implies that fatigue should not be so great that the intention of the movement can no longer be achieved. Performing strength exercises in series is therefore a learning mechanism in itself. Variation occurs without the risk of the environment needing to change. In other words, many of the main features of specificity, such as the sensory environment or the result of the movement, can be left intact in training, and yet the sensorimotor sets can be varied by creating fatigue. This puts fatigue in a different light. It is no longer a barrier to repeating a movement more often and so making it more familiar – on the contrary, it is a driving force in the learning process.

Of course, fatigue can be used as a learning resource in many more ways than just repeating an exercise until performance starts to decline. For example, 'sensorimotor chaos' can be created by first fatiguing part of the body in a strength exercise and then immediately practising a complex movement pattern that is to be improved. The organism must then respond to muscles that together will produce an output never previously generated by these particular commands, and the neural system will have to adapt the commands so that the result (the main orientation point, e.g. the end point of the movement) is still achieved. In a subsequent set, another part of the body will first be fatigued, and then the same complex exercise will be performed with yet another sensorimotor set for the same result of the movement, and so on. Local fatigue can thus be used to improve complex patterns and make the underlying basic components (attractors) more generally valid and hence more stable. Of course, the complex exercise that is to be improved need not be a strength exercise, but may also be a sporting movement (e.g. a tennis or volleyball service, a swimming start, a boxing combination and so on).

Additional information

Suppose a baseball player has to improve his batting technique. The mistake he makes is to over-adjust his arm swing to match the trajectory of the ball. However, rather than adapting the swing to make contact with the ball, for the movement to be performed correctly the adjustments should be made in the trunk (by bending sideways) and the legs (by slightly flexing or extending the front knee) rather than the arms. The arm swing (an attractor) should always make the same movement and always end up in the same position in the same way (the end point). To learn this, the organism must learn not to aim for a single familiar sensorimotor set of trunk and leg movements (always the same trunk and leg posture), but to vary these so that the sensorimotor set from the arm movements becomes more stable. So the batter first performs a number of deep single-leg squats with a heavy barbell load, which fatigue one leg. He then performs the practice strokes. Before the second series he fatigues the other leg in the same way. Before the third series he first fatigues his diagonal abdominals in the same active diagonal as in the batting movement, and so on. The body thus learns not to aim for stereotypical performance of the trunk and leg movements. Degrees of freedom are increased in the trunk and legs, and are reduced in the arms. In terms of dynamic systems theory, existing attractors (stereotypical ways of using the trunk and legs) are first perturbed, so that new attractors can develop (a stable bat swing). Batting technique improves. Fatigue is thus used in a targeted way to improve errors in contextual movement.

Another example is a basketball player who makes too much use of the explosive properties of quadriceps by flexing the knee too far and then re-extending it when pushing off for a shot. This suppresses effective reflex patterns. In that case it may be useful to fatigue *quadriceps* before practising shooting, for example in a back squat. The dominant action of this muscle

group no longer works at push-off, and force production has to be readjusted. Dominant but incorrect patterns can thus be abandoned and replaced by other, better ones. The same can be seen when an untrained high jumper has to do six jumps, jog back immediately after landing and do the next jump without a rest period. During the first jumps the body's centre of gravity is lowered in order to produce more power at take-off; but during the last jumps fatigue has increased so much that the centre of gravity remains high even without instructions – the correct technique – and the push-off becomes more elastic. A useful way to improve jumping technique.
A training protocol based on local fatigue could also be used in swimming. The main problem within strength training for swimmers is finding a link between strength exercise coordination and swimming technique, because the sensorimotor environment is so very different in the water and on land. First local fatigue is created, for example by doing a heavy strength exercise on the edge of the pool, e.g. first an overhead press, then lateral raises, and finally a lat pull-down. This is then followed by 50 metres of butterfly, with certain technique requirements. The transfer problems may then be somewhat reduced by the differing environments in which the swimmer moves.
The same protocol can be used within strength training, for example when improving the clean. In the first set, for example, quadriceps is first fatigued in one leg by means of a hack squat, immediately followed by three repetitions of the clean with 75% of the maximal possible weight. Before the second set, the hack squat is performed with both legs. Before the third set, the back muscles are fatigued with a good morning. Before the fourth set a long series of jumps is performed, and so on. Technique should then be as good as possible when performing a subsequent clean.

The opportunities created by rapidly increasing local fatigue during strength training have rarely been described or applied in practice. It is obvious that fatigue can easily be used within strength training as a means of variation in open skill sports. In open skill competitions, skills will be required under constantly changing local fatigue conditions (a tennis player having to smash after a long rally, a rugby player having to accelerate after a scrum, and so on). In addition, the movement is not performed in as fixed a way as in a closed skill. Training can easily be made 'game-like'. Since the sporting movement is variable, it makes sense for training to be variable as well. This may be more difficult with closed skills. What is the optimal level of local fatigue for learning? Too little, and there will not enough perturbation; too much, and the movement will be performed so badly that it may not be successful. The coach has to get the balance right.

Seeking variation in fatigue means that environmental factors and the task can remain unchanged. This allows coaches to avoid an awkward strength training problem: how to ensure specificity of the movement. Once it has been decided, for example, how the clean can transfer to sporting movement in a particular environment (weight and surface), the exercise can be repeated without the effect immediately being extinguished by monotony (Figures 6.18 and 6.19).

The role of fatigue in the learning process is still largely unresearched. The learning effects of using fatigue have not yet been sufficiently studied to draw far-reaching conclusions (Kerr, 1999). Whether the effects are negative or positive appears to depend on which protocols are used. For the time being, coaches and rehabilitation therapists who want to use this to train reflexes and complex motor patterns will have to keep experimenting with different protocols and gradually discover what is good practice.

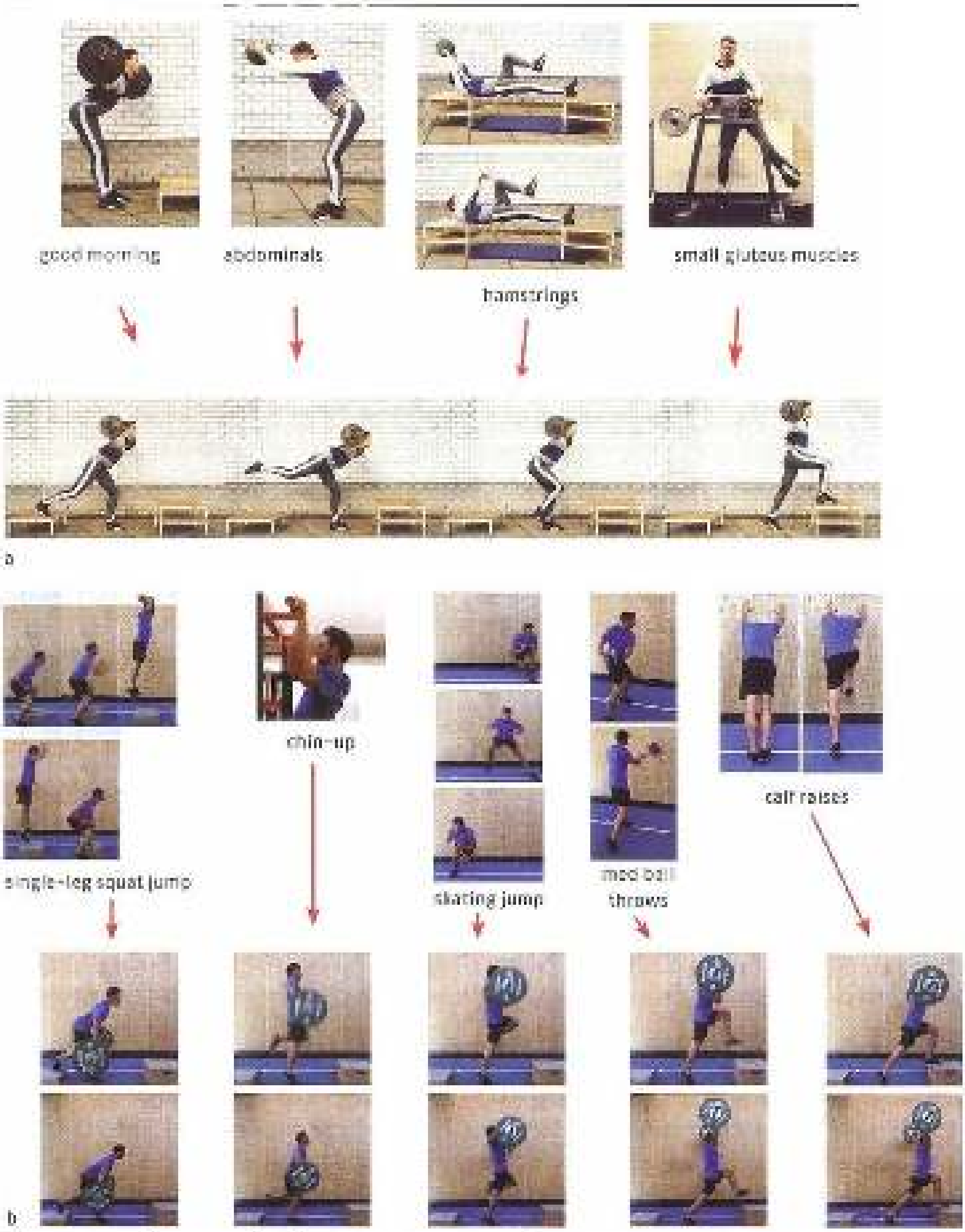


Figure 6.[illegible] Exercises to create local fatigue (variation in the organism) and three complex strength exercises that need to be technically improved.

a The movement that needs improving is a balance-to-step with a barbell load on the shoulders. First the exercise for local fatigue is performed: a good morning, bounding with a medicine ball, throwing a medicine ball with extended arms in a bridge (hamstrings) or abduction against resistance (small gluteus muscles). All the exercises are performed until fatigue occurs. The sporting movement is then immediately performed, focusing on technical performance.

b The sporting movement that needs improving is a balance-to-clean and a balance-to-snatch. First the exercise for local fatigue is performed: ten single-leg squat jumps with push-off onto a 30-centimetre box, a chin-up, twenty skating jumps, throwing a medicine ball sideways, or calf raises. All the exercises are performed until fatigue occurs. The sporting movement is then immediately performed, focusing on technical performance.

The environment can be varied in the sporting movement by moving the box and changing the barbell load.

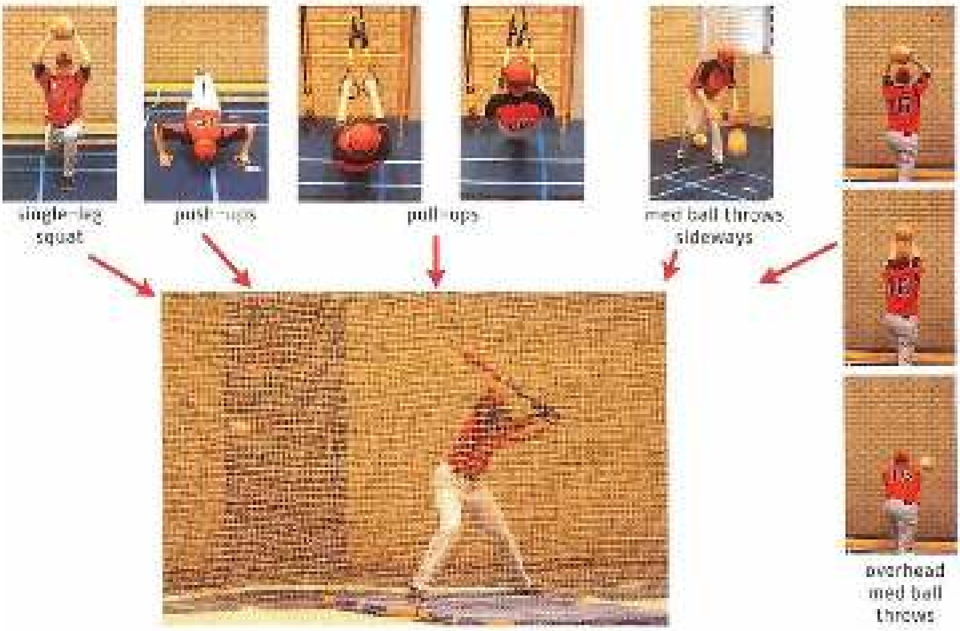


Figure 6.19 Four exercises to create local fatigue and the sporting movement that needs to be technically improved (batting in baseball). The sporting movement is performed immediately after the preparatory exercises: a single-leg squat, push-ups, a chin-up, throwing a medicine ball sideways against the ground as hard as possible, throwing a medicine ball with two hands. In the sporting movement the environment can be varied by using different baseball bats or by making the surface unstable (for instance by standing on a foam mat).

Additional information

The alpha/gamma system and fatigue

What makes movement patterns resistant to fatigue, and how is muscle control designed so that fatigue has only a limited impact on the result of the movement? Part of the answer to this question lies in the properties of the efferent structure of the central nervous system, the alpha/gamma loop, and how this contributes to intermuscular coordination.

Contextual movements always involve interaction between various active muscles; this is essential to the quality of the movement. Each of the muscles makes its own contribution to resistance against inertia and external forces. Force production and change in length may vary for each muscle during the movement. For instance, if an athlete lifts a dumbbell from the ground to as far above his head as possible, a large number of muscles are active. One muscle may act isometrically in the first part of the process, then concentrically, then eccentrically for a moment and then isometrically again. In other muscles the muscle action pattern is different (figure 6.20).

In this changing pattern of muscle actions the amount of force produced by each muscle should not be too different from what is needed to perform the whole movement properly. If one muscle does not make the required contribution because of fatigue, the movement may not end directly above the shoulder but slightly to the side, and it will be impossible to fixate the weight above the head.

Figure 6.[illegible] In complex lifting movements from various starting positions the function of a muscle may change during the movement. All the various muscle actions involved must be coordinated to result in the desired final posture.

However, output will differ from what is intended not only if a muscle is fatigued, but whenever a great deal of force has to be produced. The more force muscles have to produce, the more noise there will be in the signal to the muscles, and the greater the difference. Any incorrect output from one or more muscles will be corrected or compensated for. Key compensation systems are the preflex system and the neural gamma innervation system.

In a complex movement as described above, every active muscle must overcome external forces. This means the athlete must estimate how great those forces will be. If the signal for muscle action is only sent via the alpha path, an incorrect estimate cannot be corrected. This means that the reduced output from a seriously fatigued muscle cannot be compensated for. If a great deal of force has to be produced in a complex movement, the alpha signal containing an initial estimate of the amount of force must be accompanied by a signal via the gamma loop to monitor the length and change in length of the muscle. If the required estimate of the initial force is too low, the muscle, including the muscle spindles, may be stretched and the gamma system can compensate for the difference. With large force production in complex movements, the alpha signal should not be too far below the required level, for then the gamma signal cannot fully compensate for the difference, and not enough total force will be produced.

Satisfactory movement therefore depends on proper coordination of alpha and gamma signals. With large force production in complex movements, this is more difficult than it may seem. Not only may the alpha signal be too weak and the gamma system unable to fully compensate to the required level, but the alpha signal may be too strong. In that case there is no good way to weaken the signal. The gamma path can only strengthen the signal, and the Golgi tendon system is not innervated from the central nervous system and so cannot be made movement-specific. To perform the movement correctly in a complex strength exercise, the alpha signal must therefore always remain slightly below the actually required level, so that the gamma loop can always act correctively. This means that interaction between alpha and gamma signals in complex movements with large force production – as in most sporting movements – must be accurate, and hence must be trained. If they are correctly coordinated, the alpha signal remains within the correct limits and the muscle spindles are set so that the length and change in length of the muscles are properly monitored, the movement will be more resistant to the effects of muscle fatigue.

A good example of this within strength training is a mistake that is often made when performing the clean. Knee extension must be properly coordinated with hip extension. If the athlete tries to move too fast during the clean, the knees will extend too fast, the trunk will move anteriorly and the lever of the barbell will become so large in relation to the hip that the barbell can no longer be properly accelerated. In that case the clean will fail. The right signal must be sent to quadriceps in the first part of the movement (lifting the barbell to knee height). The alpha signal should not be too strong, for there is then no way to damper the muscle action; but nor should it be too weak, for the speed of the barbell at the end of the lifting phase is crucial, especially with a high barbell weight. If the speed is too low, the barbell cannot be accelerated, and again the clean will fail. So it is very difficult to regulate the amount of force in quadriceps correctly during the clean. Linking force regulation (alpha signal) to posture regulation (gamma signal) 'dampens' perturbations such as those caused by fatigue.

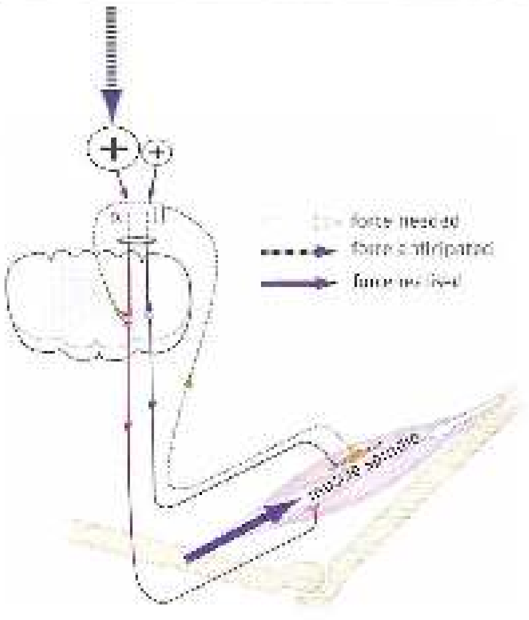


Figure 6.11 There is too much alpha activity because the force to be produced has been overestimated. The muscle action cannot be corrected for.

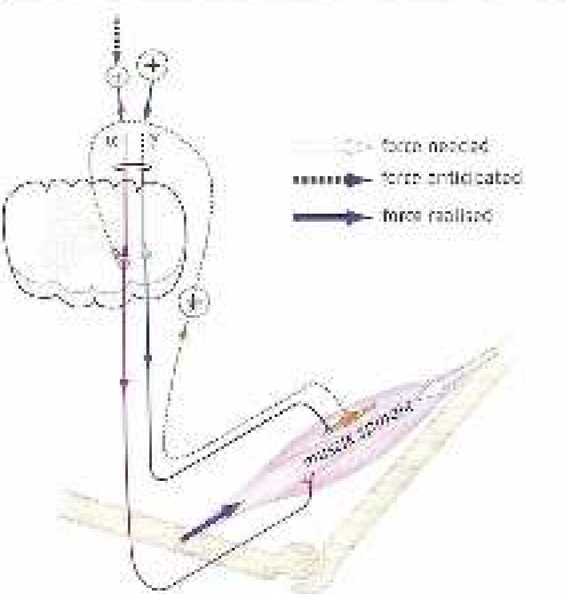


Figure 6.22 There is far too little alpha activity because the force to be produced has been grossly underestimated. Additional force is produced via the gamma loop. However, the correction is insufficient, so that the eventual level of force production remains too low.

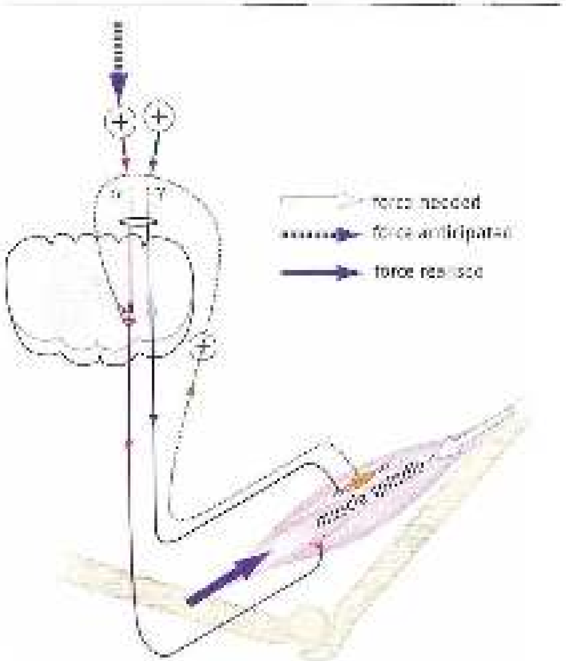


Figure 6.23 There is slightly too little alpha activity because the force to be produced has been slightly underestimated. Additional force is produced via the gamma loop, so that enough force is eventually produced.

Figure 6.24 Top: a clean with the knees extending too fast. The level of the lateral force is too great in relation to the hip, and the clean is unsuccessful (especially with heavy weights).
Bottom: a clean in which knee and hip extension are well coordinated.

Conclusion

If muscles are properly rested, it will usually not be too difficult to produce an alpha signal within the required bandwidth. However, if one or more muscle groups are fatigued, the alpha-gamma link will come under pressure, and the alertness of the system will need to be optimized in all the cooperating muscles. The signals will have to be coordinated more effectively. There is almost always a fatigue problem with large force production. Fatigue may therefore be an inherent part of learning complex motor patterns with great force production. This powerful learning aid can be used by manipulating fatigue. Confronting the organism with a different problem each time will provide a stimulus that enhances the alpha-gamma link. If the alpha and gamma signals are well coordinated, control will be more or less insensitive to perturbations, and hence to fatigue.

6.4.5 *Variation in young and inexperienced athletes*

The three types of variation generated by the constraints-led approach can of course be used either separately or in combination. Variation is the main way to achieve the almost invisible transfer to sporting movement. Which type of variation is most efficient in which situation will depend on very many different factors. Coaches cannot answer this question simply by applying knowledge – part of the answer will lie in an intuitive sense of how things work.

This is particularly true when working with inexperienced or young athletes who have yet to develop their abilities. For years, training theory was dominated by a highly reductionist approach to the parameters that influence performance and improvement in performance. This reduction of key factors appeared to make the training process controllable. However, it is now generally accepted that elite athletes' performance is not determined by a limited number of parameters. The various parameters that play a part in performance can scarcely be identified. What may be more important to performance than the mere sum total of the various qualities is the way they interact. Something that tends to be forgotten even more is how training of inexperienced young athletes may affect their long-term development. This makes it very difficult to design a development programme for any given sport simply on the basis of knowledge of adaptations. There are therefore more questions than answers when it comes to athletes' long-term development, or how broad a range of exercises should be offered to young athletes in order to optimize their abilities later on. If the basis is too narrow, development will stop earlier than necessary; if it is too broad, too much time may be spent on movement patterns that make no contribution to the eventual level of performance. Would it help a young judoka to learn various movement patterns (including types of strength exercise) based on pretensioning? Probably, for a judoka who is best at reducing muscle slack when commencing a throw has an advantage. Would learning to skate like an ice hockey player help improve his balance? That might be useful for a young tennis player, but what about a young judoka? It seems unlikely that gymnastics could help in judo, and swimming would quite certainly be a waste of time.

Training variation is always useful for young athletes. There are two key factors here:

1 Targeted training can change certain body qualities, such as muscle power, metabolism, mobility, load capacity of tendons and so on. A number of performance-developing physical properties can be identified for each sport and specifically worked on. To some extent the body can be altered. Proper identification of young athletes' potential and shortcomings is needed in order to devise an effective long-term development plan. Such plans always call for a broader approach than that required for short-term progress. This means that young athletes require a multifaceted, variable training programme, even at the expense of some specificity in relation to their particular sport. In practice we find that good juniors by no means always excel as adult athletes. Often this is because one or more facets of physical fitness have not been sufficiently developed during their early training, and this inhibits their progress later on. This inhibiting effect is not found in the short term. Coaches who interpret young athletes' short-term successes as a sign of good long-term development are therefore making a fundamental error.

2 Good motor control is based on a system of generally applicable regulating mechanisms that can be used to ensure stable yet flexible performance of movements. Such mechanisms, the building blocks of coordination, must be learned at an early age. Another reason why this is useful is that young athletes can very easily learn and improve the building blocks of movement, whereas physiological adaptations occur far less easily than when they are older. Sensitivity to training stimuli in improving the building blocks of movement indicates that movement technique can be altered. Poor technique in young athletes (such as failure to control muscle slack or to distribute abductors and extensors properly) is all too often labelled as 'personal style' that can, and should, be

left largely uncorrected. Targeted work on the building blocks of coordination at an early age by providing plenty of variation in training can have a decisive impact on the quality of technique later on.

Variation must therefore be provided at an early age, so that the athlete learns the building blocks of movement properly and can design the physical properties of his body so that later on there is no unnecessary limitation of performance and the learning system remains interested in learning. However, such variation should be provided with a view to successfully performing the movement or achieving a goal that has preferably been precisely defined. In the absence of such goals, variation may serve no purpose.

6.5 Summary

Strength training nearly always takes the form of part practice, so its specificity is limited. However, this creates an opportunity to provide overload. Overload is created by providing training stimuli that the organism is not yet equipped for, and that require adaptations. It is classically defined as a stimulus that involves a *greater* load than momentary load capacity can provide. 'Greater' implies a quantitative measure of load that can easily be linked to the supercompensation model, or single-factor model. Yet reality is more complex than the mechanism of exhaustion followed by recovery to above the initial level that this model would suggest. According to the single-factor model, many parameters that affect performance do not adapt. A two-factor model that distinguishes between acquired fitness and additional fatigue is therefore useful. In this model, overload can be measured in qualitative terms – a *different* load from momentary load capacity.

Measuring overload in qualitative terms is suitable for approaching strength training in coordinative terms. Providing overload then becomes more a question of variation – creating a sensorimotor package that the organism is not yet familiar with. This is particularly important when seeking transfer to a coordinatively more complex sporting movement. Such transfer has its own dynamics that are separate from the increase in force production in a strength exercise.

The central/peripheral model indicates the relationship between specificity and overload. High specificity rules out large overload, and vice versa. Even types of training that appear to combine large overload with high specificity, such as running uphill and downhill, turn out on closer inspection to be no different. There are no 'magic' exercises. Depending on the situation, coaches must develop strategies for selecting exercises from the central/peripheral continuum, with the focus on individualization of the training design.

To determine whether a strength exercise can provide enough quantitative overload, the associated sporting movement must be carefully analysed. In running and jumping sports, for instance, barbell training cannot be used to create overload for every joint. If abdominals are elastically loaded in the sporting movement, quantitative overload within strength training can only be achieved by elastic loading. The muscular pain that can be caused by training, say, calf muscles and abdominals should be seen not as quantitative overload, but as a sign that the muscles are not being used properly.

Strength training protocols are usually described in terms of classical physics. However, transfer takes place according to laws of motor learning, so such protocols cannot be used to identify adaptations in the sporting movement. Besides a quantitative approach, a more qualitative one is needed in order to grasp transfer processes.

The constraints-led approach brings together hierarchical schema theory, direct perception theory and dynamic patterns theory. It does so by approaching the development of movements in terms of the constraints that the task, the environment and the organism impose on performance of the movement. Such thinking in terms of constraints imposed by the task, the environment and the organism is an appropriate way to design a system for identifying qualitative overload in strength exercises. Each of the three components can be varied to create ever-new sensorimotor patterns in exercises, and hence provide learning stimuli.

Varying the environment traditionally involves varying strength exercise resistance. Yet there are other ways to do this, as long as the movement is contextual, sufficiently similar to the key components of the sporting movement, and safe.

Varying the task usually involves making special demands on the way the exercise is performed; this always has the drawback that attention may be directed inwards. The task can also be varied by performing it in random-practice exercises, for example by alternating it with highly contrasting exercises.

Varying the organism is still a largely unresearched area of motor learning. The organism can be rapidly altered by creating fatigue. Fatigue can develop very quickly within strength training, and this makes strength training particularly suitable for differential learning through fatigue. A key advantage of using fatigue is that the task and the rules for performing the movement correctly can be left unchanged. This makes specificity easy to control. Strategic use of fatigue creates new, unfamiliar sensory links that cause chaos and hence encourage learning. Fatigue may be a powerful tool for learning, for instance by finding a better link between the alpha and the gamma signal that is resistant to fatigue and hence allows more generally applicable control of movement. Local fatigue, for instance in one arm or one leg, can also be used to directly improve components of the sporting movement.

7
Sport-specific strength training in practice

7.1 Body-part and contextual approaches to strength training

7.1.1 *Practical criteria for good sport-specific strength training*

Sport-specific strength training should be related to all other components of training in support of sports performance. This means it should have two basic features:

1. It should contribute to the level of performance in the sporting movement.
2. It should cause the athlete as little physical stress as possible.

Sport-specific strength training should be both effective and efficient. The first criterion is largely met if the training is carried out in accordance with the laws of motor learning and motor control; the second if 'collateral damage' is kept to a minimum. It is an illusion to think that the effects of strength exercises will all enhance performance. Every strength exercise creates loads that are not relevant to execution of the sporting movement, and this additional stress is often actually damaging to performance. Stressors that do not enhance performance include axial load on the spinal column when carrying a weighted barbell, eccentric load on the muscles (and the resulting muscle damage) when the barbell is repositioned after the exercise, strain on joints and so on. Apart from these orthopaedic stressors, there are also coordinative ones that do not contribute to performance and may have a negative impact. For instance, there is much debate about the pointless stress created when catching the barbell weight in the clean. What makes this debate so complicated is that omitting this (by performing a 'high pull') would detract from the intention of the exercise. The initial posture in the clean is also a matter of debate. Many athletics coaches advise against starting the movement with the barbell on the ground, for such a posture, with highly flexed knees, is not found anywhere in athletics. That is why the clean is usually performed from a box. In the case of athletics, this makes obvious sense. What is interesting is how the same argument is applied to speed skating. The knee angle at the start of the push-off is deeper than in athletics and is a crucial component of skating technique. Sitting too low is no good, but nor is sitting too high. So should skating strength coaches pay close attention to the knee angle at the start of the clean?

The transferable and non-transferable adaptations that result from strength training should therefore be analysed, and exercises with a minimum of non-transferable components should be looked for. The transferable exercises should be further analysed for adaptations that result in positive and negative transfer to the sporting movement, and movements should be chosen with a maximum of components that produce positive transfer to the sporting movement.

7.1.2 *Methodical or adaptive classification of strength*

There are two main lines of thinking in standard strength-training practice:

1 The (methodical) body-part approach, which focuses on physiological aspects of strength training.
2 Contextual strength training, which focuses on coordinative adaptation.

The body-part method

Influences from bodybuilding play a major part in the body-part method. Training is focused on individual muscle groups, especially the ones that are also heavily loaded in the sporting movement. Further integration of movement patterns (the step from part practice to whole practice) is not pursued within strength training, so only simple movement patterns occur there. Within these simple patterns, the limits of physiological and other aspects of fitness are sought in order to maximize the training effect. The body-part approach is therefore marked by a high amount and level of loading.

In this approach, strength training is carried out with very heavy weights. Within overall sport-specific training, strength training based on the body-part approach accounts for a large proportion of the overall training stress. The overall training stress must therefore be managed carefully to ensure that total load does not become so excessive as to cause injury and overtraining. It is extremely difficult to find the right limit, for it is very hard to quantify the various types of stimuli within a single block of training. How do you add up running training, strength training and tactical training?

For this and other reasons, body-part strength training is now carefully focused on delayed and residual training effects, for example in the work of Verkhoshansky, Issurin and Counsilman (see Counsilman & Counsilman, 1991). What this means for strength training is that the training effect does not occur immediately but only after some time, and then persists for some time without further strength training (Figure 7.1).

The delayed training effect phenomenon is the main reason for the emergence of periodization models based on blocks of training. Training is then arranged to focus on varying aspects of training at particular times of the year. The training schedule includes strength-training blocks timed so that the delayed training effect coincides precisely with the most favourable moment for its application, e.g. during the competition period.

The mechanism behind the delayed training effect is not yet fully understood. The system may need more time because 'adaptation energy' is limited and it therefore takes longer for the strength parameters to be adapted. A simpler explanation is that the extensive strength training required in the body-part approach causes great cumulative fatigue, which must first wear off before the positive effects of strength training become apparent.

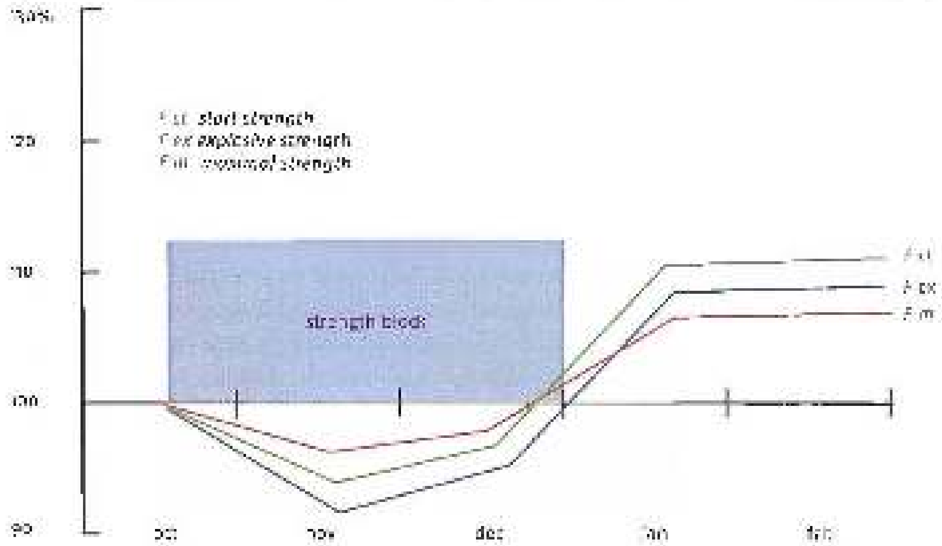


Figure 7.1 Planning a strength-training block on the basis of the delayed training effect (Verkhoshansky). Strength training in such methods is usually based on quantitative overload.

If the delayed training effect of strength training could mainly be ascribed to increasing fatigue, this would mean that in a different approach to strength training – the contextual approach – the delayed training effect would be far less evident, owing to the smaller amount of training. The training effect would then occur far sooner after the training sessions. Evidence of this can be seen in the training practice now currently adopted by many leading coaches. In contextual strength training, which places less stress on the athlete, organization into blocks is no longer needed, for the fatigue that occurs is not too great or counterproductive. Strength training is therefore maintained throughout most of the year, even until just before the competition period. It therefore seems that the delayed training effect is above all due to cumulative fatigue.

In the body part approach it is assumed that strength is a more or less separate entity and hence is more or less independent of the context in which it is applied. It is therefore assumed that transfer from the strength exercise to the sporting movement takes place without having to satisfy too many conditions. In this way of thinking, transfer is not such a great problem, and so it is not the main focus when designing strength training. The problem that must be solved within strength training is optimal improvement of the strength values in the strength exercises, because in this way of thinking increased strength values are the most efficient way to improve performance in the sporting movement.

Thinking in terms of basic motor properties is basically reductionist. Complex biological systems (dynamic patterns) theory shows that such reductionism is incompatible with the mechanisms that are really involved. It is therefore wrong to assume automatic transfer between strength training and the sporting movement. In inexperienced athletes there will always be an improvement in the sporting movement, because almost any training will help them improve. But in well-trained competition athletes transfer will only occur under certain conditions which, among other things, are part of the central/peripheral model. These conditions must therefore be taken into account. The organization of movement patterns in a highly developed athlete, and hence the interrelationships and transfer between the various patterns, depends on such a subtle matrix of rules that the boundary between positive and negative transfer is soon crossed. We have already seen how strength training adversely affects control of muscle slack. In elite athletes the positive effect of an improvement in force production can easily be cancelled out by the increased muscle slack that occurs as a side effect of strength training. In elite athletes the exact tipping point between the positive and negative effects of strength training must therefore be sought. The grey area between enough and too much strength training gets smaller as the athlete's ability increases, and the tipping point is different for each individual.

Besides the criticism that the body-part approach neglects specificity and mechanisms of transfer, there are objections to the way in which overload occurs. In body-part training, the catalogue of exercises is always limited, especially in sensorimotor terms. Variation, as a key condition for adaptation, will mainly have to be sought in changes to the resistance used. This means that strength training is coordinatively monotonous and one-sided, which inevitably means that adaptations are reduced as monotony increases. Greater stressors (i.e. resistances and weights) will therefore have to be used to achieve adaptations, and this usually has an adverse impact on the already limited specificity of the exercises.

Contextual strength training

Contextual, coordinative strength training does not presuppose automatic transfer from the strength exercise to the sporting movement, because optimal and positive transfer depends on numerous conditions being met. The most important of these is specificity. According to this approach, the transfer and hence contextuality of the movement should be the central problem when thinking about and designing sport-specific strength training. As a result, strength exercises are strongly focused on improving intermuscular coordination, preferably in overall movement patterns, and preferably with the same solutions to movement problems as when executing the sporting movement. This basically amounts to reducing and learning to control degrees of freedom. In contextual training athletes therefore train as little as possible in situations in which degrees of freedom have already been eliminated, e.g. when training on weight machines. The only built-in restriction on degrees of freedom is in the interests of health and safety, which is mainly important when training with heavy weights.

Learning to control degrees of freedom depends on variation in exercises. Contextual strength training therefore involves a large number of different exercises, with overload mainly sought in variation.

Opponents of coordinative strength training tend to claim, as an argument for rejecting contextual training, that high specificity is hard to achieve especially in explosive sports, because the speed of movement within strength training is always lower than in the sporting movement. However, there is little conclusive evidence to support this idea. Moreover, in motor learning theory (see, for example, Schmidt's concept of variant parameters of generalized motor programmes: Schmidt & Wrisberg, 2005); there are suggestions that speed of performance can vary within limits without an excessive decrease in specificity between executions of the movement. Speed of movement is then only a serious problem in strength exercises that call for elastic muscle action. If movement patterns are too slow in elastic patterns, elasticity is no longer possible and execution of the movement does indeed become unspecific (which is why strength exercises with an elastic effect are always performed with low resistance).

However, this does not mean that body-part strength exercises should be avoided in sport-specific training. The use of part practice alongside whole practice can usually result in an efficient training approach. In particular, training muscle groups that are suitable for maximal strength training can have a useful effect, such as training the hamstrings for running. Once again, the right balance between the two concepts will vary considerably from individual to individual, and finding that balance is part of what makes a good coach – 'coaching is an art' (Figure 7.2).

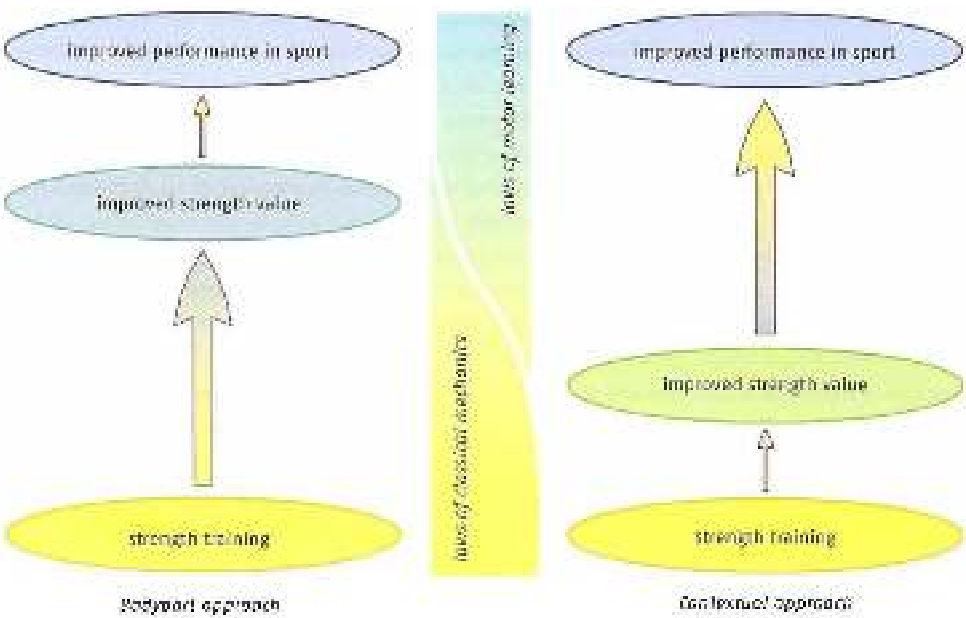


Figure 7.2 The body-part approach focuses on increasing strength values. How these contribute to a better sports performance is seen as a less relevant, or quite simply unsolvable, problem. In contrast, contextual strength training focuses on the issue of transfer, and assumes that this is the real problem in sport-specific strength training. The body-part approach mainly focuses on the laws of classic mechanics; contextual strength training mainly focuses on the laws of motor learning.

7.1.3 *Terminology used in methodical thinking*

Because the traditional body-part approach assumes that mechanical properties of muscle action are automatically transferred regardless of the sensorimotor and intentional context, strength training has become very methodical. This has distracted attention from the content of the movement patterns that are practised within strength training, and has shifted it to the organization of the exercises. Training protocols describe the number of repetitions, the number of sets, the rest periods between exercises, the weight to be lifted each time and so on, but rarely describe the adaptations, and above all coordinative adaptations, that are required. But if transfer is not automatic, organization should not be based on methodical aspects but on the coordinative and other adaptations that occur – in other words, *on the main problem to be solved*. When designing a training system, we should therefore start by asking which coordinative adaptations occur, and how they can be optimized in the exercises.

Because traditional strength training has focused on method, its terminology is not very precise in describing its effects. The terms used in classifying the various types of strength training should be as one-dimensional as possible. They should unambiguously indicate which adaptations can be expected as a result of the training. Terms such as 'a-cyclic' and 'cyclic' starting power, 'jump power', 'general power' and 'specific power', 'explosive power' and 'power irritation' do not describe precise categories of movement, and do not clearly indicate which adaptations will occur. These are umbrella terms in which numerous poorly defined effects of training can be included – it is not clear which components are part of them and which are not. The resulting training design may look very goal-oriented and systematic, but it is not, and stressors that make no contribution to the intended adaptations cannot be ruled out.

To give strength training a meaningful place in the overall training plan, it is therefore important to define the contextuality of the exercises, in order to understand how strength training can improve execution of the sporting movement. A number of analyses will then need to be made – starting with a thorough analysis of the sporting movement, which depends on knowledge of anatomy, biomechanics and motor control, as well as insight into which adaptations occur as a result of strength training. With some types of strength training the adaptations are quite easy to identify. Maximal strength training teaches athletes to make simultaneous use of more motor units ('recruitment'). Following hypertrophy training the adaptation mainly occurs in the muscle fibres; extra protein production increases the size of the muscles. Within other types of strength training it is less clear which adaptations actually occur. For example, when training with light weights, in which the speed of execution is high, it is difficult to describe the neural adaptation. Classifying types of strength training according to adaptations allows more systematic identification of how strength training contributes to the sporting movement. Finally, it is important to analyse the similarity between the sporting movement and the adaptations resulting from strength training. Such analyses can be used to determine which types of strength training make a useful contribution, and which should preferably be avoided. Unambiguous terminology is a key step in this process.

Strength endurance

'Strength endurance' is a frequently used term in sport-specific strength training. It refers to the athlete's ability to sustain, or repeat, high force production over a long period. It would primarily refer to the application of types of training in which adaptations can result in increased generic force production and longer generic endurance. However, both effects cannot be achieved with a single type of training — such a goal is 'schizophrenic'. High force production requires training with large resistance, which can only be repeated a few times. Improving endurance requires numerous repetitions, and hence low resistance. Generic improvement of both force production and endurance in a single exercise is therefore out of the question.

Besides this more practical problem there is the more theoretical problem that strength endurance can only be a workable strategy if there is more or less automatic transfer between all the various movement patterns in terms sustaining high force production. Training of strength endurance in the gym should then lead to improvement of that property in numerous other movements, such as the sporting movement. There is no reason to assume that such automatic transfer will occur. Strength endurance cannot escape the laws of coordinative similarity as a condition for transfer, and hence must meet the criteria of specificity and overload in order to achieve the intended transfer. This means that the concept of strength endurance is only workable in controlled efforts to achieve training adaptation if it is not generic, i.e. if strength endurance training is situation-dependent — in other words, if the concept of strength endurance is movement-specific. Here again, this particularly applies to training of elite athletes with a long training history, who find it far harder to improve their performance than inexperienced and developing athletes. This means that endurance should be trained in movement patterns that are similar to the sporting movement and that the resistance should not be very different from the resistance in the sporting movement. There is thus no justification for the use of the term 'strength endurance' as an independent mechanism within strength training. The point of using slightly increased resistance in a movement pattern that is close to the sporting movement is therefore much closer to technique training in conditions of fatigue (which in turn raises numerous questions and conclusions).

Application

In running, swimming, rowing and numerous other sports, the sporting movement is sometimes trained against resistance. This is done by running with a sled, and in swimming or rowing with an object that is pulled through the water. In the traditional approach such exercises are often referred to as 'specific strength training' or 'strength endurance training'. If they really did have such training effects, one would expect coaches to start with a low resistance and then gradually switch to higher and higher ones. In practice, however, coaches who see such exercises as strength training do not do this. Instead, they always reduce resistance from high to low, because they know from experience that this works better. This can be explained by seeing such types of training not as strength training but as coordination training, in which there is no longer any difference in basic motor properties. Running, swimming and rowing with additional resistance is then a way to simplify technique and so achieve a better learning effect. The additional resistance means that the movements are executed more slowly; this makes the athlete more aware of the horizontal component of force production in the stance phase when running, or the pressure on the water when swimming and rowing, so that the movement can be executed more effectively. It then makes more sense to shift from a higher to a lower resistance, because the exercises then shift from difficult to easy. This brings theory and practice more into line.

Figure 7.9
Running with a sled: strength training or technique training?

Explosive power

One of the magic terms in classic strength training is 'explosive power' (Hamann et al., 2016), which means training with light weights. When training with light weights, the speed of execution is as high as possible. The training focuses on the part of the force-velocity curve in which the power produced is the result of the link between low force production and high speed of muscle action. The term can be analysed in the same way as 'strength endurance'. The first question is whether generic speed and generic strength can be trained simultaneously (assuming there are such things as generic speed and generic strength). Once again, the answer is no. Increasing generic speed means using low resistance; increasing generic strength means using high resistance. In the force-velocity curve, that would mean being in two places at once.

Traditional theories claim that explosive power training bridges the gap between speed training and strength training. The function of training with light weights is therefore sought in the supposed transfer between heavy strength training and the sporting movement. This very much assumes that basic motor properties are independent entities. By using low barbell weights, the improved force production as a result of the training with heavy barbells is linked, as it were, to the basic motor property of speed – in other words, explosive power training can supposedly incorporate the effects of heavy training into the sporting movement.

So the second question is whether explosive power can escape the laws of specificity and the properties can simply be transferred between unrelated forms of movement. Yet again, the answer is no. This means that explosive power is only workable in a movement-specific context, and that the claim that it could bridge the gap between general strength values and speed – i.e. convert strength into speed – is unrealistic. This is, moreover, backed up by insights from research into motor learning, which indicate that transfer between part practice and whole practice may be greatly overestimated. Just like the concept of strength endurance, the concept of explosive power is thus not an independently workable entity in training theory, and the exercises from both categories must be assessed in terms of specificity to determine how they help to improve the sporting movement.

That 'explosive power' is not an independent component in overall force production by muscles is also evident from the fact that there is no clear definition of the term in the literature. In the German literature, explosive power is even dismissed as a 'verbal mixture of various components' (speed, maximal strength technique and willpower: [illegible], 1979).

7.2 Division of strength training based on the adaptations that occur

When planning sport-specific strength training it is useful to think in terms of expected adaptations. In technically complex sports the focus is on contextual, coordinative adaptations. The adaptations discussed here are related to the aspects of force production examined in Chapter 2. This can then serve as a basis for classifying the exercises in terms of specificity later in this chapter.

7.2.1 *Hypertrophy*

Hypertrophy training originates in bodybuilding, with above all the body-part approach being adopted in current sport-specific strength training. The focus in bodybuilding is on myofibrillar hypertrophy. The methods used are designed to increase muscle mass. Increased muscle mass in myofibrillar hypertrophy can be accounted for by an increase in the number of myofibrils, which means an increase in actin and myosin filaments. This allows more cross-bridges to be formed, and the number of parallel sarcomeres increases. Both effects increase the maximal strength of a muscle fibre. The number of myofibrils is increased making the muscle do as much mechanical work as possible. This means that hypertrophy exercises must be performed with large ranges of motion, because doing work (through concentric muscle action) requires far more energy than producing force (via isometric muscle action; see Section 4.1.7). The more energy is required for mechanical work, the less is available for protein synthesis. To protect the system against future shortages when work has to be done, the body will produce myofibrils over and above the original level in the recovery phase after training. The muscle will become thicker.

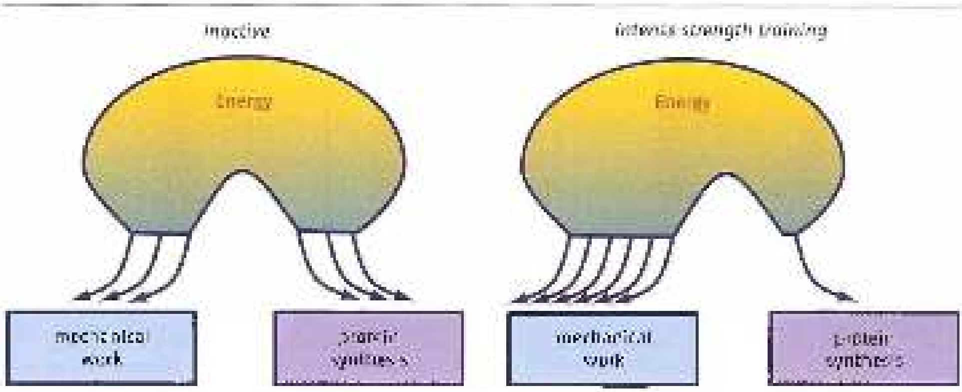


Figure [illegible]: The energy available for anabolic processes at rest and under load. In the rest situation more energy is available in the muscle cell for anabolic processes, such as protein synthesis or building up muscles.

Hypertrophy can occur in both Type I (slow-twitch, ST) and Type II (fast-twitch, FT) muscle fibres. In types of training with a limited number of repetitions and great resistance, the greatest changes are observed in FT muscle fibres. Of the FT muscle fibres, the oxidative FT fibres display the most hypertrophy at submaximal resistance. Near-maximal resistance with one or two repetitions only creates hypertrophy in glycolytic FT muscle fibres. ST muscles fibres are fatigue-resistant, and hence hard to exhaust. This makes it very difficult to create hypertrophy in such fibres (Zatsiorsky, 1995). The endurance of motor units with Type I, IIa and IIb muscle fibres may range from one to a hundred seconds. With twelve repetitions at submaximal load, performed once a second, the various types of motor unit are recruited and exhausted in a fixed pattern. In the first repetition, motor units with Type I and a number of Type II muscle fibres are recruited. After six repetitions the recruited motor units with an endurance of less than six seconds are exhausted (Type II fibres). As various motor units become exhausted, new ones must be recruited to generate the required force. The newly recruited motor units are Type II muscle fibres (according to the size principle), which are also exhausted after the last repetition. This makes it clear that hypertrophy is hard to achieve in Type I muscle fibres, since they seem insusceptible to fatigue. The following method is often used for this purpose. After the last repetition at submaximal load, when all the FT fibres are exhausted, several more repetitions are performed with a lighter load; this further exhausts the ST fibres, which are the only ones left to recruit.

This process of exhausting muscle fibres with maximal or submaximal loads shows that muscle fibres are not exhausted at random, but along corridors in the continuum from Type I to Type II fibres (Figure 7.5).

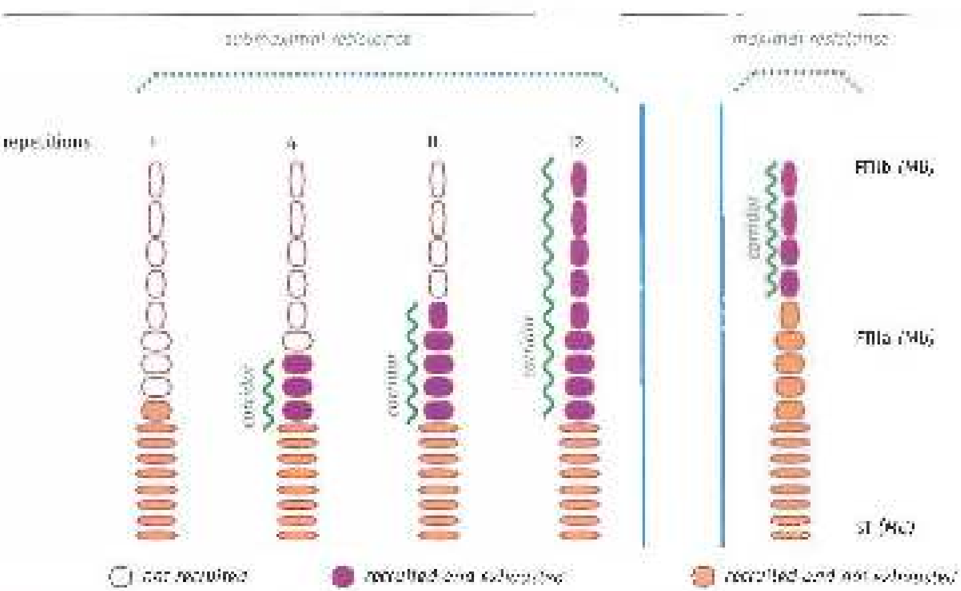


Figure 7.5 Recruitment of different types of motor unit with submaximal and maximal loads. Shows the recruitment of types of motor unit at submaximal and maximal weight. At submaximal weight, the recruitment and the degree of exhaustion of motor units consisting of slow-twitch or fast-twitch muscle fibres are determined after one, four, eight and twelve repetitions. The recruited and subsequently exhausted motor units are known as the 'corridor'. At maximal load only the largest recruited muscle fibres are exhausted.

Muscle hypertrophy will be greatest if the corridor of muscle exhaustion is maximized. Strength exercises with a submaximal weight such that exhaustion occurs after eight to sixteen repetitions creates the largest quantity of exhausted muscle fibres. That is why training with eight to sixteen repetitions to the point of exhaustion is known as the 'hypertrophy-sensitive zone'.

Hypertrophy training is going out of fashion in sport-specific training. Hypertrophy that is aggressively imposed by strength training has an adverse impact on coordination and eventually impairs performance (see Section 5.5.4). This particularly applies to movements that are very much designed by the self-organizing ability of cocontractions, such as throwing. Deterioration in these essential components of coordination as a result of hypertrophy training can be explained by the fact that hypertrophy training changes the body 'roughly', but quickly and effectively. Since the changed body no longer fits into the sensorimotor matrix created in the past, interaction between the body and the environment (contextual coordination) will be less efficient.

Sometimes a strategy is applied in which the athlete first does hypertrophy training and then attempts to improve intramuscular and intermuscular coordination. This roundabout approach will not lead to optimal improvement of performance either, and is used less and less in practice. Hypertrophy training as a key part of sport-specific strength training and hypertrophy training as a lead-up to improvement of overall patterns are routes that should probably be avoided in training.

7.2.2 *Maximal strength*

The maximal amount of force an athlete can produce can be limited in several ways. The limit can be sought in anatomical properties such as muscle structure (cross-section, force/length and force/velocity relationship and so on), neuromuscular transition, excitation and inhibition processes at spinal cord level and the brain (see Chapter 2). It can also be sought in the fundamental way in which contextual movement patterns arise under the influence of cocontractions and muscle slack (see Section 4.3). It is not known which factor is the actual limit on maximal force production – indeed, this may vary from situation to situation. What is clear is that maximal force production is highly dependent on the situation (see Section 5.2.1). This specificity of maximal strength is a key protective mechanism for the system, and without training the maximal voluntary contraction that a muscle can produce will therefore not exceed about 75% of the available muscle fibres (see Chapter 2).

This percentage can be increased by training. The standard method is of course strength training with heavy barbell weights, but the limit can also be raised in other ways. There are indications that, in basic contextual patterns such as running and jumping, muscles can produce greater forces than they can achieve with regular strength training. This puts the function of strength training in a different light. For calf muscles it is clear that barbell training does little to increase force production in athletes who already run and jump in their particular sport. When running, calf muscles absorb external forces of up to four times the athlete's body weight; in jumping sports, the figure is much higher still. Barbell training cannot possibly create even greater external forces (see Section 6.2.1). The same may be true of hamstrings. Force production in high-speed running may be higher than what can be achieved in the gym. Strength training may then seem pointless for this muscle group. However, it should be realized that high-speed running does not automatically lead to optimal development of the hamstrings. This is evident from practical experience – for instance in Australian rules football, in which the players do a lot of running, often at high speed, without this guaranteeing strong hamstrings. For instance, if a runner rotates his pelvis too far forwards when pushing off from the ground, the pelvis will have to rotate back again in the next phase, when the hamstring in the leading leg is eccentrically loaded, and this will counteract the loading of the hamstrings (Figure 7.6). As a result of this common error, the hamstrings will never be properly loaded when running, and top speed will be lower. Mechanisms that may keep the hamstrings too weak may therefore occur when running. Besides technique training to control pelvic rotation, improvement of maximal hamstring strength is in practice one of the most effective ways to correct such errors. The muscle is then, as it were, reactivated and encouraged to become active in the running cycle.

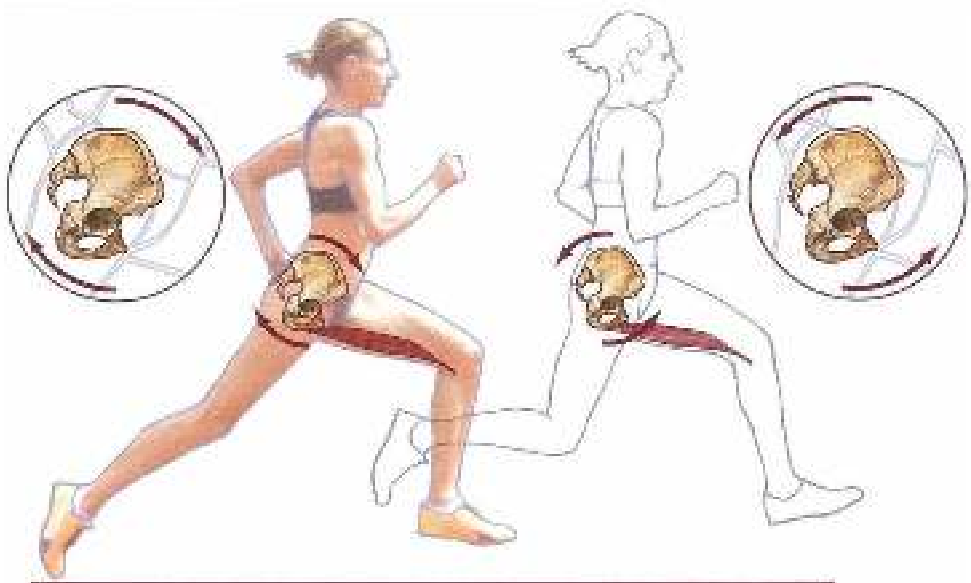


Figure 7.6 If the stance phase takes too long and the stance leg therefore remains a long way behind (left), the pelvis will have to rotate a long way forward. In the next flight phase (right), it will then have to rotate back again to its neutral position.

The hamstring example shows that it is useful to approach maximal strength training from more angles than just quantitative overload (greater loading than in the sporting movement). It also shows the importance of striking a balance between improvement of force production in the sporting movement and in the strength exercise. Strength training then serves to get force production in the more complex patterns of the sporting movement 'under way'. That is why experienced coaches are right to describe strength levels in the gym as 'good enough'. Instead of pushing for higher and higher maximal strength levels, they are satisfied with a reasonable level, for this may ensure that force production will develop further when executing the sporting movement. A good example is the standard for the single-leg Roman chair hamstring exercise (see Figure 5.17), with an extra 60% of the athlete's body weight to prevent injury, and an extra 90% of it to achieve good sprinting speeds.

Improved maximal strength is the result of better neural control. Training with a barbell weight so heavy that the exercise can only be repeated once to five times will increase the percentage of muscle fibres that can be recruited. The movements are controlled and executed slowly to ensure as strong an isometric muscle action as possible. There are two key principles here:

1 The change in posture because of the slowly executed movement serves to increase the lever of the external load in relation to the intended joints, so that the torque of the external load reaches the maximal torque that the muscles can produce in the opposite direction relative to the joint (Figure 7.7). In a back squat, the torque of the external load increases relative to the knee until the *quadriceps* reach their maximal torque for extension. In a single-leg good morning, the torque of the external load increases relative to the hip joint until the hip extensors reach their maximal torque. When working in chains, it is hard to predict where in the body the maximal possible force production of the resisting muscles will first be reached: when performing a double-leg squatting movement, will it be in the knee extensors, the hip extensors or perhaps *erector spinae*? Being able to analyse where muscles will reach their limiting force production is a basic skill for strength coaches.

Figure 7.7 The torque of a load relative to a joint depends not only the weight of the load, but also on the lever relative to the joint.

2 In most movement patterns in sport there are so many degrees of freedom of movement that a body posture can be sought in which the main muscles that are loaded with a large opposing torque can act close to their optimal length. It is only in exceptional cases that this does this not occur. In a bobsleigh push-off, for example, the athlete has so many different ways of varying his body posture that the torque in muscles can be optimized. One of the few movement patterns in which this may not be possible is a

skeleton push-off. The handle is no more than twenty centimetres above the ground and must be held during the push-off. This means the muscles may have to act beyond their optimal length (so a good starter in athletic sprint events is not automatically a good starter in skeleton).

To enhance this self-organized optimization of the force/length relationships of muscles, it is useful to build degrees of freedom of movement (wherever possible) into maximal strength exercises that are performed in kinetic chains. In exercises in which this is not possible, such as bench pressing or squats, the coach must attempt to estimate the muscle length at which maximal force production is required. If this is at a length that is irrelevant to application in the sporting movement – for instance if *pectoralis major* is at an excessive irrelevant length when the bar touches the chest during bench pressing – consideration can be given to changing the exercise, in this case for example by placing a 10-cm block on the chest. The barbell will then not descend so far, and the muscle will be loaded closer to its optimal length, allowing a heavier barbell weight to be used. The same applies to squats. Deeper does not necessarily mean better. The barbell load should therefore be chosen so that the movement is restricted to a relevant knee and hip angle, with the required monoarticular muscles at a length that is relevant to the sporting movement.

There are not so very many sports with static postures and great force production. One of the few requiring static maximal strength is gymnastic ring exercises. Yet producing great force in isometric conditions is a key aspect of movement, especially in sporting movements involving elastic muscle action. The opposing torque of external forces can only be optimally stored in elastic stretch if the contractile parts of the muscle do not yield, i.e. if they remain in isometric condition. In sporting movements the peak forces associated with elastic muscle action are often very great, whereas the time between loading and unloading of elastic energy is very short. Throwing, running and bouncing are therefore based on elastic muscle action; the start and turn in swimming, and footwork in the shot put are not. Since elastic muscle action is optimized by better recruitment of the contractile elements, maximal strength training is particularly suitable for creating conditions for improving elastic movements.

The great advantage of training maximal strength is that it mainly focuses on central nervous system processes and hence creates less stress on muscle fibres. In advanced athletes, maximal strength training creates little or no hypertrophy, and recovery time after training is short compared with hypertrophy and power training (see also Section 7.2.5). This means that strength training can easily be fitted into the overall training plan. However, given the great external resistance that is used, close attention must be paid to the orthopaedic stress that occurs, especially the axial load on the spinal column. Carrying heavy barbell weights on the shoulders requires good barbell technique, and a rigid trunk through good cocontractions of the muscles involved.

7.2.3 *Power*

The force/velocity curve for muscles shows that force production and rapid shortening are incompatible. A muscle that has to shorten quickly cannot do so with much force. Producing power (force × velocity) is therefore a problem for muscles, and requires a fair amount of energy. Another problem here is that ability to produce power is movement-specific. The following needs to be taken into account as regards specificity of power production between the strength exercise and the sporting movement (see Section 5.2.2):

- In some sporting movements power (in particular) must be produced at a given speed of movement. In other sports the speed of movement varies so much that none predominates. This makes it difficult to determine the right link between force and velocity in power training.
- In most sporting movements, power is produced in complex movement patterns with cooperation between power-producing and energy-transporting muscles (Figure 7.8). This makes it hard to identify the limiting factor for power production.

Figure 7.8 Skeleton start. Not only do the power-producing muscles have to act well beyond their optimal length in a very unusual pattern of intermuscular cooperation, but they also have to produce power at a higher and higher speed of muscle action as the push-off proceeds. Power training is therefore a problem in this sport.

- The rate of force development is greatly influenced by muscle slack. Muscle slack may be substantially altered by the barbell load, perhaps even resulting in negative transfer between the strength exercise and the sporting movement.
- Deceleration of the movement is movement-specific. Like muscle slack, the barbell weight may facilitate this.
- Another aspect of specificity that affects the specificity of power production, in addition to these mechanical factors, is similarity in intention of movement. In power training this can be translated into discrete movements (with a clear beginning and end) according to similarity in the final posture of the movement. It is preferable here to translate the final posture into a spatial target, such as ending with the dumbbell that is to be accelerated upwards touching a suspended target. The further outside the body the target is, the more natural the intentional link will be. When designing power exercises, it is therefore worth choosing an exact final posture that matches the final posture in the sporting movement. In the interests of variable or differential learning, it is also useful to choose several initial postures from which to arrive at the same highly specific final posture. Linking a variable initial posture to an exact final posture creates a powerful learning situation that fosters self-organization of movement patterns and eventually development of efficient technique.

From all this we may conclude that it is not a good idea to focus solely on power measurement figures, but that it is also necessary to look for sufficiently specific exercises.

In many sports both maximal strength and power should be trained, for both are part of the sporting movement. In sports in which only power production matters (swimming, rowing, cycling etc.) it may be useful to keep a small part of strength training for training of maximal strength. In movements that do not involve power production (long jumping, baseball pitching) power production can occasionally be included within strength training. Such a change of stimulus may serve a purpose. However, it is not advisable to include both maximal strength and power production in a single strength exercise, e.g. a deep squat with a heavy barbell weight followed by extension at maximal speed. The required technical control is not really possible in such heavy training, and movement technique will be impaired, with various consequences (including risk of injury).

7.2.4 *Reflex training*

The concept of explosive power is not sufficient to determine the value of training with light barbells. The gap between the characteristics of maximal power training and the sporting movement cannot simply be bridged by training with light barbells, overlooking the need for coordinative similarity. Power production by the individual muscles that are suitable for this can perhaps be integrated into the sporting movement without much difficulty. The real problem is how to adapt intermuscular cooperation for optimal execution of the high-intensity sporting movement. Training with light weights is a way to do this. This means that training with light barbells must be done in movement patterns that in any case are specific to the generic aspects of the movement patterns in the sporting movement. Training with light barbells therefore makes sense if it is done in as contextual patterns as possible.

The same aspects of specificity play a part in these contextual patterns with light barbells as in power training in simpler movements with heavier barbells. The rate of force development through control of muscle slack and the way in which the movement is decelerated at the end are also important when training with light barbells. Since the barbell weight is low, the rate of force development will be less facilitated by external resistance than when training with a heavy barbell (see Figure 5.3). The way in which muscle slack is shortened at the start of the movement by using cocontractions will then be a key aspect of working with light barbells.

Sporting movements are usually based on movement patterns that are largely controlled by basic rhythm and reflex-supported muscle control. Patterns based on central pattern generators (CPGs) and other fixed neural links, such as stretch reflexes, thus form the generic basis on which the sport-specific movement patterns are built. These patterns are controlled in the central nervous system, using 'highways' to transmit the signals to the muscles. These are preferential movements. The resulting output may therefore be assumed to be greater than the output of movements that are not reflex-supported and have to be developed by unaccelerated control ('ordinary roads'). Training with light barbells should make as much use as possible of the rapid neural paths that predominate in the sporting movement. This means that the resulting output will be great and above all that the resulting fatigue will be typically neuromuscular, i.e. fatigue that develops extremely fast after several repetitions but then fades just as fast. A rule of thumb is then that an exercise with a light barbell in an overall, contextual movement pattern should be performed as explosively as possible, with fatigue occurring suddenly after five to eight repetitions and then fading rapidly after the exercise. Such fatigue usually makes it impossible to keep performing a technically complex exercise correctly, so the number of repetitions seldom exceeds the aforementioned five to eight.

It is useful to summarize contextual movement patterns with low barbell weights as 'reflex training'. This term reflects the notion that training with light weights is useful if done in preferential reflex-controlled movements. Exercises that are not based on these more primary movement patterns will do much less to improve the sporting movement.

Sporting movements such as running and single-leg jumping follow fixed, reflex-based patterns, such as the stumble reflex and the crossed extensor reflex. In such patterns, larger peak forces can be achieved in muscles such as *gastrocnemius* and the hamstrings than in movements in which reflex patterns do not play a primary role. The overload that can be achieved in reflex-supported strength training is therefore greater than in movements without reflex support. Owing to the link between large neuromuscular overload and sufficient specificity in reflex patterns, types of training that include reflex movements acting as triggers of the movement are therefore extremely effective in improving patterns based on running and single-leg jumping (Figure 7.9).

Figure 7.9 Reflex patterns built into a [illegible] exercise.

Movement patterns in the lower limbs are less complex and variable than those in the upper limbs. That is why the basic reflex components of leg movements are easier to recognise than those in the arms. A boxer punching with his right hand will tend to flex his left arm and move his elbow backwards (Figure 7.19). A similar left- and right-arm link can also be observed in throwing. However, movement patterns in the upper limbs, where basic reflex components play a key role, are highly variable. This means that when designing functional strength exercises for the arms, unlike with the legs, it is not a good idea to think in terms of too standardized movements. Given the close link between reflex movements, low energy costs and self-organization of movement patterns, reflex movements can be approached in terms of sufficient degrees of freedom of movement, with variable exercises and where possible a clear, precisely determined end point. The sudden occurrence and subsequent rapid fading of fatigue may be a useful indication here that the exercises are truly reflexive exercises.

Figure 7.19 Explosive extension of the left arm is accompanied by flexing and posterior movement of the right arm.

7.2.5 *Maximal strength and reflex strength in perspective*

Large peak forces occur in both maximal strength training and reflex strength. This immediately raises the question of whether, besides this similarity, there are also differences in loading between the two types of strength training, and what the resulting differences in adaptation are. The choices made between the two types of training when designing strength training can be determined by the differences in the adaptations to be expected.

As we have seen (Section 4.4.2), corticospinal activity increases with variation and decreases with monotony in training. If the possibilities of varying both sensory and motor information in the repertoire of maximal strength exercises are compared with the possibilities of variation in reflex training, maximal strength training is more likely to lead to monotony, for the exercises produce less sensory information than in reflex strength training and less variation is possible in motor patterns. Numerous different initial postures, surfaces and even asymmetrical barbell loads that would be dangerous in maximal strength training because of the heavy barbell load are possible in reflex strength training because the barbell load is lighter. Apart from monotony, there are other effects that differ between the two types of training. The most significant differences in adaptation are due to differences in impulse and rate of force development.

'Impulse' can be defined as ground-reaction force multiplied by 'time under tension'. Ground reaction forces are the forces that occur in the support base during the exercise. In the clean these are the forces acting on the feet, in bench pressing the forces acting on the thoracic back, and so on. Time under tension corresponds to the length of the exercise: how long the body has to produce large forces because of the external load. When producing large forces, time under tension is more directly related to fatigue than the total power produced (Cronin & Crewther, 2003; Tran, Docherty & Behm, 2006). From this we can conclude that impulse is a useful indication of fatigue. Evidence for this is found in data gathered during sessions with similar activation (EMGs) but different overall impulse. In maximal strength training, the total impulse of the exercise is substantially greater than in reflex strength training. Not only is the barbell load heavier, but above all the length of the exercise is a multiple of the length of a reflex strength exercise. Fatigue due to maximal strength training is therefore greater than in reflex strength training. Such fatigue may occur in the central nervous system ('central fatigue'), in the transition between the central nervous system and the muscle (neuromuscular fatigue) and in the muscle.

Central fatigue only occurs in efforts with constant maximal activation levels. Because of the great impulse and great activation, it therefore also occurs in maximal strength training. However, recovery from this fatigue occurs just a few minutes after the session (Taylor et al., 1996). Muscle fatigue also occurs in maximal strength training, and recovery from this is much slower. Such neuromuscular fatigue as does occur may only involve the first hundred milliseconds of the muscle action. This means that there is not too much pressure on neuromuscular transition in maximal strength training, owing to the rapid increase in central and muscle fatigue – the number of sets that can be performed is too small. This is particularly true of strength training designed as part of a training plan in which other (technical and tactical) types of training play an important part. In such a training plan, total fatigue due to strength training must not be followed by too long a recovery time. As a rule of thumb, in high-impulse maximal strength training there is a reduced rate of force development (the start of the muscle action) and fatigue is still limited after a total of five sets of all the exercises taken together, whereas after ten sets muscle fatigue is so great that recovery takes at least 24 hours. To keep total fatigue low and so avoid clashes with other types of training within 24 hours after strength training, training sessions must therefore be limited to about five sets, owing to the rapidly increasing overall impulse of the sets. This substantially limits the number of times that the rate of force development, and hence the neuromuscular system, will come under pressure during those five sets.

Central and muscle fatigue play less of a part in reflex training, because the total impulse is low. Furthermore, the training is performed in preferential patterns. Owing to reflex support, the neuromuscular transition can be put under greater pressure without creating too much central or muscle fatigue. The stimulus to the neuromuscular system will therefore be greater, possibly with improved adaptation. The rule of thumb here is about twelve sets can be performed in a single reflex strength training session for all the exercises taken together. In that case recovery will be so rapid that other types of training will not be affected.

In twelve sets of five to six repetitions, the rate of force development will therefore come under pressure far more often than in maximal strength training. In explosive sports, in which rapid build-up of force is a major performance-determining factor, this will have a major impact on the effectiveness of strength training.

The advantages and disadvantages of maximal strength training and reflex training been summed up as follows by Raphael Brandon (Brandon, 2011):

Maximal strength training:
- High impulse; significant muscle fatigue (with slow recovery) and significant central fatigue (with rapid recovery).
- Limited training of the rate of force development.
- Useful if there is sufficient recovery time.
- Suitable for developing a high level of strength.

Reflex training:
- Low impulse and high power or high force production; minimal central and muscle fibre fatigue.
- Useful when training the rate of force development.
- Compatible with other types of training on the same day in the micro and meso cycle.
- Suitable for maintaining a high level of strength.

A choice between maximal strength training and reflex training can be made in the light of these features. During a competition period, for instance, preference may be given to reflex training, because of the highly specific rate of force development and limited fatigue after training. At other times of the year, when there is sufficient recovery time, the focus may be shifted to features that will improve well in response to maximal strength training, such as high neuromuscular activation and recruitment. It should be noted here that there has been relatively little research into these mechanisms, and that more is needed in order for them to be considered well demonstrated.

7.3 Exercises approached in terms of coordination

Contextual movements always consist of a combination of fixed, stable components (attractors) and changeable, unstable ones (fluctuators). Attractors are low-energy, and fluctuators are high-energy. Attractors are needed to keep the number of degrees of freedom of movement controllable. Fluctuators are needed to adapt the movement pattern so that the movement is executed in accordance with the changing demands of the environment. Especially in open skills, these adaptations may cause great differences in the execution of the generic movement pattern.

Attractors are unchangeable, and can therefore be seen as the basic components or building blocks of the movement. Strength training is an important way to improve the most elementary, generic building blocks. Carefully designed strength training can be used to further deepen the attractor wells, in which the elementary principles of movement are located (Figure 5.4), while guaranteeing the specificity of the exercises in relation to the features of high-intensity movement.

The organization of a movement pattern into stable and unstable components may suddenly change ('phase transition'). As a result, specificity between low-intensity and high-intensity movements is no longer guaranteed. Movements would then always have to be high-intensity to guarantee sufficient specificity with the sporting movement. This usually means that the exercise would also have to be executed at high speed, making it very difficult to learn complex patterns in stages. Strength training is appropriate for learning and improving certain components of a movement pattern, because the increased resistance allows the characteristics of the high-intensity movement to be maintained without the movement having to be executed at high speed. The relatively slow, controllable movements within strength training therefore simplify the learning process. Strength training thus complements technique training. In high-intensity movement, for instance, core stability through cocontractions is important. In low-intensity movement without extra resistance, cocontractions are not so important. In such low-intensity exercises, deliberately tensing all the trunk muscles is highly academic and can hardly be transferred into high-intensity movement. At the same time, high speed of movement makes it difficult to time, and hence control, the cocontractions. When working with a barbell load in, say, a step-up, speed of movement will be low, and core stability will simultaneously have to be controlled by cocontractions. This provides a very useful intermediate stage in the progression from easy to difficult application of cocontractions in core stability (Figures 7.11 and 7.12).

Figure 7.7
Core stability through cocontractions in a step-up in series. Attractors in the running movement are trained against resistance.

Figure 7.12
Attaching two extra weights (such as 5- or 10-kilogram barbell plates) to the barbell with chains makes the load variable and unpredictable. Before the athlete runs up the steps, the weights are made to swing, placing greater demands on co-contractions in the trunk. In addition, the feet will be planted from higher up to compensate for unexpected perturbations. This will train the foot plant from 'above' (the hammer principle; see Figure 3.9).

7.3.1 *The three-layer model and strength training*

To design useful sport-specific strength training, it is first necessary to identify the generic building blocks of the sporting movement, so that a useful selection of exercises can then be made. The three-layer specificity model (Figure 7.13) is suitable for making such an analysis. In this model, intramuscular, intermuscular and external attractors are particularly important when designing strength training. Wherever possible, an attempt is also made to match the intention of the exercises to the sporting movement. Similarity in sensory input is the hardest to achieve, so it will usually be disregarded. It is important to ensure that the layers are suitably linked. In the self-organization of a movement pattern, stable action of individual muscles is the basis for organizing a stable intermuscular pattern, and this intermuscular organization in turn influences external features of the movement. In javelin throwing after a run-up, for example, the abdominals can only perform their elastic energy-transporting action (whip action) in a satisfactory robust manner if the muscle action is performed isometrically and close to optimal length. The attractor state of the abdominals has implications for the timing of the muscle action in the intermuscular pattern when bracing the front leg. If the abdominals are tensed too late after the hip action, opposing forces will make the length too great for optimal force production. This has implications for the outer layer, namely the position of the pelvis in relation to the trunk when bracing the front leg (rotation and lateroflexion of the spinal column).

This close connection between the three layers creates opportunities for systematic action within strength training. A choice can be made between exercises that focus on stable high-intensity muscle action (such as elastic loading of the isometrically working abdominals in throwing sports, or isometric maximal strength training for hamstrings in jumping sports), on fixed patterns of cooperation between muscles (such as throwing a medicine ball with extended arms for abdominal muscle function in the kinetic chain in throwing sports, or step-up variants with horizontal movement for hamstring function at take-off in jumping sports) or on correct ranges of motion when executing the movement (such as a smaller backswing when throwing medicine balls in throwing sports, or the optimal pelvic position at the end of a step-up in jumping sports). In combination with a clear intention of the exercise (such as fast, accurate throwing in throwing sports, or a step-up without anterior or longitudinal rotations in jumping sports) this provides a systematic division of the exercises that makes it easy to navigate within the central/peripheral model and links up key aspects of training.

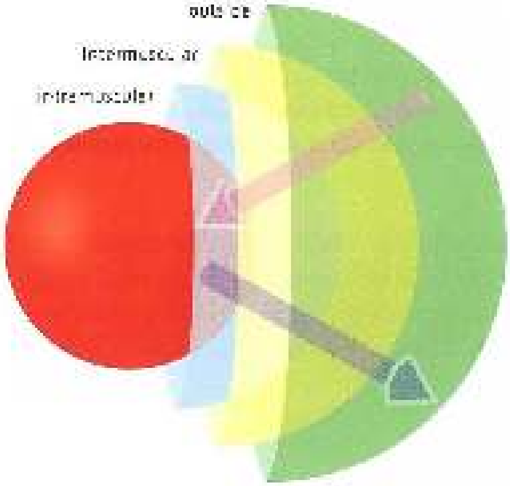


Figure [illegible] Specificity can be analysed in terms of intention, intramuscular patterns, intermuscular patterns and external appearance. The various aspects should be logically connected. The better the match between performance of exercises from the various layers, the more coherent the training will be.

7.3.2 *Intramuscular attractors*

As we have seen, self-organization of movement patterns is an essential aspect of intensive sporting movements. It is determined by stability of muscle action and by energy costs. Obviously, physical and reflex muscle properties will greatly affect how self-organization takes place. Interaction between muscles and the influence of external forces ensure that muscles link stable action to stability in the joints within a movement pattern. The principle of dividing the movement over as many joints as possible (as in a tennis service) is an example of self-organization based on stable muscle action and stable joint positions.

Stability of muscle action is largely determined by muscle architecture. Elasticity, pennation angle and changing levers ensure that some muscle actions are more resistant to perturbations than others. Since muscle anatomy varies, so does the stability of muscle action. This means that muscles behave differently in overall contextual movements (Biewener & Roberts, 2000).

When considering strength training for individual muscles, it is therefore important to know the stable attractor state of the various muscles. This can be taken as a starting point for training; it should then be determined whether the muscle should only be trained in stable muscle action, or also in variations. The centrifuge model for muscle action (see Section 5.5.2) indicates the sports in which muscles behave in a specialist way and hence the attractor state of the muscles should be deepened within strength training. This basically means all sports in which speed of movement is helps determine performance. In these sports it may also be assumed that training muscles in their attractor state is useful, although working in chains is preferable. Muscle anatomy is the guide for determining which type of strength training is appropriate for individual muscles.

However, this is not to say that muscle architecture does not matter in sports with a slower speed of movement. Although there is more flexibility in the way in which a muscle can be used in a movement pattern, it is still advisable not to train muscles beyond their anatomical specialization unless they are also clearly used beyond their specialization in the sporting movement.

7.3.3 *Individual muscles and their suitability*

Reasoning on the basis of anatomy, we can determine for individual muscles which load within strength training corresponds to contextuality in sporting movements.

The calf muscles

The *gastrocnemius-soleus* complex muscles have a highly pennate structure and long elastic tendons. At peak loads they act above all isometrically in contextual movements. This makes them suitable for maximal strength training. However, since within strength training it is not possible to give muscles qualitative overload in isometric conditions, there is little point in training with a barbell load (see Section 6.2.1). It is therefore better to avoid strength training of calf muscles through isolating exercises for running and jumping sports, except in a rehabilitation setting. It makes more sense to exercise the right timing of *gastrocnemius* action in overall intermuscular patterns, such as a single-leg pretension hang clean, and then focus on technique training in the sporting movement.

Abdominals

Owing to their architecture, abdominals also have a narrow force/velocity range. They also have a very large lever relative to the joints of the lumbar spine that are to be moved – more than 10 centimetres, whereas most muscles have levers of a few centimetres at most. This means it is no easy matter to maintain the length of the abdominals when there are great opposing forces. If the upper body rotates only a small number of degrees in relation to the pelvis or bends sideways only a very little, for instance in the backswing for a smash, there will be a relatively large change in muscle length because of the large lever. Together with the narrow force/velocity range of the muscle, this means that the abdominals can easily move beyond their attractor state (the optimal length). So there is only a relatively narrow bandwidth of ranges of motion in which the abdominals can act properly. In movement patterns in which the abdominals easily move beyond this range, such as fast bowling in cricket, which involves extreme lateroflexion at the point of release, it is useful to pay a great deal of attention to the organization of the attractor landscape, in which other attractors of importance at the point of release help optimize self-organization of the length of the abdominals.

In contextual movements that require the abdominals to absorb large opposing forces (running, throwing, jumping), they often act elastically. This makes them suitable for maximal strength training. However, muscle action lasting just a few seconds cannot be maintained, owing to the resulting abdominal pressure and blood congestion. That is why it is better to focus on training intramuscular coordination through brief elastic loading. As well as by throwing medicine balls, this can be done by quickly increasing and then reducing the lever of the external load (a barbell plate). This 'thrusting' of the plate can be easily linked to end-point orientation, for instance by touching a target with the plate (Figure 7.14). In these exercises the initial posture is lying on the back with the knees pulled up (at 90°) and the shoulder blades off the ground. The feet are fixated and an extra weight is held in front of the chest. The athlete then does a half sit-up while thrusting the weight upwards and backwards as fast as possible. The weight is then pulled back in front of the chest as fast as possible, and the trunk returns to the initial posture. The weight can be thrust upwards and sideways rather than straight upwards and backwards, and the surface can be inclined downwards rather than flat. The movement should be executed more and more vigorously and faster and faster, with the focus on the weight being pulled back. This 'thrusting' principle can be performed in various versions and with various weights, such as a ball partly filled with water.

Figure [illegible] Training [illegible] by elastic loading.

Erector spinae

Because of the passive tissues between the muscle fibres, the back muscles, like the abdominals, have very narrow force/velocity characteristics. In high-intensity contextual movements they therefore also stay close to their optimal length. This means there should not be any significant trunk flexion with heavy loads; if there is trunk flexion, for instance when speed skating, the back muscles will be greatly lengthened and their contractile force will therefore be low (no more than 40% of MVC at push-off; Roelands & Van Kempen, 2003). Because of its force/length characteristics, *erector spinae* is therefore suitable for maximal strength training.

In strength training this mainly has implications for the execution of squats. Deep double-leg squats are often said to produce particularly favourable strength adaptations, and attempts are made to flex the knees more than 90°. However, if the athlete moves down from an upright position, the pelvis will start to rotate posteriorly at some point. This movement will cause a change of position in the spinal column that resembles trunk flexion. When the pelvis rotates backwards, *erector spinae* lengthens and force is rapidly reduced. 'Flexion relaxation', in which substantial flexion causes the EMG signal to stop, plays a part here. For this and other reasons the spinal column is no longer well protected. In squats with a barbell load it is therefore advisable not to go deeper than can be achieved without posterior pelvic rotation. Not only is the value of deep squats questionable, but so is the claim that double-leg squats are particularly suitable for improving strength in the legs. Strength in the back muscles may be the limiting factor, rather than strength in the legs, and so double-leg squats may in fact be a maximal strength exercise for the back muscles. A better way to train the back muscles with a maximal load is the 'good morning exercise'. Here again, there should be no trunk flexion of the spinal column (which will occur if the athlete bends too far forwards). The best way to avoid this is to use a heavy barbell load, which will prevent the trunk from moving so far forwards (Figure 7.15).

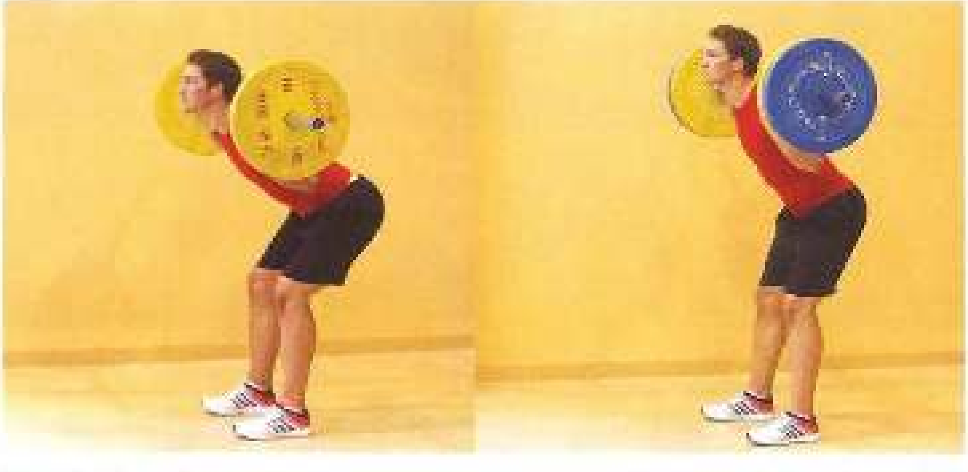


Figure 7.15 Left: a good morning with slightly flexed knees and slightly too little weight. The trunk flexes, possibly so far that the pelvis is unable to rotate far enough and spinal column flexion occurs. Right: the maximal possible range of motion with a heavy barbell weight; the pelvis can rotate far enough, and the spinal column will remain extended.

Apart from the good morning, there are other alternatives for maximal strength training of *erector spinae*, such as lifting a barbell off the ground with extended arms and two fixated legs on a Roman chair (Figure 7.16).

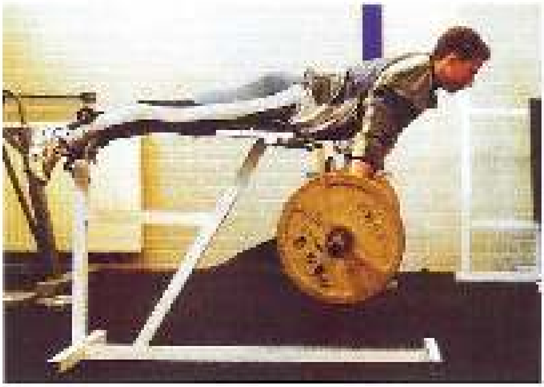


Figure 7.16
The exercise is performed in the same way as the hamstring exercise in Figure 7.12, but now both legs are fixated, so that erector spinae limits performance and hence receives the overload with a large barbell load.

Latissimus dorsi and pectoralis major

Latissimus dorsi and *pectoralis major* play a key role in pulling movements, pushing/thrusting movements (*pectoralis major*) and throwing. In pulling and pushing movements, the muscles act in opposing ways, and their force production is important. In throwing, they work together to shift the abducted arm from external to internal rotation, acting elastically (Figure 7.17). In sports that focus on throwing it is therefore useful to train the muscles mainly by maximal strength training and elastic loading.

These two muscles are unusual in that they play a key role in two different movement patterns, requiring essentially different muscle action. Strength training for boxers or shot putters will therefore differ from strength training for baseball pitchers or tennis players. Not only will different exercises be selected, but the strategy for linking specificity to overload (see Section 6.1.5) will be substantially different. In the case of pulling and pushing/thrusting movements, it is usually possible to execute movements under heavy resistance that outwardly resemble the sporting movement. In the case of throwing this is not possible, and when working under heavy resistance the movement will have to differ from the specific throwing movement. Furthermore, as regards force production for pulling and pushing movements it is possible to vary the resistance within the intended movement pattern and so choose from the whole central/peripheral continuum. In the case of throwing, the resistance must remain close to the resistance in the sporting movement (the weight of the ball or javelin) if the exercise is to be very movement-specific. This means that exercises at either end of the central/peripheral model will mainly be chosen (very specific with little overload, or unspecific with a lot of overload). Among other things, this means that opinions differ regarding optimal training protocols for, say, baseball pitchers (Derenne *et al.*, 2001; Van den Tillaar, 2004). The two different, unrelated functions of *latissimus dorsi* and *pectoralis major* in high-intensity movements are an example of a phase transition in self-organizing synergies (see Section 7.3.4). There is no useful intermediate form between thrusting and throwing.

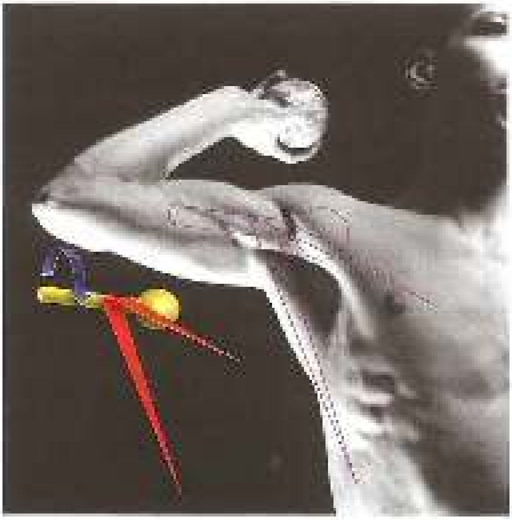


Figure [illegible] Both pectoralis major and latissimus dorsi are muscles that provide internal rotation. When throwing, they are both eccentrically loaded when the humerus comes into play; unloading of the stored elastic energy then causes internal rotation of the arm.

Iliopsoas and gluteus maximus

Iliopsoas and *gluteus maximus* are the counterparts round the hip of *latissimus dorsi* and *pectoralis major*. Both muscles produce only power and should be trained on this basis.

It is somewhat difficult to train *iliopsoas* in isolation because the muscle always cooperates with the abdominals, so exercises must be found in which the abdominals are not the limiting factor. At the same time, the importance of *iliopsoas*, particularly in the running movement, tends to be underestimated. *Iliopsoas* plays a key part in the anterior swing of the free (swing) leg and in loading the hamstrings with elastic energy (see Figure 5.9), and is often underdeveloped in sprinters and players of ball sports, in whose performance running plays a key part.

Quadriceps femoris

Although *quadriceps* is pennate, it has few elastic structures. This means that in sporting movements it is mainly a power-producing muscle, although it often also acts to maintain the knee angle. The changing axis of the movement in knee flexion and extension gives the muscle group the greatest torque in relation to the knee when the knee is flexed.

Of all the muscles, *quadriceps* is the fastest to adapt physiologically, in terms of both increase and decrease in force and size. Hypertrophy occurs faster than in other muscles, and when the knee is immobilized muscle size decreases faster than in other muscles.

In contextual movements *quadriceps femoris* produces its power in a closed chain. Power training should therefore be done in a closed chain, never in open-chain situations such as a knee-extension machine.

Adductors

Their structure makes adductors particularly suitable for producing power. In contextual movements such as running, they remain active throughout the cycle. Within strength training for the adductors, consideration should be given to whether the muscle group should be loaded in unspecific movement patterns. In running, the adductors are part of a complex movement pattern in which movements in the sagittal plane are coordinated with internal and external hip rotation. Training through movements in the frontal plane (adduction) could adversely affect coordination in complex sporting movements. What is more, training in contextual patterns, for example in step-ups and running, usually creates a load that is hard to equal within strength training involving adduction movements.

Biarticular muscles

Biarticular muscles are energy transporters. In contextual dynamic closed-chain movements, lengthening of the muscle in one of the joints it crosses is always accompanied by shortening in the other joint. This keeps the muscle in the isometric attractor state. Even in an open chain the pennate biarticular muscles absorb large external forces isometrically – for example, the hamstrings in running during the lower leg pendular motion in the flight phase. Isometric muscle action may be accompanied by loading and unloading of elastic components of the muscle.

Biarticular muscles can therefore best be trained for maximal strength, for this is most compatible with the isometric muscle action in the sporting movement. In addition, strength training through elastic loading is specific for sporting movements in which large opposing forces have to be absorbed elastically.

Both timing of the activity and the amount of force produced by the biarticular muscles in contextual patterns are crucial to performance. If too much or too little force is produced, or if it is produced too early or too late in the overall pattern, performance will deteriorate, even though the muscles' preflex properties within this complex intermuscular cooperation will damp down any errors (Jacobs et al., 1996; Van Soest & Bobbert, 1993). Since the biarticular muscles are so crucial to intermuscular coordination, it is a good idea to train them in contextual patterns wherever possible. Muscles should therefore only be isolated in strength exercises if this will achieve results that are cannot readily be achieved in complex patterns. Each muscle or muscle group has its own specific factors here:

* Hamstrings: as we have seen, more force is produced in the hamstrings when running at high speed than can be achieved in maximal strength training with external resistance. The logical conclusion is that maximal strength training serves no purpose for hamstrings, the level of strength being guaranteed by sufficient sprint training. However, maximal strength tests in athletes who do plenty of high-speed running (such as elite rugby players) show that sprinting does not guarantee sufficient hamstring strength. It seems that the hamstrings gradually withdraw from the running cycle, their hip-extension task possibly being taken over by the adductors (or other muscles) and their task in decelerating knee extension by gastrocnemius. Running technique is somewhat altered, top speed decreases and there is less hamstring recruitment during the running cycle, so that passive tissues in the hamstrings receive fewer adaptation stimuli and the risk of injury increases.

  The best way to get athletes who have developed such a pattern back to the desired pattern is maximal strength training in combination with technique training, focusing on control of anterior pelvic rotation, short ground contact times and correct body posture. Given the great influence of hamstrings in the running cycle, it is advisable to do regular maximal strength training and elastic loading with barbells, in order to maintain the intended 'good enough' level.

- *Rectus femoris*: it is not clear whether *rectus femoris* also tends to withdraw during the running cycle; but it seems less likely, since no muscle group is particularly suitable for taking over the function of *rectus femoris* in decelerating hip extension. This means there is no obvious need for additional maximal strength training. It is also questionable whether the muscle is loaded to the limits of its capacity when running; *rectus femoris* has to absorb greater forces in a single-leg push-off. In any case it is difficult to find strength exercises that can load *rectus femoris* to the limits of its capacity. The best way to train *rectus femoris* for force production is therefore probably jump exercises.
- *Triceps brachii*: this muscle has two monoarticular heads and a single biarticular one. In sport-specific strength training, the three heads are preferably trained together. This can be done by combining elbow extension with flexion in the shoulder joint. Exercises such as 'dipping' seek to concentrate the load in that pattern mainly on *triceps brachii* and so create overload (Figure 7.18). Isolating the short heads, a bodybuilding practice, does less to improve coordination.

Figure 7.18
A basic exercise for triceps brachii: dipping

- *Biceps brachii*: in most contextual movements the biceps flexes the elbow, while decelerating extension in the shoulder joint. Muscle action remains close to isometry. In such movement patterns it is difficult to make this muscle group the limiting factor in the exercise.

7.3.4 *Attractors in smaller intermuscular cooperation systems*

Self-organisation of intramuscular attractors has implications for the intermuscular patterns that make up high-intensity contextual movements. The preferential muscle actions create larger sets of cooperating muscles which have attractor features and hence are larger building blocks of the sporting movement. They can be seen as large modular components of the movement which – just as in housing – are used to create an end product (in this case the sporting movement) as quickly as possible. Such large sets of attractors are a crucial means of controlling movements. If a muscle is stabilized in its attractor state, there is only one way it can influence other muscles (its immediate neighbours). This will make cooperation between the muscles involved less flexible, with only a limited number of possible cooperation arrangements, which in turn will determine what is possible in overall patterns. This will greatly reduce the number of degrees of freedom in high-intensity movement, which is necessary in order to keep the movement controllable. To repeat the housing metaphor, if a house is built out of separate bricks, almost any design is possible, but the house will take a long time to build. If large modular components are used, there will be fewer alternative designs, but the house will be built faster. Small, self-organizing systems such as muscle action are the building blocks for self-organizing larger units such as the mutual influence of the muscles round the shoulder. These larger systems are in turn building blocks for the self-organization of even larger systems, and ultimately the rhythm-controlled systems of coordinated arm and leg movements. In Bernstein's terminology, these larger units of cooperation between muscles are known as 'synergies' (Bernstein, 1967; Latash, 2008). Synergies are essential to high-intensity movement, allow the movement to be controlled and include mechanisms such as cocontractions, which reduce the risk of injury. There is also evidence that synergies between muscles make the locomotor system resistant to the impact of fatigue (Singh & Latash, 2011). This again supports the somewhat speculative idea that fatigue is a key instrument in the learning process (see Section 6.4.4).

If the implications of the centrifuge model are extended to the larger systems of cooperating muscles, we can conclude that intermuscular cooperation in high-speed movements may be less flexible than in low-speed movements. Muscles act closer to their specialization in high-speed movements, which in turn rules out many types of intermuscular variation. For example, the hamstrings, *erector spinae* and *gluteus maximus* will always cooperate in the same way in high-speed movements. The same applies to, for example, the abdominals and *iliopsoas*. In the shoulder girdle, alternative ways of cooperation in high-speed movements will also be limited. Throwing, in particular, will be based on a fixed set of synergies. This constraint makes it possible to select such patterns and make use of them within strength training.

Strength training is therefore an effective way to improve high-intensity intermuscular patterns. Intermuscular coordination tends to be overlooked within strength training; instead, the main focus is on training individual muscles and overall patterns that resemble the outward form of the sporting movement. The result is an approach that merely attempts to load muscles at the same joint angles as analysed in the sporting movement. This can easily lead to training movement patterns that do not fit into the synergy matrix, and hence are unique and not readily transferable to other movements such as the sporting movement.

Since intermuscular patterns are largely self-organizing, it is useful to provide sufficiently varied exercises for the selected patterns, while maintaining the basic principles for intermuscular patterns. This contribution of strength training to deeper attractor wells may be crucial to transfer of strength training to the sporting movement.

In the interests of consistency it is useful to start by identifying the smaller links between muscles. These are the building blocks for the larger overall movement patterns.

The trunk

If the trunk muscles are loaded with large opposing forces in contextual movements, they can only function in a more or less neutral position of the spinal column, given their narrow force/length range. In trunk flexion (cycling, speed skating) the abdominals are too short and the back muscles are too long to generate much force. Even with plenty of hyperextension or torque, the length of the abdominals will be insufficient, especially if the movement is vigorous and the muscle is not prestretched. In sporting movements involving large opposing forces, these will always be abrupt. This means that sensory feedback will take place under time pressure. Reflex control through cocontractions will therefore be important as an addition to sensory feedback, especially in open-skill situations and situations in which control is a very critical factor. Cocontractions of the trunk muscles with rapid build-up of force are therefore a major attractor in contextual sporting movements.

Various core exercises based on cocontractions

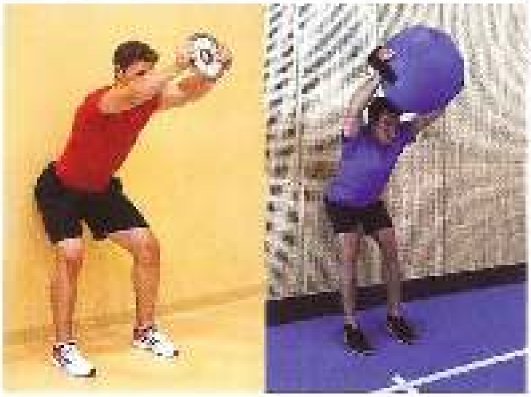


Figure [illegible]
Basic exercise for core stability through cocontractions of all the muscles that influence the spinal column (see also figure [illegible]). The barbell plate may be replaced by a ball partly filled with water; the movement of the water will make greater demands on the cocontractions.

Cocontractions in the trunk

In figure [illegible] the trunk is bent forwards by 30–45° and the spinal column is extended. The cocontraction is created by thrusting the weight as far away as possible. This posture is always maintained for a few seconds. The thrusting can be combined with torque, in which both the weight and the shoulders should turn. In this exercise the diagonal abdominals are stretched over one diagonal. This makes the exercise substantially harder, and the capabilities of the muscles' force/length characteristics are explored.

Figure 7.20
Cocontractions in a squat.

The cocontractions can be performed in numerous body postures, such as a squat or a split squat. The basic principle remains the same: pushing the weight as far upwards as possible increases the tension on the trunk muscles (Figure 7.20).

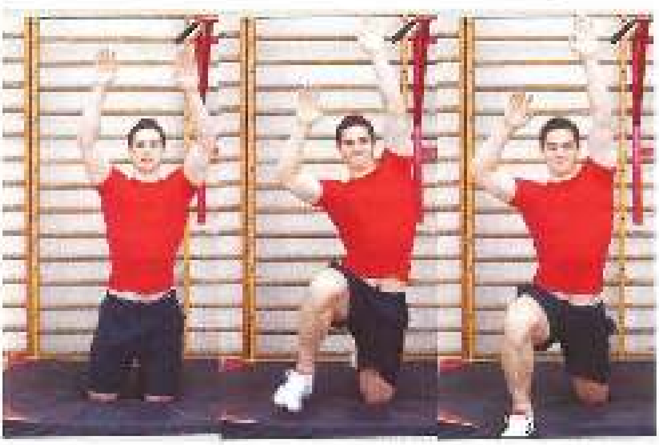
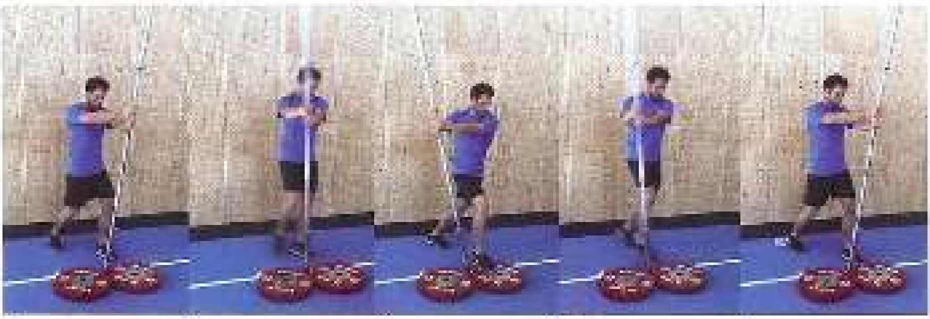


Figure 7.21 Cocontractions round the hip. A leg that is moved forwards from a kneeling position should move as little as possible from abduction to flexion. In addition, the foot should not touch the ground when moving forwards. While the leg is being moved, the hand on the supporting side should be pushed as far upwards as possible.

Cocontractions in dynamic situations

Cocontraction technique can be put under greater pressure by using it when moving rather than in a static posture (Figure 7.21) – for instance by kneeling with a barbell plate, a stick or a light barbell held above the head, then lifting each leg off the ground in succession and moving it anteriorly. At the same time the free (swing) hip should be moved upwards. Dynamic alternatives include cycling on a home trainer with a stick held above the head, or lunges in various directions with a barbell plate held above the head, and the plate rotating on landing so that there is torque towards the leading leg. Performing the changes in posture in the dynamic situations faster and more vigorously can put greater pressure on the timing of the cocontractions.

a b c



d e f g h

Figure 7.22 Barbell rotations with the hands at shoulder height and so much weight that rapid movements can only just be made.

Cocontractions while standing

From the initial posture in Figure 7.22 (b) the athlete rotates as fast as possible to the left or right and then back to the initial posture (a and c). The movement should be reversed abruptly and the sideways movement should not take place in the shoulder joint, but through rotation in the trunk (the shoulders turning at the same time). An alternative is to step forwards with the leg on the side the athlete turns to. If the athlete turns to the left, the left leg steps forwards and the foot is planted next to the right-hand end of the barbell. When turning back, the athlete returns to the initial posture. This again explores the limits of the abdominals' force/length capabilities. The timing of the cocontraction can be put under greater pressure by placing the right foot in front and the left foot behind in the initial posture, slightly more than a foot-width apart. While the athlete performs a rapid barbell rotation to the left, the left foot jumps forwards and the right foot backwards, landing with a bounce (with the left foot next to the barbell) and immediately jumping back to the initial posture.

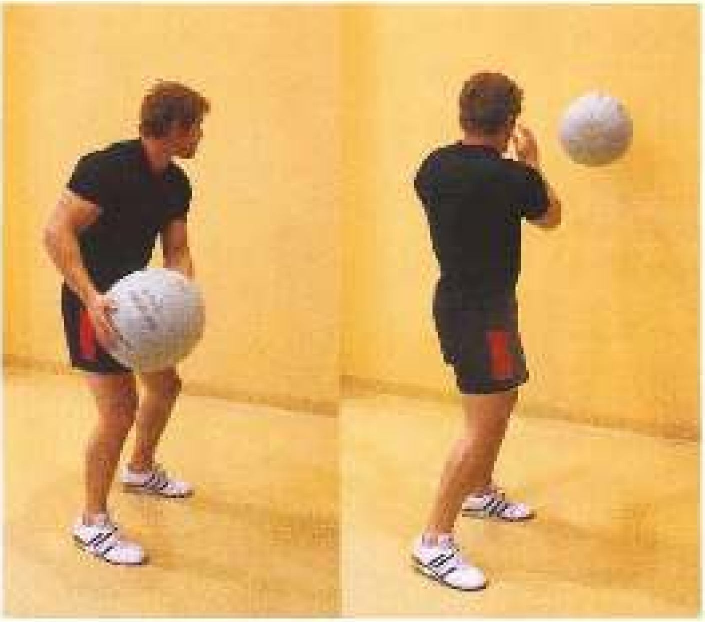


Figure 7.23 Core exercise with medicine ball.

Cocontractions in sideways throws

The medicine ball is thrown sideways against a wall and then caught sideways (Figure 7.23). The movements should be brief and vigorous, the feet far enough apart and the trunk lengthened.

Figure 7.24 Alternative core exercise.

Cocontractions while lying on the side

The movement from (a) to (b) to (c) in Figure 7.24 is executed as rapidly as possible. The posture in photograph b is then fixated for two seconds. Lying on the side allows external forces to be controlled so that the focus in the contractions is on cooperation between the diagonal abdominals and unilateral action of *erector spinae*.

Lower limbs

To identify the intermuscular attractors in the lower limbs, it is first important to distinguish between open- and closed-chain movements, for the effects of cooperative muscle action are quite different in each case. Since the foot is fixated in a closed chain, the torque that a muscle produces on a joint will have a different effect on the eventual movement than in an open chain. This means that the self-organization of intermuscular cooperation will also be different.

Cooperation between hamstrings and back muscles and types of strength training for basic cooperation between the two muscle groups have already been discussed in Section 5.6. This interplay is crucial to the functioning of the hamstrings, and applies to both open- and closed-chain high-intensity movements.

Open chain

There are few open-chain sporting movements in which force production in the muscles of the lower limbs comes under such pressure as to require separate strength training for the lower limbs. However, such high-load open-chain patterns do occur in high-speed running. These are to some extent performance-determining, and attention should be paid to them within strength training. The intermuscular open-chain patterns that can limit performance are cooperation between *iliopsoas* and the abdominals (when leaving the ground) and cooperation between the back muscles, the hamstrings in the leading leg and *iliopsoas* in the trailing leg during the lower leg pendular motion (somewhat later in the flight phase). For players of ball sports who not only have to run but also shoot a ball hard, strength training to support running performance will also be enough for shooting a ball, so no additional strength training will be required. For all other land sporting movements, force production in open-chain situations is lower than what is provided simply by training for top-speed running. For open-chain movements in the water, such as the breaststroke and in water polo, force production may be a crucial factor, although the eventual decision about whether or not to do additional contextual strength training on land for the lower limbs will be difficult, given the aforementioned problems of transfer between land training and movement in the water.

Exercises

Back muscles and hamstrings in an open chain

There are no good types of strength training for basic open-chain cooperation between back muscles and hamstrings during the lower leg pendular motion. It is therefore useful to train this cooperation entirely in technique training for the sporting movement and related exercises.

Figure [illegible]
Training of iliopsoas (in cooperation with the abdominals).

Iliopsoas and abdominals

In contextual movements *iliopsoas* always cooperates with the abdominals. When these two muscle groups come under pressure (in the transition from the closed to the open chain in high-speed running), this makes great demands not only on power production by *iliopsoas*, but also on force production by the abdominals, which have to limit anterior pelvic rotation. Both aspects of performance – force production and timely use of abdominals and power production by *iliopsoas* – can be combined into one exercise by starting with the thigh horizontal in the exercise on the hip-flexing machine, then extending the leg as fast as possible in the hip, and then flexing in the hip as fast as possible to return to the initial posture (Figure [illegible]). There should not be a rest period at the point of reversal, and the knee of the free (swing) leg should not end up behind the knee of the stance leg. The abdominals play a key part in absorbing this opposing force, and *iliopsoas* plays a key part in producing power in order to move the weight back upwards. The stance-leg hamstrings are also active, to prevent the activity of *iliopsoas* from making the pelvis rotate anteriorly.

Closed chain

In the closed chain, force production in muscles is mainly under pressure while extending. There are three main forms of organization of cooperating muscles in the extension movement:

1. Moving the free (swing) side of the pelvis upwards at the end of a single-leg extension (lock position).

2 Coordinating the various joint torques in the flexed-leg position, i.e. coordinating the knee and hip angles.
3 Extension by returning elastic energy from bouncing.

If complete single-leg extension is required, it is always important to focus on the final extension posture. In double-leg extension the focus has to be elsewhere, for instance on timely (early) energy transport from the knee to the ankle, i.e. well-timed extension of the ankle. It is therefore advisable to distinguish between types of strength training that focus on cooperation between muscles in the last part of single-leg extension and those that seek good coordination between the knee and hip angle in a flexed position:

- The last part of single-leg extension. When extending on one leg, the last part of single-leg extension in the sagittal plane must be combined with elevating the free (swing) hip. The forces that muscles produce on each other are therefore very different in single-leg and double-leg extension. The final posture with the free (swing) hip elevated (see Sections 1.3.3 and 3.2.6) is the result of cocontraction of muscles round the hip, and is a key component of a single-leg push-off, both when starting and when accelerating.

Exercises

From the initial posture with one leg on a box and the barbell against the wall, the stance leg (Figure 7.26) extends and the hip is placed in the lock position. The barbell weight can be varied and even carried asymmetrically, with the greatest force in resistance on the stance-leg side.

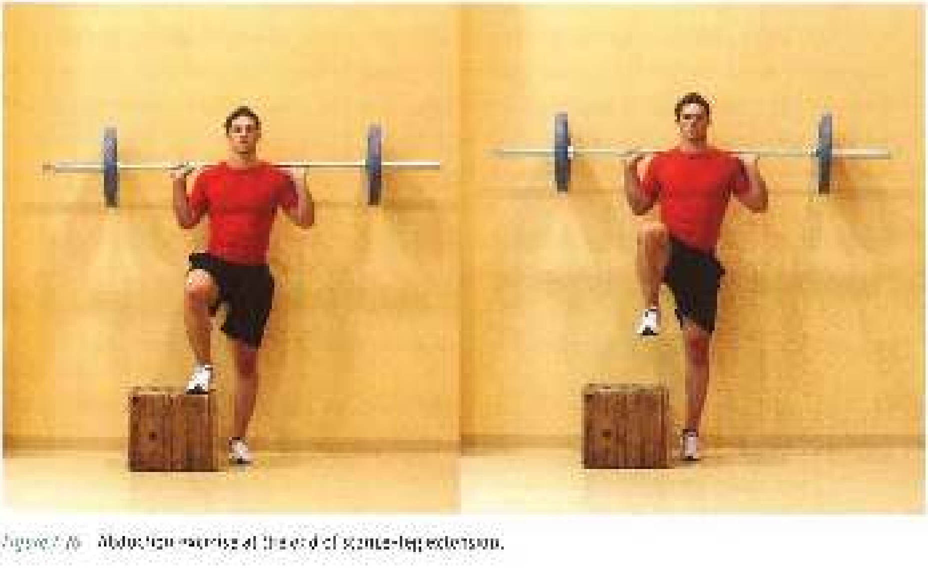


Figure 7.26 Abduction exercise at the end of stance-leg extension.

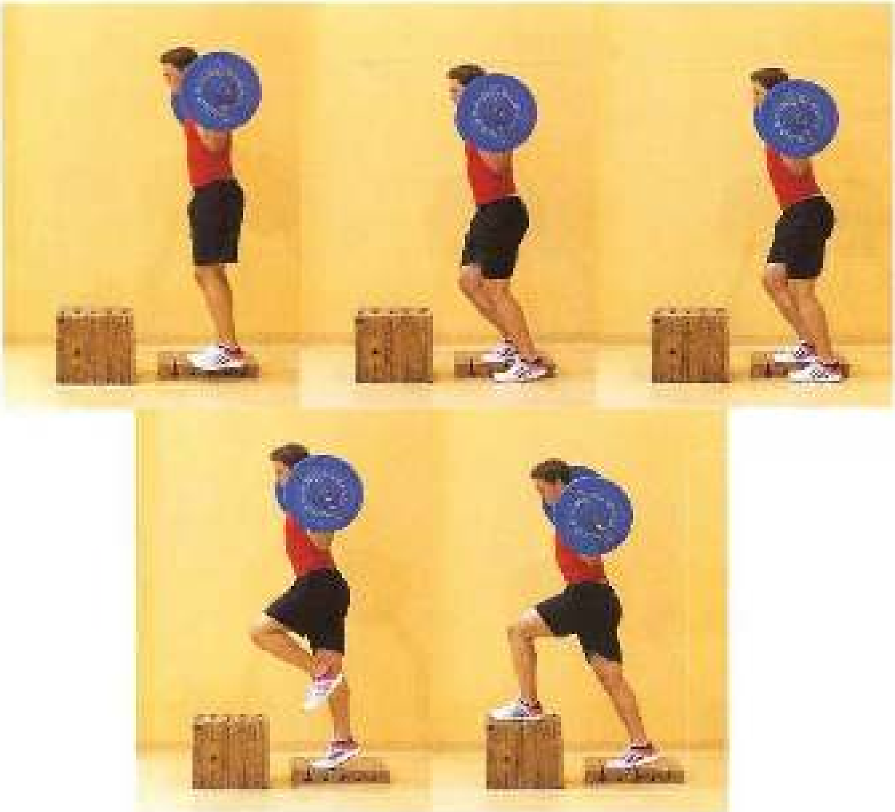


Figure 7.37 Abduction exercise at the end of stance-leg extension, with the swing leg explosively tucked in (see also Figure 2.45).

Standing on one leg on a 10- to 15-cm box, the stance leg flexes slightly (Figure 7.37). The ball of the free (swing) leg foot taps the ground, and the stance leg then extends. The swing leg is flexed at the knee and hip, and the free (swing) hip moves forcefully upwards. The swing-leg foot then lands on a box.

- Coordination between knee and hip angle. With the knee highly flexed, there are more alternative ways to combine various knee angles with various hip angles than in a more extended position (with the knee in front of or behind the foot and the pelvis rotated anteriorly or backwards). These additional degrees of freedom must be controlled. This is relevant in, for example, abrupt deceleration in soccer or a deep squat in speed skating. The optimal posture (with the knee angle linked to the hip angle) depends on the properties of the active muscles. The muscles involved here, especially the hamstrings and *rectus femoris* must act as close as possible to their optimal length. This optimal balance of the knee and hip angle must be achieved in the technique of lunges, double-leg and single-leg squats and step-offs against resistance (see also Distribution of pressure when decelerating; Section 3.2.6).

Exercises

A step-off followed by a step-up must be performed so that all the weight is not placed on the trailing foot, but so that the foot only touches the ground briefly and lightly and the step-up movement immediately follows (Figure 7.28). Lunges, squats or step-off movements must be performed with free barbell weights to provide sufficient freedom for optimal coordination of the knee and hip angle. A well-extended back is essential in all these techniques.

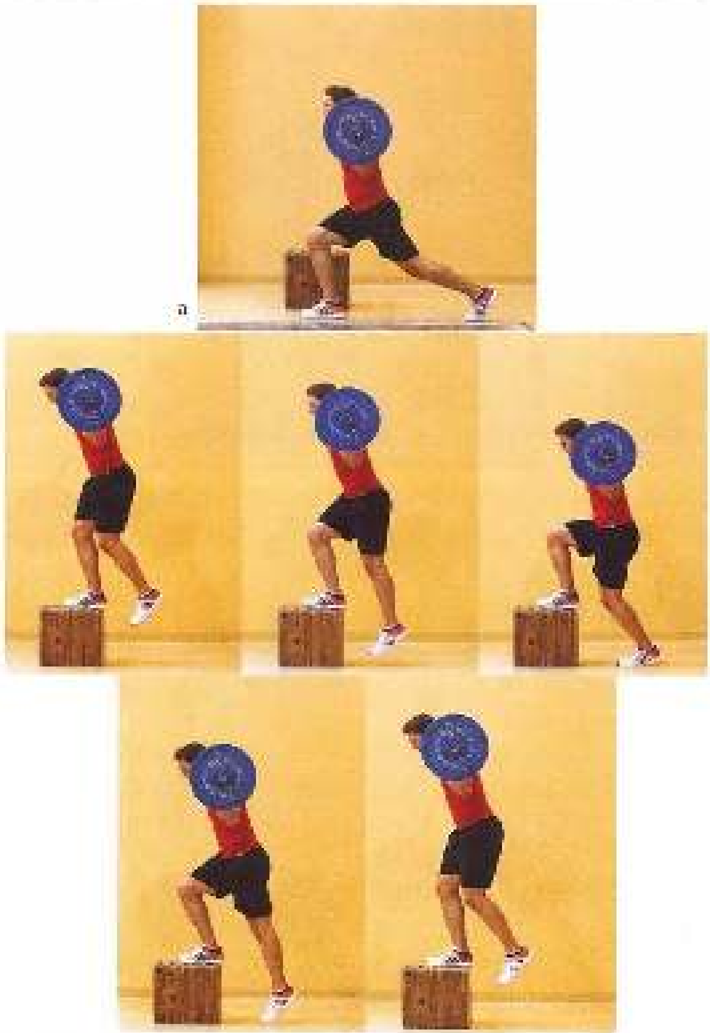


Figure 7.28: Lunge (a) and step-off (b).

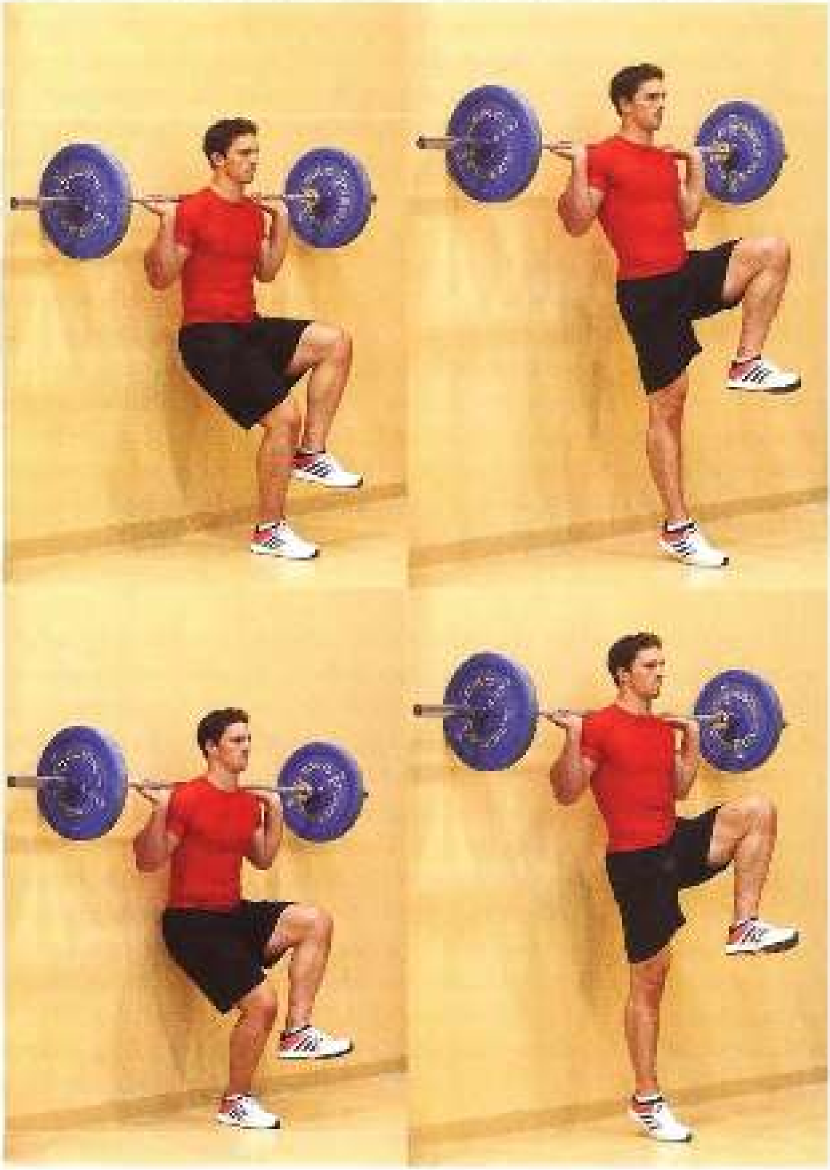


Figure 7.29 A single-leg hack squat against the wall, combining energy transport and the final posture.

Varying the distance between the stance leg and the wall requires different timing of the extension each time (Figure 7.29). Extension of the leg against resistance operates according to the proximal/distal principle. Timing of the order – hip extension, then knee extension and finally ankle extension – has a major impact on performance. This timing can be improved by a single-leg hack squat against the wall, with Olympic barbell plates rolling up and down as the leg is flexed and extended. In the initial posture the knee is flexed between 30° and 90°, and in the final posture all three joints are fully extended. Placing the foot closer to or further from the wall each time (compare top and bottom) requires the extension to be timed slightly differently. Of course it also makes sense to require a good final posture in executing the movement, with the free (swing) side of the pelvis moved upwards correctly. Especially if the barbell weight is not too high, or the knee flexes only a little in the initial posture, the exercise is suitable for this purpose.

Elastic muscle action

Bounces are often performed in series with a light barbell load to improve the loading and unloading of elastic energy. Here it is wrongly assumed that this will create overload in comparison with jumping without a barbell. The energy that can be stored elastically depends on the total weight and the drop height. If the weight is increased by carrying the barbell, the drop height will be reduced and thus there will be no overload. But bouncing with a barbell weight will increase the ground-contact time, and may have an adverse impact on optimal technique. Bouncing with a barbell load should perhaps therefore be omitted, or else executed with a barbell weight so light that technical execution remains within the requirements of elastic jumping (mainly reflected in brief contact time and the small change in knee angle) and there is still qualitative overload.

Upper limbs

The distinction between an open and closed chain is not so helpful when describing the contextuality of movement patterns in the shoulder girdle and arm. It makes more sense to divide movements into two-arm and single-arm movements. In two-arm movements the hands are stably linked via the surface, for instance in push-ups on the parallel bars (so a push-up on the rings is not a two-arm movement), or the resistance to be moved, such as the barbell in bench pressing. The division is meaningful because in two-arm exercises stability in the shoulder joint is largely created by the reduction in degrees of freedom resulting from the link between the two hands. In single-arm exercises this link is absent and there are more degrees of freedom to be controlled. The loading within strength training may therefore be different than in two-arm movements. Since the stability of the shoulder joint is guaranteed somewhat better in two-arm movements, large power-producing and energy-transporting muscles will reach their limits more easily within strength training than in single-arm exercises. These muscles include *pectoralis major* and *triceps brachii* in pushing movements and *latissimus dorsi* and *biceps brachii* in pulling movements.

Two-arm exercises

Two-arm explosive exercises such as bench pressing and bench pulls are suitable for training power with a heavy weight and for training maximal strength. Such exercises will always involve energy transport from the shoulder joint to the elbow joint. *Triceps brachii* (the long head), which ensures this, will do so isometrically. The key factor in two-arm exercises is therefore what happens in the power-producing muscles. In bench pressing, for instance, *pectoralis major* will have to act at a very great length when the barbell descends to the chest. It must be wondered whether such a length is contextual (i.e. also occurs in the sporting movement). If not, it is better not to let the barbell descend so far, and perhaps to work with a heavier barbell load. The same applies to other two-arm exercises.

In bench pressing with dumbbells there is no link between the two hands, and stability of the shoulder matters more. The exercise is often perceived to be harder than one with a fixed barbell. Besides the additional demands on stability, this is because the dumbbells can descend below chest level and *pectoralis major* has to act at an even greater length than in bench pressing with a fixed barbell. Here again it must be wondered how contextual such exercises are.

Single-arm exercises
In single-arm exercises there are three main organizational forms of intermuscular cooperation that are worth being trained within strength training:

- Preflex control of the muscles round the shoulder. Since the shoulder joint is a very unstable ball-and-socket joint, it is advisable for athletes who place heavy loads on their shoulders to pay particular attention to the use of cocontractions as controlling mechanisms.
- Power production in pulling and pushing movements.
- Elastic storage and return of energy in movement patterns that focus on internal and external rotation.

Single-arm preflex exercises
Single-arm exercises mainly put pressure on the muscles that run from the trunk to the shoulder girdle and the muscles that run from the shoulder girdle to the arm and help stabilize the shoulder girdle and the shoulder joint. That is why they are suitable for improving preflex control of the shoulder girdle. There are two ways they can do this:

1. Varying the posture in which the resistance must be fixated will constantly change the torque that the resistance produces on the shoulder girdle. If these postures are difficult enough, or have to be adopted fast enough, this can only be controlled by preflex control using cocontractions.
2. Using an unstable resistance (equipment that moves back and forth) will require constant cocontractions to control the unpredictable forces.

Both of these can of course be combined. The organization of preflex control in single-arm strength exercises requires a precise final posture.

Exercises

Snatch pulls are performed with one hand and a dumbbell (Figure 7.39). When catching the weight with an arm extended above the head, the weight must be caught suddenly, which can only be done if perturbations are absorbed by cocontractions. The posture in which this is done may be varied, for instance by making a lunge or placing the feet close together, so it is harder for the athlete to keep his balance and there is less scope for correcting errors by means of cocontractions.

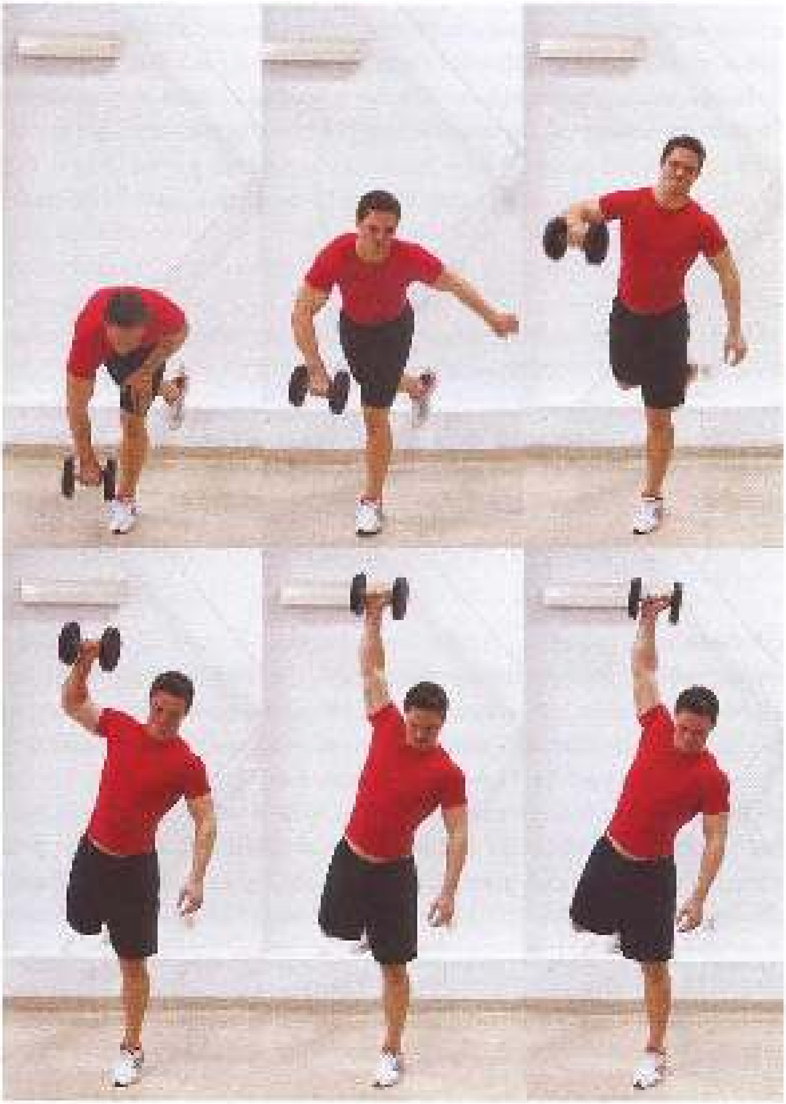


Figure [illegible] Preflex training exercise for pushing and pulling/thrusting movements (single-arm pull [here on one leg]).

The same single-arm movement can be performed in numerous different initial postures, for instance not upright but in a prone plank, supported by the surface and the dumbbell and ending in a side plank with the dumbbell held in the hand of the arm that is extended vertically upwards. The dumbbell must be kept still (for, say, two seconds) in the final posture, making the exercise result-oriented. The free hand can also be placed on an unstable surface, such as a physio ball.

Figure 7.31 Reflex exercise for throwing. After the swinging movement from in front to behind, the weight is abruptly fixated in the initial posture for the throwing action.

The throwing arm with the dumbbell swings backwards and upwards, and is abruptly fixated in the release posture (external rotation and abduction in the shoulder) (Figure 7.31). The exercise can be made harder by, for instance, standing on a tumbling mat. The abrupt fixation is brought about by cocontractions of muscles round the shoulder.

Single-arm power production exercises

Single-arm power production exercises (pushing and pulling movements) can be improved by self-organization of intermuscular cooperation round the shoulder. For optimal self-organization the exercises should satisfy two criteria:

1. They should be varied by changes in equipment and resistance (dumbbells, kettlebells and so on), surface and initial posture.
2. The final posture should be determined in advance.

Here again, influences such as reflexes may be able to play a part in how the movement is organized by precisely determining the final posture and varying the start of the exercise. In pushing/thrusting movements, the focus in the final posture is on stability in the shoulder joint. With rapid movements, and hence low resistance, the final posture in the shoulder joint will preferably be close to 90° abduction. This means it must be possible to execute the movement with rotation of the trunk round the *longitudinal axis*. In exercises in which this 90° angle is not feasible, such as vertical pushing movements, as much freedom of movement as possible should be allowed in the trunk to optimize the final posture.

Exercises

For the purposes of self-organization it may be useful to combine a pulling movement in one arm (Figure 7.32) with a pushing/thrusting one in the other. Self-organization of the movement should then take place in both shoulder girdles at once, and may produce an additional learning effect.

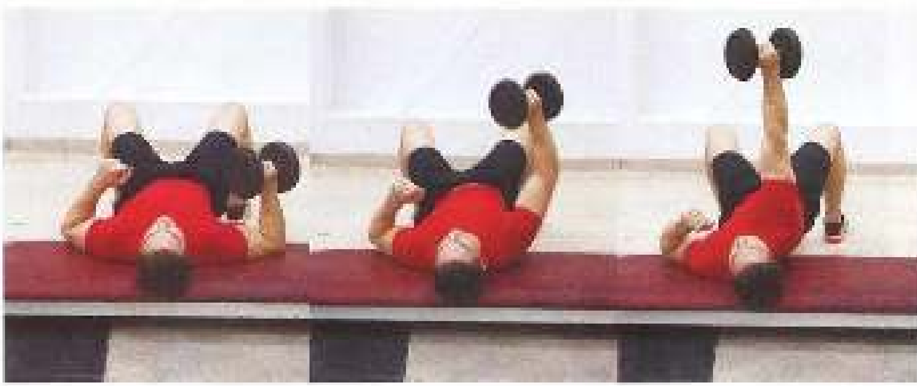


Figure 7.31: Single-arm push-out with rotation round the longitudinal axis so that the movement ends in 90° abduction. Performing the exercise on an unstable surface will allow coordination of the movements in the shoulder girdle to be practised variably.

Figure 7.32: Single-arm rowing movement.

In the initial posture the supporting arm is slightly flexed and the arm holding the dumbbell is extended (Figure 7.33). When the dumbbell is raised, the free shoulder also moves upwards. The movement ends with the shoulder of the supporting arm at about 90° abduction. When heavy loads are lifted, the various sub-movements will be coordinated.

Figure 7.34 Combined pushing and pulling exercise.

The pushing and pulling movement is performed with an elastic band and combined with torque round the longitudinal axis by jumping from one split stance into the opposite split stance during the pulling movement (Figure 7.34). When the right arm flexes and the elastic band stretches, the left arm extends powerfully and the upper body rotates. At the same time the left foot jumps forwards and the right foot backwards. When the elastic band relaxes and the right arm extends, the athlete jumps back to the initial posture. The pulling and jumping movements are rapid. Pulling a strong elastic band makes great demands on the arm muscles; with a weaker elastic band the higher speed makes great demands on the abdominals. The combination of pushing and pulling movements with the action of the abdominals is a key movement pattern for many contextual movements, such as running and combinations of punches in boxing.

Elastic storage and return of energy

Strength exercises involving elastic muscle action, e.g. in throwing training, can only be performed with one arm if a relatively light weight is used. The great demands made on stability of the shoulder joint in single-arm throwing rule out great variation in weights. In javelin throwing and baseball, for instance, opinions differ as to how much the weight of the javelins and balls being thrown should differ from the competition weight. Throwing with a correct weight overload is said to improve performance, but slightly too much weight overload is said to cause injury.

If the athlete still wants to throw with one arm and a heavier weight, stability of the shoulder joint should be more effectively guaranteed, for instance by throwing with a shorter backswing and the arm more extended. The shoulder girdle is then actively lifted, the amount of external rotation is limited and the external forces are easier to control.

When throwing with even greater resistance, the athlete can choose to use both arms, likewise extended. Two-arm throwing guarantees stability of the shoulder more effectively. Throwing with heavier weights and a shorter backswing is a good way to learn pretensioning in the shoulder girdle.

Besides throwing exercises, strength training for throwers also includes exercises that do not imitate elastic muscle action but do imitate the powerful movement from external to internal rotation. To avoid shoulder joint injuries, every effort should be made to perform such exercises safely. The focus should be on stopping the movement in the right final posture with the throwing shoulder turned forwards, so that the shoulder ends in more of an abduction position, with internal rotation in the shoulder and pronation in the forearm. This final posture ensures that the passive structures of the arm are well protected by muscle action.

Exercises

The value of movements that imitate the movement from internal to external rotation with high resistance for throwers is not clear. There is probably no quantitative overload, for the forces produced when actually throwing are extremely large. The exercise may help to imitate the extreme ranges of motion involved in throwing.

Figure 7.35 Single- and two-arm throwing with a heavy weight; pretensioning should be built up around the shoulder girdle. The heavier the projectile, the fewer degrees of freedom there should be and the more the throwing arm should be extended. If the resistance is too heavy for single-arm control, the throw will have to be executed with two extended arms.

Figure [illegible] Internal and external rotation in the shoulder joint. With a narrow grip and the upper arms more parallel, there will be more rotation in the shoulder joint. When movement is limited, as in the illustrations, the posture will have to be adjusted.

7.3.5 Attractors in overall patterns

The smaller intermuscular coordination systems are the building blocks for the overall contextual patterns that are executed against resistance. In overall patterns specificity in outward structure between the strength exercise and the sporting movement can be further increased, as can sensory specificity and specificity in intention. That is why it is useful to execute overall patterns against resistance, provided these are based on relevant intramuscular and intermuscular coordination patterns. In many sports, however, it is hard to find strength exercises in overall contextual patterns that are very specific for the sporting movement. Of course, this is especially true of open-skill sports, but it is also true of many closed-skill ones.

Near-total specificity in a strength exercise is impossible to achieve, even with low resistance. That is why maximizing specificity may not be such an effective strategy. Part of the effectiveness of a strength exercise will also have to come from overload and variation. For a strategy in which some specificity in the overall movement is linked to variation, the concept of attractors and fluctuators may be useful. In such an overall movement against resistance, the basic attractors in the movement must be guaranteed and variations on the sporting movement (or parts of it) performed against resistance. Constant variation in these exercises that are outwardly specific to the sporting movement is the key to effective transfer.

In open-skill sports this is self-evident, for an open skill always consists of parts that have to remain unchangeable in order to keep the movement controllable and parts that clearly fluctuate in response to changing conditions. In judo, for example, the basic throwing technique depends on the environment, and part of the technique must be adapted to changing external influences. The unpredictable forces that act on the judoka can be more or less imitated by also making the forces in the strength exercises unpredictable. For example, a throw can be performed with a dummy that is partly filled with water. During the movement the water will move uncontrollably inside the dummy, so that unexpected forces have to be controlled in the intended movement pattern. If properly used, perturbations in the movement pattern due to unpredictable loads can create a more effective movement pattern and hence a more effective distribution of attractors and fluctuators.

Such a strategy is also useful in closed skills. Deliberate perturbations will make some parts of the movement pattern more flexible, and others more stable. Causing perturbations can thus deepen attractors even here. In this way components that are essential to the closed-skill pattern can be trained by means of variation. The system of variability based on changing the task, the environment and the organism (see Section 6.4) is suitable for changing overall movement patterns within strength training.

Effective variation of components of the sporting movement within strength training requires not only sufficient creativity but also thorough knowledge of the sporting movement. In the settings in which specialized strength coaches work, input from the people who provide technical training for the sporting movement is therefore essential.

7.3.6 *From basic exercise to overall pattern: an example*

The extension of the classic clean to a complex strength exercise that is specific to running is presented as an example of development of movement-specific overall patterns.

This starts with the classic clean, in which intermuscular systems such as cooperation between muscles (e.g. back muscles and hamstrings) must be optimally executed. The double-leg exercise can be changed so that it becomes more and more specific for running and jumping. The first change is to start the exercise from a more extended rather than squatting posture. The second change is to make it a single-leg rather than double-leg exercise. Components of the single-leg clean are then changed to make the exercise even more specific for running and jumping. If specificity can no longer be usefully increased, perturbations of the movement pattern can be introduced, for instance by using unstable resistance.

Figure 7.37 Classic technique clean from the ground

Classic clean

The classic clean starts with the barbell bar 30–50 centimetres above the ground (Figure 7.37). The back is extended, the hands grip the barbell at shoulder width, the arms are extended, the knees are close to the barbell and the seat is as low as possible. The barbell is lifted off the ground, with the arms and back extended as far as possible. The knee and the hip extend at the same time. The barbell moves upwards just in front of the body, and the movement accelerates. When the barbell reaches the hips, it is lifted further by the arms, and the elbows are raised up high to the sides. At this point the athlete starts catching the barbell. The elbows now point forwards and the movement ends in an upright posture, with the barbell in front of the chest.

Figure 7.33 Double-leg hang clean with plate stacking.

Hang clean with pretensioning

Initial posture: upright, with the barbell in front of the body (Figure 7.38). The hands are at shoulder width. The upper body then bends forwards to about 45°, with the spinal column extended and the knees extended or slightly flexed. From this posture the body moves back very slowly to the more or less extended posture. As the athlete straightens up, the lever of the barbell load is reduced in relation to the hip joint and the joints of the spine. The muscles then have to produce less and less force, but the athlete tries to keep body tension as high as possible. Tensioning is followed by the clean. Owing to the almost upright posture, the clean can only be performed after pretensioning the muscles. There is very little time to transport energy from the hip to the ankle, and this provides good training for energy transport during the brief ground contact when running, especially energy transport from the knee to the ankle.

The difficulty of the exercise mainly depends on how erect the body is before the clean. In this version of a hang clean, well-trained athletes can manage a maximal barbell weight of about 85% of their maximum in the classic performance of the clean.

The exercise is more specific for running and jumping from a run-up than a classic clean, for the rate of force development is under much greater pressure and the knee and hip angles at the start of the movement are close to those in the sporting movement.

Single-leg clean with pretensioning

The initial posture in the single-leg exercise is the same as in the double-leg pretension hang clean (Figure 7.39). The free (swing) leg is placed next to the stance-leg foot with minimal pressure on the surface, and only helps the athlete to keep his balance. The pretensioning action is the same as in the double-leg pretension clean. During the acceleration in the clean the free (swing) leg leaves the ground, and the foot lands on a box in front of the stance leg. It is harder to maintain pretensioning on one leg than on two. The maximal barbell weight is usually just over half of the maximum in the classic clean.

Because this is a single-leg exercise, erector spinae is no longer the limiting factor. This will increase pressure on the energy-transporting action of the stance leg and hence create greater overload. Greater demands will be made on the movement of the swing leg, which will have to start as early as possible; the hip and knee will have to keep flexing as long as the stance leg extends. This will provide reflex support for the overall movement pattern (the crossed extensor reflex). Both single-leg loading and the presence of the crossed extensor reflex are more specific than double-leg versions of the clean.

Balance-in-clean

The balance movement adds the stumble reflex and elastic loading of the hamstrings to the relevant components were already present in the single-leg clean. The overall movement is complex and includes many components that are specific for running and single-leg jumping from a run-up. Execution of the movement can be controlled and corrected in a number of ways. The maximal barbell weight will be somewhat lower than in exercises without the balance movement. During the exercise the body's centre of gravity will shift forwards and the movement will end in a lunge with the front foot landing on a box (Figure 7.40).

Figure 7.[illegible] Single-leg clean with pretensioning. The movement is based on the same basic pattern as the step-up exercise in Figure 7.45.

Figure 7.[illegible] Balance-to-clean. The movement is based on the same basic pattern as the exercise in Figure 7.[illegible].

The anterior shift is only possible if knee extension is postponed slightly longer during the clean. This means the hamstrings will play a somewhat more dominant part in the overall pattern than in a more vertical clean. They play a similar part in the running movement, postponing knee extension and directing the push-off force further backwards. It is important to include these versions of the clean in the training. This will improve use of the hamstrings in the stance phase when running.

If a heavier barbell load is to be used, the clean can be omitted by carrying the barbell on the shoulders. After the bending movement, the legs will follow a pattern identical to the one in the version including a clean.

Balance-to-clean movement with perturbation

The clean and catch can be replaced by arm and upper body movements that may perturb the leg pattern. In the balance movement a barbell plate, a sandbag or a half-filled water bag is moved to the stance-leg side, creating additional torque in the trunk. In the anterior swing movement, the load is vigorously thrust towards the swing-leg side as the crossed anterior movement is executed, and then pulled back towards the chest (Figure 7.41). The arm movement is thus a distractor that perturbs the scissor movement of the legs. However, the leg pattern must remain unchanged, and compensation must be sought elsewhere in the body, for instance in external and internal rotation of the stance-leg hip. The exercise also makes

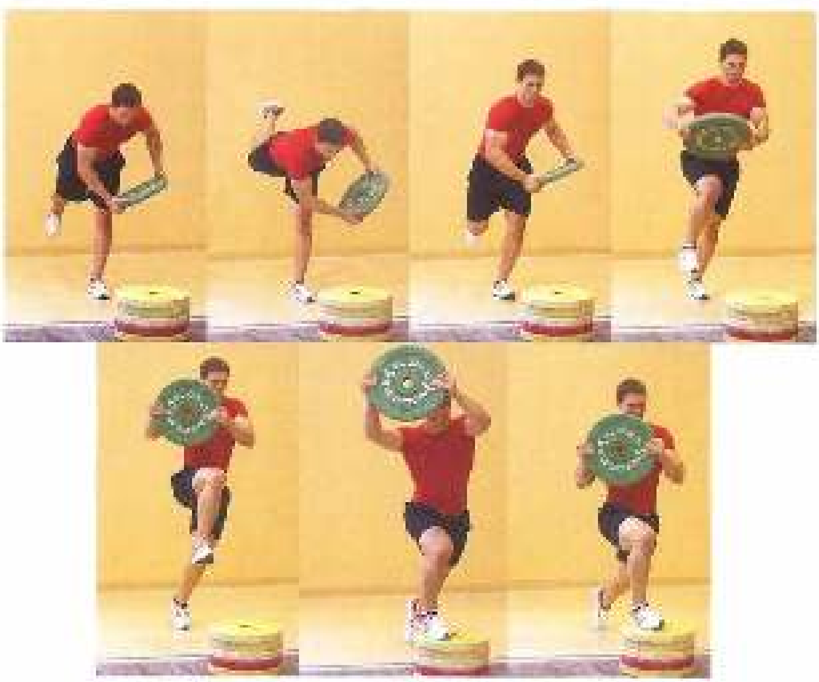


Figure 7.41 Version of the balance movement with a barbell plate; the balance movement may be followed by a step-up, possibly with an unstable surface and an unstable load.

great demands on stability in the trunk (see also 'Extending the back while rotating' in Section 3.2.6).

Balance-to-step-up

The balance-to-clean exercise is performed with a half-filled water bag that is held in front of the chest in the initial posture and above the head with extended arms in the final posture. In the final posture the unpredictable movement of the water makes great demands on the cocontractions in the trunk.

The exercise can be made more difficult by landing on a somewhat unstable surface rather than a stable box, or by making a step-up movement after landing on the box. Such a final posture will make great demands on the fluctuating components of the movement, and it is important to ensure that the leg movements are performed correctly throughout.

Figure 7.[illegible] Some sporting movements involve pushing off with the arms in extended position (preparation for [illegible] in basketball, a block in volleyball and so on). In the strength exercise the clean is performed while jumping and the [illegible] ends with the feet on a bouncing [illegible].

Clean and jerk

In sport-specific training for running and jumping, the emphasis in the clean is on the last part of the extension, which largely determines the quality of the push-off. Following the clean with a barbell jerk will put pressure on the last part of the extension (Figure 7.43). Landing on one leg with a flexed swing leg not only imitates a single-leg push-off, but also includes the result of information about balance provided by correct execution of the exercise.

Working with overall movements blurs the boundary between strength training and technique training. Whether this is a technique exercise or a strength exercise is then no longer relevant. The aforementioned debate about the long toss in baseball and running with a sled, usually focusing on whether or not the exercises count as specific strength training, should therefore be replaced by the question of which links between specificity and overload are the most efficient in training. Translated to the central/peripheral model, this means asking where in the continuum from high specificity to large overload the athlete should work. This may involve factors that are not directly related to the intended transfer. One option is then to work solely at either end of the central/peripheral model, because exercises in the middle of the model may, for example, create too many organizational problems, or may prove too complex and hence too great a mental burden for the athlete. A strategy mainly focusing on technique in training of the sporting movement will therefore be more successful. With athletes who are mentally equipped for this, it may be a good idea to focus on technique in all types of training, for this may make the difference between winning and losing. Being able to provide the individual athlete with tailor-made training is a coach's most important skill, besides expert knowledge and being able to adapt to the demands of the moment.

7.4 Summary

Sport-specific strength training should not only provide transfer but also, where possible, avoid loading that does nothing to improve performance. In an approach based on the body part method, which mainly focuses on increasing strength values within strength training, such additional loading is unavoidable because of the limited variation in exercises. In the contextual approach, which focuses on transfer of strength values to the sporting movement, the exercises are much more varied and additional loading can be avoided; overload can then be qualitative as well as quantitative.

The standard terminology used to describe strength training fails to meet these criteria, for it is based on a methodical classification rather than one based on adaptations. This results in such terms as 'strength endurance' and 'explosive power', which are not one-dimensional – and hence are unworkable – in terms of transfer.

The value of the various types of strength training can be analysed in the light of the adaptations that occur:

- Hypertrophy training: this focuses on increasing the cross-section of the muscle by making it do as much mechanical work as possible. The disadvantage is a substantial decline in coordination, so hypertrophy training is not advisable in sports in which coordination is crucial.
- Maximal strength training: improved recruitment is useful in sports involving isometric muscle action and elastic muscle action. Biarticular muscles usually work isometrically, especially in sports in which abrupt high impact has to be absorbed and converted into movement.
- Power training: the link between force and muscle-shortening speed is highly movement-specific. Rate of force development and deceleration of the movement are often overlooked.
- Reflex training: working with light weights in contextual reflex-supported movement patterns loads and develops the neuromuscular system.

Since large peaks of force occur in both maximal strength and reflex strength training, it is useful to know the benefits and drawbacks of both. Decisions can then be made about when to include which type of training in the training plan.

Strength training should mainly focus on the stable components of the movement in order to be transferable to high-intensity movement. Intermuscular attractors in rapid movements can largely be determined from muscle architecture. The natural suitability of the muscle for certain types of muscle action can be used to identify larger sets of fixed intermuscular cooperation, which can in turn become building blocks for overall contextual patterns. These are important in keeping movements controllable. This approach largely blurs the boundary between technique and strength training.

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Figure 1.22	Photographs: with thanks to the Flic Flak gymnastics club and Christian Selk
Figure 3.1	Photograph: Rob Jelsma
Figure 3.7	Photographs: Rob Jelsma
Figure 4.21	Photographs: with thanks to Christian Selk
Figure 7.10	Photograph: Loek Ehlen

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Training theory has traditionally distinguished between strength, speed, agility, stamina and coordination – basic motor properties that have been seen as more or less separate factors. Frans Bosch's book *Strength Training and Coordination: An Integrative Approach* claims that this distinction is questionable in both theory and practice. In particular, transfer of training cannot be understood if the five properties are treated as separate factors. This is a highly original and scientifically substantiated viewpoint that has never before been presented in a textbook.

This book does not approach strength training in terms of its mechanical manifestations; instead, the author presents a model based on what is known about the underlying – especially neurological – processes. Specificity between strength exercises and athletic movement is thus identified, and the term 'overload' acquires a new meaning. Dynamic systems theory is used to create a link between motor learning and strength training. In addition, theory is constantly translated into guidelines for practice. In this book, sport-specific strength training means coordination training under increased resistance.

Strength Training and Coordination: An Integrative Approach is the translation of the fully revised second edition of the Dutch-language book *Krachttraining en coördinatie, een integratieve benadering*, which originally appeared in 2012. It will be of interest not only to students of sport science and physiotherapy, but also to sport and other physiotherapists, and to movement specialists and other coaches who want to do more than just strengthen the musculoskeletal system. This book is about contextual strength training.

FRANS BOSCH is a lecturer in motor learning and training theory at the Fontys sports college in the Netherlands. He also works as a consultant and specialist coach for the Welsh and Japanese national rugby teams and West Ham United Football Club. Together with **RONALD KLOMP** he has previously written the book *Running: Biomechanics and Exercise Physiology in Practice*, which was published by Churchill Livingstone in 2004.

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